Capability Matters

A CASEBOOK

Learning Engineering for Next-Generation Systems

Capability

Matters

EDITED BY

William Gray-Roncal, PhD
James Diamond, PhD

JOHNS HOPKINS UNIVERSITY

LEARNING DESIGN AND TECHNOLOGY

FIRST EDITION · REVISED MAY 2026

An ongoing artifact of the LENS Practice Flywheel — Identify, Activate, Prototype, Analyze, Transition.

DEDICATION

To everyone who believes in a dream and refuses to give up.

COLOPHON

Learning Engineering for Next-Generation Systems: Capability Matters
A Casebook.
First edition, 2026.
Edited by William Gray-Roncal, PhD and James Diamond, PhD.
Compiled for the LENS specialization within the Learning Design and Technology program at the Johns Hopkins University School of Education.
Set in Instrument Serif and DM Sans.
Printed via Lulu at 8 × 10 in, perfect-bound.

HOW THIS VOLUME WAS MADE

This casebook was assembled using AI tools as part of an iterative learning-engineering process — the same Practice Flywheel (Identify → Activate → Prototype → Analyze → Transition) that the volume's case studies argue for. The methodology treats AI as the dual entity the curriculum names it: a creative partner that accelerated drafting, cross-referencing, layout, and citation lookup, and an epistemic risk that had to be hand-checked against the record. Every case in the book was reviewed by the editors and hand-checked by students for accuracy: confirming that the sources cited exist and are findable, that the quoted attributions are fairly represented, and that the case studies are accurate accounts of the incidents and the investigations they draw on. Items where the source could not be confirmed are marked "Paraphrasing…" so the attribution is honest about what is the author's reconstruction and what is verbatim from the record.

The volume is not finished. It is the first iteration of a reference dataset that LENS will continue to develop. We will revise the book based on reader feedback, reviewer findings, and our own ongoing practice of learning engineering — adding new cases as the operational record grows, correcting our record where it can be improved, and updating the curriculum crosswalk as the program itself iterates. Corrections, new cases, and reader feedback are welcomed.

SOURCES, ATTRIBUTION & LEGAL NOTICES

All cases reference publicly reported incidents and published investigations. Quoted material is attributed to source investigations, court documents, and academic publications. Reflection questions are original to this volume. The audit record of citations checked and changes made is maintained alongside the source repository.

References to convictions, settlements, and findings reflect the public record at the time of writing. Where individuals are named, they are named only as participants in investigations, court proceedings, or public inquiries already in the public domain. The casebook makes no legal characterization beyond what those records establish.

Corrections, requests for clarification, and notice of any inaccuracy can be sent to LDT / LENS at the Johns Hopkins University School of Education. Substantive corrections will be incorporated in subsequent printings and noted in the casebook's public errata.

WELCOME

Learning engineering is a toolbox, not a single tool.

At its center it is a dialogue between two traditions — the engineering disciplines that design, build, and sustain complex systems, and the learning sciences and education that study how people come to know and do. Around that dialogue gather many other fields: cognitive and human-factors engineering, measurement, implementation science, design, data and computation, and the operational domains themselves. The work is plural by design — many disciplines at once — and it draws on domain knowledge, coding, theory, systems thinking, design, teaching, and analysis.

It welcomes everyone who brings a real tool to it. It is not, for that, a free-for-all: learning engineering stands in an intellectual tradition with its own methods, evidence standards, and lineage, and it holds the work accountable to them. No one carries every tool. This is a team sport — capability gets built at the seams between people who each see part of the system clearly and trust the others to see the rest.

So you don't need to know everything before you start. You need to know what you bring, stay honest about what you don't, and find the people who carry the tools you're missing. What you bring is yours to decide. The hundred cases that follow are an invitation to find your part in the work.

— THE EDITORS

INTRODUCTION

Capability is a system parameter.

Every complex system depends on what people can do inside it. That dependency is measurable, designable, and too important to leave to chance.

> **01.** Systems don't fail because the technology breaks. They fail because someone couldn't do what the system required.
>
> **02.** We engineer every parameter of a system except the people.
>
> **03.** Capability is not a soft problem. It is a systems engineering problem.
>
> **04.** Decision-grade evidence. Not opinions about training.
>
> **05.** Capability without agency is automation. The goal is operators who can perform, adapt, and lead — not comply.
>
> **06.** Sometimes the constraints make the goal unreachable. Saying so early is a result, not a failure.

The cases in this book are not, individually, hard to understand. A pilot recovered from a stall the wrong way. A nurse could not stop a doctor who skipped a step. A safety operator was watching television on her phone when a pedestrian crossed the road. A captain shot down a civilian airliner because the radar data and the operational framing pointed in opposite directions, and the framing won. The proximate cause is usually obvious. The first investigator on the scene almost always finds it within hours.

What is hard to understand — and what this book exists to make legible — is the system that produced the gap into which each incident fell. In every case the proximate actor was operating inside an architecture of training, procedure, authority, measurement, and incentive that had, for years, been quietly degrading. The pilot had never trained to recover a stall at cruise altitude. The nurse had no procedural path to intervene. The safety operator had been hired into a role nobody had figured out how to make performable. The radar operator had been trained to a tempo and a presumption that the system above him no longer shared with the system around him.

Each of these architectures was designed. Each was funded. Each was reviewed. Each carried, written somewhere in its specifications, the assumption that the people inside it would be able to do what the system required of them when the system required it. That assumption is what this book calls the **capability parameter**. It is a property of the entire sociotechnical system, not of any individual operator. Capability lives at the **interface** between what a system requires of its operators and the impact the system has to deliver — and when that interface is wrong, the gap may originate in human development, in system design, in organizational performance, or in the interaction among them. Distinguishing which is itself a measurement and engineering problem; the casebook calls it **gap attribution**, and treats it as the diagnostic move that the rest of the discipline depends on. When the interface fails — wherever the source — the consequences are paid in lives, in dollars, in trust, and in time the institution will never recover.

The premise of the Johns Hopkins School of Education's Learning Design & Technology program — and of the Learning Engineering for Next-Generation Systems specialization that has grown out of it — is that the capability parameter is engineerable. Not by accident. Not by declaration. By the same kind of evidence-grounded, methods-grounded, cross-domain discipline that built modern reliability engineering, modern epidemiology, and modern systems safety. The discipline does not yet exist at the scale the evidence demands. LDT and LENS are organized to help build it.

I · THE COST OF THE GAP

The Institute of Medicine's **To Err Is Human** (1999) estimated that preventable medical error caused between forty-four thousand and ninety-eight thousand deaths a year in the United States — already, at the lower bound, a top-ten cause of death. The report catalyzed a generation of patient-safety work: surgical checklists, central-line bundles, team training, computerized order entry. Some of those interventions are documented in Part II of this book as successes. And yet in 2016 Makary and Daniel returned to the question from the same institution that produces this casebook and concluded that medical error remains the third leading cause of death in the United States, killing more than two hundred fifty thousand people a year [1].

The interval between the IOM report and the Makary update is approximately the canonical gap implementation science has identified between research evidence and clinical practice: seventeen years [2]. The average. The median. Only about fourteen percent of research findings ever reach practice at all [3]. The system to surface them, vet them, deploy them, sustain them, and measure their effect at the scale at which they would matter has never been built as an institution. It has been built as a series of grants. The difference is the point.

This is not only a healthcare problem. The U.S. Navy lost seventeen sailors in two avoidable destroyer collisions in 2017 because in 2003 it had cut a sixteen-week classroom-and-simulator course down to a set of CD-ROMs [4]. The world's most expensive aircraft program is operating at half of its design readiness in 2026 because the platform was fielded faster than the maintainer pipeline could be built [5]. One hundred million dollars of educational infrastructure dissolved in fourteen months because the institution building it did not engineer the governance and trust that the technology presupposed [6]. A predictive grading algorithm downgraded a third of an entire nation's high-school students in a single afternoon, and was withdrawn within a week, because nobody inside the agency could be specifically held to account for what the measurement instrument was measuring [7].

None of these failures is a failure of effort. None is a failure of intelligence. Each is a failure of the capability infrastructure to match the system it was operating. In each case, the gap was diagnosable in advance. In most cases, it was actually diagnosed in advance, by someone — and the diagnosis did not produce the remediation. The cost of the unrepaired diagnosis is what **capability** matters.

II · WHAT AN ENGINEERABLE DISCIPLINE LOOKS LIKE

Crew Resource Management did not exist as a discipline in 1976. In March 1977 two 747s collided on a foggy runway at Tenerife and 583 people died. Within five years United Airlines had operationalized the first CRM curriculum [8] within twenty years the Commercial Aviation Safety Team had built the closed-loop hazard-identification system that lets the FAA know whether CRM is still working [9]. The result over the next two decades was an eighty-three percent reduction in U.S. commercial aviation fatality risk and a ninety-five percent reduction in fatalities per hundred million passengers [10]. CRM works not because it taught individual airmanship — pilots were already excellent — but because it redesigned the **system of interaction** in the cockpit: the authority gradient, the communication protocol, the standard brief and debrief. The discipline treated team coordination as an engineerable property of the system.

The same pattern shows up in every domain where the cost of failure has been high enough to fund the discipline. After Three Mile Island the U.S. nuclear industry stood up the Institute of Nuclear Power Operations within nine months — before the Kemeny Commission's report was even finalized — because every utility had recognized that an accident at any single plant would affect every operator's license to operate [11]. INPO and the National Academy for Nuclear Training, founded in 1985, have presided over more than four decades of zero significant releases at U.S. commercial reactors. The comparison with the surface Navy, which cut its training during the same era, is the cleanest available test of what happens when capability is engineered versus when it is deferred.

Healthcare has its own version of this arc, but only partially. Peter Pronovost's central-line checklist, paired with the nurses' authority to enforce it, drove bloodstream infections to near zero across 103 Michigan ICUs and saved an estimated fifteen hundred lives in eighteen months [12]. Atul Gawande's nineteen-item surgical safety checklist, paired with three required pauses in the operative timeline, halved the surgical mortality rate across eight hospitals in eight countries [13]. TeamSTEPPS, jointly developed by AHRQ and DoD, translated five decades of high-reliability research from aviation and the military into clinical practice in years rather than decades — because the implementation infrastructure was funded as part of the intervention rather than as an afterthought [14].

These are not stories about individual hospitals or individual ships. They are stories about the difference between a domain that has organized itself to engineer capability and one that has not. The pattern is consistent across all of them: a catastrophe makes the capability gap visible; an institution is built that treats the gap as the institution's responsibility; the institution funds the measurement architecture that lets it know whether the gap is closing; and twenty years later the death rate is half what it was. The intervention is always paired. A technical artifact — a checklist, a brief, a cord — combined with a cultural artifact — protected authority to use it, a no-blame protocol, a credible governance body. Neither alone moves the curve. Both together move it dramatically.

III · THE DISCIPLINE WE HAVE, THE DISCIPLINE WE NEED

The intellectual material to build this discipline already exists. The learning sciences have produced four decades of converging evidence on how skill, knowledge, and judgment develop and decay [15]. Cognitive engineering has produced rigorous methods for analyzing human-machine systems [16]. Human factors and ergonomics have produced an evidence base for interface and workflow design [17]. Implementation science has produced frameworks for taking effective interventions to scale [18].

Systems safety engineering has produced both diagnostic tools — Reason's swiss-cheese model, Rasmussen's migration-to-the-boundary, Leveson's STAMP — and control-theoretic methods for design [19]. High-reliability- organization theory has documented what the institutions that have solved the capability problem actually do [20]. The discipline is not missing its evidence base. It is missing its integration.

Learning engineering is the integration. The term, in its modern usage, traces to Bror Saxberg and was elaborated by Goodell, Kolodner, and the IEEE Industry Connections Industry Consortium on Learning Engineering (ICICLE) [21]. The IEEE Learning Engineering Body of Knowledge (LEBoK) and its associated process model are the most developed available specification of what the discipline does [22]. The premise is that the work of building, deploying, and sustaining capability at scale is an engineering activity: it requires explicit requirements, evidence-based design, instrumented measurement, and feedback-driven iteration. It is not exclusively a research activity. It is not exclusively a managerial activity. It is engineering, and it can be taught and practiced as engineering.

LENS — Learning Engineering for Next-Generation Systems — is a specialization within the Johns Hopkins School of Education's Learning Design & Technology program that operationalizes this premise for high-consequence operational domains. Its core competencies — capability analysis and requirements, evidence and analytics, human-AI teaming, knowledge transfer, bias and governance, computational and AI methods — are organized to produce graduates who can do the work the cases in this book required, and who can build the institutions that the success cases in Part II had to invent. The curriculum threading commitments — implementation science as a through-line, equity as a design commitment rather than an audit, decision-grade evidence as a deliverable rather than a report, communication and system-of-systems integration as engineerable properties of the system, the disciplined use of learning- technology aids (from XR to adaptive platforms) as instruments whose evaluation is part of the work, and **AI as both a creative partner and an epistemic risk** — leveraging AI's generative reach while guarding against offloading the thinking — are drawn directly from the patterns the cases reveal.

Three of those commitments deserve a moment of their own — they are the substrate the rest of the curriculum runs on. The first is **communication, translation, and integration within and across disparate systems**. A striking number of cases in this book are, at their proximate cause, translation or integration failures across boundaries — failures of language, convention, unit, role, agency, or discipline to carry meaning reliably from one part of a system to another, or failures of subsystems built to different assumptions to compose into a working whole. Mars Climate Orbiter was a metric-versus- imperial translation failure (Case 54). Tenerife was a translation failure at the takeoff-clearance boundary between two cockpits and a tower under fog and time pressure (Case 12). The Patriot at Dhahran failed because the manufacturer's assumption about run time and the operator's actual run time were on opposite sides of a documentation boundary that had not been engineered to be crossed (Case 19). Eagle Claw failed because the rotary-wing, fixed-wing, ground, and command components were assembled across service boundaries that had no shared operating practice (Case 46). 9/11 was a cross-agency translation failure measured in thousands of dead (Case 86). Boeing 737 MAX was, at one level, an integration failure: a single-string sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together (Case 2). The successful cases — AlphaFold (Case 98), MICrONS, CRM (Case 12 paired with Case 23), Keystone (Case 14), the paired-intervention examples in Chapter 8 — are, equivalently, cases of disciplined translation and integration: biology into computation, science into operational practice, technical reform into cultural reform, multiple subsystems into a working whole. Engineering capability across these boundaries — language, discipline, agency, subsystem-to-subsystem, system-of-systems — is itself a designable target. The LENS curriculum treats it as such.

The second is **technology aids and learning platforms**. The discipline has at its disposal a fast-evolving toolkit: extended- reality (XR) simulation environments — VR, AR, and mixed-reality systems used for procedural training, spatial-reasoning practice, and high-fidelity rehearsal under conditions that would be unsafe or impossible in the field — together with learning- management systems, adaptive learning platforms (the lineage from Cognitive Tutor and ASSISTments to current ITS work; Cases 42, 38), intelligent tutoring frameworks such as GIFT, learning- analytics infrastructures such as xAPI and the Total Learning Architecture (Case 40), game-based learning environments, LLM-augmented tutors and authoring tools, and high-fidelity simulators in aviation, surgery, defense, and process industries. These tools are not the discipline. They are the discipline's instruments — and the cases in this book show, repeatedly, that powerful instruments do not by themselves engineer capability (Cases 8, 21, 25, 38, 96, 99). The LENS curriculum teaches the practitioner to choose, configure, evaluate, and govern these tools with the same evidentiary discipline applied to any other intervention, and to recognize when the binding constraint is the surrounding architecture rather than the tool itself.

The third — and the design constraint on all the others — is **human agency**. The capability discipline can be misread as optimization: produce operators whose behavior matches a specification. That reading collapses the very property the cases reveal as load-bearing. Every successful institution in Part II — INPO operators trained to challenge their reactor team, Keystone ICU nurses authorized to halt a procedure (Case 14), the Nuclear Navy's questioning attitude (Case 28), the first-officer authority CRM built into the cockpit (Case 12) — engineered for capability **and** for the operator's capacity to exercise independent judgment, to adapt to conditions the system was not designed for, and to shape the surrounding architecture rather than only execute inside it. Capability without that capacity is automation. LENS treats the preservation and development of agency — the room the system leaves its operators to think, decide, and lead — as a design constraint on every intervention, including those in which AI is the partner. The paired risk runs the same direction: an AI collaborator can accelerate the flywheel or quietly offload the thinking, and which one it does is a design and governance decision, not a property of the tool.

IV · LENS AS AN ITERATIVE LEARNING-ENGINEERING PROCESS

Learning engineering is not a one-time act of design. Its defining feature, as articulated in the IEEE Industry Connections Industry Consortium on Learning Engineering's process model, is iteration: understand the operational context, specify capability requirements, prototype an intervention, instrument it, measure, learn, redesign. The work cycles. The cases in this book that produced sustained outcomes — CRM and CAST, INPO, Keystone ICU, the WHO Surgical Safety Checklist, TeamSTEPPS — all share that loop structure. They were not built once and left alone. They were built, measured, rebuilt, remeasured, and rebuilt again. The institutional capability to sustain the loop **is** what distinguishes them from interventions whose effects decayed.

The loop also has a legitimate negative result, and the discipline has to be honest enough to reach it. Capability engineering always operates inside constraints — budget, schedule, regulation, institutional will, politics, ethics, public trust — and many of the binding ones are not technical. Part of the work is **managing those constraints and the risk they carry**: reading the constraint space accurately enough to recognize when external, non-technical factors have narrowed the solution space to the point where a particular capability goal is no longer realistic to pursue. Reaching that conclusion — and documenting **why** — is itself a valid outcome of a project, not a failure of one. It redirects effort toward goals that are actually reachable, surfaces the non-technical barriers for the stakeholders who can change them, and prevents the far more expensive failure of driving an infeasible target all the way to the operational record before it collapses there. A number of cases in this book are, in part, stories of institutions that could not say "not like this, not yet" in time.

LENS is organized as that loop made teachable. Required coursework carries students through the cycle: framing the system whose capability is at stake, eliciting requirements from operational analysis, building the evidence architecture that lets an institution know whether its interventions worked, examining the human and AI roles inside the designed system, and running the loop end-to-end on a real operational problem in an integrated capstone studio. Electives extend the loop into governance and risk, knowledge transfer and organizational learning, and computational methods. The program graduates a practitioner who has been around the loop enough times to know where it breaks. Specific course titles and sequencing evolve with the program — current titles and detailed descriptions live in the LDT/LENS catalogue; this casebook tags each case to the courses that most directly use it as a worked example so the mapping remains legible as the curriculum iterates.

The casebook is itself an iteration artifact. Each case is read, diagnosed, paired with a learning-engineering response, and tested in studio against students' own operational domains. The cases that return as success stories in Part II are the cases whose institutions completed the loop. The cases in Part I are the cases whose institutions did not — or did not loop fast enough. The pedagogical commitment of the program is that the practitioner who graduates can identify, instrument, and close the next such loop before the next case is written.

V · WHY LENS · THE FIVE PILLARS

A specialization that promises to teach a discipline still being built must show that its credibility rests on more than its ambitions. LENS stands on five pillars: **Mission Literacy**, **JHU Ecosystem**, **Intersectional Expertise**, **Capability Focus**, and **Flywheel Iteration**. Each is a real institutional asset, with a record. None of them, alone, would carry the argument; the pillars are load-bearing together.

WHY LENS — FIVE PILLARS

MISSION LITERACY

Real operational problems, real sponsors, real consequences — and the disciplined ability to read them. LENS is anchored in the conviction that the practitioner must be able to enter an operational setting, listen to the people doing the work, understand what the mission actually requires, and translate that understanding into design and measurement decisions that a serious reviewer would accept. Mission literacy is what makes the cases in this book teachable as cases and not as anecdotes.

JHU ECOSYSTEM

An institutional environment broad and deep enough to host the discipline. At Johns Hopkins this is the School of Education with its centers, doctoral programs, and faculty in learning sciences, educational measurement, and design-based research; the practice partners across the university (Armstrong Institute for Patient Safety and Quality at the School of Medicine; the Bloomberg School of Public Health's implementation-science tradition; Whiting School engineering and cognitive science; the Berman Institute's governance and ethics work); and the partnerships with sponsor and operational organizations that bring real problems through the door.

INTERSECTIONAL EXPERTISE

The distinguishing pillar — and the most demanding. The work sits at the intersection of the learning sciences, educational measurement, instructional design, design-based research, cognitive engineering, implementation science, the equity-and-governance disciplines, the relevant operational domain, and the engineering tradition that produced the artifact under study. No one of those disciplines is sufficient, and the work fails when the disciplines do not translate to one another. Intersectional expertise is what allows a single practitioner — or a single team — to hold the engineering, the learning, the measurement, the equity, and the operational context in the same conversation, and to make design decisions that survive scrutiny from any of them. Cultivating it is the central pedagogical task of LENS.

CAPABILITY FOCUS

The outcome — the central claim of the book. Capability is a designable, measurable, engineerable property of every complex system. The program produces it in graduates; the cases document where it was absent and where it was successfully engineered; the curriculum teaches how to recognize it, instrument it, and reproduce it. Capability focus is the discipline that keeps every artifact, every measurement, and every iteration accountable to what the system can actually do when the work is done.

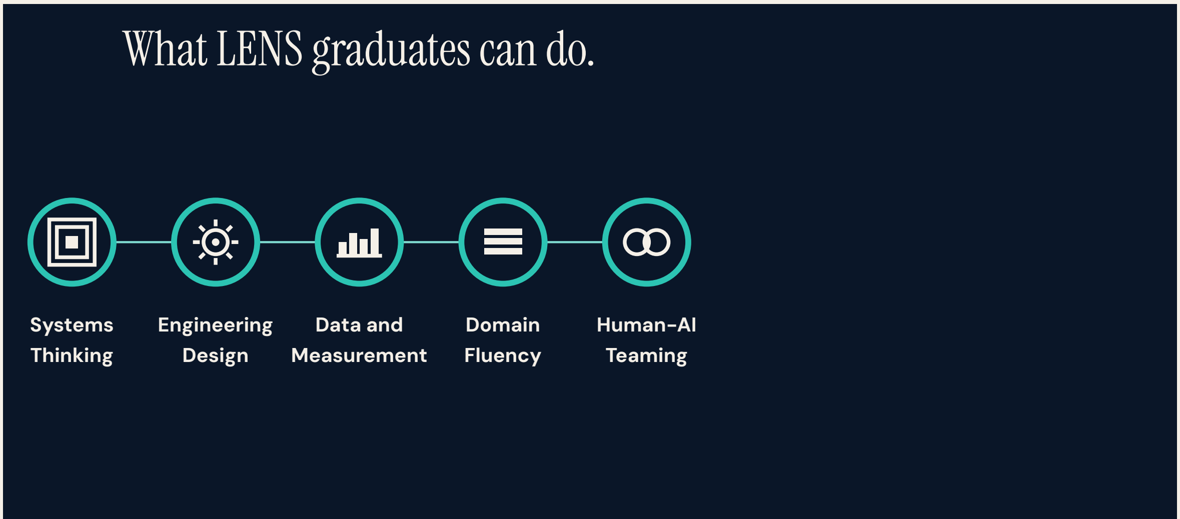
FLYWHEEL ITERATION

> The iterative method — the Practice Flywheel — that animates the other four pillars: **Identify → Activate → Prototype → Analyze → Transition**, the cycle through which mission literacy turns into capability and capability turns back into refined mission literacy. The flywheel is what prevents the discipline from being a series of unconnected projects and makes it instead a compounding institutional asset — each cohort, each case, each pilot shortening the implementation gap for the next. The casebook itself is a flywheel artifact: the work of prior cohorts produces the reference dataset for the next.

The pillars compound. Mission literacy reveals what is needed; the JHU ecosystem hosts the inquiry; intersectional expertise designs the intervention; capability focus keeps the loop closed on the outcome; the practice flywheel turns each iteration into the starting condition of the next.

WHAT LENS GRADUATES CAN DO · THE FIVE COMPETENCIES

The five pillars describe why LENS exists. The five concentration learning outcomes describe what its graduates can do. They map to the work the cases in this book required.



WHAT LENS GRADUATES CAN DO — FIVE COMPETENCIES

> ### 1 · SYSTEMS THINKING AND ANALYSIS
>
> Decompose system performance requirements into measurable human capability requirements; apply systems-engineering lifecycle models to scope, sequence, and evaluate capability interventions **within and across disparate systems**; analyze and communicate the interdependencies between human performance and system performance to predict operational impact at scale.

2 · LEARNING ENGINEERING DESIGN AND IMPLEMENTATION

Design capability-development interventions that integrate learning-sciences principles with measurable outcomes and system-design constraints, and that function at **speed and scale** in operational environments; construct and communicate implementation plans that address adoption, sustainment, and lifecycle integration across organizational and system boundaries.

3 · DATA, MEASUREMENT, AND EVALUATION

Design ethical instrumentation strategies that produce measurable evidence in regulated or high-consequence environments; construct **decision-grade evidence** artifacts — dashboards, reports, governance documentation — that link learning outcomes to operational impact; differentiate capability gaps from system-design and organizational- performance problems using mixed-methods **gap attribution**.

4 · CONTEXT AND DOMAIN FLUENCY

Analyze the regulatory, organizational, and cultural constraints that shape capability development in healthcare, defense, or education contexts; apply human-systems-integration frameworks to evaluate the measurable impact of capability approaches on operational environments; synthesize and communicate stakeholder requirements across disciplinary and institutional boundaries into coherent design specifications.

5 · HUMAN-AI TEAMING AND ADAPTIVE SYSTEMS

Design human-AI teaming configurations that **leverage AI as a creative partner** in capability development while preserving human agency and judgment; evaluate the measurable impact of AI-mediated learning systems on human performance — both the gains AI enables and the risks it introduces, including automation bias and cognitive offloading; communicate evidence-based recommendations for when AI augmentation improves versus degrades capability outcomes.

Three phrases recur across those five competencies and across the curriculum that produces them: **within and across disparate systems** (the scope at which capability is actually engineered), **speed and scale** (the operational tempo the discipline must match), and **gap attribution between learning, system design, and organizational performance problems** (the diagnostic that decides what kind of intervention will actually move the outcome). They are the working language of the program. They appear, repeatedly, in the cases that follow.

The remainder of this section names the three operational anchors the pillars are currently producing — the CIRCUIT–MERIT–COMPASS lineage, the Practice Flywheel in action across LE pilots, and the studio and applied-project model — as proof of work the pillars have already delivered.

The first operational anchor is the **CIRCUIT model, MERIT framework, and COMPASS content** — a more-than-decade record originating at the JHU Applied Physics Laboratory. CIRCUIT (Cohort-based Integrated Research Community for Undergraduate Innovation and Trailblazing) was built as the best operational solution to the IARPA MICrONS connectomics program's proofreading requirement, and was subsequently extended to additional sponsor problems. The **scientific and engineering impact** the trained cohorts produced — peer-reviewed contributions to the MICrONS reconstructions in **Nature**, the open-data infrastructure of BossDB and CAVE, the NeuVue proofreading framework — is the headline result; the student development was a deliberate and well-engineered byproduct of the mission solution, making doing well for the sponsor compound with doing good for the students. MERIT (Mentoring Exceptional Researchers to Innovate and Thrive) is the generalization of the CIRCUIT model into a replicable framework: holistic recruiting, the CIRCUIT eight pillars, the CCR competency framework, and the explicit "hidden curriculum" literature. **COMPASS** is the layer of targeted mentor interventions that operates MERIT inside a specific institutional setting — designed to help students uncover the **tacit and explicit knowledge gaps** that block independent research practice. COMPASS is not bound to a fixed workshop count; the COMPASS-Core curriculum is one instance, with an example set of workshops covering orientation and goal-setting, resilience and mindset, research-navigation, networking and faculty engagement, scientific reading and writing, presentations and figures, metacognition and advanced study, professional conduct and ethics, STEM identity, and long-term career and funding strategy. The CIRCUIT → MERIT → COMPASS lineage is documented in the peer-reviewed engineering-education literature [24]. Section VII expands on this evidence.

The second operational anchor is the **Practice Flywheel**: the iterative cycle — **Identify → Activate → Prototype → Analyze → Transition** — that has now been demonstrated across multiple LE pilots run by the LENS partner team at JHU/APL. Each iteration produces working artifacts, sponsor-visible evidence, and curriculum material for the next cohort. The flywheel is what prevents the program from being a series of unconnected projects and makes it instead a compounding institutional asset — each cohort shortening the implementation gap for the next. The casebook itself is a flywheel artifact: the work of prior cohorts produces the reference dataset for the next.

The third operational anchor is **integrated student work** — the studio, practicum, and applied-project model that ensures every LENS graduate has, in addition to coursework, a portfolio of capability engineering deliverables produced inside real operational systems. The Summer 2025 PISTA incubator (Section VII) is the most recent example: a JHU/APL cohort of forty-four students collaborated across projects, with nine student fellows specifically shipping a working CBRNE (chemical, biological, radiological, nuclear, and explosives) decision-support prototype to DTRA leadership in ten weeks, with three full redesigns inside that window. The cases in this book are not the only operational record the program produces. Every graduate carries an addition to it. The credibility of the program is, in the end, the credibility of the artifacts its graduates have produced in the field.

VI · WHY THE SCHOOL OF EDUCATION

A reasonable reader asks: if learning engineering is engineering, why does this program live in a School of Education rather than a School of Engineering? The answer is that the binding constraint on the discipline is not the engineering, which is mature. It is the learning sciences, the educational measurement, the implementation science, and the equity-and-governance machinery — which live in Schools of Education and which the cases in this book establish as the load-bearing elements when capability interventions actually scale.

The learning sciences are the discipline that knows how skill, knowledge, and judgment develop, decay, and transfer across contexts. CRM was not an engineering achievement; the engineering of the cockpit was already mature in 1976. CRM was a **learning** achievement — a redesign of how aircrews learn to operate as a coordinated team, drawn from psychological research on group performance under stress. The same is true of TeamSTEPPS, the WHO Checklist's pause protocol, INPO's accreditation regime, and the Rickover oral examination tradition. The interventions that move the curves are interventions in how capability is learned, retained, and transferred — not in how the hardware is built.

Educational measurement is the discipline that knows how to construct the instruments that determine, validly and reliably, whether a person can do something. The cases in this book that turn on measurement failures — the VA wait-time scandal, Atlanta Public Schools, LIBOR, Wells Fargo, the F-35 readiness reporting — are measurement-instrument failures at heart. The capability to design measurement instruments that survive contact with the institutions measuring themselves is an educational-measurement competence. It lives in Schools of Education.

Implementation science — the field that studies the gap between evidence and practice — was developed largely in public-health schools and schools of education. Its frameworks (Fixsen et al., Aarons et al., Damschroder et al.) are the operational backbone of any serious capability-engineering effort, and Schools of Education have been their institutional home for two decades. A School of Engineering can teach the artifact. A School of Education teaches the artifact, the measurement instrument that decides if the artifact works, and the implementation pathway by which the artifact reaches scale. The casebook's central argument — that capability is a system parameter — requires all three.

Equity, governance, and ethics in education at scale are also, evidently, education problems. The inBloom, A-Level, COMPAS, Robodebt, and educational-predictive-analytics cases in this book are not failures of engineering competence. They are failures of the disciplines that know how to engineer measurement instruments, credentialing systems, and decision regimes that do not encode structural inequity. Those disciplines live in Schools of Education. The School of Engineering can build the model. The School of Education is where the construct gets validated, the consent framework gets designed, and the equity audit gets conducted.

The case for housing LENS in the Johns Hopkins School of Education is, finally, a case for treating capability engineering as the applied form of the educational sciences — engineering applied through learning, measurement, and equity, not engineering applied to learning as content. That is a real distinction. Schools of Engineering routinely build artifacts; the cases in this book show that the artifacts then need disciplines that those schools are not organized to provide. The placement is not a default. It is the shortest path between the discipline as it is and the discipline as it needs to be.

THE RECORD AT JOHNS HOPKINS

The argument above would be theoretical if Johns Hopkins did not already have an operating record. It does — and the deeper, longer half of that record is the School of Education's. The SoE has been studying the learner, designing the assessment, and engineering the intervention since long before the discipline of learning engineering had its current name. The disciplines this casebook argues are load-bearing — learning sciences, educational measurement, design-based research, implementation science, and equity-and-governance work — are disciplines the school's faculty have published in, taught, and applied at scale, in K–12, higher-education, clinical, military, and corporate settings, for decades.

Three durable lines of work make the point. The Center for Talented Youth, founded in 1979, has built one of the longest continuous evidence bases in U.S. education on the identification, assessment, and accelerated instruction of academically advanced learners; its measurement and program-evaluation work has been the empirical foundation of an entire sub-discipline. The Center for Research and Reform in Education has, since 2004, run large-scale evaluations of educational programs against rigorous evidence standards (Every Student Succeeds Act tiers, What Works Clearinghouse review), publishing the Best Evidence Encyclopedia and synthesizing what is known about which interventions actually work for which learners under which conditions. The Institute for Education Policy carries the same evidence-first orientation into the governance, curriculum-policy, and reform-design work that the case studies in Chapter 5 of this book name as the binding capability in modern education systems. These are not adjacent to the LDT program; they are the institutional environment that taught the program how to be rigorous.

The Learning Design and Technology program sits inside that environment and inherits it. For two decades the LDT M.Ed. and EdD tracks have graduated cohorts who entered as practitioners and left as designers and evaluators of learning systems — instructional designers and architects of learning experience, learning-analytics specialists, technology-mediated-learning researchers, professional developers, evaluators, and program leaders across the same K–12, higher-education, military, clinical, and corporate settings the cases in this book draw on. The program's intellectual core has always been the learning-engineering question — what does the system have to do in order to produce reliably capable people? — and its pedagogical core has always been design-based research: put the artifact in front of real learners, measure what happens, iterate, and publish what was learned about both the artifact and the learner. LENS is the formalization of that orientation as a specialization, not a departure from it.

Alongside that academic record runs a complementary thread of applied collaboration. Faculty and graduates of the LDT program have, over the past decade, partnered with operational organizations — including teams at the Johns Hopkins Applied Physics Laboratory — to study capability in real mission settings. Programs such as CIRCUIT (the workforce arm of the IARPA/MICrONS consortium, whose connectomics result appeared in Nature in 2025), the precision-medicine clinical-decision-trainer pilot reported in 2024, the DARPA AIE-supported TITAN work, and the DTRA-supported PISTA incubator in 2025 are useful proving grounds for the learning-engineering question because the deliverables are evaluated by sponsors who care whether the system worked. These collaborations have informed the program's case-based pedagogy and contribute several of the operational vignettes that follow. They are not the institutional home of the discipline; the School of Education is. They are how the school's disciplines meet operational settings, and how operational settings show up in the curriculum.

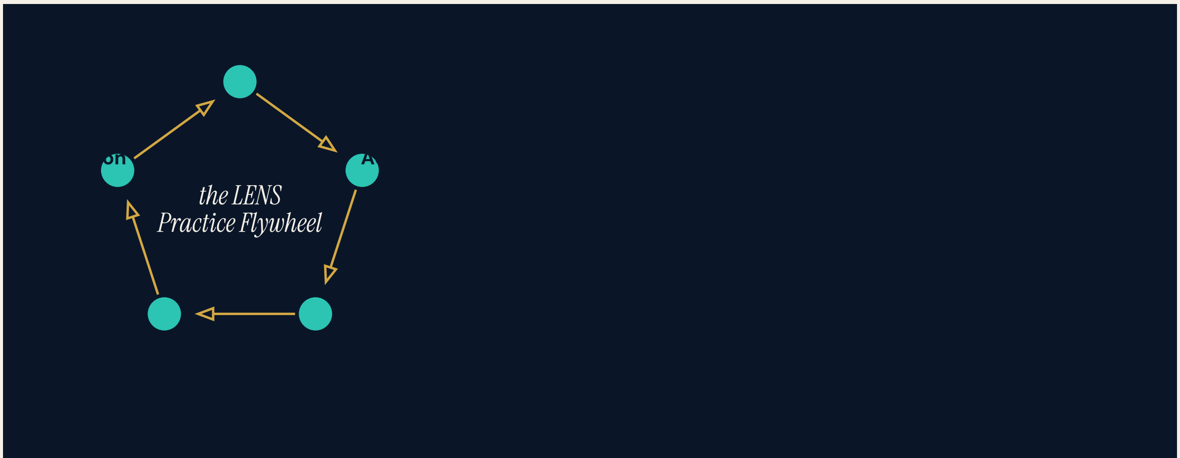
Other parts of the Johns Hopkins ecosystem do adjacent work the program draws on. The Armstrong Institute for Patient Safety and Quality at the School of Medicine — the institutional descendant of the Keystone ICU work foregrounded in Chapter 8 — is the most visible example: a continuing program of capability-engineering in clinical settings, with measurement and implementation methods that the casebook treats as canonical. The Bloomberg School of Public Health contributes the implementation-science tradition the program rests on, with departments and centers that have spent decades studying how interventions reach scale. The Whiting School of Engineering contributes human-systems engineering, AI methods, and cognitive-engineering research relevant to the human-AI teaming cases in Chapter 9. The Krieger School's Department of Cognitive Science and adjacent psychology programs contribute the learning- science methods the LDT program teaches. The Berman Institute of Bioethics contributes the governance-and-ethics framing that the cases in Chapter 5 require. None of these are formal partners in this volume; all of them are part of the institutional environment in which the discipline this casebook describes is recognizably at home.

That meeting is also embodied in the editors. One of us has spent a career in applied research environments thinking about how to make scientific and engineering teams reliably capable; the other has spent a career in the School of Education thinking about how to design, study, and evaluate the learning of learners. Both of us have, in practice, lived in the other half. The casebook is one product of that doubling. The discipline it describes is not new. The institutional home that lets it be taught at scale — the School of Education at Johns Hopkins, with its faculty, centers, doctoral programs, and the practitioner partnerships those carry — is where the work continues.

VII · PATHWAYS EVIDENCE · LE PILOTS

A program built on the claim that capability is engineerable should be able to point to the institutional evidence that it has done that work. LENS draws on a documented record of learning-engineering pilots — operated at the Johns Hopkins University Applied Physics Laboratory and across collaborating sites — that demonstrate the iterative LE process at operational scale before LENS itself enrolled its first cohort. The pilots span the same three domains the casebook treats as co-equal — science (instantiated here as connectomics and precision medicine), defense, and education — and they each carry independent evidence of the loop closing.

The Practice Flywheel



THE LENS PRACTICE FLYWHEEL — EACH ITERATION GENERATES EVIDENCE, ARTIFACTS, AND REFINED PRACTICE

The pilots share an operating cycle: **identify** a real challenge in an operational setting, **activate** a cohort of practitioners and the tools they need, **prototype** a working artifact under realistic constraints, **analyze** the evidence the prototype produces, and **transition** validated solutions to the sponsor or institution that asked the original question. Each iteration generates artifacts, evidence, and refined practice; each cohort inherits the prior cohort's tools and lessons. The flywheel makes the program compounding rather than serially episodic.

Pilot · CIRCUIT, MERIT, COMPASS — scientific and engineering impact at MICrONS / BRAIN CONNECTS scale

CIRCUIT — the **Cohort-based Integrated Research Community for Undergraduate Innovation and Trailblazing** — was built at JHU/APL as the best operational solution to a sponsor problem: the IARPA MICrONS connectomics program needed a proofreading workforce that could exercise expert biological judgment at scale, and CIRCUIT was the architecture that delivered it. The headline result is scientific. The trained cohorts contributed materially to the MICrONS reconstructions now published in **Nature**, with downstream contributions to FlyWire and H01 datasets, the BossDB and CAVE open-data platforms, and the NeuVue proofreading framework [23]. These are not auxiliary outputs of a workforce program. They are the foundational datasets and infrastructure on which contemporary connectomics depends.

The student training was a **byproduct** of the mission solution — but a deliberate, well-engineered byproduct that turned an operational requirement into a sustained opportunity to develop the next generation of computational neuroscientists. Doing well for the sponsor and doing good for the students compounded rather than competed. That structural alignment between scientific mission and human-capability development is the pattern LENS treats as the canonical positive case for how the Practice Flywheel operates at the boundary between science and workforce.

The model that produced those outputs was then **generalized into MERIT (Mentoring Exceptional Researchers to Innovate and Thrive)** — a replicable research-training framework built on holistic recruiting, the CIRCUIT eight pillars, the CCR competency framework, the hidden-curriculum literature, and the DoD KSAT alignment vocabulary [24]. MERIT operates across five developmental stages — Holistic Recruiting, Orientation and Research Foundations, Guided Research Practice, Independent Research with Mentorship, and Leadership / National Impact — and has been instantiated in computational neuroscience, AI, cybersecurity, threat assessment, and pandemic modeling. **COMPASS** is the layer of targeted mentor interventions that operates MERIT inside a given institutional setting — designed to help students uncover the **tacit and explicit knowledge gaps** that separate competent execution from independent research practice. COMPASS is not bound to a fixed workshop count. The COMPASS-Core curriculum is one well-documented instance, with an example set spanning orientation and SMART goal-setting, resilience and growth mindset, research navigation, networking and faculty engagement, scientific reading and writing, figures and oral presentations, metacognition and study skills, professional conduct and research ethics, STEM identity and belonging, and long-term career and funding strategy. The casebook treats the trio (CIRCUIT model, MERIT framework, COMPASS mentor interventions) as the load-bearing instance of the Practice Flywheel: requirements elicited from operational science, instrumented measurement of cohort progress, paired technical and cultural scaffolding, and an explicit iteration cycle that has now produced multiple cohorts and hundreds of student-authored contributions to the peer-reviewed scientific literature.

Pilot · TITAN (DARPA AIE)

TITAN — a DARPA Artificial Intelligence Exploration program — was implemented and explored by junior staff and student fellows when senior staff did not take it up, and produced both technical artifacts and a demonstrated proof of concept for the cohort-based incubator model the program now treats as a primary delivery mechanism. The pattern — talent identification, rapid prototyping under sponsor constraints, transition planning — is the same Practice Flywheel the casebook describes. TITAN is the earliest evidence that the model scales beyond the connectomics domain in which it was first piloted [25].

Pilot · PISTA (DTRA, Summer 2025)

PISTA — the **Perceptual Inference System for Tactical Agentic AI** — was a ten-week incubator delivering tactical AI decision support for the Defense Threat Reduction Agency's chemical, biological, radiological, nuclear, and explosives mission. Nine student fellows shipped PISTA, drawn from the broader Summer 2025 cohort of forty-four students at the JHU/APL AI Pathfinding Lab; all forty-four benefited from the shared cohort infrastructure, mentor pool, and cross-project collaboration even when working on other sponsor problems. The PISTA team delivered a working perception-reason-report agentic pipeline on a COTS stack (Gemini Flash, GPT-4o, Firebase) with deployed sensors on UGVs and small UAS. Three full redesigns inside ten weeks. Live demonstration to DTRA leadership. Follow-on funding under exploration. The pilot demonstrates the Practice Flywheel at compressed timescale in a defense context [26].

Pilot · ECHO (healthcare, culturally responsive provider training)

ECHO is an AI simulator for limited-English-proficiency patient encounters — a population of more than twenty-five million Americans whose access to safe care is structurally constrained by language barriers. The system pairs scenario-driven practice with rubric-based assessment and real-time coaching, designed for implementation inside existing medical-curriculum infrastructure. ECHO is the healthcare instance of the casebook's central argument: health equity is a **design commitment**, not an audit. The pilot shows the same flywheel operating in the clinical-training context.

Pilot · TeachMe EDDIE (methodology)

TeachMe EDDIE is the methodological instance of the pilot portfolio: an AI-augmented knowledge engineering system built around a 100-plus-node causal knowledge graph, a dual-axis learner model (declarative and procedural), and Mentor and Narrator agents bounded by a curated corpus. EDDIE operationalizes the difference between **retrieval** and **learning** — between an LLM that can produce a fluent answer and an instructional system that can move a learner from competence in one context to competence in another. Transfer is the north star. EDDIE is the LENS instance of the learning-engineering question that the rest of the dataset asks at scale.

Pilot · Precision-medicine community detection (clinical decision support)

Precision-medicine clinical decision tools represent the program's pilot in the application of evidence-based methodology to a high-consequence operational domain. Students and staff at APL contributed to community-detection methods for precision-medicine cohort identification — the kind of work that combines computational rigor with the clinical interpretive judgment LENS is organized to teach. The preprint is publicly available on medRxiv [27]. The same pattern recurs across the pilots: the technical artifact is necessary, the institutional scaffolding to use it under operational constraints is what makes it capability.

What the pilots demonstrate together

No single pilot would settle the LENS proposition. The cumulative record is what matters. Across pilots, the same set of LE practices recurs: explicit capability requirements; structured pedagogical scaffolding; instrumented measurement linked to operational outcomes; paired technical and cultural interventions; cohort-based delivery with deliberate inheritance of tools and lessons across cohorts; and an explicit Identify → Activate → Prototype → Analyze → Transition cycle. The same artifacts produced in those pilots — the rubrics, the prompts, the dashboards, the after-action reports, the published papers — become the curriculum material for the next cohort. The program is, in this sense, the institutional flywheel its case studies argue every high-consequence domain requires.

VIII · HOW TO READ THIS BOOK

Part I, "The Failure Modes," organizes one hundred cases under six recurring patterns: training gap, capability designed out, normalization of deviance, interface failure, governance failure, and knowledge loss. A seventh chapter — the Evidence Gap — anchors the measurement question. The taxonomy is not a theory. It is a finding: six categories that account for almost every well-documented case in the literature of preventable failure in complex sociotechnical systems, and that recur across healthcare, defense, education, energy, transportation, and government.

Part II, "What Works," collects the cases in which the capability parameter was engineered rather than left to drift. Every success case in the dataset shares a structural feature: a **paired intervention.** A technical artifact paired with a cultural authority. A measurement instrument paired with a governance body that listens to it. A curriculum paired with the institutional architecture to sustain it. Neither half alone moves the curve. The pair is irreducible. This is the strongest empirical pattern in the book and it directly informs the LENS pedagogical commitment to co-optimization across technical and human design.

Part III, "The Frontier," concerns the cases where the discipline is being asked to do work it has not done before — particularly the human-AI teaming cases that anticipate the next decade of capability engineering. The cases are deliberately less settled. The discipline is being asked to specify what good looks like before the catastrophe arrives. That is the work the program exists to teach.

Each case occupies a two-page spread. The left page tells what happened — narrative, evidence, attribution. The right page is the **Learning Engineering Lens:** a synthesis of what the case teaches, how the LENS curriculum addresses the pattern, and reflection questions designed for studio discussion. The casebook is built to be taught from, read straight through, or consulted as a reference. The curriculum crosswalk in the closing matter maps each case to the specific courses in the program that use it as a worked example.

IX · COURSE COVERAGE AND PEDAGOGICAL GAPS

The one hundred cases in this volume are mapped to the LENS courses they most directly inform. The distribution is itself diagnostic. The strongly covered topics are the ones the existing case literature most readily supports: bias and governance (LEN 7, 49 cases) and knowledge transfer (LEN 8, 45 cases); capability analysis (LEN 5, 36 cases) and evidence and measurement (LEN 4, 31 cases); human-AI teaming (LEN 2, 23 cases) and principles of learning engineering (LEN 1, 21 cases). These are not arbitrary numbers — they describe a dataset saturated with cases of **what goes wrong when ethics, governance, evidence, and institutional learning are not engineered.**

The thin coverage is also informative. The computational-methods topic (LEN 9, 10 cases) has the failure modes of algorithmic systems well-represented but the computational tools for capability instrumentation under-represented. The capstone (LEN 10, 19 cases) maps to fewer cases than the number of studio projects each LENS cohort undertakes. The systems-integration topic (LEN 3, 18 cases) and applied problem-solving topic (LEN 6, 13 cases) required focused retagging to reach defensible coverage; the work surfaced that operational systems-engineering content is **under-represented in the published case literature** relative to its curricular centrality, and that stakeholder-analysis and problem-framing have not historically been packaged as case material by other disciplines. These are pedagogical gaps the LENS studio sequence is organized to close — not by adding more failure cases, but by having students produce the missing artifacts in real operational settings and entering them into the dataset for the next cohort.

> *Capability is a designable, measurable, engineerable property of every complex system. The strongest argument for the discipline of learning engineering is the cumulative record of what its absence has cost. The strongest argument for its possibility is the cumulative record of what its presence has produced. Both records are in this book. The work of LDT and LENS at the Johns Hopkins School of Education is to add to the second record and shorten the first.*

HOW TO USE THIS BOOK

A field manual for capability engineering.

This book collects one hundred real incidents in which capability — what people could or could not do inside a complex system — determined whether the system worked. Some are failures: lives lost, systems wrecked, billions written off. Some are successes: deliberate interventions that engineered capability into the architecture of the work. Together they form an evidence base for the argument that capability is a designable, measurable property of every complex system.

Each case occupies a two-page spread. The left-hand page tells the story: what happened, what investigators found, and the documentary record. The right-hand page is the **Learning Engineering Lens** — what the case teaches about capability as a system parameter, how the LENS curriculum addresses the pattern, and a short set of reflection questions designed for studio discussion.

The Lens page also carries a **Who Builds This** note: the mix of expertise and tools a team would bring to engineer the fix, read off the case's failure modes. A Training-Gap case pulls in learning scientists and instructional designers; an Interface case, human- factors and interaction designers; a Governance case, policy, ethics, and measurement. Most cases need several at once, held together by domain experts and a learning engineer. No single discipline carries a case alone — reading the modes as a staffing list is itself a LENS skill, and what **you** bring is yours to decide.

READING THE TAXONOMY

Each case is tagged with one or more failure modes. Most cases involve several. The primary mode is listed first; contributing modes follow.

T Training Gap
Personnel were never taught what they needed to know.

D Designed Out
Human capability was deliberately removed from the system.

N Normalization of Deviance
Known risks were accepted as routine.

H Human-System Interface
Correct performance was made unreasonably difficult.

G Governance & Trust
Ethics, accountability, and stakeholder trust were after-thoughts.

K Knowledge & Institutional Memory
Organizational knowledge degraded or was never captured.

LENS COURSE CODES

Each case maps to one or more courses in the LENS specialization:

SHARED LDT FOUNDATIONS · ADDITIONAL CONTEXT

F1	**Learning Sciences Studio.** How learning, motivation, and skill develop — applied to technology-mediated design.
F2	**Critical Perspectives on Educational Technology.** How power, equity, and society shape — and are shaped by — learning tech.

LENS-DESIGNED COURSES

LEN 1	Principles of LE for Complex Systems
LEN 2	Human-AI Teaming and Adaptive Systems
LEN 3	Learning Engineering Systems
LEN 4	Evidence, Analytics, and Measurement
LEN 5	Human Capability Analysis and Requirements
LEN 6	Applied Problem Solving in LE
LEN 7	Bias, Risk, and Governance

LEN 8	Knowledge Transfer and Organizational Learning
LEN 9	Computational and AI Methods
LEN 10	Learning Engineering Project (capstone)

> *The strongest cases pair a catastrophic failure with a systematic capability intervention. Together they show both the cost of the gap and the tractability of the solution. The discipline that closes that gap is learning engineering.*

THE CASE MATRIX

One hundred cases at a glance

Gold numbers indicate failures and systemic conditions; teal numbers indicate paired-intervention successes and the open closing case.

1	USS Fitzgerald & McCain 2017	T·K·N	1·5·8
2	Boeing 737 MAX / MCAS 2018–19	D·T·H	1·2
3	Air France Flight 447 2009	T·H	1·5·2
4	Deepwater Horizon 2010	T·N·K	5
5	Three Mile Island 1979	T·H	1·5
6	Challenger & Columbia 1986/2003	N·K·G	1·8
7	Therac-25 1985–87	H·D·G	7·9
8	inBloom 2014	G	7
9	USMC INDOPACOM Training ongoing	T·K	5
10	Healthcare.gov 2013	K·T·G	7
11	V-22 Osprey 1991–	T·H·N	5
12	CRM + CAST 1981–	T·H·N	2
13	WHO Surgical Checklist 2008–	T·N	10
14	Keystone ICU (Pronovost) 2004–	T·N	4·10
15	Navy SWO Reform 2018–	T·K·N	10
16	INPO / Nuclear Training 1979–	T·K·G	8
17	Bhopal 1984	T·K·N·G	5·7
18	USS Vincennes 1988	H·T	2
19	Patriot Missile 1991	D·H·K	5·9

20	Grenfell Tower 2017	G·T·K·N	7
21	Summit Learning Rollout 2014–19	G·T·K	7·10
22	Tennessee Pre-K Study 2009–18	G·D	4·7·10
23	Korean Air Transformation 2000–	T·N	8
24	Toyota Andon Cord 1950s–	N·G	8
25	EHR/CPOE Implementation 2005–	H·D·G	7·9
26	F-35 Sustainment ongoing	T·K·D	5
27	TeamSTEPPS 2006–	T·N	8
28	Rickover Nuclear Navy 1954–	T·K·N	8
29	Uber ATG / Tempe 2018	T·N·G·H	2
30	Kegworth / BM 92 1989	T·H·K	5
31	Medical Errors (Makary) ongoing	T·H·N·K·G	4
32	VA Wait-Time Scandal 2014	G·K·N	4
33	Military Fratricide 1991–	T·H·K	5
34	ACGME Duty-Hour Reform 2003–17	T·K·N	5·4·10
35	UK A-Level Algorithm 2020	G·H·D	4·7
36	Australia Robodebt 2016–20	G·D·H	7·2
37	Algorithmic Bias in Ed ongoing	G·H·D	4·7·9
38	GIFT and the Adoption Gap 2012–	K·G·N	1·10·8
39	Georgia State Univ. 2012–	T·K	4
40	xAPI / TLA Gap ongoing	K·G	4·8
41	17-Year Implementation Gap ongoing	K·G·N	1·10·8
42	Cognitive Tutor 1990s–	T	1·4·9
43	Colgan Air 3407 2009	T	4·5·8

44	Asiana Airlines 214	2013	T·H	5·2
45	Mark 14 Torpedo	1941–43	T·K·G	10·7·8
46	Operation Eagle Claw	1980	T·K	5·8
47	Helios Airways 522	2005	T·H	5·2
48	AeroPerú 603	1996	H·T	5·2
49	Atlas Air 3591	2019	T·K	4·8
50	TransAsia 235	2015	T·H	5·2
51	Ford Pinto	1971–78	D·G	1·7
52	Takata Airbags	2008–23	D·G	4·7
53	GM Ignition Switch	2002–14	D·G	4·7·8
54	Mars Climate Orbiter	1999	D·K	5·8
55	Knight Capital	2012	D·K	5·9
56	Texas City BP Refinery	2005	N·T·K·G	4·7
57	Davis-Besse Reactor	2002	N·K·G	7·8
58	Mid Staffordshire NHS	2005–09	G·N·K	4·7
59	Sago Mine	2006	N·T·K	5·8
60	Upper Big Branch Mine	2010	N·G·K	4·7
61	Fukushima Daiichi	2011	N·G·K	7·8
62	Northeast Blackout	2003	H·K	2·8
63	Eastern 401	1972	H·T	5·2
64	Boeing 737 Rudder	1991, 1994	H·D	5·7
65	CrowdStrike Outage	2024	D·K·G	5·2
66	Petrov / 1983 False Alert	1983	H·T	2
67	TSB Bank IT Migration	2018	H·G	7·8

68	UK Post Office Horizon	1999–2015	G·H·K	7·2
69	Theranos	2003–18	G·D	4·7
70	Wells Fargo Accounts	2011–16	G·N	4·7
71	VW Dieselgate	2015	D·G	4·7
72	Cambridge Analytica	2014–18	G	1·7
73	Equifax Breach	2017	G·K	5·7
74	Hyatt Regency Walkway	1981	D·G	5·7
75	FIU Pedestrian Bridge	2018	D·G·N	5·8
76	Camp Fire / PG&E	2018	G·N·K	7·8
77	Texas Grid Freeze	2021	G·K·N	7·8
78	Saturn V Documentation	1972–	K	1·8
79	Boeing Starliner	2019–24	K·D	5·8
80	Ariane 5 Flight 501	1996	D·K·H	5·8
81	Tacoma Narrows Bridge	1940	D·K	1·8
82	Aliso Canyon	2015–16	G·N·K	7·8
83	LIBOR Manipulation	2003–12	G·N	4·7
84	Atlanta Schools Cheating	2009–15	G·N	4·7
85	Madoff / SEC Failure	1992–2008	G·K·N	4·7
86	9/11 Intel Sharing	1996–2001	G·K	7·8
87	Vioxx	1999–2004	G·D	4·7
88	Tylenol Recall	1982	G·N	10·7
89	ASRS	1976–	T·K·N	4·8
90	Bristol Heart Reform	1991–	G·N	4·7
91	Singapore Airlines	1980s–	T·N	8

92	Tesla Autopilot	2016–	T·N·G·H	7·2
93	Cruise Robotaxi	2023	G·D·H	10·7
94	COMPAS Recidivism	2016–	G·H·D	4·7·9
95	Radiology AI Miscalibration	2018–	H·K·D	4·7·9
96	LLMs in Healthcare	2023–	H·D	10·7·2
97	Predictive Policing	2011–	G·H·D	7·9
98	AlphaFold	2020–	T	1·7·9
99	AI-Augmented Coding	2021–	T·H	10·2·8
100	The Discipline We Build Next	ongoing	all	1·10·8

Modes. T Training Gap · D Designed Out · N Normalization of Deviance · H Human-System Interface · G Governance & Trust · K Knowledge & Institutional Memory

LEN. The LDT / LENS courses each case is most relevant to. LEN 1 Foundations · LEN 4 Evidence & Measurement · LEN 10 Studio · LEN 5 Capability Analysis · LEN 7 Bias, Risk & Governance · LEN 2 Human-AI Teaming · LEN 8 Knowledge Transfer · LEN 9 Computational & AI Methods.

Parts. Part II — **What Works** — opens at Chapter 8 with the paired-intervention successes. Part III — **The Frontier** — closes the volume with the human-AI teaming cases and the open question at Case 100.

Chapter 1

Training Gap

They weren't taught what they needed to know.

Individuals and groups of individuals could no longer recognize that the processes designed to identify and communicate readiness were no longer working.

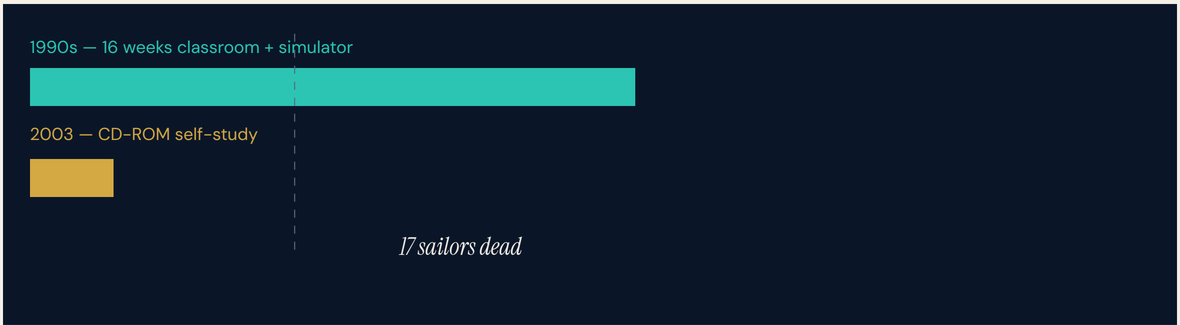
PARAPHRASING THE U.S. NAVY STRATEGIC READINESS REVIEW, 2017

CASE 1 DEFENSE 2017

USS Fitzgerald & USS John S. McCain

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT 17 sailors killed in two avoidable destroyer collisions in the Western Pacific within nine weeks



SURFACE WARFARE OFFICER TRAINING, 2003 → 2017

BACKGROUND

The two destroyers that collided in the summer of 2017 belonged to the U.S. Navy's forward-deployed forces in Japan — the Seventh Fleet ships kept at the highest operational tempo in the surface fleet. For more than a decade that tempo had been paid for out of training. In 2003 the Navy disestablished the in-person Surface Warfare Officers School division-officer course that had taught new officers seamanship, navigation, and the international rules of the road, and replaced it with a set of self-study CD-ROMs — known in the fleet as "SWOS in a Box" — that officers were expected to complete on their own at sea.[0] The change saved schoolhouse time and money; it also removed the structured setting in which the fundamentals had been taught, practiced, and tested. Forward-deployed ships compounded the problem: unlike vessels based in the United States, which rotate through a defined cycle of maintenance, training, and deployment, the Japan-based ships were treated as continuously available, and the training time that cycle protects was the first thing spent when demand rose.

By 2017 the erosion was measurable. The Government Accountability Office reported that, as of June 2017, 37 percent of the warfare certifications held by the cruiser and destroyer crews based in Japan — including the certifications for basic seamanship — had expired, a fivefold increase since 2015.[1] Ships were operating in some of the world's busiest waterways with lapsed qualifications because the schedule never left room to renew them, and expired certifications had become so routine that they were waived rather than corrected.

WHAT HAPPENED

On 17 June 2017 the USS Fitzgerald, steaming off the Izu Peninsula south of Tokyo Bay, crossed into the path of the container ship ACX Crystal and was struck on the starboard side; seven sailors drowned in their flooding berthing compartment. The Fitzgerald was the give-way vessel in a crossing situation and had the clearer picture, yet its bridge team took no early or substantial action to avoid the much larger ship, and an unexplained course change in the final minutes turned the destroyer directly into the collision.[2] The container ship, many times the destroyer's displacement, opened the hull below the waterline and flooded two berthing compartments where the crew was sleeping; only the damage-control efforts of the surviving sailors kept the Fitzgerald from sinking.

Nine weeks later, on 21 August 2017, the USS John S. McCain collided with the tanker Alnic MC in the Singapore Strait; ten sailors died and dozens more were injured. In the minutes before impact the watch team believed it had lost steering. It had not. While shifting control of the throttles between consoles, a watchstander had unintentionally transferred steering to a different station — steering was available the entire time, but no one on the bridge understood the ship-control console well enough to recognize what had happened or to take it back.[3] The destroyer swung across the strait and into the path of the tanker, which struck it on the port side at the berthing compartments, repeating in nine weeks the same lethal geometry.

THE INVESTIGATION

The Navy's own Comprehensive Review, completed in November 2017, judged both collisions avoidable and traced them to deficiencies in the most basic skills of mariners — watchstanding, navigation, and the operation of the ships' own equipment.[4] The National Transportation Safety Board, investigating independently, arrived at the same place by a different road: its report on the McCain found the probable cause to be a lack of effective operational oversight by the Navy that produced insufficient training and inadequate bridge operating procedures.[5] The board also faulted the console itself. The McCain's integrated bridge used a touch-screen steering and throttle system the Navy had adopted across the destroyer fleet, and its design made the unintended, unilateral transfer of control not merely possible but easy — a human-factors trap waiting for an under-trained crew.[6] The two investigations reinforced each other. On the Fitzgerald the NTSB found a bridge team that did not use its radars or lookouts effectively and did not communicate as the situation developed; on the McCain it found watchstanders who had never been drilled on the casualty they now faced. In both cases the crews were fatigued, working watch rotations that left little room for rest, and in both cases the proximate errors sat on top of the same foundation — sailors asked to do, in the dark and at speed, things their training had never reliably given them.

THE CAPABILITY GAP

What makes Fitzgerald and McCain the canonical training-gap case is that the gap was invisible from inside the system. The Strategic Readiness Review, delivered in December 2017, described risks that had accumulated over time and done so insidiously: the dynamic environment of the Western Pacific had normalized to the point where individuals and groups could no longer recognize that the very processes meant to surface readiness shortfalls had themselves stopped working.[7] Stated readiness and actual readiness had diverged for more than a decade, and the instruments that should have caught the divergence kept reporting that all was well. The divergence was not a secret kept from leadership so much as a fact the system had stopped being able to see: commanders who asked for more time to train were told the mission came first, and the readiness reports continued to register green. The cost of the gap was not paid in a failed inspection. It was paid in seventeen lives, two ships, and two months.

AFTERMATH & REFORM

The response was the most far-reaching surface-warfare reform in a generation. The Navy reconstituted formal, in-person officer training, rebuilding and lengthening the junior-officer pipeline it had dismantled in 2003; stood up a Ready-for-Sea Assessment and tighter oversight of forward-deployed readiness; and adopted circadian watchbills to manage the crew fatigue the reviews had documented.[8] Acting on the NTSB's findings, the Navy also moved to replace the touch-screen throttle and helm controls with physical throttles and a traditional wheel — an admission that the interface, not only the people, had been part of the failure. Commanding officers and senior watchstanders were relieved and disciplined, and the Navy stood up a dedicated command in Japan to oversee the readiness of its forward-deployed ships, so that certification and training could no longer be quietly traded away for operational tempo. The case opens this book because it shows both how far a capability system can erode before anyone notices and what it costs to rebuild one after the fact rather than before — when the bill arrives not as a budget line but as a casualty list.

REFERENCES

1. T. C. Miller, M. Faturechi & R. Rotella, "Years of Warnings, Then Death and Disaster: How the Navy Failed Its Sailors," ProPublica (2019) — documents the 2003 shift from the in-person SWOS course to self-study CD-ROMs.
2. U.S. Government Accountability Office, Navy Readiness: Actions Needed to Address Persistent Maintenance, Training, and Other Challenges Affecting the Fleet, GAO-17-809T (Sept. 2017) — 37% of Japan-based cruiser/destroyer warfare certifications expired as of June 2017.
3. National Transportation Safety Board, Collision between USS Fitzgerald and MV ACX Crystal, NTSB/MAR-20/02 (2020), case DCA17PM018 — sequence and probable cause.
4. National Transportation Safety Board, Collision between USS John S. McCain and Tanker Alnic MC, NTSB/MAR-19/01 (2019), case DCA17PM029 — sequence of events.
5. U.S. Navy, Comprehensive Review of Recent Surface Force Incidents (Nov. 2017) — both collisions judged avoidable; deficiencies in seamanship, navigation, and equipment operation.
6. NTSB/MAR-19/01 (2019) — analysis and probable cause: insufficient training and inadequate bridge operating procedures stemming from a lack of effective Navy operational oversight.
7. R. Rotella et al., "The Navy Installed Touch-Screen Steering Systems to Save Money. Ten Sailors Paid With Their Lives.," ProPublica (2019) — the McCain's ship-control console design.
8. U.S. Navy, Strategic Readiness Review (Dec. 2017) — systemic erosion of readiness; risks that "accumulated over time and did so insidiously."
9. U.S. Navy, Comprehensive Review (2017), corrective actions; NTSB/MAR-19/01 safety recommendations; USNI News coverage of the reforms (2017–2020).

THE LEARNING ENGINEERING LENS

> *The risks that were taken in the Western Pacific accumulated over time and did so insidiously.*
>
> U.S. NAVY STRATEGIC READINESS REVIEW, 2017

LE INSIGHT

Fitzgerald/McCain is the canonical Training Gap case because the gap was invisible from inside the system. Operational tempo and self-study "training" combined to produce a fleet whose stated readiness and actual readiness diverged for more than a decade. Seven sailors died on Fitzgerald; ten on McCain. The cost of the divergence was paid in lives long after it could have been measured in dollars or inspections.

LENS APPROACH

LENS treats this as the worked example of LEN 5 (Human Capability Analysis and Requirements) interacting with LEN 4 (Evidence and Measurement). Students reconstruct the capability requirements for underway watchstanding from operational analysis and design the evidence system that would have surfaced the gap before the collisions. LEN 8 covers the institutional dynamics of deferred training.

REFLECTION QUESTIONS

1. The Navy replaced classroom and simulator instruction with CD-ROM self-study in 2003. What measurement would have detected the capability shortfall before 2017?
2. The Strategic Readiness Review found the readiness-reporting system had itself stopped working. Design a capability-evidence pipeline that would not normalize its own gaps.

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards

FURTHER READING

• GAO-17-809T, Navy Readiness (2017)
• Rasmussen (1997), "Risk management in a dynamic society"
• Snook, Friendly Fire (2000)

LENS COURSES LEN 1 LEN 5 LEN 8 LEN 3

CASE 3 **AVIATION** 2009

Air France Flight 447

[T] Training Gap [H] Human-System Interface

IMPACT 228 killed; the deadliest accident in Air France's history

54 SECONDS: STALL WARNING SOUNDED, RESPONSE NEVER CAME

BACKGROUND

Air France Flight 447 left Rio de Janeiro for Paris on the night of 31 May 2009 with 228 people aboard an Airbus A330 — a highly automated, fly-by-wire airliner with one of the best safety records in the sky. Its route crossed the Intertropical Convergence Zone, the band of towering equatorial thunderstorms that long-haul crews routinely pick their way around. What the crew could not see was a vulnerability already known to the manufacturer and the airline: the aircraft's pitot probes, which measure airspeed, were prone to brief blockage by high-altitude ice crystals. A series of unreliable-airspeed events on A330s and A340s had already prompted a program to replace the probes, and Air France had begun retrofitting its fleet — but the accident aircraft had not yet been modified.[0]

WHAT HAPPENED

A few hours into the flight, at 35,000 feet, the pitot probes iced over. The airspeed readings became invalid and the systems that depend on them responded as designed: the autopilot and autothrust disconnected and handed the aircraft to the crew, with the flight-control law degraded so that the envelope protections the pilots normally relied on were no longer there. The pilot flying responded with sustained nose-up inputs. The aircraft zoom-climbed, bled off speed, and entered an aerodynamic stall from which it never recovered, descending some 38,000 feet into the ocean in about four and a half minutes. The captain was on a scheduled rest break when the trouble began; the two first officers left at the controls could not reconcile the contradictory indications, and when the captain hurried back into the cockpit moments into the emergency, the attitude, the alarms, and the readouts gave him no quick way to see that his airplane was stalling. For much of the descent neither pilot seems to have registered that the other was holding the sidestick nearly full aft. The stall warning sounded for roughly fifty seconds; then, at the extreme angles of attack the aircraft reached, the airspeed data fell below the threshold the warning logic trusted and the alarm went silent — only to sound again each time the nose was lowered, which was the one input that would have begun a recovery. The system, in effect, warned against the correct action.[1]

THE INVESTIGATION

The Bureau d'Enquêtes et d'Analyses (BEA), France's accident investigator, published its final report in July 2012. The probable cause was a chain: the temporary loss of valid airspeed from pitot icing, the crew's inappropriate control inputs that destabilized the flight path, and — decisively — the crew's failure to recognize that the aircraft had stalled and to apply the standard recovery.[2] The pilots were not incompetent; they were operating outside anything their training had prepared them for, and the cues that should have told them they were stalling never registered. The BEA stated the training failure in institutional terms: "the combination of the warning system ergonomics, the conditions under which pilots are trained and exposed to stalls during their professional and recurrent training, did not result in reasonably reliable expected behaviour patterns."[3]

THE CAPABILITY GAP

The gap was specific and, in hindsight, glaring. Airlines trained stall recognition and recovery at low altitude, in the takeoff and landing regime where stalls were thought to be the real risk. The flight simulators of the era could not faithfully reproduce how an airliner behaves in a fully developed stall at cruise altitude, so that scenario was simply never practiced. Crews were trained to **prevent** stalls and to manage an automated aircraft; they were not trained to **recover** a large jet hand-flown into a stall at 35,000 feet, because the industry's training environment could not produce the situation. The trained capability envelope and the operational capability envelope had quietly diverged, and AF447 fell precisely into the gap between them.[4] There was a deeper version of the same gap. Years of reliable automation had let routine hand-flying skills atrophy across the profession; pilots spent their careers managing systems rather than flying raw, and the rare moment when the automation handed control back — at night, in weather, with instruments in disagreement — was the moment they were least equipped to meet. The capability the airline assumed its pilots held was a capability the operation itself had slowly eroded.

AFTERMATH & REFORM

The BEA's final report carried dozens of safety recommendations whose reach went far beyond one airline. The vulnerable pitot probes were replaced across the fleet. The recommendations pressed for training in manual flying at high altitude, in approach-to-stall and full-stall recovery, and in handling unreliable airspeed, and for better angle-of-attack information in the cockpit.[5] Most consequentially, the accident reshaped the global training standard: regulators moved to require Upset Prevention and Recovery Training — explicit instruction in manual handling and in recovering an aircraft from unusual attitudes and stalls — for airline pilots, closing at the regulatory level the gap that had been invisible at the airline level.[6] AF447 is in this book because the crew performed exactly as trained. The training was the wrong training.

REFERENCES

1. Bureau d'Enquêtes et d'Analyses (BEA), Final Report on the accident on 1 June 2009 to the Airbus A330-203 registered F-GZCP operated by Air France flight AF447 (July 2012) — full report: flight, aircraft, and the known pitot-icing vulnerability with retrofit pending.
2. BEA, Final Report AF447 (2012) — accident sequence: autopilot/autothrust disconnection, degraded control law, sustained nose-up input, stall, and the stall-warning logic dropping out at high angle of attack.
3. BEA, Final Report AF447 (2012) — probable cause: airspeed inconsistency from pitot icing, inappropriate control inputs, and failure to recognize and recover from the stall.
4. BEA, Final Report AF447 (2012) — finding on warning-system ergonomics and stall-training conditions (quoted).
5. G. Palmer / E. Strickland, "Air France Flight 447 Crash Caused by a Combination of Factors," IEEE Spectrum (2014) — analysis of the divergence between the trained and operational envelopes.
6. BEA, Final Report AF447 (2012), safety recommendations — pitot-probe replacement, manual high-altitude flying, approach-to-stall and stall recovery, unreliable-airspeed procedures, and angle-of-attack indication.
7. European Union Aviation Safety Agency, Upset Prevention and Recovery Training (UPRT) requirements for airline pilots (phased in by 2019); ICAO, Manual on Aeroplane Upset Prevention and Recovery Training (Doc 10011).

THE LEARNING ENGINEERING LENS

> *The crew never understood that they were stalling and consequently never applied a recovery manoeuvre.*
>
> BEA, FINAL REPORT ON AIR FRANCE FLIGHT 447, JULY 2012

LE INSIGHT

AF447 is the canonical case of training that matched one envelope of operation perfectly and another not at all. Stall **prevention** training was excellent. Stall **recovery from cruise altitude** training did not exist because the simulators could not produce it. The pilots performed exactly as trained. The training was the wrong training.

LENS APPROACH

LENS uses AF447 in LEN 5 to teach the difference between the trained capability envelope and the operational capability envelope, and in LEN 2 to introduce human-AI handoff as a capability problem: the autopilot disconnection was a transition the crew had been trained to avoid rather than to handle. The case maps directly to canonical problem type three — Capability Cliff at Automation Boundary.

REFLECTION QUESTIONS

1. The simulators of 2009 could not produce high-altitude stall behavior. What is the equivalent gap in your domain — the operational regime your training environment cannot reproduce?
2. Design the recurrent-training curriculum that would have caught the AF447 gap. Be specific about cost, evidence, and what makes the curriculum falsifiable against the operational record.

WHO BUILDS THIS learning science & instructional design · human factors & interaction design — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs

FURTHER READING

- Dismukes, Berman & Loukopoulos (2007), The Limits of Expertise
- Casner & Hutchins (2019), “What Do We Tell the Drivers?”
- Sarter & Woods (1995), “How in the World Did We Ever Get into That Mode?”

LENS COURSES LEN 1 LEN 5 LEN 2

CASE 5 ENERGY 1979

Three Mile Island

T Training Gap H Human-System Interface

IMPACT Partial meltdown of a Babcock & Wilcox PWR; minimal off-site dose; most serious accident in U.S. commercial nuclear history; catalyst for industry-wide reform

A MINOR COOLING FAULT, AN AMBIGUOUS INDICATOR, A MELTDOWN

BACKGROUND

Unit 2 at Three Mile Island, near Harrisburg, Pennsylvania, was a Babcock & Wilcox pressurized-water reactor run by operators whose training emphasized large, fast, design-basis accidents — the worst-case ruptures the industry believed it most needed to guard against. The slow, ambiguous, small-break sequence that actually unfolded was not the scenario they had drilled.[0] It should not have been a surprise. Eighteen months earlier, in September 1977, a nearly identical stuck-open relief valve had occurred at the Davis-Besse plant, another B&W reactor; operators there caught it, but neither the utility, the vendor, nor the Nuclear Regulatory Commission grasped the event's significance or pushed its lesson out to the fleet. The warning that might have prepared TMI's crew never reached them.[1] Many of the operators had come up through the Navy's nuclear program and were capable, disciplined people; what they had been given was a body of training and a set of procedures built around the accidents the industry expected, not the one that the plant would actually hand them at four in the morning.

WHAT HAPPENED

Early on 28 March 1979 a minor disturbance in the secondary cooling system tripped the reactor. A pilot-operated relief valve opened to bleed off pressure, as designed — and then stuck open, quietly draining coolant from the core. The control room gave the operators no honest picture of this: the indicator light reported the **command** to close the valve, not its actual position, so the panel said “closed” while the valve stayed open. Reading the rising pressurizer level as a sign the system was about to go “solid” with water, the operators throttled back the high-pressure injection that was the one thing feeding the starving core — cutting the coolant precisely when more was needed. About half the core melted. The crisis then deepened for days before it was understood: a hydrogen bubble in the reactor vessel raised fears of an explosion, conflicting information reached officials and the public, and the governor advised pregnant women and pre-school children within five miles to leave. The reactor was eventually brought to cold shutdown and Unit 2 was destroyed; off-site radiation release, as it turned out, was minimal.[2]

THE INVESTIGATION

The President’s Commission on the Accident at Three Mile Island — the Kemeny Commission — reported in October 1979. Its central finding inverted the industry’s assumptions: “the fundamental problems are people-related problems and not equipment problems.” The equipment, it judged, was good enough that, but for the human failures, the accident would have been a minor incident. The criticism reached past the operators to plant management, the utility, and the NRC itself — an institution-wide mindset that treated serious accidents as effectively impossible.[3] The commission was specific about where capability had failed: operator training was too narrow and too oriented to theory rather than to the diagnosis of real, messy transients; the control room was a wall of nearly identical gauges and annunciators that buried the few signals that mattered; and procedures gave little help in an event that did not match the script.

THE CAPABILITY GAP

The gap was not a deficit of intelligence or diligence but of the right capability for the situation that arrived. Operators trained to recognize design-basis ruptures had no rehearsed model for an ambiguous cascade, and a control room whose indicators reported commands rather than states made the correct diagnosis nearly impossible to assemble in real time. Layered beneath that was a second-order failure: the capability to **learn** — to turn the Davis-Besse near-miss into fleet-wide knowledge — had itself broken down. The accident sat at the intersection of training, interface, and institutional memory, which is why it resists any single-cause reading.[4]

AFTERMATH & REFORM

TMI did not produce another reactor accident. It produced a system of capability reform. Within nine months the industry created the Institute of Nuclear Power Operations (December 1979) to set operating standards, evaluate plants against them, and — directly addressing the Davis-Besse failure — force the systematic sharing of operating experience among utilities.[5] The NRC overhauled operator licensing and required plant-referenced simulators and human-factors review of control rooms; it stationed resident inspectors at every plant, rebuilt off-site emergency planning, and tied operator training to a national accreditation standard so that the diagnosis of ambiguous transients — not just textbook ruptures — became something crews actually practiced. TMI is paired later in this book with INPO (Case 16) as its strongest argument: the same incident that exposed how training, interface, and institutional memory can fail together also generated the most durable example of engineered reform in the book.[6]

REFERENCES

1. Kemeny Commission, Report of the President's Commission on the Accident at Three Mile Island (Oct. 1979) — operator training oriented to large design-basis accidents.
2. M. Rogovin & G. T. Frampton, Three Mile Island: A Report to the Commissioners and to the Public, NUREG/CR-1250 (U.S. NRC, 1980) — the September 1977 Davis-Besse stuck-PORV precursor and the failure to disseminate its lessons.
3. U.S. Nuclear Regulatory Commission, Backgrounder on the Three Mile Island Accident — accident sequence, misleading PORV indication, throttled high-pressure injection, 50% core damage, minimal off-site dose.
4. Kemeny Commission (1979) — central conclusion that the fundamental problems were people-related, not equipment-related (quoted).
5. C. Perrow, Normal Accidents (1984); J. Reason, Human Error (1990) — why ambiguous cascades defeat design-basis training and command-not-state interfaces.
6. Institute of Nuclear Power Operations, "Our History" — INPO established December 1979 to set standards and force sharing of operating experience; see also NEI, "Lessons from the 1979 Accident at Three Mile Island."
7. J. V. Rees, Hostages of Each Other: The Transformation of Nuclear Safety since Three Mile Island (Univ. of Chicago Press, 1994) — TMI as the origin of systematic capability reform.

THE LEARNING ENGINEERING LENS

> *The fundamental problems are people-related problems and not equipment problems.*
>
> KEMENY COMMISSION REPORT ON THREE MILE ISLAND, 1979

LE INSIGHT

TMI is the textbook moment when an industry discovered that its capability assumptions did not survive contact with operational reality. Training to design-basis events left the operators blind to the genuinely ambiguous failures that complex systems actually produce. The human element is not a residual risk but the dominant one.

LENS APPROACH

LENS uses TMI in LEN 1 as the foundational problem-framing case: when trained scenarios and operational scenarios diverge, the divergence is the real risk. LEN 5 examines the control-room interface and the gap between what indicators displayed and what operators inferred; the pairing with INPO (Case 16) is the Failure-to-Reform arc this book treats as its strongest argument.

REFLECTION QUESTIONS

1. TMI operators were trained for worst-case scenarios but failed in an ambiguous one. What is the equivalent training gap in your domain between the trained case and the messy case?

2. The Kemeny Commission called the human element the dominant risk. What evidence would you need to demonstrate the same conclusion in your own domain?

WHO BUILDS THIS learning science & instructional design · human factors & interaction design — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs

FURTHER READING

- Walker, Three Mile Island: A Nuclear Crisis in Historical Perspective (2004)
- Perrow, Normal Accidents (1984)
- Reason, Human Error (1990)

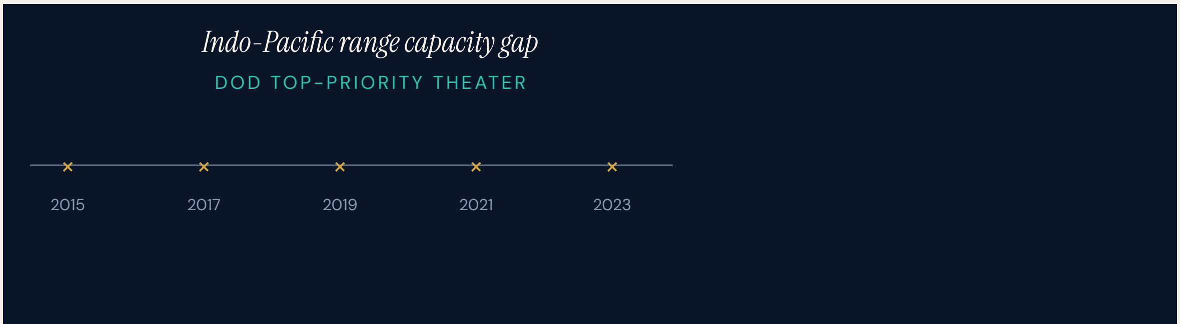
LENS COURSES LEN 1 LEN 5

CASE 9 DEFENSE ONGOING

Marine Corps Training in the INDOPACOM AOR

T Training Gap K Knowledge & Institutional Memory

IMPACT Structural readiness gap in DoD's stated top-priority theater



DECADE OF UNMET TRAINING IN DOD'S PRIORITY THEATER

BACKGROUND

The 2022 National Defense Strategy names the Indo-Pacific the Department of Defense’s priority theater and the People’s Republic of China its “pacing challenge” — the single competitor against which the U.S. military is meant to measure and build itself.[0] Set against that declaration is an awkward fact about the theater so designated: it has among the least mature live-training infrastructure in the force. The place the strategy calls most important is, in terms of where forces can actually rehearse the fight, one of the least built out.[1]

WHAT HAPPENED

This case has no single date because the failure is a persistent condition rather than an event. For nearly a decade the U.S. Marine Corps has been unable to meet its training requirements at ranges in the Indo-Pacific Command area of responsibility. The Corps has not simply absorbed the shortfall; it has papered over it with workarounds — rotating units back to the continental United States to use ranges that exist there, substituting virtual and simulated training for live events, and pressing multinational exercises into proxy duty for capabilities that have no home range in theater. The workarounds keep units busy and partially trained; the structural gap they are compensating for does not close.[2]

THE INVESTIGATION

In May 2024 the Government Accountability Office examined military readiness across the services and documented the Marine Corps strand in detail: the requirements unmet, the decade over which they had gone unmet, and the workarounds standing in for the real thing. GAO recommended that the Marine Corps complete a formal analysis of its unmet training requirements and develop a plan to identify and remediate them at Indo-Pacific ranges; the Department concurred.[3] The report was not the first warning. GAO has pressed the same family of readiness recommendations across successive reviews and in testimony to Congress, and the limits of overseas training — and their absence from readiness reporting — were flagged in its work two decades earlier.[4] What the workarounds cannot reproduce is the very thing the ranges exist to provide: the chance to rehearse, at scale and against a realistic threat, the specific operations the theater would demand. Rotating a unit back to the continental United States trains it on American terrain and range geometry, not the distances, basing, and electromagnetic environment of the Western Pacific. Virtual and constructive simulation fills part of the gap but cannot fully stand in for live, instrumented, force-on-force events. Multinational exercises build relationships and interoperability but are scheduled around partners' priorities, not around the Marine Corps' unmet requirements. GAO's recommendation was therefore not simply to train more but to first establish what, precisely, the unmet requirements are — to convert a decade of acknowledged shortfall into a documented, prioritized list against which progress could be measured and funded. Until that analysis exists, the Corps cannot say with rigor how large the gap is, which makes it easy for the gap to persist, unmeasured, behind the reassuring activity of the workarounds.

THE CAPABILITY GAP

What makes INDOPACOM diagnostically useful is the inverse correlation between stated priority and engineered priority. A theater can be named the pacing challenge in every strategy document and briefing while the training infrastructure to operate in it remains structurally short for ten years — and that is not a contradiction the system experiences as urgent, because capability follows where resources and construction actually flow, not where strategy points. Declared priority is cheap; engineered priority — ranges, instrumented airspace, basing, the pipeline of certified units — is expensive and slow, and the gap between the two is the real measure of intent. For a decade that measure has pointed the same way. Building a theater's training capability is not a briefing-slide commitment but a construction program — ranges with the space and instrumentation to rehearse modern fires and maneuver, airspace and sea space cleared for it, host-nation agreements, basing, and a steady pipeline of units that have actually trained there and been certified. None of that materializes from a document's wording; all of it competes for the same dollars and political capital as the deployments that consume the force day to day. The Marine Corps' workarounds are evidence of professionals doing their best inside that constraint — and evidence that best effort inside a structural gap is not the same as closing it.

AFTERMATH & REFORM

The reform, such as it is, remains mostly prospective. The GAO recommendations were open as of the 2024 report: an analysis of unmet requirements to be completed and a remediation plan to be built and funded.[5] Whether the gap closes will be decided not by another strategy document but by whether the Marine Corps and the Department convert the recommendation into programmed ranges, dollars, and schedule. The case sits at the front of this book as the live, ongoing counterpart to its historical failures: a capability shortfall that is fully recognized, repeatedly documented, and still — at the time of writing — not engineered away.

REFERENCES

1. 2022 National Defense Strategy of the United States of America (U.S. Department of Defense, 2022) — the Indo-Pacific as priority theater and China as the "pacing challenge."
2. U.S. Government Accountability Office, Military Readiness: Actions Needed for DOD to Address Challenges, GAO-24-107463 (May 2024) — the least-mature training infrastructure in the priority theater.
3. GAO-24-107463 (2024) — the Marine Corps unable to meet INDOPACOM training requirements for nearly a decade, and the CONUS-rotation, virtual, and multinational-exercise workarounds.
4. GAO-24-107463 (2024) — recommendation that the Marine Corps analyze unmet training requirements and develop a remediation plan for Indo-Pacific ranges; DoD concurrence.
5. GAO, Military Readiness testimony before the Senate Armed Services Committee (D. C. Maurer) — the sustained pattern of readiness recommendations; see also GAO-02-525, Military Training: Limitations Exist Overseas but Are Not Reflected in Readiness Reporting (2002).
6. GAO-24-107463 (2024) — recommendation status (open as of 2024); the gap not yet remediated.

THE LEARNING ENGINEERING LENS

> *Without meeting these requirements, the Marine Corps cannot ensure that its forces will be ready for combat.*
>
> PARAPHRASING GAO-24-107463, MILITARY READINESS, 2024

LE INSIGHT

INDOPACOM illustrates the difference between declared priority and engineered priority. A theater designated as DoD's pacing challenge while the training infrastructure to operate in it remains structurally short is not a resourcing oversight — it is a capability architecture failure. Capability follows where resources actually flow, not where briefings name as critical.

LENS APPROACH

LENS uses INDOPACOM in LEN 5 to teach the gap between stated requirements and engineered requirements, and in LEN 8 to examine the organizational dynamics that allow declared priorities to coexist with unfunded capability gaps for a decade. The case is also a live test for any student's claim about how to engineer cross-service capability at scale.

REFLECTION QUESTIONS

1. In your domain, what is the gap between **declared** priority and **engineered** priority? How would you measure it?
2. Construct the capability requirements artifact for a theater you do not currently operate in. What would it cost, and who would sign for it?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines

FURTHER READING

• GAO-24-107463 and predecessor reports
• Fixsen et al. (2005), Implementation Research
• Rasmussen (1997), "Risk management in a dynamic society"

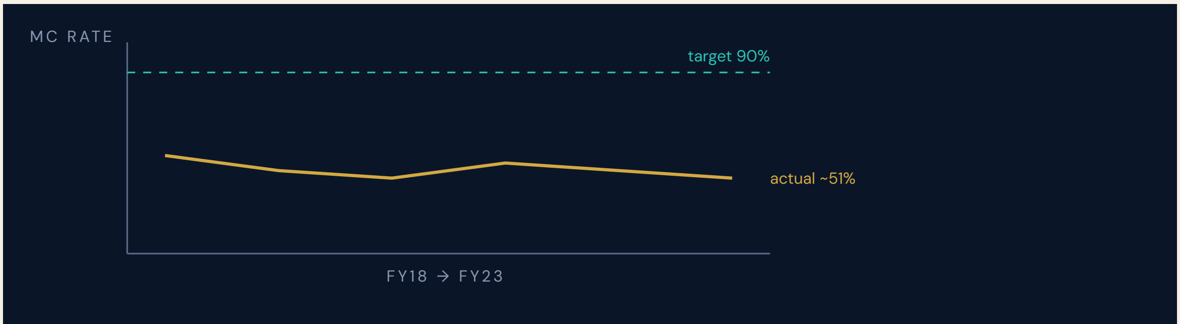
LENS COURSES LEN 5 LEN 8

CASE 26 DEFENSE ONGOING

F-35 Sustainment & Maintainer Shortage

T Training Gap K Knowledge & Institutional Memory D Designed Out

IMPACT Fleet mission-capable rate about 55% (March 2023), far short of program goals; lifecycle cost exceeds $1.7T, with ~$1.3T in operating and support; maintainer, technical-data, and depot shortfalls are the binding readiness constraint (GAO-23-105341)



MISSION-CAPABLE RATE VS. FLEET TARGET

BACKGROUND

The F-35 is the most expensive weapons program in history. The Department of Defense plans to buy nearly 2,500 of the jets, and the program's estimated lifecycle cost now exceeds $1.7 trillion — of which roughly $1.3 trillion is not the aircraft at all but the decades of operating and sustaining them.[0] That ratio is the whole story in miniature: the flyaway jet is the visible, finite part; keeping a global fleet of them mission-ready — the maintainers, the technical data, the repair depots — is the open-ended part, and it is the part that has lagged from the beginning.

WHAT HAPPENED

Like the INDOPACOM case, this failure is a standing condition rather than an event. As of March 2023 the F-35 fleet's mission-capable rate stood at about 55 percent — the share of aircraft able to perform at least one tasked mission — far short of the program's goals. Behind that number sat a maintenance system that could not keep up: more than 10,000 components were waiting in the repair queue, above desired levels, and the service depots were taking, on average, about 72 days to turn a component around. The depots were themselves behind schedule in standing up the capacity to do the repairs at all, which left the fleet leaning on a pipeline that could not meet demand.[1]

THE INVESTIGATION

GAO's September 2023 review carried an unusually blunt title: the Department and the services **need to reassess the future sustainment strategy**. The report traced the readiness shortfall to a cluster of human-capability and information problems — shortages of trained maintainers, the military's lack of access to the technical data needed to do its own repairs, and the resulting dependence on the prime contractor for work the services cannot perform themselves.[2] None of this was a surprise. GAO has reported on F-35 sustainment year after year, and successive reviews have repeated the same diagnosis even as procurement continued and the readiness numbers stayed flat.[3] The information problem had a name. For years the fleet's maintenance ran through the Autonomic Logistics Information System, a centralized software backbone so cumbersome and error-prone that maintainers improvised workarounds simply to keep aircraft moving; the program eventually committed to replacing it, but the replacement has come slowly, and a maintenance enterprise can only be as good as the system that schedules its parts, tracks its components, and authorizes its work. Layered on top were the maintainer shortfalls — too few trained people for a fleet still growing — and depots that could not absorb the repair demand. The reports return to the same structural point year after year: the sustainment model was built around assumptions about cost, contractor support, and data access that have not held, and adjusting it at the margins has not changed the trajectory.

THE CAPABILITY GAP

The F-35 is the cleanest modern example of a platform fielded faster than the capability infrastructure to sustain it. Aircraft were delivered on a production schedule; the maintainers to keep them flying, the technical data to repair them, and the depots to overhaul them were treated as follow-on costs to be sorted out later rather than as deliverables that had to arrive with the jets. The result is a fleet that exists physically but cannot be kept ready, because the human and informational half of the system was never engineered to match the hardware half — and that half does not catch up on its own. The hardest part of the gap is less the headcount of maintainers than the access to information. Much of the technical data needed to diagnose and repair the aircraft has stayed controlled by the prime contractor rather than delivered to the government, so the services cannot freely write their own procedures, stand up and qualify their own depots, or compete the work — a dependency that is merely expensive in peacetime and genuinely dangerous in a conflict that would strain the contractor's own supply chain. Fielding the jets without first securing that data was a choice whose cost arrives every day an aircraft sits grounded for want of a repair the military is not permitted to perform itself.

AFTERMATH & REFORM

GAO recommended that the Department reassess its entire sustainment strategy — how it secures government access to technical data, builds depot capacity, and grows the maintainer workforce — rather than keep patching a model that was not delivering.[4] Those recommendations remain a work in progress: the program's later reviews show costs continuing to climb while mission-capable rates stay below goal.[5] The F-35 sits in this book as the live argument for treating capability infrastructure — people, data, and the means to sustain them — as a fielding gate rather than an afterthought. The bill for skipping that gate does not disappear; it compounds.

REFERENCES

1. U.S. Government Accountability Office, F-35 Aircraft: DOD and the Military Services Need to Reassess the Future Sustainment Strategy, GAO-23-105341 (Sept. 2023) — 2,500 planned aircraft and a lifecycle cost exceeding $1.7 trillion, $1.3 trillion of it in operating and support.
2. GAO-23-105341 (2023) — 55% fleet mission-capable rate (March 2023); over 10,000 components awaiting repair; 72-day average depot turnaround; depot stand-up behind schedule.
3. GAO-23-105341 (2023) — sustainment shortfall traced to maintainer shortages, lack of military access to technical data, and contractor dependency.
4. GAO, F-35 Sustainment: DOD Faces Several Uncertainties..., GAO-22-105995 (2022), and the broader GAO F-35 series — the recurring, year-over-year diagnosis.
5. GAO-23-105341 (2023) — recommendation that DOD reassess the future sustainment strategy.
6. GAO, F-35 Sustainment: Costs Continue to Rise While Planned Use Has Decreased, GAO-24-106703 (2024) — costs rising while readiness stays below goal.

THE LEARNING ENGINEERING LENS

> *Organizational-level maintenance has been affected by a number of issues, including a lack of technical data and training.*
>
> PARAPHRASING GAO-23-105341, F-35 AIRCRAFT, 2023

LE INSIGHT

F-35 is the live, current example of fielding a platform faster than its capability infrastructure can be built. The aircraft is the easy part; the maintainers are the hard part. A decade of program schedules treated maintainer training as a follow-on cost, not a fielding gate. The fleet is now operating at half of its design readiness, and the program of record is more than a trillion dollars over its 2018 estimate.

LENS APPROACH

LENS treats the F-35 in LEN 5 as the canonical case of **Capability- System Misalignment at Transition**, the program's first canonical problem type. Students design the capability infrastructure that should have accompanied each lot of aircraft delivered, including the training pipeline, technical-data deliverables, and depot establishment.

REFLECTION QUESTIONS

1. Pick a current technology platform in your domain. Estimate the capability infrastructure that must field with it. What happens if half of that infrastructure is years behind the hardware?
2. The F-35 program treated maintainer training as follow-on cost. Design a fielding gate that would prevent that decision being available to a future program manager.

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · task analysis · design prototyping

FURTHER READING

- GAO F-35 series, 2014–present
- Fixsen et al. (2005), Implementation Research
- Augustine, Augustine's Laws (1986)

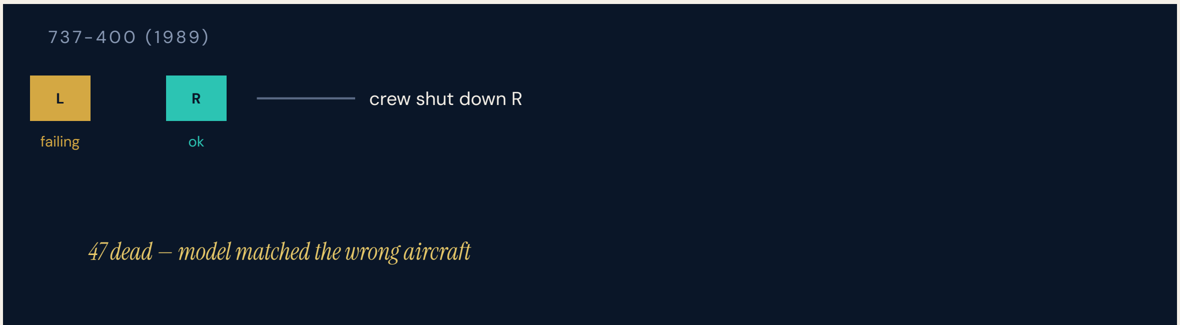
LENS COURSES LEN 5 LEN 8 LEN 3

CASE 30 AVIATION 1989

Kegworth / British Midland 92

T Training Gap H Human-System Interface K Knowledge & Institutional Memory

IMPACT 47 killed and 74 seriously injured when the crew shut down the wrong engine



CREW MENTAL MODEL FROM OLDER VARIANT; CONFIGURATION CHANGED

BACKGROUND

On 8 January 1989 British Midland Flight 92, a nearly new Boeing 737-400 registered G-OBME, left London Heathrow for Belfast with 126 people aboard. The 737-400 was a recent variant — a stretched fuselage, uprated CFM56 engines, and a partly redesigned cockpit with new electronic engine instruments in place of the older round dials. The crew were experienced 737 pilots, but their conversion onto the new variant had been brief and largely self-directed; the differences from the airplane they had been flying were communicated as a matter of reading, not of practiced retraining.[0] The variant had entered service only months earlier, and British Midland was among its first operators; the type's introduction had outpaced the depth of crew familiarity that comes from years on an airframe. Pilots converting from the older 737-200 received a conversion course measured in days, much of it covering the new electronic instrument displays as information to be absorbed rather than as a perceptual change to be drilled. By every formal measure the crew were qualified on the aircraft they were flying that night. What they did not yet have was the kind of familiarity that turns the right response into a reflex when an engine tears itself apart at altitude and the cabin fills with smoke.

WHAT HAPPENED

Climbing through about 28,300 feet, a fan blade fractured in the left engine, producing heavy vibration, a burning smell, and smoke in the cabin. The crew had seconds to decide which engine was failing. Reasoning from the older 737s they knew — on which the air-conditioning bleed drew from the right engine — they concluded the right engine was the source of the smoke and throttled it back. The symptoms eased, which seemed to confirm the diagnosis, and they shut the right engine down. It was the one still working. Believing they had isolated the trouble, they did not restart it; the vibration had eased when they throttled the right engine back — partly because the still-active autothrottle had reduced power on the failing left engine at the same moment — and that easing read to them as confirmation. They began a diversion to East Midlands Airport, and for some minutes the crippled left engine ran well enough to sustain the descent while the crew worked checklists, radios, and a frightened cabin, the window to reconsider slipping past. On final approach, when they called for more thrust, the damaged left engine failed completely — and with the good engine already shut down, the aircraft had no power at all. It struck the embankment of the M1 motorway short of the runway, breaking apart just short of safety. Forty-seven people were killed and seventy-four seriously injured.[1]

THE INVESTIGATION

The Air Accidents Investigation Branch traced the disaster to the shutdown of the serviceable engine, and to why two competent pilots made that choice. The decisive factor was a mental model carried from earlier 737 variants and applied to a -400 on which Boeing had changed the bleed-air configuration; the difference training had never disturbed the old model.[2] The new electronic engine displays did not help: the secondary instruments, including the vibration indicator that would have pointed straight at the failing engine, were harder to read at a glance than the dials they replaced.[3] And although cabin crew and passengers could see flames from the left engine, that observation never reached the flight deck — a crew-resource-management gap of exactly the kind aviation would later train hard against.[4]

THE CAPABILITY GAP

The Kegworth crew were skilled, and their mental model was correct — for the airplane they had flown the week before. The -400 had been treated, for training purposes, as an incremental change to a familiar type; under the time pressure of an in-flight emergency it behaved as a categorical one. The training system had not engineered the difference with enough force to overwrite a deeply practiced prior model, and the cockpit redesign had quietly removed some of the perceptual cues that a startled crew relies on. Capability had degraded under system change without anyone noticing, because nothing had failed until the day it did.[5] It is worth being precise about what “training” would have had to do. Reading a manual that notes the -400′s revised bleed-air arrangement is not the same as building, through practice under realistic pressure, a new reflex that fires before the old one. The crew did not reach for the wrong control out of carelessness; they reached for the model their thousands of prior hours had built, and the conversion course had given them nothing strong enough to override it in the seconds that mattered. The gap was not in the pilots’ diligence but in the design of the transition itself.

AFTERMATH & REFORM

The AAIB issued thirty-one safety recommendations, reaching across conversion and difference training, engine-instrument and vibration-display design, communication between cabin and flight deck, and occupant crashworthiness.[6] The accident became a standard teaching case in human factors and a reference point for how a transition program should be built — so that the differences between an old system and a new one are made hard to overlook rather than easy, and so that the people closest to the evidence have a path to the people making the decision.

REFERENCES

1. Air Accidents Investigation Branch, Report on the accident to Boeing 737-400 G-OBME near Kegworth, Leicestershire, on 8 January 1989, AAIB Aircraft Accident Report 4/90 (1990) — aircraft, route, and the limited conversion training; see also U.S. FAA, Lessons Learned: G-OBME.
2. AAIB Report 4/90 (1990) — accident sequence: left-engine fan-blade failure, shutdown of the serviceable right engine, and total power loss on approach; 47 killed, 74 seriously injured.
3. AAIB Report 4/90 (1990) — the older-737 bleed-air mental model misapplied to the 737-400′s changed configuration.
4. AAIB Report 4/90 (1990) — the new electronic engine instrumentation and the readability of the vibration indicator.
5. AAIB Report 4/90 (1990) — the cabin-to-flight-deck communication gap (crew resource management).
6. AAIB Report 4/90 (1990) — the 31 safety recommendations on difference training, instrument design, CRM, and crashworthiness.
7. J. Reason, Human Error (Cambridge Univ. Press, 1990) — Kegworth as a case in the misapplication of a correct mental model to a changed system.

THE LEARNING ENGINEERING LENS

> *The crew's mental model of the older 737 was inappropriately applied to the 737-400 on which they were operating.*
>
> PARAPHRASING AAIB AIRCRAFT ACCIDENT REPORT 4/90 ON BRITISH MIDLAND 92, 1990

LE INSIGHT

Kegworth is the textbook example of Capability Degradation Under System Change. The pilots were excellent. Their mental model was excellent — for the airframe they had flown the previous week. The difference training had not been engineered to disturb the prior model with enough force to keep it from being misapplied. The new airframe was treated as an incremental change. The crew's response revealed it was a categorical one.

LENS APPROACH

LENS treats Kegworth in LEN 5 as the canonical **system-change** capability problem and in LEN 8 as the institutional version: how knowledge about what has changed travels from engineering to operations and what gets lost in transit. The case sits alongside Patriot (Case 19) in the canonical problem-type pair for system transition.

REFLECTION QUESTIONS

1. The Kegworth crew's mental model was right for the previous variant. What change in your domain currently risks an analogous misapplication?
2. Difference training is a generic deliverable in transition programs. Design the artifact that would make differences hard to overlook rather than easy.

WHO BUILDS THIS learning science & instructional design · human factors & interaction design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs · knowledge bases · data pipelines

FURTHER READING

• Wickens, Engineering Psychology and Human Performance (multiple eds.)
• Sarter & Woods (1995), "How in the World Did We Ever Get into That Mode?"
• Boeing 737 NG/MAX difference-training literature (post-2020)

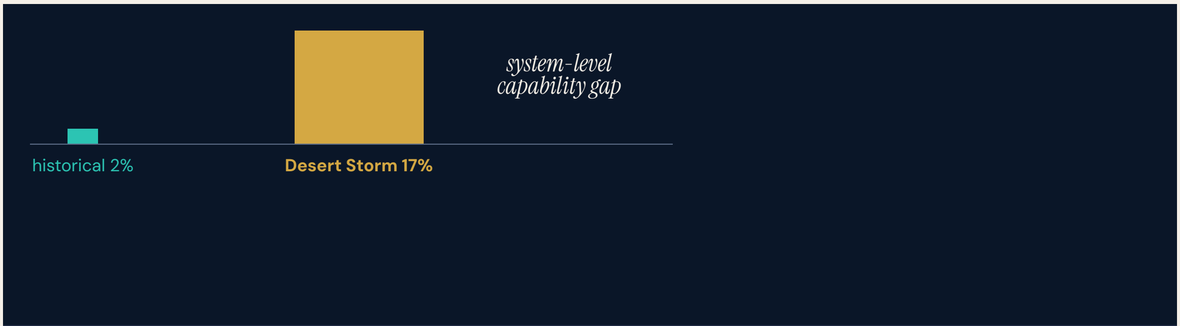
LENS COURSES LEN 5 LEN 8

CASE 33 DEFENSE 1991 – 2014

Military Fratricide — Desert Storm to Afghanistan

T Training Gap H Human-System Interface K Knowledge & Institutional Memory

IMPACT 24% of U.S. KIA in Desert Storm from friendly fire (35 of 148, per the DoD friendly-fire report) — an order of magnitude above the historical baseline



FRIENDLY-FIRE RATE, U.S. CASUALTIES, DESERT STORM VS. HISTORICAL

BACKGROUND

Friendly fire is as old as organized war, but its true rate has always been hard to pin down. The most-cited historical estimate came from Charles Shrader, whose 1982 study Amicicide examined 269 incidents and concluded that fratricide accounted for something under two percent of battle casualties. That figure later proved fragile: subsequent analysts argued the real historical rate was closer to ten or fifteen percent, and Shrader himself stepped back from the two-percent number. The disputed baseline matters because it became the yardstick against which a modern war would be measured.[0]

WHAT HAPPENED

The 1991 Gulf War made the question concrete and grim. Of 146 U.S. service members killed in action, 35 — about 24 percent — died by friendly fire; of 467 wounded, 72 (about 15 percent) were hit by their own side.[1] The toll was not abstract. An A-10 strike on U.S. LAV-25 light-armored vehicles near Khafji killed seven Marines after a coordinated engagement went wrong, and a single A-10 attack on British Warrior fighting vehicles killed nine soldiers. Whatever the right historical baseline, a quarter of American combat deaths coming from one's own forces was a number no one could explain away as the ordinary friction of battle.[2] The incidents shared a grim signature: the weapons worked exactly as designed, and the people firing them were competent and trying to do the right thing. What failed was the chain that lets a shooter know, in the seconds available, whether the vehicle or the position in the sights is friend or foe — a forward controller's read of a fluid ground picture, the coordination between aircraft and the units below them, the markings and signals meant to say "us." In the desert, at night, in fast-moving armored battle, that chain broke repeatedly, and it broke not because any one part was obviously deficient but because the parts did not hold together under the speed and confusion of combat.

THE INVESTIGATION

The post-war reviews converged on a familiar set of causes: the chaos and confusion of combat, inadequate situational awareness, weak adherence to fire-control measures, and combat-identification failures — crews and controllers unable to tell friend from foe fast enough under fire.[3] One finding cut deeper than the rest. Analysts noted that the military lacked a comprehensive, accessible record of fratricide incidents, which meant the institution could not study its own pattern rigorously. The capability to **learn** from friendly fire was itself missing — a second-order gap sitting beneath the first.[4] That second-order gap mattered as much as any single engagement. Without a rigorous, shared record, the services could not separate the incidents that arose from training from those that arose from doctrine, equipment, or the irreducible chaos of battle; they could not tell whether a given fix was working or whether an apparent improvement was simply a quieter war. Each service investigated its own tragedies, often thoroughly, but the lessons stayed local, and the pattern across them — the part that would have told the institution where to invest — went largely unassembled. A problem that is not measured well is a problem that is easy to declare solved and easy to keep having.

THE CAPABILITY GAP

Fratricide is not one problem but the failure of several systems to integrate: situational awareness, fire-control discipline, combat identification, and joint coordination each became the subject of dedicated programs, yet the rate did not fall to anything like a negligible level in the conflicts that followed. In Afghanistan a satellite-guided bomb strike went fatally wrong after a controller's device reset its coordinates following a battery change, and in 2014 a B-1's targeting pod could not detect the infrared strobes U.S. troops were wearing to mark themselves, and the strike killed five. Each contributor had been worked on; the integration **across** them — the property that actually keeps friendly forces from killing each other — is the hardest thing to engineer by program, and it is where the capability kept failing.[5] This is why fratricide resists the single fix that intuition reaches for. A better radio, a new transponder, a tracking display — each addresses one contributor, and each has been bought; but the instant that decides whether a strike kills the enemy or a friend is the instant all of them must come together correctly, fast, under stress, across services and sometimes across allies. Integration is not a box that any one program office owns, and it does not appear on a procurement schedule. It has to be trained, exercised, and measured as a capability in its own right — which is far harder than fielding any of its parts.

AFTERMATH & REFORM

The response was a generation of combat-identification investment: better identification-friend-or-foe systems, position-reporting and blue-force-tracking networks, and changes to fire-control doctrine and training. The improvements were real, and yet fratricide never dropped to a confidently low, stable rate, and even measuring it remained contested.[6] That is the lesson the case carries into this book: where capability is an emergent property of many systems working together, no single program — and no single technology — closes the gap, and progress has to be measured against an honest baseline rather than a convenient one.

REFERENCES

1. C. R. Shrader, Amicicide: The Problem of Friendly Fire in Modern War (U.S. Army Combat Studies Institute, 1982) — the under-2% estimate from 269 incidents, a baseline later challenged as too low (with estimates nearer 10–15%).
2. "Friendly Fire: Facts, Myths and Misperceptions," U.S. Naval Institute Proceedings (June 1994) — Desert Storm: 35 of 146 KIA (24%) and 72 of 467 wounded (15%) by friendly fire; critique of the 2% "historical norm."
3. Khafji and Warrior fratricide incidents (Feb. 1991) — see USNI Proceedings (1994) and Time, "They Didn't Have to Die". (Per-incident casualty figures vary across sources; see AUDIT.)
4. U.S. GAO, Operation Desert Storm fratricide investigations — Apache incident (OSI-93-4) and Army fratricide investigation (OSI-95-10) — causes and the Army's reviews.
5. Post-war combat-identification reviews — leading causes (situational awareness, fire-control measures, combat ID) and the absence of a comprehensive incident database. (Synthesized from secondary analyses; see AUDIT.)
6. Later-conflict recurrences (2001 coordinate-reset strike; 2014 B-1 infrared-strobe strike) and S. Snook, Friendly Fire (Princeton Univ. Press, 2000) on the 1994 Black Hawk shootdown — the systems-integration archetype.
7. "IFF Update: Stalled Again," U.S. Naval Institute Proceedings (June 1994) — the combat-identification and identification-friend-or-foe programs pursued after Desert Storm, and how slowly they matured.

THE LEARNING ENGINEERING LENS

> *The lack of a comprehensive and accessible automated database prevented thorough examination of the problem.*
>
> PARAPHRASING POST–DESERT STORM FRATRICIDE REVIEWS, C. 1993

LE INSIGHT

Fratricide is a multi-decade capability problem that resists single- intervention solutions. Each of the contributing causes — situational awareness, fire-control discipline, combat identification, joint coordination — has been the subject of dedicated programs. The rate persists because the integration across the contributors is itself the capability, and integration is the hardest property to engineer by program.

LENS APPROACH

LENS uses fratricide in LEN 5 to teach systems-of-systems capability analysis and in LEN 2 to introduce combat identification as a Human-AI Teaming problem (operators rely on automated IFF systems whose limitations are not in their training). LEN 8 examines why decades of awareness have not produced a sustained reduction.

REFLECTION QUESTIONS

1. Identify a capability problem in your domain that has been the subject of repeated programs without sustained improvement. What is the integration gap that the programs do not address?
2. Design a measurement system that would distinguish a temporary improvement in fratricide rate from a structural one.

WHO BUILDS THIS learning science & instructional design · human factors & interaction design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs · knowledge bases · data pipelines

FURTHER READING

• Shrader, Amicicide (1982)
• Snook, Friendly Fire (2000) on Black Hawk shootdown over Iraq
• Joint and Coalition Operations Analysis literature

LENS COURSES

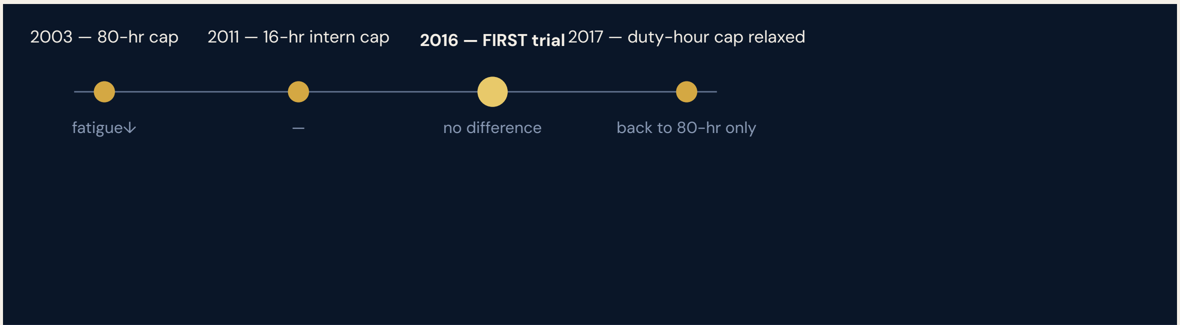
LEN 5 LEN 2 LEN 8

CASE 34 HEALTHCARE 2003–2017

ACGME 80-Hour Resident Duty-Hour Reform

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT ACGME capped U.S. resident physician work hours at 80/week to reduce fatigue-related errors; subsequent RCTs (FIRST, iCOMPARE) showed mixed effects on patient outcomes and increased hand-off-related errors



ACGME DUTY-HOUR REFORM — CAPPING HOURS DID NOT, ON ITS OWN, ENGINEER THE CAPABILITY

BACKGROUND

In 1984 Libby Zion, an 18-year-old patient, died at a New York teaching hospital in a case her family blamed on overworked, under-supervised residents. The death became the focal point of a decades-long argument about resident-physician fatigue: trainees routinely worked punishing hours with little sleep and thin supervision, and the worry was that exhausted doctors made errors that rested ones would not. New York convened the Bell Commission, whose recommendations led the state in 1989 to limit residents' hours and strengthen on-site supervision — the first such rules in the country. For more than a decade those limits remained a largely local experiment while the national accreditation system weighed whether to follow; accumulating evidence on sleep deprivation and a rising public appetite for patient-safety reform eventually carried the argument, and the pressure for a national standard built from there.[0]

THE INTERVENTION

In 2003 the Accreditation Council for Graduate Medical Education capped resident work at 80 hours per week, averaged over four weeks. In 2011 it tightened the rules again, limiting first-year residents to shifts of no more than 16 hours. The logic was clean and intuitive: fatigue causes error, so reduce the hours and the errors should follow. It was, in capability terms, a single-variable intervention — change one input and expect the outcome to move.[1]

HOW IT WORKED

Hours, though, were one input to a system with many. Cutting them did not remove the work; it redistributed it, and it multiplied the number of patient hand-offs between residents — themselves a well-documented site of error, where information falls through the gaps between shifts. Continuity of care suffered, and so did residents' exposure to the long arc of a case and to procedures they needed repetitions to master. In surveys, many residents reported feeling less prepared rather than better rested. The cap had changed the input without engineering the rest of the system around it.[2] There was a quieter loss as well. The long shift had been doing work no one had fully accounted for: it was where a resident watched a patient's course unfold from admission through crisis to resolution, where supervision happened in the moment rather than across a hand-off note, and where the sheer volume of repetitions built procedural skill. Capping the hours removed those by-products along with the fatigue, and nothing was put in their place — the reform subtracted an input without redesigning the system that input had quietly been sustaining.

THE EVIDENCE

Two large randomized trials put the policy to a real test. FIRST (Bilimoria et al., NEJM 2016), in surgical-training programs, found that flexible, less-restrictive duty-hour policies were non-inferior to the strict caps on patient outcomes and no worse for residents' well-being and education.[3] iCOMPARE (Silber et al., NEJM 2019), in internal-medicine programs, reached a parallel result: flexible first-year policies were associated with non-inferior patient outcomes and no significant difference in residents' satisfaction with their well-being and education.[4] Neither trial found the patient-safety gain the strict cap had been imposed to deliver. In 2017 the ACGME relaxed the 16-hour intern limit.[5] It is worth being careful about what the trials did and did not establish. They did not show that fatigue is harmless, nor that hours never matter; they showed that, within the range tested, trading the strict cap for flexibility did not worsen patient outcomes or resident well-being — which means the strict cap's promised benefit failed to appear where it had been most confidently predicted. The result reframed the question. The issue was never whether tired residents err, but whether reducing one input, by itself and without rebuilding supervision, hand-offs, and continuity around it, reliably produces a safer system. On the evidence, it did not.

WHAT TRANSFERRED

Duty-hour reform is the clearest case in this book of a single-variable intervention into a multi-variable capability system — and of the evidence eventually catching up with a well-intentioned policy. It is worth reading against the successes elsewhere in these pages. The Keystone ICU project, crew resource management, and the surgical safety checklist worked because they engineered supervision, hand-off, and measurement architecture **together with** the behavioral change, rather than adjusting one input and assuming capability would follow.[6] The lesson is not that resident fatigue does not matter; it is that capability is a property of the whole system, and a reform that moves one variable while leaving the others untouched will be measured, in the end, by what it actually produced. The field eventually learned the constructive version of the lesson. The most durable patient-safety advances of the era were not single prohibitions but engineered bundles — structured hand-off protocols, checklist-driven processes, and supervision redesigned around the points where information was being lost — that changed several variables together and measured the result. Duty-hour reform earns its place in a training-gap chapter precisely because it began as the intuitive move, the one that felt obviously right, and because the evidence then did what evidence is for: it told the field that feeling right was not the same as being effective, and sent it back to design the harder, whole-system answer.

REFERENCES

1. B. H. Lerner, The Libby Zion Case and the Reform of Medical Education (2006); and the 1989 New York State (Bell Commission) duty-hour regulations — the origin of the reform.
2. Accreditation Council for Graduate Medical Education, Common Program Requirements (2003 and 2011 revisions) — the 80-hour weekly cap and the 16-hour first-year shift limit.
3. D. A. Asch et al., “Resident Duty Hours and Medical Education Policy,” NEJM 370: 1671–1673 (2014); Institute of Medicine (Ulmer, Wolman & Johns, eds.), Resident Duty Hours: Enhancing Sleep, Supervision, and Safety (2008) — hand-offs and continuity trade-offs.
4. Bilimoria, Chung, Hedges et al., “National Cluster-Randomized Trial of Duty-Hour Flexibility in Surgical Training” (FIRST), NEJM 374: 713–727 (2016). doi:10.1056/NEJMoa1515724.
5. Silber, Bellini, Shea et al., “Patient Safety Outcomes under Flexible and Standard Resident Duty-Hour Rules” (iCOMPARE), NEJM 380: 905–914 (2019). doi:10.1056/NEJMoa1810641.
6. ACGME, Common Program Requirements (2017 revision) — relaxation of the 16-hour first-year shift limit.
7. P. Pronovost et al., Keystone ICU intervention, NEJM 355: 2725–2732 (2006), doi:10.1056/NEJMoa061115 A. B. Haynes et al., surgical safety checklist, NEJM 360: 491–499 (2009), doi:10.1056/NEJMsa0810119 — integrated interventions that engineered the surrounding architecture.

THE LEARNING ENGINEERING LENS

> *Flexible, less-restrictive duty hour policies for first-year residents were associated with non-inferior patient outcomes and no significant difference in residents' satisfaction with overall well-being and education quality.*
>
> PARAPHRASING ICOMPARE TRIAL (SILBER ET AL., NEJM 2019)

LE INSIGHT

The clearest healthcare case in the dataset of a single-variable intervention into a multi-variable capability system. Pairs with Case 33 (fratricide) and Case 11 (V-22). The success cases — Keystone (14), CRM (12), Korean Air (23) — engineered supervisory, hand-off, and measurement architecture **together with** the behavioral change.

LENS APPROACH

LENS uses duty-hour reform in LEN 5 as the foundational capability-system case (what was the long-hour regime **producing** that was lost when the input was capped?), in LEN 4 to discuss measurement architecture (what FIRST and iCOMPARE measured, and what they did not), and in LEN 10 as a studio prompt for the integrated resident-training redesign that the reforms did not deliver.

REFLECTION QUESTIONS

1. What capability is the long-hours / heavy-workload regime in your domain currently producing — supervisory exposure, continuity, procedural reps, tacit-knowledge transfer — that a simple cap would lose?
2. Design the integrated redesign — supervision, hand-off, measurement, exposure — that would substitute for the capability the input cap removes.

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards

FURTHER READING

- Bilimoria et al. (2016), FIRST trial; Silber et al. (2019), iCOMPARE trial
- Lerner (2006), The Libby Zion Case and the Reform of Medical Education
- Ulmer, Wolman & Johns, eds. (2008), Resident Duty Hours: Enhancing Sleep, Supervision, and Safety, Institute of Medicine

LENS COURSES LEN 5 LEN 4 LEN 10 LEN 8

CASE 43 AVIATION 2009

Colgan Air Flight 3407

T Training Gap

IMPACT 50 killed near Buffalo; precipitated the FAA's 1,500-hour rule and the Pilot Records Database

50

KILLED, CLARENCE CENTER NY

the data was in the system; the data flow was not

COLGAN 3407 — AND THE REFORM THAT FOLLOWED

On approach to Buffalo-Niagara International, the captain of Colgan Air 3407 responded to the activation of the aircraft's stick shaker by pulling back on the control column rather than lowering the nose. The Bombardier Q400 stalled and crashed into a house in Clarence Center, killing all forty-nine people on board and one person on the ground. The NTSB found the captain had a documented history of training failures earlier in his career, and that the first officer had been ill, fatigued, and was paid roughly sixteen thousand dollars a year. The airline knew about the training history. The hiring pipeline knew. The regulator did not.

Family members of the victims organized one of the most effective aviation-safety lobbying efforts in a generation. The result was the Airline Safety and Federal Aviation Administration Extension Act of 2010, which raised the minimum experience requirement for Part 121 first officers from 250 to 1,500 hours and established the Pilot Records Database. The 1,500-hour rule remains controversial; the PRD has fundamentally restructured how pilot-history information flows between carriers.

The capability gap was visible to everyone **except** the system that licensed the pilots.

THE LEARNING ENGINEERING LENS

> *The captain's inappropriate response to the activation of the stick shaker led to an aerodynamic stall from which the airplane did not recover.*
>
> NTSB AIRCRAFT ACCIDENT REPORT AAR-10/01, PROBABLE CAUSE, 2010

LE INSIGHT

Colgan is the canonical case for the gap between the information an institution holds about its operators and the information that reaches the decisions about whether those operators should be in that seat. The data was already in the system — multiple failed training events. The data flow that would have made that data actionable at the hiring decision did not exist.

LENS APPROACH

LENS uses Colgan in LEN 4 as a case for evidence-system design (the PRD as a designed information flow) and in LEN 8 as a case for advocacy-driven institutional change. Studio projects examine how families of victims became the load-bearing element of a reform the industry had resisted for years.

REFLECTION QUESTIONS

1. What information about operators in your domain exists somewhere in the system but does not flow to the decisions that depend on it?
2. Design the data-flow architecture that would make a Colgan-equivalent visible **before** the accident rather than after.

WHO BUILDS THIS learning science & instructional design — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models

SOURCES

→ NTSB Aircraft Accident Report NTSB/AAR-10/01 (2010)
→ Airline Safety and Federal Aviation Administration Extension Act of 2010, Public Law 111-216
→ GAO-12-203, Pilot Records Database Implementation (2012)

FURTHER READING

• NTSB AAR-10/01 (2010)
• Families of Continental Flight 3407, public-facing campaign archive
• FAA Pilot Records Database technical documentation

LENS COURSES LEN 4 LEN 5 LEN 8

CASE 44 AVIATION 2013

Asiana Airlines Flight 214

T Training Gap H Human-System Interface

IMPACT 3 killed, 187 injured; Boeing 777 struck the seawall short of SFO runway 28L

EXPECTED	ACTUAL
137 kt	103 kt

the autothrottle was not maintaining the speed the crew believed it was

ASIANA 214 — APPROACH AIRSPEED AT THE SEAWALL

On a clear afternoon at San Francisco, the crew of Asiana 214 allowed the aircraft to slow well below approach speed without recognizing that the autothrottle was not in the mode they expected. The Boeing 777 struck the seawall short of runway 28L, killing three and seriously injuring forty-nine of the 304 on board. The NTSB found inadequate monitoring of airspeed and confusion about autothrottle behavior when the autopilot pitch channel was in FLCH SPD: after the pilots took the autopilot offline and pulled thrust to idle, the autothrottle reverted to HOLD and did not wake to maintain selected speed. The captain later said he believed the autothrottle would maintain speed; in that configuration, it does not.

NTSB identified Boeing's design of the autothrottle and autopilot flight-director systems as a contributing factor — the documentation, training, and interface together did not make the system's actual state salient to the crew. The proximate cause was the crew's failure to monitor airspeed; the system-level cause was that the automation could silently leave the crew responsible for a parameter they believed it was holding.

Asiana 214 is the most prominent recent case of automation surprise on a Western wide-body airliner.

THE LEARNING ENGINEERING LENS

> *Complexities of the autothrottle and autopilot flight director systems contributed to the accident.*
>
> NTSB AIRCRAFT ACCIDENT REPORT AAR-14/01, CONTRIBUTING FACTORS, 2014

LE INSIGHT

Asiana 214 is the aviation case for the LENS Human-AI Teaming proposition that automation transparency is a capability deliverable. The mode the autothrottle was actually in and the mode the crew believed it was in diverged silently — a textbook automation surprise. The training, the documentation, and the interface together did not surface the divergence.

LENS APPROACH

LENS treats Asiana 214 in LEN 2 as a worked case for automation transparency requirements. Students design the interface and training intervention that would have made the mode disconnect salient to the crew at the moment the aircraft slowed.

REFLECTION QUESTIONS

1. Identify a mode in an automated system you work with where the operator's mental model and the system's actual behavior can silently diverge. How would the operator know?

2. Design the interface change that would have alerted the Asiana crew to the mode mismatch before the airspeed decayed.

WHO BUILDS THIS learning science & instructional design · human factors & interaction design — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs

SOURCES

→ NTSB Aircraft Accident Report NTSB/AAR-14/01 (2014)
→ Boeing 777 Flight Crew Operating Manual, autothrottle and autoflight sections
→ Casner, Geven & Williams (2013), "The Effectiveness of Airline Pilot Training for Abnormal Events," Human Factors

FURTHER READING

• Sarter & Woods (1995), "How in the world did we ever get into that mode?"
• Casner et al. (2014) on automation surprises
• Parasuraman & Manzey (2010), automation complacency

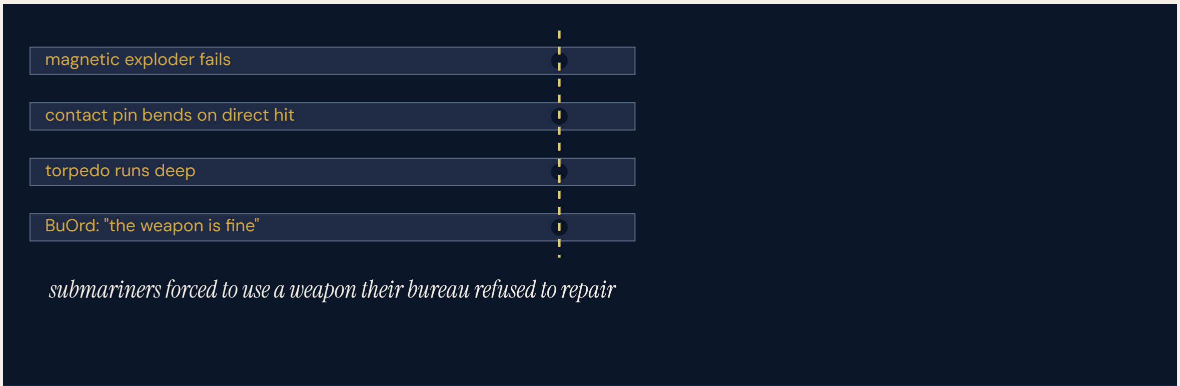
LENS COURSES LEN 5 LEN 2

CASE 45 DEFENSE 1941 – 1943

Mark 14 Torpedo Failures

T Training Gap K Knowledge & Institutional Memory G Governance & Trust

IMPACT Persistent torpedo failures across the first ~20 months of the Pacific War; resolved only after fleet-level testing forced acknowledgment of the defects



MARK 14 — MULTIPLE DEFECTS, SINGLE INSTITUTIONAL REFUSAL

The U.S. Navy entered World War II with the Mark 14 torpedo, which had been so expensive to test in peacetime that the Bureau of Ordnance had effectively forbidden live trials. Submarine crews in 1942 reported torpedo after torpedo running deep, running shallow, failing to detonate, or detonating prematurely. The Bureau of Ordnance insisted the weapon worked and that the operators were at fault. Submarine captains who reported failures faced career consequences.

It took eighteen months and the personal intervention of Admiral Charles Lockwood (COMSUBPAC) — who ordered fleet-level testing, including a famous live-fire trial against the Kahoolawe cliff face and, separately, the USS Tinosa’s July 1943 attack on the Tonan Maru in which fifteen hits produced a single detonation — to confirm that the weapon was running ten feet too deep, that its Mark 6 magnetic exploder failed routinely, and that even its contact firing pin crushed on perpendicular impact. The fixes were straightforward once the existence of the problem was acknowledged. The institutional resistance was the binding constraint.

The Mark 14 is a foundational case in U.S. military capability history for what happens when the institution responsible for a weapon refuses to believe the operators using it.

THE LEARNING ENGINEERING LENS

> *The Bureau certified the weapon; field reports of failure were treated as evidence of operator error.*
>
> PARAPHRASING BLAIR, SILENT VICTORY, 1975

LE INSIGHT

The Mark 14 is the canonical Navy case for the institutional refusal to believe operator feedback. The capability that was missing was not at the front. It was at the bureau that owned the weapon. The cost of the refusal was paid by the submariners forced to use it until the bureau yielded. The eventual fixes — re-setting depth calibration, replacing the magnetic exploder, redesigning the contact pin — were technical and small. The reform was institutional and slow. The capability that had to be built was the channel by which the bureau could be made to hear what the boats already knew.

LENS APPROACH

LENS uses the Mark 14 in LEN 7 as a governance failure case and in LEN 8 as an organizational-learning case in which the operators had the diagnosis and the institution lacked the structure to receive it. Studio projects (LEN 10) examine what the equivalent operator-to-institution feedback channel should look like.

REFLECTION QUESTIONS

1. Identify a system in your domain whose owners are institutionally insulated from the operators using it. What feedback would they refuse to hear, and what would it cost?
2. Design the operator-feedback channel that the U.S. Navy Bureau of Ordnance should have had in 1941. Who signs, who receives, what triggers action?
3. The USS Tinosa fired fifteen torpedoes for one detonation before the bureau accepted the diagnosis. What is the operator-evidence threshold in your domain that would force the equivalent institutional acknowledgment — and how would you make sure it is reached before the cost is paid?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · governance frameworks · bias & evidence audits

SOURCES

→ Blair, C. (1975), Silent Victory: The U.S. Submarine War Against Japan
→ Rowland, B. & Boyd, W. (1953), U.S. Navy Bureau of Ordnance in World War II
→ Naval History and Heritage Command, Mark 14 torpedo files

FURTHER READING

• Blair (1975), Silent Victory
• Rowland & Boyd (1953)
• Edmondson (2018), The Fearless Organization

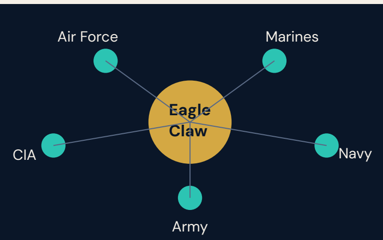
LENS COURSES LEN 10 LEN 7 LEN 8

CASE 46 DEFENSE 1980

Operation Eagle Claw

T Training Gap K Knowledge & Institutional Memory

IMPACT 8 servicemembers killed in Iran; mission to rescue 52 American hostages aborted; catalyst for the founding of U.S. Special Operations Command



five organizations · no joint command · no shared training

THE ABSENCE OF AN INSTITUTION THAT COULD OWN THE MISSION

The mission to rescue fifty-two American hostages from the U.S. embassy in Tehran was aborted at a desert refueling point in eastern Iran when three of eight helicopters were lost to mechanical or weather failures. During the withdrawal a helicopter collided with a C-130, killing eight servicemembers. The hostages remained in Iranian custody for another nine months.

The Holloway Commission identified the mission's "ad hoc-ery": each service contributed its own units, equipment, command relationships, and communications systems; the aircrews had not trained together; the RH-53D Sea Stallions had been chosen for range and a minesweeping cover story rather than fitness for a desert rescue. There was no standing joint special-operations command to own the mission. The capability had to be improvised, and the improvisation failed.

Eagle Claw produced the Holloway Commission, JSOC (1980), and ultimately the Nunn-Cohen Amendment establishing USSOCOM (1987); Goldwater-Nichols (1986) drew on it alongside Beirut and Grenada in mandating joint command structures across the U.S. military.

THE LEARNING ENGINEERING LENS

> *The mission was ad hoc — assembled from units, equipment, and command relationships that had never operated together.*
>
> PARAPHRASING THE HOLLOWAY SPECIAL OPERATIONS REVIEW GROUP, 1980

LE INSIGHT

Eagle Claw is the canonical case in U.S. defense for the absence of institutionalized cross-service capability. Each service was competent inside its own boundary; the integration across them did not exist as a deliverable — until the reform that followed built the institution the mission had needed and not had.

LENS APPROACH

LENS uses Eagle Claw in LEN 5 as a worked case for cross- organizational capability requirements and in LEN 8 for the institutional reform that followed; it pairs with INPO (Case 16) as the defense analog of failure-driven institution building. The seven-year arc to Goldwater-Nichols (1986) and USSOCOM (1987) is itself a measurement: the institutional response time when the fix requires statutory action.

REFLECTION QUESTIONS

1. Where in your domain does a cross-organizational capability exist on paper but not in practice? What would force its institutionalization?
2. Eagle Claw produced USSOCOM and Goldwater-Nichols six years later. Sketch the institutional design that an equivalent failure in your domain would force into existence.
3. The Holloway Commission named the mission's ad-hoc-ery as the diagnosis. What standing capability — institution, command, training pipeline — does your domain currently lack that an Eagle-Claw-class failure would force into existence?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines

SOURCES

→ Holloway Special Operations Review Group, Rescue Mission Report (1980)
→ Goldwater-Nichols Department of Defense Reorganization Act of 1986
→ Locher, Victory on the Potomac (2002)

FURTHER READING

• Locher (2002), Victory on the Potomac
• Holloway Commission report (1980)

LENS COURSES LEN 5 LEN 8

CASE 47 AVIATION 2005

Helios Airways Flight 522

T Training Gap H Human-System Interface

IMPACT 121 killed; cabin failed to pressurize after a maintenance setting was left on "manual" and the crew misread the warning

ACTUAL	INTERPRETED
cabin alt.	T/O config.

one horn, two meanings · the training had drilled only one

HELIOS 522 — A CUE CARRYING TWO MEANINGS WITHOUT DIFFERENTIATION

The crew of Helios 522 ignored a warning horn shortly after takeoff from Larnaca because the same intermittent horn used on the Boeing 737 to indicate incorrect takeoff configuration on the ground sounds, in flight, when cabin altitude exceeds approximately 10,000 feet. The pressurization selector had been left in “manual” after a pressurization leak check the previous night, and the cabin failed to pressurize. By the time the aircraft reached altitude the crew was hypoxic; by the time it neared Athens, only one cabin attendant remained conscious. The aircraft circled Athens on autopilot until fuel exhaustion and crashed into a hillside.

The accident investigation found the single warning horn carried different meanings on the ground and in flight, with no differentiation in the cue. The training had drilled the ground-configuration meaning. The in-flight meaning was an edge case that the airline’s training had not made salient. The crew spent twenty minutes troubleshooting a problem they had misidentified, by which time it was already too late.

THE LEARNING ENGINEERING LENS

> *The crew's interpretation of the cabin-altitude warning horn as a takeoff-configuration warning was a critical and irrecoverable error.*
>
> PARAPHRASING THE HELLENIC AAIB FINAL REPORT ON HELIOS 522, 2006

LE INSIGHT

Helios 522 is the textbook example of a single cue carrying two meanings without differentiation. The interface lied by ambiguity. The training filled in only one meaning. The combination produced a twenty-minute window in which the crew was solving the wrong problem.

LENS APPROACH

LENS uses Helios in LEN 5 as a case in cue ambiguity and in LEN 2 as a Human-AI Teaming case where the interface failed to communicate which of two possible system states it was warning about. Students redesign the cue.

REFLECTION QUESTIONS

1. Identify a cue in your domain that carries two meanings in different operational contexts. How would the operator know which?
2. Redesign the Helios configuration warning to distinguish ground- and in-flight meaning without adding cognitive load.

WHO BUILDS THIS learning science & instructional design · human factors & interaction design — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs

SOURCES

→ Hellenic AAIB Aircraft Accident Investigation Report 11/2006
→ Boeing 737 Flight Crew Operations Manual, configuration warnings
→ Reason (1990), Human Error — Helios 522 as worked case

FURTHER READING

• Hellenic AAIB Report 11/2006
• Endsley (1995), situation awareness theory
• Sarter, Woods & Billings (1997), automation surprise literature

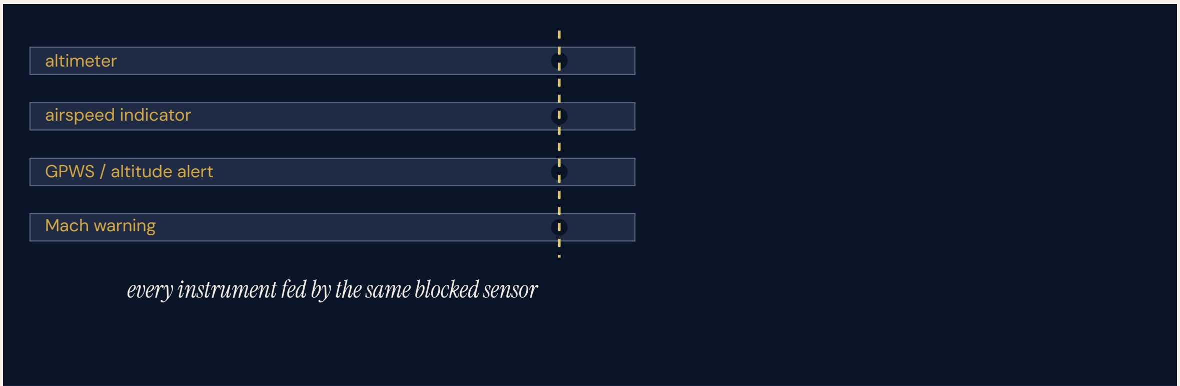
LENS COURSES LEN 5 LEN 2

CASE 48 AVIATION 1996

AeroPerú Flight 603

H Human-System Interface T Training Gap

IMPACT 70 killed; Boeing 757 crashed into the Pacific after maintenance tape was left over static ports



AEROPERÚ 603 — REDUNDANCY THAT WAS NOT REDUNDANCY

Maintenance staff in Lima taped over the Boeing 757's static ports while polishing the fuselage and failed to remove the tape before flight. With the static ports blocked, the aircraft's altimeters, airspeed indicators, and altitude alert system all read inconsistently. The crew received contradictory warnings — overspeed, stick shaker, ground proximity — and could not determine the aircraft's actual state. They believed they were higher than they were. They flew into the ocean.

The Peruvian accident report identified the maintenance error as the primary cause but emphasized that the cockpit had no independent reference to detect the inconsistency. Every instrument that should have flagged the failure depended on the same blocked sensor. The crew's training did not include a procedure for "everything you see is wrong."

AeroPerú 603 sits at the intersection of maintenance procedure, cockpit interface design, and the limits of crew training under instrument failure.

THE LEARNING ENGINEERING LENS

> *The crew received contradictory indications they could neither reconcile nor override.*
>
> PARAPHRASING THE PERUVIAN DGAC / COMISIÓN INVESTIGADORA DE ACCIDENTES DE AVIACIÓN REPORT ON AEROPERÚ 603, 1996

LE INSIGHT

AeroPerú 603 is the case for redundancy that is not redundancy. Every cockpit indicator depended on the same physical sensor. The apparent redundancy in the cockpit was an illusion at the source. The training did not include the failure case because the failure case had been assumed not to occur.

LENS APPROACH

LENS uses AeroPerú in LEN 5 to teach the difference between apparent and actual redundancy and in LEN 2 for the human role when all interface data is unreliable. Studio projects examine the "trust nothing" procedure design.

REFLECTION QUESTIONS

1. Identify a redundant interface in your domain whose redundancy depends on a common upstream component. What is the operator procedure when the upstream component fails?
2. Design the "trust nothing" procedure for the AeroPerú crew. What information remains reliable when all instruments are corrupted?

WHO BUILDS THIS human factors & interaction design · learning science & instructional design — plus domain experts and a learning engineer to integrate.

TOOLS usability testing · cognitive walkthroughs · scenario simulation · competency models

SOURCES

→ Peru Dirección General de Aeronáutica Civil (DGAC), Comisión Investigadora de Accidentes de Aviación, accident report on AeroPerú 603 (1996)
→ Boeing 757 Flight Crew Training Manual
→ Wiener, Human Factors of Cockpit Automation (NASA, 1989)

FURTHER READING

• Peru DGAC / CIA report on AeroPerú 603 (1996)
• Leveson (2011), STAMP — common-cause failure
• Dismukes et al. (2007), The Limits of Expertise

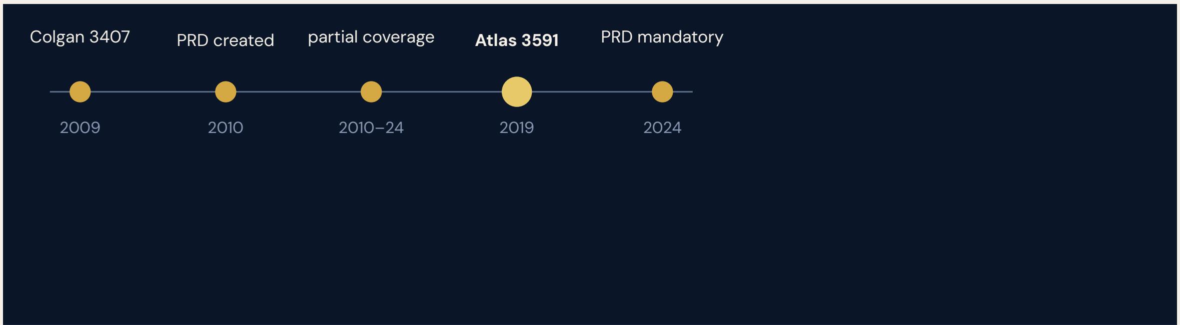
LENS COURSES LEN 5 LEN 2

CASE 49 AVIATION 2019

Atlas Air Flight 3591

T Training Gap K Knowledge & Institutional Memory

IMPACT 3 killed; Boeing 767 cargo flight crashed in Texas after the first officer mistakenly activated go-around mode



THE COST OF PARTIAL IMPLEMENTATION BETWEEN COLGAN AND THE PRD'S FINAL COVERAGE

The first officer of Atlas Air 3591 — Conrad Aska, an Amazon Prime Air cargo flight — inadvertently activated the TOGA (go-around) mode during a turbulent descent, then, in a somatogravic illusion that the aircraft was pitching up, pushed the nose down sharply in response to a perceived stall, and crashed the Boeing 767 freighter into Trinity Bay, Texas. The NTSB found that the first officer had a documented history of training failures spanning multiple prior carriers and had not disclosed those failures on his Atlas Air application. Atlas could not detect the omissions because the Pilot Records Database — directed by Congress after Colgan Air 3407 (Case 43) — was not yet operational at the time of hiring; Atlas relied on the older PRIA records system, which only required five years of training history.

The case is the live, current version of the Colgan story. The information existed. The data-flow system had not yet been built. The first officer had been advanced past failures that should have ended his career. The reform that Colgan's families had won was not yet complete enough to stop the next iteration.

The PRD final rule was published in 2021 and became effective for Part 121 carriers in June 2022, with full historical coverage phasing in through 2024.

THE LEARNING ENGINEERING LENS

> *Atlas Air did not have access to portions of the first officer's training record that would have informed its hiring decision.*
>
> PARAPHRASING NTSB AIRCRAFT ACCIDENT REPORT AAR-20/02, 2020

LE INSIGHT

Atlas Air 3591 is the iteration test of the Colgan reform. The reform partially worked: the PRD exists. The reform was incomplete: not all carriers were covered. The case is a cautionary worked example of why partial implementation of a regulatory remedy leaves the original capability gap partially open.

LENS APPROACH

LENS uses Atlas Air 3591 in LEN 4 as a case for measurement-system coverage (the PRD had been built but its mandatory coverage was incomplete), and in LEN 8 for the iteration cycle of reform: an intervention that leaves an aperture creates a measurable harm inside the aperture.

REFLECTION QUESTIONS

1. Identify a regulatory remedy in your domain whose coverage is partial. What harm is occurring inside the aperture?
2. Design the deliverable that would close the PRD coverage gap in advance of a future Atlas Air 3591.

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines

SOURCES

→ NTSB Aircraft Accident Report AAR-20/02 (2020)
→ FAA Pilot Records Improvement Act of 1996; updates 2010, 2024
→ New York Times coverage of Atlas Air 3591

FURTHER READING

• NTSB AAR-20/02
• GAO reports on the Pilot Records Database, 2014–2024
• Fixsen et al. (2005), implementation fidelity

LENS COURSES LEN 4 LEN 8

CASE 50 AVIATION 2015

TransAsia Airways Flight 235

T Training Gap H Human-System Interface

IMPACT 43 killed in Taiwan; crew shut down the only working engine after the other failed

RIGHT

auto-feathered

LEFT

shut down by crew

the working engine; misidentified in ~15 seconds under stress

TRANSASIA 235 — THE SAME WRONG-ENGINE PATTERN AS KEGWORTH (CASE 30)

Forty-three seconds after takeoff from Taipei Songshan, the ATR 72's right engine auto-feathered following a sensor malfunction. The crew, working from memory under stress, shut down the **left** engine — the one still producing thrust — and committed the aircraft to a glide for which it was much too low. The aircraft clipped a taxicab on the Huandong Viaduct and crashed into the Keelung River.

The Aviation Safety Council (Taiwan) found the captain had failed proficiency checks twice in the year before the accident and had not been retrained against the specific failure scenario the aircraft actually presented. The crew did not use the published engine-shutdown checklist; they responded from memory. The misidentification took place in roughly fifteen seconds.

TransAsia 235 sits in the same conceptual family as Kegworth (Case 30): a crew shut down the wrong engine because the mental model they applied did not match the configuration they were in.

THE LEARNING ENGINEERING LENS

> *The flight crew was unable to identify which engine had failed and shut down the operating engine in error.*
>
> PARAPHRASING THE AVIATION SAFETY COUNCIL (TAIWAN) FINAL REPORT ON TRANSASIA 235, 2016

LE INSIGHT

TransAsia 235 is the modern recurrence of the Kegworth pattern under stress. Crews under acute time pressure default to memory; if the memory is incomplete or wrong, the action follows the memory. The capability gap was the same as in 1989. The intervention pattern needed is also the same.

LENS APPROACH

LENS uses TransAsia 235 in LEN 5 as a recurrence case for well-known capability problems (wrong-engine shutdown) and in LEN 2 to teach the difference between checklist-driven response and memory-driven response under stress.

REFLECTION QUESTIONS

1. Identify a procedure in your domain that is supposed to be checklist-driven but is actually memory-driven under stress. What is the gap?

2. Why has the wrong-engine shutdown pattern persisted across 25 years of aviation reform? What intervention has not yet been engineered?

WHO BUILDS THIS learning science & instructional design · human factors & interaction design — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs

SOURCES

→ Aviation Safety Council (Taiwan) Final Report (2016)
→ Reason (1990), Human Error — wrong-engine cases
→ Dismukes, Berman & Loukopoulos (2007), The Limits of Expertise

FURTHER READING

• Taiwan ASC report (2016)
• Dismukes et al. (2007)
• Klein (1998), Sources of Power

LENS COURSES LEN 5 LEN 2

Chapter 2

Designed Out

Capability was removed to save cost.

If we emphasize MCAS is a new function there may be greater certification and training impact.

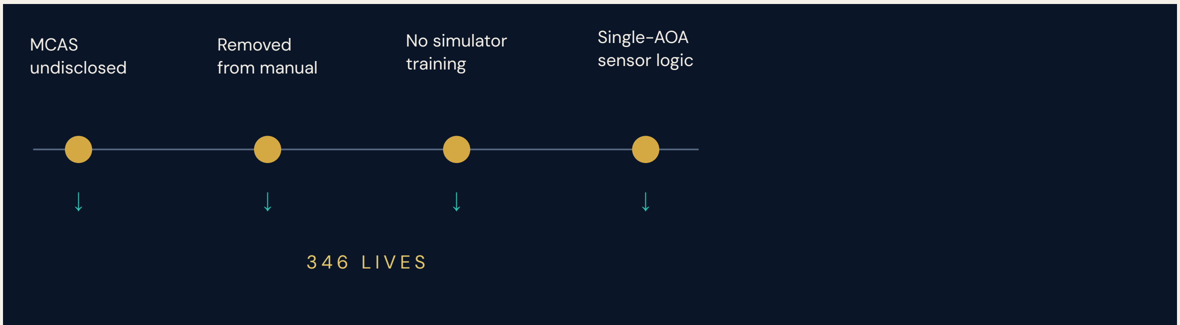
BOEING INTERNAL MEETING MINUTES, 2013

CASE 2 AVIATION 2018 – 2019

Boeing 737 MAX / MCAS

D Designed Out T Training Gap H Human-System Interface

IMPACT 346 killed across two crashes — Lion Air 610 and Ethiopian Airlines 302



MCAS — CAPABILITY ENGINEERED OUT AT FOUR DECISION POINTS

The Maneuvering Characteristics Augmentation System was not in the flight manual. It was not in pilot training. The acronym itself was kept out of customer-facing materials. Internal meeting minutes recovered by Congressional investigators showed Boeing employees planning the omission in writing: emphasizing MCAS as a new function would trigger "greater certification and training impact." Boeing had reportedly promised Southwest Airlines a one-million-dollar rebate per aircraft if the MAX did not require simulator training.

When MCAS misfired on Lion Air Flight 610 in October 2018, the crew had no prior knowledge the system existed, no procedure for disabling it, and no simulator experience of its failure mode. One hundred eighty-nine people died. Five months later Ethiopian Airlines Flight 302 repeated the sequence almost exactly. One hundred fifty-seven more.

A 2016 internal Boeing survey had already found 39% of employees conducting certification reviews perceived "undue pressure" from management. The House Transportation Committee's 2020 report concluded that Boeing's assumption that simulator training was unnecessary "diminished safety, minimized the value of pilot training, and inhibited technical design improvements." This is the inverse of the capability-engineering thesis: human capability was not overlooked, it was engineered out of the system to avoid the cost of training that would have created it.

THE LEARNING ENGINEERING LENS

> *Boeing's assumption that simulator training was unnecessary diminished safety, minimized the value of pilot training, and inhibited technical design improvements.*
>
> HOUSE TRANSPORTATION COMMITTEE REPORT, 2020

LE INSIGHT

The 737 MAX is the documentary record of a design choice to remove human capability to avoid the regulatory and commercial cost of sustaining it. The pilots were not undertrained by accident — they were undertrained as a **requirement**. The omission of training was a contract deliverable. Once that is understood, the accident becomes legible as an engineering decision rather than a training failure.

LENS APPROACH

LENS treats capability requirements as system requirements: derived from operational analysis, traceable to design decisions, falsifiable against operational outcomes. A LENS-trained engineer in the MAX certification process would have produced a capability-requirements artifact mapping pilot tasks to system behaviors — and flagged the proposed manual omission as an unmitigated requirements gap.

REFLECTION QUESTIONS

1. If a customer contract required removing a training requirement you judged necessary, what artifact would you produce, who would you route it to, and what would you do if the routing failed?

2. Reconstruct the MAX accident as a **capability-requirements** failure: what was the elided requirement, at what life-cycle stage was it elided, and who had the authority to insert it?

WHO BUILDS THIS systems & human-factors engineering · learning science & instructional design · human factors & interaction design — plus domain experts and a learning engineer to integrate.

TOOLS task analysis · design prototyping · scenario simulation · competency models · usability testing · cognitive walkthroughs

SOURCES

→ House Transportation Committee, The Boeing 737 MAX Aircraft: Costs, Consequences, and Lessons from Its Design, Development, and Certification (2020)
→ Department of Transportation Office of Inspector General, AV2020042 (2020)
→ Herkert, Borenstein & Miller, "The Boeing 737 MAX: Lessons for Engineering Ethics," Science and Engineering Ethics (2020)
→ FAA Return-to-Service Summary (2020); Boeing internal communications released by the U.S. House (2019–2020)

FURTHER READING

• Herkert, Borenstein & Miller (2020) on certification-by-omission ethics
• Leveson, Engineering a Safer World (2011) — STAMP
• Casner & Hutchins (2019), "What Do We Tell the Drivers?"

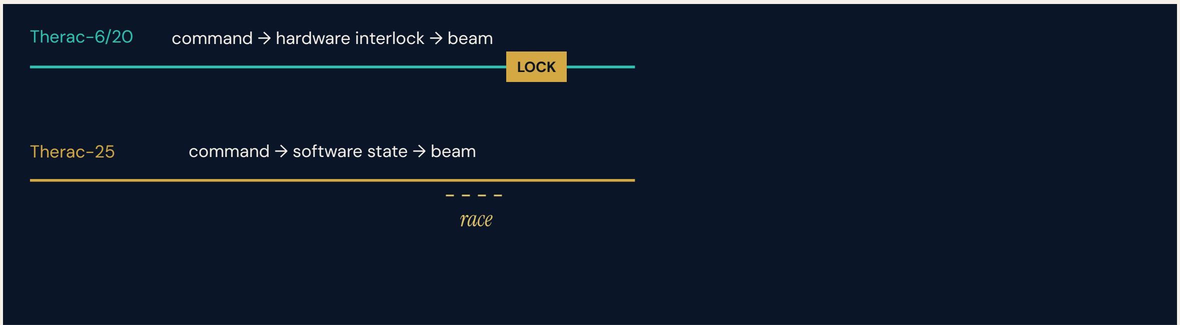
LENS COURSES LEN 1 LEN 5 LEN 2

CASE 7 HEALTHCARE TECHNOLOGY 1985 – 1987

Therac-25

H Human-System Interface D Designed Out G Governance & Trust

IMPACT Six known massive radiation overdoses; four deaths; canonical case in software safety engineering



SOFTWARE RACE CONDITION REPLACED HARDWARE INTERLOCK

Earlier machines in the Therac line had hardware interlocks: physical mechanisms that would not allow the electron beam to fire unless the metal target was in place. The Therac-25 removed them. The new design relied entirely on software to assure the beam configuration matched the prescription. Doing so was cheaper, lighter, and faster to certify. It also moved the entire safety case onto software not written for safety-critical operation.

Operators encountered a race condition the engineers did not know existed. If a technician entered a prescription, noticed an error, and corrected it within roughly eight seconds, the machine could enter a state in which the high-energy beam fired without the target. Patients absorbed up to a hundred times the intended dose. The console displayed “MALFUNCTION 54” — a code without operational meaning — and offered the option to proceed. Operators, trained that the machine was safe, proceeded.

When patients reported burning and pain, the manufacturer insisted overdose was impossible. Leveson and Turner’s 1993 analysis became the founding case study of modern software safety engineering: overconfidence in software, removal of independent hardware safeguards, error messages that communicated nothing, and a reporting architecture that absorbed incidents.

THE LEARNING ENGINEERING LENS

> *Therac-25 illustrates the dangers of relying on software safety controls without rigorous engineering practices.*
>
> PARAPHRASING LEVESON & TURNER, IEEE COMPUTER, 1993

LE INSIGHT

Therac-25 is the moment when removing the human safety margin became visible as a design choice. The hardware interlock was not redundant — it was the safety case. When it was removed, no equivalent capability was put in its place. The operator was retained in the system but stripped of the ability to detect what it was doing wrong. Capability engineering would have asked, before removing the interlock, **what function takes its load?**

LENS APPROACH

LENS frames Therac-25 as a **Design-Out** failure made visible through Interface and Governance pathologies. Studio projects in LEN 5 ask students to produce capability-load diagrams tracing every safety function to a hardware backstop, a software check, or a trained operator action with the information needed to perform it. LEN 7 examines incident reporting as governance.

REFLECTION QUESTIONS

1. Identify a system in your domain that migrated a safety function from hardware to software. Where did the human-capability load go, and who is accountable for sustaining it?

2. Therac-25 operators saw “MALFUNCTION 54” and continued treatment. Redesign that interface using LEN 2 principles so that the operator’s correct action is the easiest action.

WHO BUILDS THIS human factors & interaction design · systems & human-factors engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS usability testing · cognitive walkthroughs · task analysis · design prototyping · governance frameworks · bias & evidence audits

SOURCES

→ Leveson & Turner, “An Investigation of the Therac-25 Accidents,” IEEE Computer 26(7) (1993)
→ The Joint Commission Journal on Quality and Patient Safety analysis (2004)
→ Leveson, Safeware: System Safety and Computers (1995)
→ Ethics Unwrapped, McCombs School of Business, University of Texas at Austin

FURTHER READING

• Leveson, Engineering a Safer World (2011) — STAMP for medical devices
• Rasmussen (1997), “Risk management in a dynamic society”
• Perrow, Normal Accidents (1984)

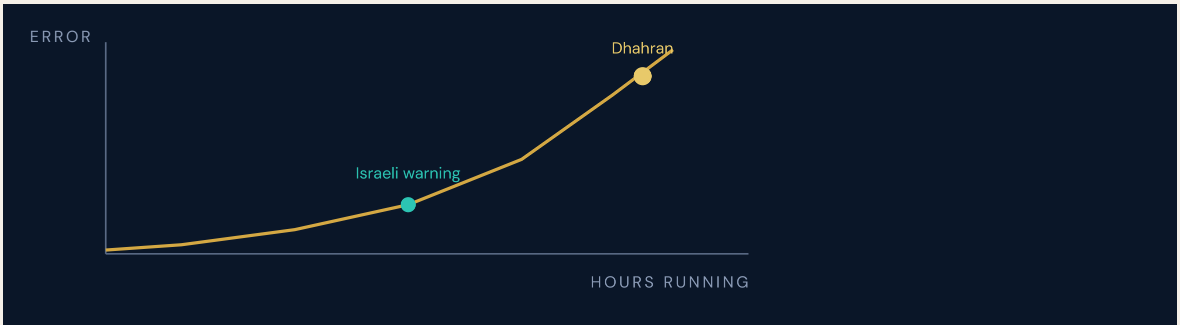
LENS COURSES LEN 5 LEN 7 LEN 2

CASE 19 DEFENSE 1991

Patriot Missile / Dhahran

D Designed Out H Human-System Interface K Knowledge & Institutional Memory

IMPACT 28 U.S. soldiers killed in their barracks; roughly 100 wounded



TRACKING ERROR GREW WITH EACH HOUR OF CONTINUOUS OPERATION

The Patriot was designed to defend Western Europe against Soviet aircraft in engagements measured in minutes. In February 1991 it was defending Saudi Arabia against ballistic missiles in batteries running more than a hundred hours without reset. The mismatch between original design context and field deployment was invisible to operators because no one had told them.

The system tracked time using a 24-bit fixed-point register. Each clock tick introduced a tiny rounding error. The error grew with every hour of continuous operation. After 100 hours the tracking gate was off by about a third of a second — far enough that an incoming Scud arrived where the radar was no longer looking. The missile that struck the Dhahran barracks on February 25, 1991 — killing twenty-eight soldiers of the 14th Quartermaster Detachment, a Pennsylvania Army Reserve unit — was not intercepted because the system was searching empty sky.

Two weeks earlier, Israeli operators had identified the drift and warned the U.S. Army. Engineers had produced a software patch. The patch arrived in Dhahran on February 26 — the day after the strike. The advisory issued to field operators told them to reboot the system after "very long" run times but did not specify what counted as long. GAO later found that Army officials had presumed users would never run a battery continuously for so long.

THE LEARNING ENGINEERING LENS

> *Army officials presumed that the users would not continuously run the batteries for such extended periods.*
>
> GAO/IMTEC-92-26, 1992

LE INSIGHT

Patriot is the canonical Designed-Out case from defense. The capability to maintain accuracy under sustained operation was omitted because the original concept of operations did not anticipate it. When the concept changed, the assumption did not travel. The system did not fail; the transition from one operational context to another did, and there was no capability infrastructure to catch it.

LENS APPROACH

LENS treats Patriot as the textbook example of **Capability Degradation Under System Change**. LEN 5 methods require operators of any transitioning system to have current capability-relevant documentation specifying what assumptions of the original design have changed. LEN 8 examines how organizational knowledge about design constraints travels from engineering to operations — and why, here, it arrived a day late.

REFLECTION QUESTIONS

1. The Patriot's design assumed short engagements. What assumption in a current system you work with would become lethal if the operational context shifted, and how would operators learn of the shift?

2. Construct the capability-transition artifact that should have accompanied the redeployment of Patriot batteries from Europe to Saudi Arabia. What would it have said, and who would have signed it?

WHO BUILDS THIS systems & human-factors engineering · human factors & interaction design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS task analysis · design prototyping · usability testing · cognitive walkthroughs · knowledge bases · data pipelines

SOURCES

→ U.S. General Accounting Office, GAO/IMTEC-92-26, Patriot Missile Defense: Software Problem Led to System Failure at Dhahran, Saudi Arabia (1992)
→ Barr Group, "An Update on the Patriot Missile Software Problem"
→ Skeel, University of Minnesota analysis of the Patriot software defect
→ Blair, Nikiforuk & Gertz, primary reporting in 1991–1992

FURTHER READING

• Skeel, "Roundoff Error and the Patriot Missile," SIAM News (1992)
• Reason, Human Error (1990) — swiss-cheese model
• Fixsen et al. (2005), Implementation Research

LENS COURSES LEN 5 LEN 8 LEN 9 LEN 6

CASE 51 INDUSTRIAL 1971 – 1978

Ford Pinto Fuel Tank

D Designed Out G Governance & Trust

IMPACT Fatalities attributed to rear-impact fuel-tank fires; foundational case in U.S. engineering ethics

$11

PER CAR · THE FIX FORD DECLINED

an internal cost-benefit calculated against projected wrongful-death settlements

PINTO — THE PRICE OF A DESIGNED-OUT SAFETY CAPABILITY

Ford engineers identified the Pinto's rear-impact fuel-tank vulnerability during pre-production testing. An eleven-dollar modification — a polyethylene shield between the tank and the differential — would have largely eliminated the failure mode. Internal cost-benefit calculations, later subpoenaed by Congress, found the modification more expensive than the projected wrongful- death settlements and chose not to implement it. The calculation was not secret to its authors; it was secret only to the public and to Pinto buyers.

The Pinto case became the founding case study in U.S. engineering ethics curricula. It is the most direct documentary evidence in the industrial dataset of a manufacturer pricing human life against an engineering change and choosing against the change. The lawsuits that followed produced the largest punitive damages of the era and contributed to the modern regulatory consumer-protection regime.

The capability that was designed out of the Pinto was the capability to survive a rear-impact crash. The reason it was designed out was institutional, not technical.

THE LEARNING ENGINEERING LENS

> *Ford's internal calculation found the cost of the fix higher than the projected cost of the deaths it would prevent.*
>
> PARAPHRASING DOWIE, MOTHER JONES, "PINTO MADNESS," 1977

LE INSIGHT

The Pinto is the canonical Designed-Out case in industrial engineering ethics. The fix was known. The cost was small. The decision was a documented institutional choice not to fix it. The case anchors the LENS argument that "Designed Out" is rarely an oversight — it is almost always a decision someone signs.

LENS APPROACH

LENS uses the Pinto in LEN 7 as the foundational engineering-ethics case and in LEN 1 as a problem-framing case for what happens when safety capability is treated as a cost line item.

REFLECTION QUESTIONS

1. Identify a current product in your domain where a safety capability is being priced against expected litigation. Whose signature would close that decision?

2. Design the regulatory deliverable that would have made the Pinto cost-benefit calculation unworkable in 1971.

WHO BUILDS THIS systems & human-factors engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS task analysis · design prototyping · governance frameworks · bias & evidence audits

SOURCES

→ Dowie, M. (1977), "Pinto Madness," Mother Jones
→ Grimshaw v. Ford Motor Co. (1981), 119 Cal. App. 3d 757
→ NHTSA Pinto investigation files (1978)

FURTHER READING

• Dowie (1977), Mother Jones
• Birsch & Fielder (eds.) The Ford Pinto Case (1994)
• Vandenberg, The Engineering Ethics casebook

LENS COURSES LEN 1 LEN 7 LEN 6

CASE 52 INDUSTRIAL 2008 – 2023

Takata Airbag Inflators

D Designed Out G Governance & Trust

IMPACT More than 30 deaths and hundreds of injuries linked to inflator ruptures; largest automotive recall in history

100M+

INFLATORS RECALLED · 19 AUTOMAKERS

the desiccant competitors used was the designed-out capability

TAKATA — THE LARGEST AUTOMOTIVE RECALL ON RECORD

Takata's ammonium-nitrate-based airbag inflators were designed without the desiccant that competitors used to stabilize the propellant against humidity and temperature cycling. Under hot, humid conditions over years of use, the propellant degraded; on deployment, the inflator could rupture and send metal fragments into the cabin. Takata's internal documents, later subpoenaed, showed engineers had raised the concern repeatedly. The company's own testing data showed the failure. The data was reported to regulators as anomalous rather than as a class.

The recall — when it finally came — covered more than a hundred million inflators across nineteen automakers and remains the largest in automotive history. Takata filed for bankruptcy. The parent company was sold. The death count continues to rise as older affected vehicles remain in service.

The desiccant was the capability that was designed out. The reason was cost. The fact that competitors used it was visible to anyone who looked.

THE LEARNING ENGINEERING LENS

> *Takata engaged in a sustained pattern of misrepresenting inflator safety data to its automaker customers and to regulators.*
>
> PARAPHRASING THE U.S. DEPARTMENT OF JUSTICE DEFERRED-PROSECUTION AGREEMENT, 2017

LE INSIGHT

Takata is the modern Pinto: an engineering choice to omit a stabilizing component, internal data showing the consequence, and a regulatory architecture that treated the manufacturer's representation as authoritative rather than as one input. The capability gap was at the regulator's evidence pipeline as much as at the manufacturer's bench.

LENS APPROACH

LENS uses Takata in LEN 4 as a measurement-system case (the manufacturer's data was reported but not interpreted as a class) and in LEN 7 as an industrial-governance failure spanning manufacturer, regulator, and customer auditors.

REFLECTION QUESTIONS

1. Where in your domain does a regulator receive manufacturer test data without an independent verification pipeline? What is the load-bearing trust assumption?
2. Design the verification regime that should have surrounded ammonium-nitrate inflator testing. Who funds it, who runs it, and what does it produce?

WHO BUILDS THIS systems & human-factors engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS task analysis · design prototyping · governance frameworks · bias & evidence audits

SOURCES

→ DOJ settlement with Takata Corporation (2017)
→ NHTSA Takata Airbag Recall coordination documents
→ Reuters and New York Times investigative reporting on Takata internal documents

FURTHER READING

- DOJ deferred-prosecution documents (2017)
- NHTSA Takata recall coordination archive
- Birsch & Fielder, engineering-ethics casebooks

LENS COURSES LEN 4 LEN 7

CASE 53 INDUSTRIAL 2002 – 2014

GM Ignition Switch

D Designed Out G Governance & Trust

IMPACT 124 deaths attributed; ~30M vehicles recalled; $900M federal settlement; criminal prosecution under wire-fraud statute

GM IGNITION SWITCH — EIGHT YEARS BETWEEN THE FIX AND THE RECALL

A faulty ignition switch in several GM compact-car models could rotate from “run” to “accessory” while driving, disabling power steering, power brakes, and — critically — the airbag system. GM engineers identified the problem in 2002 and approved a fix in 2006 **without changing the part number**, meaning the fix did not propagate as a recall would have. Cars with the faulty switch continued to be built and sold. The internal practice of changing a part without changing the part number violated GM’s own engineering standards. Internal investigations later called it “the GM nod” and “the GM salute.”

The Valukas Report commissioned by GM after the recall identified cultural patterns at the company that suppressed escalation of safety issues. Mary Barra, who became CEO during the recall, used the report as the basis for the company’s restructuring of safety decision-making.

The capability that was designed out of GM was the institutional pathway by which a safety problem reached the decision to recall.

THE LEARNING ENGINEERING LENS

> *There was a fundamental failure to use the formal escalation processes that GM had established.*
>
> PARAPHRASING THE VALUKAS REPORT TO THE GM BOARD OF DIRECTORS, 2014

LE INSIGHT

The GM ignition switch case is the canonical example of a corporate organizational structure that suppressed the upward flow of safety information by procedural design. The fix existed for eight years before the recall. The institutional path between the fix and the recall did not.

LENS APPROACH

LENS uses GM in LEN 7 as a corporate-governance failure case, in LEN 8 to study Valukas-style retrospective institutional analysis, and in LEN 4 for the part-number-as-measurement-instrument story (changing the part without changing the number suppressed the signal).

REFLECTION QUESTIONS

1. What information channel in your organization carries the same load that GM's part-number system did? Could it be silently bypassed?
2. Design the escalation deliverable that would have moved the GM ignition switch fix to a recall in 2006.

WHO BUILDS THIS systems & human-factors engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS task analysis · design prototyping · governance frameworks · bias & evidence audits

SOURCES

→ Valukas Report to GM (2014)
→ NHTSA Investigation of GM Ignition Switch (2014)
→ DOJ Deferred Prosecution Agreement with GM (2015)

FURTHER READING

- Valukas (2014), report to GM
- Hetu et al. (2016), corporate-governance analysis of GM
- Edmondson (2018), The Fearless Organization

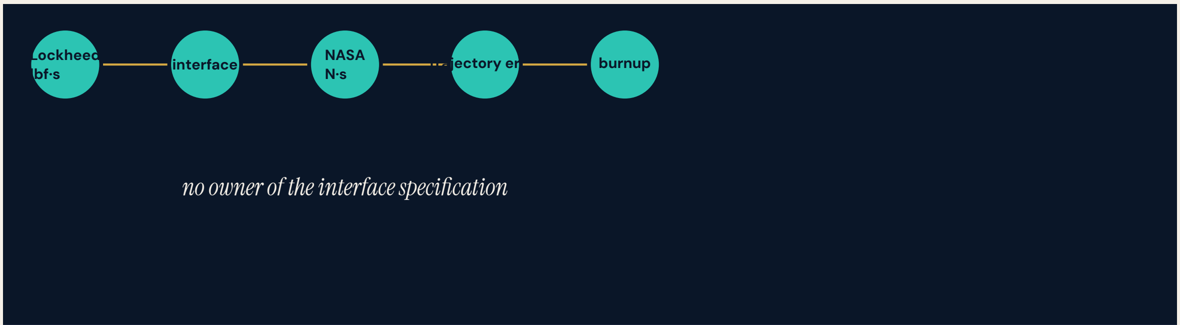
LENS COURSES LEN 4 LEN 7 LEN 8 LEN 6

CASE 54 SPACE / AEROSPACE 1999

Mars Climate Orbiter — Unit Mismatch

D Designed Out K Knowledge & Institutional Memory

IMPACT $327M spacecraft lost on approach to Mars; ground software produced thrust output in pound-force while navigation expected newtons



MARS CLIMATE ORBITER — THE UNOWNED INTERFACE

Lockheed Martin's ground software produced thrust outputs in pound-force seconds. NASA's navigation system expected newton- seconds. The unit conversion was not performed at the interface. As a result, every trajectory correction the spacecraft executed was off by a factor of about four and a half. The Mars Climate Orbiter arrived at Mars on a trajectory that took it too close to the planet's surface. It burned up in the atmosphere.

The Mars Climate Orbiter Mishap Investigation Board found no individual fault. The fault was a missing interface specification between two contractors and an institutional review process that had not caught the gap. Both contractors had done their internal jobs correctly; the boundary between them was the missing deliverable.

The case is canonical in software engineering for what happens when interface specifications are treated as documentation rather than as engineering deliverables.

THE LEARNING ENGINEERING LENS

> *The root cause was the failed translation of English units to metric units in a segment of ground-based, navigation-related mission software.*
>
> PARAPHRASING THE NASA MARS CLIMATE ORBITER MISHAP INVESTIGATION BOARD, 1999

LE INSIGHT

Mars Climate Orbiter is the textbook case for interface boundaries as engineering deliverables. The contractors did their work. The boundary between them did not have an owner. The capability that was missing was the interface-specification verification step.

LENS APPROACH

LENS uses Mars Climate Orbiter in LEN 5 to teach interface-as- requirement and in LEN 8 to discuss cross-contractor capability boundaries. The case is the foundational software-engineering parallel to Patriot (Case 19).

REFLECTION QUESTIONS

1. Identify a contractor-to-contractor interface in your domain whose specification ownership is unclear. What would the equivalent unit-mismatch look like there?

2. Design the interface-verification deliverable that would have caught the pound-force / newton boundary before launch.

WHO BUILDS THIS systems & human-factors engineering · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS task analysis · design prototyping · knowledge bases · data pipelines

SOURCES

→ NASA, Mars Climate Orbiter Mishap Investigation Board Report (1999)
→ Sauser, B. et al. (2009), retrospective on Mars Climate Orbiter
→ Software-engineering literature on interface-specification failures

FURTHER READING

• NASA MCO MIB Report (1999)
• Leveson (2011), Engineering a Safer World
• Brooks, The Mythical Man-Month (1995)

LENS COURSES LEN 5 LEN 8

CASE 55 TECHNOLOGY 2012

Knight Capital Trading Loss

D Designed Out K Knowledge & Institutional Memory

IMPACT $440M loss in 45 minutes; firm forced to seek emergency capital and was effectively acquired within months

$440M

LOSS IN 45 MINUTES · 7 OF 8 SERVERS UPDATED

dead code on the eighth server, repurposed flag, no deployment verification

KNIGHT CAPITAL — THE COST OF AN UNVERIFIED DEPLOYMENT

On the morning of August 1, 2012, Knight Capital deployed a software update to seven of eight servers in its order routing system. On the eighth, dead code from a previous release that had never been removed was still active. When the new flag was set in a configuration file, the dead code began executing buy orders at a rate of millions of shares per second. Knight's loss reached one hundred seventy-two million dollars in the first forty-five minutes of trading. By close, the loss exceeded four hundred forty million dollars. The firm was acquired within months.

The SEC investigation found that Knight's deployment procedure did not include a verification step to confirm that all eight servers held identical code. The dead code was a Chekhov's gun left in the repository years earlier — a function that had been retired but not removed. The flag that triggered it was repurposed for a different feature without auditing what else still listened to it.

Knight Capital is the canonical case for what happens when deployment procedure is not an engineering deliverable.

THE LEARNING ENGINEERING LENS

> *Knight did not have written procedures requiring a second technician to verify the deployment.*
>
> PARAPHRASING THE SEC ORDER INSTITUTING ADMINISTRATIVE PROCEEDINGS AGAINST KNIGHT CAPITAL, 2013

LE INSIGHT

Knight Capital is the financial-industry version of Mars Climate Orbiter (Case 54): a small, unspecified boundary inside a large system that took the institution down. The capability that was missing was deployment verification. The dead code was the proximate trigger; the absent procedure was the cause.

LENS APPROACH

LENS uses Knight Capital in LEN 5 to teach deployment-as-capability and in LEN 9 for the technical-debt argument: every line of dead code carries an option on a future failure. Students design the deployment deliverable that would have caught the eighth server.

REFLECTION QUESTIONS

1. Identify a deployment procedure in your domain whose verification step depends on convention rather than on a designed check. What is the eighth server?
2. Design the deployment deliverable that would prevent a Knight Capital-equivalent loss in your organization.

WHO BUILDS THIS systems & human-factors engineering · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS task analysis · design prototyping · knowledge bases · data pipelines

SOURCES

→ SEC Order Instituting Administrative Proceedings, Knight Capital (2013)
→ Knight Capital 8-K filing (2012)
→ Software-engineering literature on deployment safety

FURTHER READING

- SEC Knight Capital order (2013)
- Allspaw & Hammond, Web Operations (2010)
- Beyer et al. (eds.), Site Reliability Engineering (2016)

LENS COURSES LEN 5 LEN 9

Chapter 3

The Slow Erosion

Known risks became routine.

NASA culture allowed flying with flaws when problems were defined as normal and routine.

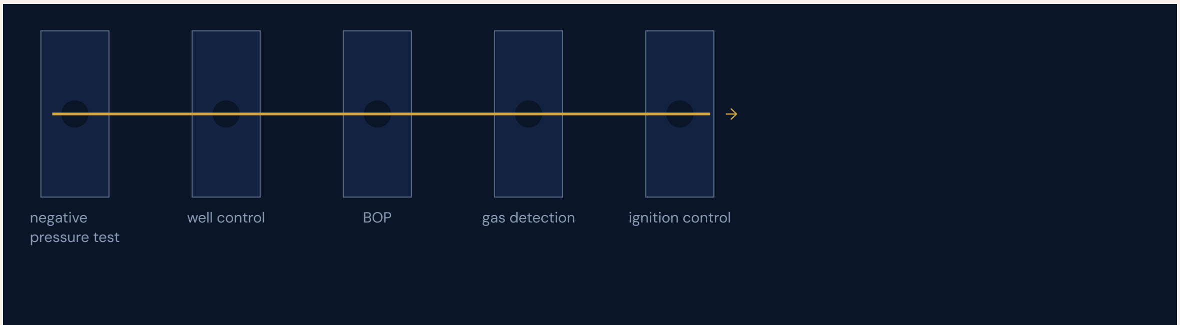
COLUMBIA ACCIDENT INVESTIGATION BOARD, 2003

CASE 4 ENERGY 2010

Deepwater Horizon

T Training Gap N Normalization of Deviance K Knowledge & Institutional Memory

IMPACT 11 killed; largest marine oil spill in U.S. history; $65B+ in damages



CASCADING DEFENSES — EACH FAILED IN TURN

The National Commission concluded that "human error, rather than mechanical failings, was the root cause." The rig crew misinterpreted the negative pressure test — the primary safety check for well integrity — and proceeded with displacement operations on a well that was no longer sealed. The blowout that followed killed eleven men and released roughly five million barrels of oil into the Gulf of Mexico.

BP and Transocean training had not adequately covered the well-control situations the crew actually faced. The National Academies workshop found "a lack of preparation and experience and an unclear chain of command prevented key decisions at every step." Every one of the rig's many defenses failed — some never engaged, some engaged too late, some not as designed: "a complex and interlinked series of mechanical failures, human judgments, engineering design, operational implementation and team interfaces."

THE LEARNING ENGINEERING LENS

> *The immediate causes of the Macondo well blowout can be traced to a series of identifiable mistakes ... that reveal such systematic failures in risk management that they place in doubt the safety culture of the entire industry.*
>
> NATIONAL COMMISSION, DEEP WATER (REPORT TO THE PRESIDENT), 2011

LE INSIGHT

Deepwater Horizon is the canonical multi-layer failure: training, procedure, supervision, contractor-coordination, and equipment all drifted independently until none caught the others. The negative- pressure test was the proximate trigger, but the system as a whole had been operating with accumulated procedural debt for years. The capability to recognize an unsafe well-state simply was not present at the moment it was needed.

LENS APPROACH

LENS uses Deepwater Horizon in LEN 5 as the canonical case for multi-layer capability analysis. Students reconstruct each defense layer, identify the deviation accumulated in each, and design the measurement system that would have surfaced the drift before April 20, 2010. LEN 8 examines contractor-coordination as a capability boundary problem.

REFLECTION QUESTIONS

1. Identify a multi-defense process in your domain. For each layer, what is the procedural debt that has accumulated since it was designed?

2. The negative-pressure test was the proximate trigger. Design a capability check that would have caught the misinterpretation in real time.

WHO BUILDS THIS learning science & instructional design · safety science & organizational analysis · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · incident analysis · safety dashboards · knowledge bases · data pipelines

SOURCES

→ National Commission on the BP Deepwater Horizon Oil Spill (2011)
→ BOEMRE/Coast Guard Joint Investigation Report (2011)
→ National Academies Workshop on Deepwater Drilling Safety (2018)
→ Leveson analysis of Deepwater Horizon

FURTHER READING

• National Commission (2011), Deep Water: The Gulf Oil Disaster
• Reason, Human Error (1990) — swiss-cheese model
• Vaughan, The Challenger Launch Decision (1996)

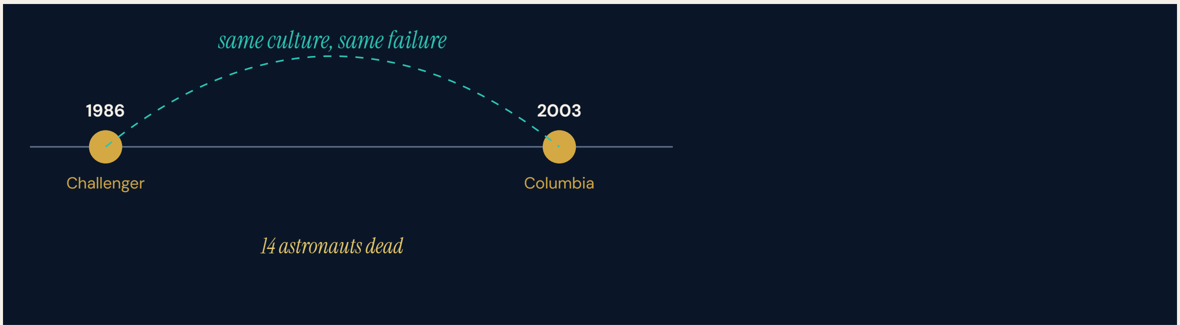
LENS COURSES LEN 5 LEN 8 LEN 3

CASE 6 SPACE / AEROSPACE 1986 / 2003

Challenger & Columbia

N Normalization of Deviance K Knowledge & Institutional Memory G Governance & Trust

IMPACT 14 astronauts killed (7 per accident); 17 years between identical organizational pathologies



17 YEARS, SAME PATTERN — CHALLENGER TO COLUMBIA

The Columbia Accident Investigation Board concluded plainly: “The NASA organizational culture had as much to do with this accident as the foam.” Seventeen years after Challenger, CAIB found the same cultural patterns intact: a Shuttle Program in which flying with flaws was permissible when the flaws were defined as routine, and in which the organizational structure suppressed the upward flow of critical information.

Diane Vaughan’s concept of “normalization of deviance” — where deviations from safety protocols become ingrained through production pressures, poor communication, and workplace culture — was developed from the Challenger investigation and then validated, against the intent of its author, by Columbia. Two crews paid the price for an organizational learning failure diagnosed in 1986 and never repaired — the strongest single-institution evidence that culture is engineerable, and that leaving it unengineered is a choice with consequences.

THE LEARNING ENGINEERING LENS

> *These repeating patterns mean that flawed practices embedded in NASA's organizational system continued for 20 years and made substantial contributions to both accidents.*
>
> COLUMBIA ACCIDENT INVESTIGATION BOARD, 2003

LE INSIGHT

Challenger/Columbia is the strongest documentary evidence that organizational culture is an engineerable property of a system, and that diagnoses without engineered remediations decay. The same pathology, twice, seventeen years apart, in the same institution, with the same human cost — and with the diagnosis already on the record. Capability engineering treats culture as a deliverable.

LENS APPROACH

LENS uses the pair in LEN 1 to introduce normalization of deviance as a systems concept, in LEN 8 to examine why organizational learning failed across a 17-year interval, and in LEN 7 to address the governance failure that allowed a known pathology to persist.

REFLECTION QUESTIONS

1. What is the equivalent "diagnosed but not repaired" pathology in your domain? What evidence would close the loop?
2. The CAIB called culture engineerable. Sketch the engineering deliverable for the cultural intervention you would propose in your domain — including its measurement signal.

WHO BUILDS THIS safety science & organizational analysis · knowledge management & data engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS incident analysis · safety dashboards · knowledge bases · data pipelines · governance frameworks · bias & evidence audits

SOURCES

→ Rogers Commission, Report on the Space Shuttle Challenger Accident (1986)
→ Columbia Accident Investigation Board, CAIB Report Vol. I (2003)
→ Vaughan, The Challenger Launch Decision (1996)
→ ScienceDirect, organizational analysis of NASA culture

FURTHER READING

• Vaughan (1996), The Challenger Launch Decision
• Starbuck & Farjoun (eds.) Organization at the Limit: Lessons from the Columbia Disaster (2005)
• Weick & Sutcliffe, Managing the Unexpected (2007)

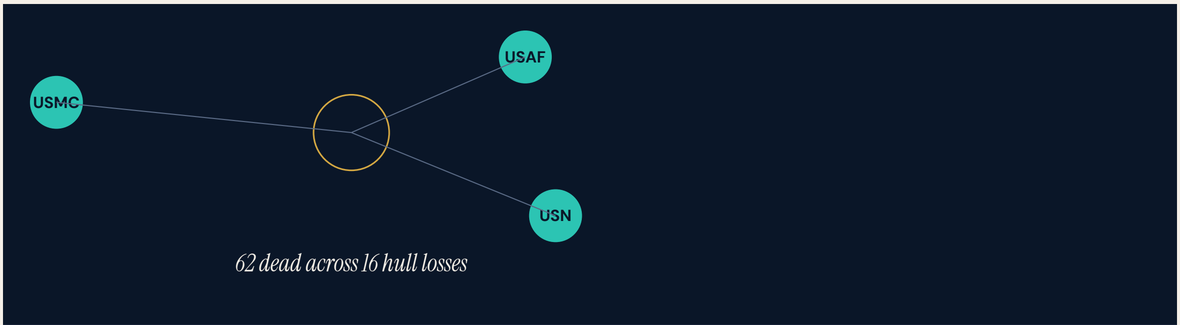
LENS COURSES LEN 1 LEN 7 LEN 8 LEN 3

CASE 11 DEFENSE 1991 – PRESENT

V-22 Osprey

T Training Gap H Human-System Interface N Normalization of Deviance

IMPACT 62 killed across 16 hull-loss incidents (through 2023); mission-capable rates ~50–60%



PERSISTENT CAPABILITY GAPS ACROSS THREE SERVICES

A 2025 NAVAIR review and accompanying GAO report found that "materiel failure and human-error factors were the most frequently cited causal factors in serious Osprey accidents." The 2000 Marana crash that killed nineteen Marines occurred during testing in which the program had cut corners on trials to keep pace with schedule. Poorly understood maintenance procedures and lack of communication across the three services operating the platform were identified as persistent factors. Mission-capable rates remain at half of design.

The V-22 is structurally instructive because the platform has survived for thirty years in a state of normalized capability shortfall. The shortfall has been documented in successive GAO reports without producing the readiness regime that the platform requires. Each crash carries the same set of contributing factors; each is followed by service-level adjustments that do not converge across the three services that fly it.

The capability problem here is not that no one knows it exists. It is that the system designed to remediate it has accepted its own incompleteness as the cost of operating the airframe at all.

THE LEARNING ENGINEERING LENS

> *Materiel failure and human-error factors were the most frequently cited causal factors in serious Osprey accidents.*
>
> PARAPHRASING THE NAVAIR INDEPENDENT SAFETY REVIEW OF THE V-22, 2025

LE INSIGHT

V-22 demonstrates the steady-state version of normalization of deviance: a platform whose shortfall has been documented, reviewed, and acted on across multiple administrations without producing sustained improvement. The capability gap has itself been normalized. Each incident is treated as an event rather than as a sample from a distribution the program continues to draw from.

LENS APPROACH

LENS treats V-22 in LEN 5 as a multi-service capability-coordination problem and in LEN 8 as a study in long-cycle organizational learning failure. Students design the evidence system that would distinguish a true reduction in mishap rate from the natural variation of a chronically marginal platform.

REFLECTION QUESTIONS

1. Identify a platform or process in your domain that has been operating in a documented shortfall for years. What measurement would have to change for the shortfall to become unacceptable?
2. The V-22's three services do not converge on remediation. Design the governance structure that would force convergence.

WHO BUILDS THIS learning science & instructional design · human factors & interaction design · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs · incident analysis · safety dashboards

SOURCES

→ NAVAIR Independent Safety Review of the V-22 (2025)
→ GAO V-22 Sustainment Report (2025)
→ USNI News, V-22 mishap coverage (2023–2025)
→ PBS NewsHour V-22 investigative reporting

FURTHER READING

• Rasmussen (1997), “Risk management in a dynamic society”
• Weick, “Organizational Culture as a Source of High Reliability” (1987)
• Snook, Friendly Fire (2000)

LENS COURSES LEN 5 LEN 8 LEN 3

CASE 56 ENERGY INDUSTRIAL 2005

Texas City BP Refinery Explosion

N Normalization of Deviance T Training Gap K Knowledge & Institutional Memory G Governance & Trust

IMPACT 15 killed, 180 injured at BP's Texas City refinery; CSB found systemic safety-culture decay

TEXAS CITY — THE WRONG MEASUREMENT, REPORTED AS A WIN

A startup overfilled a distillation tower because the operators were working from instruments that had been malfunctioning for years. Vapor accumulated, found an ignition source — an idling pickup truck — and exploded. Fifteen contractors in temporary trailers parked next to a process unit were killed instantly. The U.S. Chemical Safety and Hazard Investigation Board (CSB) found accumulated safety-culture decay, deferred maintenance, broken instruments tolerated as routine, and a corporate cost-cutting program that had been celebrated internally for the savings it produced.

The CSB's Texas City report became the most influential CSB report in the agency's history. It elevated the distinction between **process safety** and personal safety — developed over decades in the CCPS literature and codified in OSHA's 1992 PSM standard — into mainstream U.S. industrial-regulation practice after 2005. Personal-safety metrics at BP Texas City had been among the best in the industry. Process-safety metrics, had anyone been measuring them, would have shown the opposite.

The case is the canonical evidence that the wrong measurement is worse than no measurement.

THE LEARNING ENGINEERING LENS

> *Indicators of personal safety are not indicators of process safety.*
>
> U.S. CHEMICAL SAFETY BOARD, BP TEXAS CITY FINAL INVESTIGATION REPORT, 2007

LE INSIGHT

Texas City is the foundational evidence that organizations can be measuring the wrong thing and reporting excellent performance while their actual capability gap widens. Personal-safety metrics had no information about process-safety state. The signal regime was blind in the dimension that killed people.

LENS APPROACH

LENS uses Texas City in LEN 4 as the canonical wrong-measurement case and in LEN 7 for the governance failure that allowed years of cost-cutting to be reported as wins. Studio projects design the process-safety measurement deliverable.

REFLECTION QUESTIONS

1. Identify a “personal safety vs. process safety” equivalent in your domain. What capability gap is invisible to the metric your institution currently reports?
2. Design the process-safety dashboard that BP Texas City’s executives should have been receiving in 2003.

WHO BUILDS THIS safety science & organizational analysis · learning science & instructional design · knowledge management & data engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS incident analysis · safety dashboards · scenario simulation · competency models · knowledge bases · data pipelines · governance frameworks · bias & evidence audits

SOURCES

→ CSB Investigation Report 2005-04-I-TX, Refinery Explosion and Fire (2007)
→ Baker Panel Report on BP North American Refineries (2007)
→ Hopkins, Failure to Learn: The BP Texas City Refinery Disaster (2008)

FURTHER READING

• CSB Texas City Report (2007)
• Hopkins (2008), Failure to Learn
• Reason (1997), Managing the Risks of Organizational Accidents

LENS COURSES LEN 4 LEN 7 LEN 3

CASE 57 ENERGY 2002

Davis-Besse Reactor Head Corrosion

N Normalization of Deviance K Knowledge & Institutional Memory G Governance & Trust

IMPACT Football-sized cavity discovered in the reactor pressure-vessel head; near-miss; ~$600M recovery and extended outage

1/4″

OF STAINLESS CLADDING REMAINED

between a 6-inch corrosion cavity and reactor coolant at 2,200 psi

DAVIS-BESSE — THE POST-TMI NEAR-MISS

During refueling at the Davis-Besse nuclear plant in Ohio, workers discovered a cavity in the reactor pressure vessel head about the size of a football, eaten through six inches of carbon steel by boric acid leaking from a cracked nozzle. Only a quarter-inch of stainless-steel cladding remained between the cavity and reactor coolant at 2,200 psi. A breach would have produced a medium-break loss-of-coolant accident whose recovery margin was significantly degraded.

The boric acid leakage had been observable for years. Inspections had been deferred. FirstEnergy had requested — and received from the NRC — a deferral of a mandatory inspection (NRC Bulletin 2001-01) to align with its February 2002 refueling outage. The NRC's Office of Inspector General later found that the deferral decision had inappropriately weighted economic over safety arguments.

Davis-Besse is the canonical post-TMI near-miss case in U.S. commercial nuclear power. INPO and the NRC restructured significant portions of their inspection regimes in response.

THE LEARNING ENGINEERING LENS

> *The NRC's actions were not adequate to ensure that safety would not be compromised.*
>
> NRC OFFICE OF INSPECTOR GENERAL, 2003

LE INSIGHT

Davis-Besse is the case for how regulator-operator dynamics can erode the capability of an industry that had previously demonstrated, through INPO, that it knew how to engineer safety. Regulatory capture is a capability failure at the institutional layer above operations.

LENS APPROACH

LENS uses Davis-Besse in LEN 7 to study regulatory-capture dynamics and in LEN 8 to compare with INPO's earlier success. The case is a reminder that institutional capability requires sustained re-engineering.

REFLECTION QUESTIONS

1. Identify a regulator-operator relationship in your domain in which the regulator may be at risk of accepting the operator's economic argument over its own safety judgment. What signal would surface it?
2. Design the institutional control that would prevent a Davis-Besse-style deferral from being granted.

WHO BUILDS THIS safety science & organizational analysis · knowledge management & data engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS incident analysis · safety dashboards · knowledge bases · data pipelines · governance frameworks · bias & evidence audits

SOURCES

→ NRC OIG, Davis-Besse Lessons Learned Report (2003)
→ Lochbaum et al. (Union of Concerned Scientists) analysis (2002)
→ Government Accountability Office, GAO-04-415

FURTHER READING

• NRC OIG Davis-Besse report (2003)
• Lochbaum (UCS) analyses
• Rees (1994), Hostages of Each Other

LENS COURSES LEN 7 LEN 8

CASE 58 HEALTHCARE 2005 – 2009

Mid Staffordshire NHS Foundation Trust

G Governance & Trust N Normalization of Deviance K Knowledge & Institutional Memory

IMPACT Excess deaths at Stafford Hospital documented across years; the Francis Inquiry produced 290 recommendations and restructured UK patient-safety governance

MID STAFFS — MEASUREMENT AND REALITY DIVERGED ACROSS YEARS

Patients at Stafford Hospital in England were left in their own excrement, denied food and water, given the wrong medication or none at all. Mortality was substantially above expected. The Foundation Trust had been pursuing — and had achieved — financial targets that depended on staffing cuts. The Healthcare Commission investigated. The Department of Health did not. The local trust board, which was supposed to be the safety conscience, was instead the principal architect of the staffing decisions that produced the harm.

The Francis Inquiry's final report ran to two thousand pages and made two hundred ninety recommendations. It identified the gap between the trust's reported performance and the patients' actual experience as the structural problem: every governance layer above the ward had received reports that the hospital was meeting its targets. None of those reports were checked against what was happening to patients.

Mid Staffs is the strongest case in the dataset for the harm produced when measurement and reality diverge over years.

THE LEARNING ENGINEERING LENS

> *The system as a whole failed in its most essential duty — to protect patients from unacceptable risks of harm.*
>
> ROBERT FRANCIS QC, REPORT OF THE MID STAFFORDSHIRE NHS FOUNDATION TRUST PUBLIC INQUIRY, 2013

LE INSIGHT

Mid Staffordshire is the British analog of the VA wait-time scandal (Case 32). Measurement and reality diverged over years; every layer of governance above the operating environment was acting on the measurement; patients paid the cost of the divergence. The Francis Inquiry recommendations changed how the NHS thinks about patient safety as an institutional capability.

LENS APPROACH

LENS uses Mid Staffs in LEN 4 for the divergence-of-measurement problem and in LEN 7 for the governance failure across multiple layers. Studio projects examine the Francis recommendations as a template for institutional reform deliverables.

REFLECTION QUESTIONS

1. Identify a multi-layer reporting chain in your domain. What would it take for the top layer to know whether the reports correspond to reality?

2. The Francis Inquiry produced 290 recommendations. Pick five that you think were most load-bearing and explain why.

WHO BUILDS THIS governance, ethics & measurement · safety science & organizational analysis · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · incident analysis · safety dashboards · knowledge bases · data pipelines

SOURCES

→ Francis QC, R. (2013), Report of the Mid Staffordshire NHS Foundation Trust Public Inquiry
→ Healthcare Commission Investigation (2009)
→ Berwick, D., A Promise to Learn — A Commitment to Act (2013)

FURTHER READING

• Francis QC (2013)
• Berwick (2013), A Promise to Learn
• Walshe & Higgins (2002) on NHS safety governance

LENS COURSES LEN 4 LEN 7 LEN 3

CASE 59 ENERGY INDUSTRIAL 2006

Sago Mine Disaster

N Normalization of Deviance T Training Gap K Knowledge & Institutional Memory

IMPACT 12 killed in a West Virginia coal-mine explosion; emergency-response failures compounded the initial event; MINER Act of 2006

SAGO — CUMULATIVE INADEQUACY ACROSS MULTIPLE DEFENSES

A lightning strike ignited methane in a sealed area of the Sago Mine in West Virginia. The seal failed, the explosion propagated, and thirteen miners were trapped behind it. Initial reports — broadcast nationally — said twelve had been found alive. The reports were wrong; twelve had died and only one (Randal McCloy Jr.) survived. The miscommunication tortured the families for hours and exposed an emergency-response architecture that had not been engineered to handle the kind of information flow a modern news cycle creates.

The MSHA investigation found the seals had been built to a design that did not meet then-current standards, that the mine's emergency plan was inadequate, and that miner training in self-rescue had not been kept current. The Sago disaster led to the MINER Act of 2006, which strengthened mine-rescue requirements and required refuge chambers underground.

The case is the modern coal-mining analog of the slow-erosion pattern: seals, plans, training, all marginally adequate for years, all simultaneously inadequate in the only minute that mattered.

THE LEARNING ENGINEERING LENS

> *The miners faced multiple equipment, training, and emergency-response shortcomings that compounded their initial trapping.*
>
> MINE SAFETY AND HEALTH ADMINISTRATION, SAGO INVESTIGATION REPORT, 2007

LE INSIGHT

Sago is the case for the cumulative inadequacy pattern in industrial accidents. No single failure caused the disaster. Each failure on its own would have been recoverable. The combination was not, and the combination had been the operating condition of the mine for years.

LENS APPROACH

LENS uses Sago in LEN 5 for cumulative-inadequacy analysis and in LEN 8 for the legislative-reform arc that followed. Studio projects compare Sago and Upper Big Branch (Case 60) as paired cases.

REFLECTION QUESTIONS

1. Identify a process in your domain that is marginally adequate across multiple parameters. What is the cumulative failure mode?
2. Sago produced the MINER Act. What legislative change would your domain require if a Sago-equivalent occurred?

WHO BUILDS THIS safety science & organizational analysis · learning science & instructional design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS incident analysis · safety dashboards · scenario simulation · competency models · knowledge bases · data pipelines

SOURCES

→ MSHA, Report of Investigation, Sago Mine (2007)
→ MINER Act of 2006, Public Law 109-236
→ West Virginia Sago Investigation report (2006)

FURTHER READING

• MSHA Sago report (2007)
• Hopkins, Failure to Learn (2008)
• West Virginia governor's independent investigation

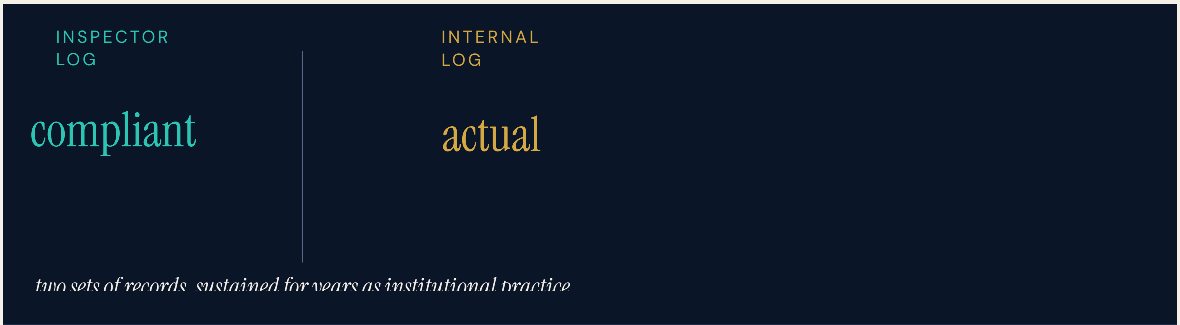
LENS COURSES LEN 5 LEN 8

CASE 60 ENERGY INDUSTRIAL 2010

Upper Big Branch Mine Explosion

N Normalization of Deviance G Governance & Trust K Knowledge & Institutional Memory

IMPACT 29 killed in West Virginia coal mine; MSHA found systematic falsification of safety records; first U.S. mining-industry CEO convicted of a federal mine-safety charge (misdemeanor)



UPPER BIG BRANCH — MEASUREMENT ENGINEERED AGAINST THE REGULATOR

Coal dust and methane ignited at Massey Energy's Upper Big Branch Mine, killing twenty-nine miners — the worst U.S. mine disaster in forty years. MSHA's investigation, and a parallel U.S. Department of Justice probe, found that Massey had maintained two sets of records: an internal log of actual conditions and a separate log for MSHA inspectors. Mine foremen were instructed to suppress methane readings, to disable monitors, to falsify pre-shift examinations. Massey CEO Don Blankenship was eventually convicted of a misdemeanor count of conspiring to willfully violate mine safety standards — the first U.S. mining-industry CEO to face a federal criminal conviction on mine-safety charges. (He was acquitted on the felony counts.)

The dual-records architecture was not exceptional behavior. It was a stable institutional practice maintained across years and multiple management layers. The capability gap was not in the miners — it was in the executive ranks that designed and operated a measurement system specifically to defeat the regulator.

Upper Big Branch is the strongest case in U.S. industrial safety for criminal accountability of corporate-officer roles in capability failure.

THE LEARNING ENGINEERING LENS

> *Massey kept two sets of books — one for inspectors, one for itself.*
>
> PARAPHRASING FEDERAL INVESTIGATORS, MSHA UPPER BIG BRANCH REVIEWS (2011 – 2012)

LE INSIGHT

Upper Big Branch is the case for deliberately engineered measurement deception. The dual-records practice was not error. It was capability design — by management, against the regulator. The case anchors the LENS argument that decision-grade evidence requires structural protection against the institution that produces it having a stake in what it reports.

LENS APPROACH

LENS uses Upper Big Branch in LEN 4 as the most explicit case for measurement-system protection and in LEN 7 as a corporate-criminal accountability case. Studio projects examine what regulator-side architecture would have detected the dual-records system earlier.

REFLECTION QUESTIONS

1. Where in your domain could two sets of records plausibly be kept? What architectural change would make that impossible?
2. What does it mean for a measurement system to be "structurally protected" from the institution that produces it?

WHO BUILDS THIS safety science & organizational analysis · governance, ethics & measurement · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS incident analysis · safety dashboards · governance frameworks · bias & evidence audits · knowledge bases · data pipelines

SOURCES

→ MSHA Internal Review of Upper Big Branch (2012)
→ United States v. Blankenship (S.D.W. Va. 2015)
→ Berkes, NPR investigative reporting on Massey

FURTHER READING

• MSHA UBB Internal Review (2012)
• Blankenship trial record
• Hopkins (2012), Disastrous Decisions

LENS COURSES LEN 4 LEN 7

CASE 61 ENERGY 2011

Fukushima Daiichi

N Normalization of Deviance G Governance & Trust K Knowledge & Institutional Memory

IMPACT Three reactor meltdowns following the Tōhoku earthquake and tsunami; ~160,000 people displaced; cleanup projected at $200B+

3 of 6

REACTORS MELTED DOWN

tsunami exceeded design basis the institutional evidence had already questioned

FUKUSHIMA DAIICHI — "MADE IN JAPAN," PER THE DIET INQUIRY

The Tōhoku earthquake of March 11, 2011, and the tsunami that followed, exceeded the design-basis assumptions of TEPCO's Fukushima Daiichi nuclear plant. The plant lost off-site power. Its emergency diesel generators were inundated by water that came in higher than the site's seawall. Cooling was lost. Three reactor cores melted down. Hydrogen explosions distributed radioactive material across the surrounding region.

The independent investigation chaired by Kiyoshi Kurokawa (NAIIC) identified the accident as "made in Japan" — the product of regulatory capture, deference to authority, and institutional reluctance to challenge utility assumptions. Tsunami evidence going back to the ninth-century Jōgan event, discussed in TEPCO's internal tsunami-assessment work c. 2008, suggested the design-basis assumptions were optimistic. The institutional path by which those assessments would have changed the seawall did not exist. The other major investigations — the Hatamura government commission (2012) and the IAEA Director General's report (2015) — emphasized under-estimation of external hazards and defense-in-depth design assumptions over the cultural critique; the Kurokawa framing is the most-cited but not the only consensus interpretation.

Fukushima is the canonical case in the post-TMI era for the failure of regulator-utility dynamics under cultural pressure against confrontation.

THE LEARNING ENGINEERING LENS

> *What must be admitted — very painfully — is that this was a disaster "Made in Japan."*
>
> NATIONAL DIET OF JAPAN FUKUSHIMA NUCLEAR ACCIDENT INDEPENDENT INVESTIGATION COMMISSION, 2012

LE INSIGHT

Fukushima is the post-TMI case that establishes that the INPO pattern (Case 16) is not self-executing. The U.S. industry built INPO; the Japanese industry did not. The cost of the difference, paid in 2011, is the strongest available evidence that capability institutions must be deliberately built, not assumed.

LENS APPROACH

LENS uses Fukushima in LEN 8 as the cross-cultural comparison to INPO and in LEN 7 for the regulator-utility dynamics study. The case is paired with INPO and Davis-Besse to triangulate what sustained nuclear-safety capability requires.

REFLECTION QUESTIONS

1. Identify a regulatory regime in your domain whose effectiveness depends on a cultural willingness to challenge authority. What if the culture changes?
2. INPO is U.S.-specific. Design the structural features that would have to be in place for a comparable institution to function in a different national context.

WHO BUILDS THIS safety science & organizational analysis · governance, ethics & measurement · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS incident analysis · safety dashboards · governance frameworks · bias & evidence audits · knowledge bases · data pipelines

SOURCES

→ National Diet of Japan Independent Investigation Commission (2012)
→ IAEA Director General Report on the Fukushima Daiichi Accident (2015)
→ Lochbaum, Lyman & Stranahan, Fukushima: The Story of a Nuclear Disaster (2014)

FURTHER READING

• Kurokawa Commission Report (2012)
• Lochbaum, Lyman & Stranahan (2014)
• Funabashi & Kitazawa (2012), Fukushima in Review

LENS COURSES LEN 7 LEN 8 LEN 3

CASE 62 ENERGY 2003

Northeast Blackout

H Human-System Interface K Knowledge & Institutional Memory

IMPACT 55 million people without power across eight U.S. states and Ontario; $6B+ economic loss; FERC Order 693 followed

NORTHEAST BLACKOUT — SILENT ALARM FAILURE ABSORBED THE OPERATOR'S AWARENESS

A tree contact in Ohio sagged a 345 kV transmission line into a branch and tripped a single transmission asset. Under normal conditions, the loss would have been absorbed by the grid. On August 14, 2003, FirstEnergy's alarm system had been silently failing for over an hour, and its operators were unaware that the line had tripped. Subsequent line trips cascaded across the Eastern Interconnection. Within minutes, fifty-five million people were without power.

The U.S.-Canada Power System Outage Task Force found that FirstEnergy operators did not have the situational awareness to detect the developing failure, that the company's alarm system had failed without notice, that vegetation management had been inadequate, and that the regional reliability coordinator did not have the authority or the information to intervene. The reforms that followed produced FERC Order 693, which made compliance with reliability standards mandatory and enforceable for the first time.

The capability gap was at the boundary between automation and operator: when the alarm system failed silently, the operators had no other signal that they were losing control of their grid segment.

THE LEARNING ENGINEERING LENS

> *FirstEnergy … did not have adequate situational awareness of conditions on its system.*
>
> U.S.-CANADA POWER SYSTEM OUTAGE TASK FORCE, FINAL REPORT ON THE AUGUST 14 2003 BLACKOUT, APRIL 2004

LE INSIGHT

The 2003 blackout is the canonical case for silent automation failure: a system that stops doing its job without telling its operators. The capability that was missing was the meta-monitor — the system that watches the monitor. The reform that followed treated grid-reliability compliance as a deliverable rather than as a best practice.

LENS APPROACH

LENS uses the 2003 blackout in LEN 2 as a Human-AI Teaming case for silent-automation-failure handling and in LEN 8 for the legislative-reform arc that produced enforceable reliability standards.

REFLECTION QUESTIONS

1. Identify an automated monitoring system in your domain whose silent failure would not be detected by its operators. How would they know?
2. Design the meta-monitor that should have been watching FirstEnergy's alarm system in 2003.

WHO BUILDS THIS human factors & interaction design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS usability testing · cognitive walkthroughs · knowledge bases · data pipelines

SOURCES

→ U.S.-Canada Power System Outage Task Force, Final Report on the August 14, 2003 Blackout (2004)
→ North American Electric Reliability Council reports (2004)
→ FERC Order No. 693, Mandatory Reliability Standards (2007)

FURTHER READING

• U.S.-Canada Outage Task Force Final Report (2004)
• Endsley (1995), situation awareness theory
• Casner & Hutchins (2019), automation transparency requirements

LENS COURSES LEN 2 LEN 8

Chapter 4

The Interface Problem

Correct performance was made impossible.

The shootdown of Flight 655 showcases the chaos of combat environments, but it also reveals lessons for technology adoption and its use in stressful situations.

PARAPHRASING U.S. NAVAL PROCEEDINGS RETROSPECTIVE ON THE VINCENNES INCIDENT, 2018

CASE 18 DEFENSE 1988

USS Vincennes & Iran Air Flight 655

H Human-System Interface T Training Gap

IMPACT 290 civilians killed — the deadliest shootdown of a commercial airliner by a military force

AEGIS RADAR DATA CORRECTLY DISPLAYED; FIVE OPERATORS READ IT WRONG

In the closing weeks of the Iran-Iraq War, during a surface engagement with Iranian patrol boats, the USS Vincennes misidentified Iran Air Flight 655 — a civilian Airbus A300 climbing on a scheduled commercial route — as a descending hostile F-14. The cruiser fired two surface-to-air missiles. Two hundred ninety civilians died.

The Aegis combat system did not malfunction. The radar data correctly showed an aircraft ascending. Five different crew members in the Combat Information Center independently believed the aircraft was **descending.** The formal U.S. Navy inquiry attributed the tragedy to “human error under extreme stress” — to confirmation bias, to “stress and unconscious distortion of data.” Operators interpreted all indications through the lens of a presumed hostile act because the framing arrived before the data did.

The Naval Proceedings retrospective in 2018 placed the case at the center of the human-AI teaming conversation: the most advanced combat system afloat failed because the interface between the system and its operators did not support correct performance under operational stress.

THE LEARNING ENGINEERING LENS

> *The shootdown of Flight 655 … reveals lessons for technology adoption and its use in stressful situations.*
>
> PARAPHRASING U.S. NAVAL PROCEEDINGS RETROSPECTIVE ON THE VINCENNES INCIDENT, 2018

LE INSIGHT

Vincennes is the canonical case for human-AI teaming under stress. The system did not lie. The operators did not act in bad faith. The interface and the operational framing combined to make a particular misreading not just possible but likely. Correct performance was possible in principle and unsustainable in practice — and that gap is the engineering problem.

LENS APPROACH

LENS uses Vincennes in LEN 2 as the foundational case for human-AI teaming under operational stress, and in LEN 5 to teach the analysis of capability requirements in conditions the original interface designers never expected. The case sits at the center of the program's argument that interface design is a capability deliverable, not an aesthetic one.

REFLECTION QUESTIONS

1. Identify a high-stress interface in your domain. What framing arrives before the data, and how does it shape what operators see?

2. Vincennes' operators acted under tunnel vision. Design the procedural intervention that would have forced one of the five to call out the contradiction.

WHO BUILDS THIS human factors & interaction design · learning science & instructional design — plus domain experts and a learning engineer to integrate.

TOOLS usability testing · cognitive walkthroughs · scenario simulation · competency models

SOURCES

→ Fogarty Investigation Report on the Vincennes Incident (1988)
→ U.S. Naval Proceedings retrospective (2018)
→ Cummings, "Human Supervisory Control of Weapon Systems" (MIT)
→ Britannica, Iran Air Flight 655

FURTHER READING

• Klein, Sources of Power (1998) on naturalistic decision-making
• Endsley, "Toward a Theory of Situation Awareness" (1995)
• Woods et al. (1994), Behind Human Error

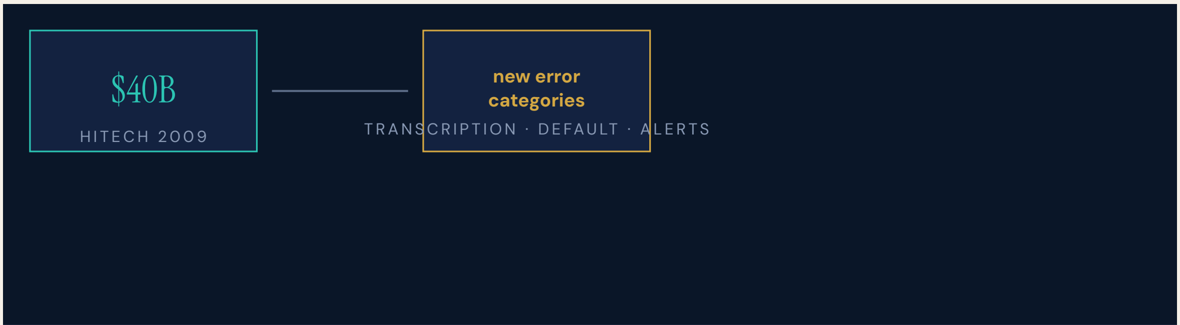
LENS COURSES LEN 5 LEN 2

CASE 25 HEALTHCARE 2005 – PRESENT

EHR / CPOE Implementation

H Human-System Interface D Designed Out G Governance & Trust

IMPACT ~$30B federal investment under HITECH; documented increase in pediatric ICU mortality post-CPOE at one institution; ongoing usability harm at scale



$40B INVESTED; NEW ERROR PATHWAYS EMERGED FROM INTERFACE DESIGN

The HITECH Act of 2009 authorized roughly thirty billion dollars in incentive payments to accelerate Electronic Health Record adoption. Adoption surged. Computerized Provider Order Entry — CPOE — was deployed widely. So were new categories of harm. Han et al. (2005) reported that deployment of a commercial CPOE system at one pediatric ICU was associated with a near-doubling of mortality (the result is from a single institution and provoked debate; later studies of other CPOE deployments showed mixed or improved outcomes). The finding remains canonical for the warning that powerful tools without surrounding workflow and human-factors integration can disrupt established practice at the moment of greatest acuity.

A 2023 KLAS Arch Collaborative survey across more than two hundred hospitals found that EHR usability is consistently the number-one complaint of physicians, nurses, and pharmacists, and that end-user experience scores correlate with patient-safety outcomes. A patient's cancer treatment was delayed by years because a default setting surfaced an old normal Pap result instead of a recent abnormal one. A baby died from a drug overdose caused by a transcription error that an automated alert would have caught had it been activated.

The systems were built to administrative and billing specifications, not to clinical workflow or human-factors specifications. There is still no regulatory framework to monitor EHR system safety.

THE LEARNING ENGINEERING LENS

> *Reports of new types of errors directly related to EHR implementation — errors that can compromise quality of care and patient safety — have emerged.*
>
> PARAPHRASING SITTIG & SINGH, EHR-RELATED SAFETY RISKS, 2013

LE INSIGHT

EHR/CPOE is the canonical ongoing case of an interface designed for one set of requirements and deployed against another. The systems work for billing and administration. They fail for clinical workflow because clinical workflow was not the design constraint. Forty billion dollars later, usability remains the single largest contributor to in-system harm in U.S. healthcare.

LENS APPROACH

LENS treats EHR/CPOE in LEN 7 as the live, ongoing example of Design-Out and Interface failure under governance opacity, in LEN 2 as a human-AI teaming problem (alert fatigue, default surfacing), and in LEN 9 as a computational systems problem (the alerting architecture itself is a learnable model). The Han 2005 pediatric ICU finding is the durable warning; the KLAS 2023 surveys across two hundred hospitals are the contemporary, longitudinal evidence that usability is now itself a patient-safety variable.

REFLECTION QUESTIONS

1. What is the equivalent system in your domain that was designed for one specification and deployed against another? How would you measure the harm?
2. Design the regulatory architecture that would surface EHR safety harms at scale. Be specific about signal, threshold, and authority.
3. The Han 2005 mortality result was disputed; the 2023 KLAS surveys are consistent across two hundred hospitals. What ongoing measurement architecture would have to exist for the equivalent emerging clinical-AI deployment in your domain to be evaluated honestly while in use?

WHO BUILDS THIS human factors & interaction design · systems & human-factors engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS usability testing · cognitive walkthroughs · task analysis · design prototyping · governance frameworks · bias & evidence audits

SOURCES

→ Han et al., "Unexpected Increased Mortality After Implementation of a Commercially Sold CPOE System," Pediatrics (2005)
→ KLAS Arch Collaborative, EHR usability and safety surveys (2023)
→ AMA / Pew / MedStar EHR Usability Report (2018)
→ AHRQ Patient Safety Network, EHR-related harm cases

FURTHER READING

• Sittig & Singh (2013), Safety-related EHR risks
• AMA / Pew / MedStar (2018), EHR Usability and Patient Safety
• Wachter, The Digital Doctor (2015)

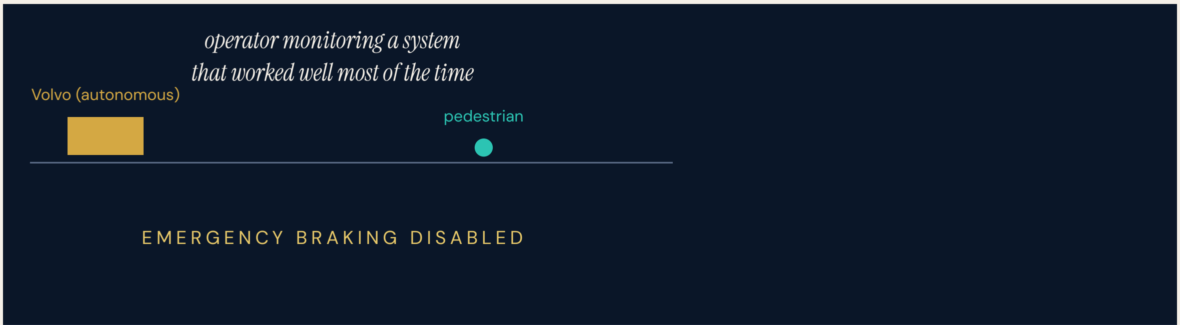
LENS COURSES LEN 7 LEN 2 LEN 9

CASE 29 AUTONOMOUS SYSTEMS 2018

Uber ATG / Tempe Fatality

T Training Gap N Normalization of Deviance G Governance & Trust H Human-System Interface

IMPACT One pedestrian killed — the first fatality involving a self-driving vehicle striking a pedestrian



DESIGNED AS PASSIVE MONITOR; EQUIPPED ONLY TO FAIL

A modified Volvo SUV operating in autonomous mode struck and killed Elaine Herzberg as she crossed a road at night in Tempe, Arizona. The probable cause, per NTSB, was the safety operator's failure to monitor the road — she was watching television on her phone. But NTSB placed heavy blame on Uber's systemic failures. The company "did not adequately recognize the risk of automation complacency and develop effective countermeasures." Training was inadequate; the company did not enforce its own anti-cell-phone policy. The system was programmed **not** to apply emergency braking when a crash was considered unavoidable — a design decision that removed the automated safety backstop. The system also could not classify an object as a pedestrian unless it was near a crosswalk.

NTSB board members stressed automation complacency is cross-domain: "most every industry has some level of automation and it often presents problems. Repeatedly, humans tend to tune out when tasked with monitoring automated systems that work well most of the time."

The human role was designed as passive surveillance, with no capability infrastructure to make that role performable.

THE LEARNING ENGINEERING LENS

> *Repeatedly, humans tend to tune out when tasked with monitoring automated systems that work well most of the time.*
>
> NTSB CHAIRMAN ROBERT SUMWALT, UBER TEMPE HEARING REMARKS, 2019

LE INSIGHT

Uber ATG is the defining case for the LENS Human-AI Teaming competency. A human was retained in the system not because the designers believed a human could meaningfully act, but because the regulatory architecture required a human be present. The role of "monitor" was assigned without the capability infrastructure to support it. The system that resulted was performable only on the assumption that the failure case would not arrive — until it did.

LENS APPROACH

LENS treats this case in LEN 2 as the live exemplar of monitoring as an unsupportable role. Students reconstruct the capability requirements for the safety operator and design the interface, training, and authority structure that would have made the role performable — or made the case for not retaining the role at all.

REFLECTION QUESTIONS

1. Identify a passive-monitor role in your domain. What evidence would tell you the role is or is not performable as designed?
2. The Tempe vehicle was programmed not to brake when a crash was unavoidable. Reconstruct the design rationale and propose the deliverable that should have prevented that decision.

WHO BUILDS THIS learning science & instructional design · safety science & organizational analysis · governance, ethics & measurement · human factors & interaction design — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · incident analysis · safety dashboards · governance frameworks · bias & evidence audits · usability testing · cognitive walkthroughs

SOURCES

→ NTSB Highway Accident Report HAR-19/03 (2019)
→ Consumer Reports coverage of the Tempe investigation
→ BBC News, Uber ATG and the Herzberg case

FURTHER READING

• Parasuraman & Manzey (2010) on complacency in automated systems
• Bainbridge (1983), "Ironies of Automation"
• Endsley (2017), "From Here to Autonomy"

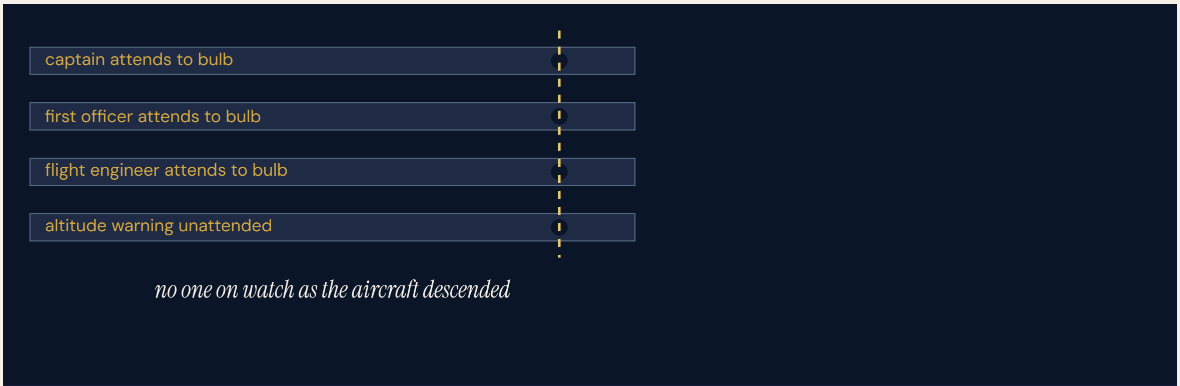
LENS COURSES LEN 2

CASE 63 AVIATION 1972

Eastern Air Lines Flight 401

H Human-System Interface T Training Gap

IMPACT 101 killed in the Everglades; the entire flight crew fixated on a landing-gear indicator bulb while the autopilot silently disengaged



EASTERN 401 — THREE CREW, ONE TASK, NO MONITOR

Approaching Miami at night, the crew of Eastern 401 noticed that the landing-gear indicator light had not illuminated. All three flight-deck members focused on the bulb. While they worked the problem, one of them inadvertently disengaged the autopilot altitude hold. The L-1011 descended slowly into the Everglades. The crew did not notice. The altitude warning chime sounded — according to the CVR — but in the auditory clutter of the cockpit no one heard it as significant.

The NTSB findings inaugurated decades of research into attention, monitoring, and the design of cockpit alerts. The accident is cited in virtually every introductory cognitive-engineering curriculum as the canonical example of how a low-priority task can crowd out a life-critical one when attention is undivided among the available channels.

Eastern 401 was one of the formative events that led to the development of Crew Resource Management (Case 12) and to modern standards for cockpit alerting systems.

THE LEARNING ENGINEERING LENS

> *The crew became so engrossed in the landing-gear difficulty that they failed to maintain altitude.*
>
> PARAPHRASING NTSB AIRCRAFT ACCIDENT REPORT NTSB-AAR-73-14 ON EASTERN 401, 1973

LE INSIGHT

Eastern 401 is the canonical attention-failure case. The capability that was missing was a designed division of attention across the crew. CRM exists because Eastern 401 happened, and the discipline of cockpit alert design exists because the altitude warning chime did not carry the priority weight it needed to.

LENS APPROACH

LENS uses Eastern 401 in LEN 5 to teach attention as a designable parameter and in LEN 2 to introduce alert prioritization as a capability deliverable. The case anchors the CRM origin story.

REFLECTION QUESTIONS

1. Identify a low-priority task in your domain that could plausibly absorb all of an operator's attention. What is the life-critical task that would be displaced?
2. Redesign the altitude warning chime of an L-1011 so that it cuts through a focused troubleshooting conversation.

WHO BUILDS THIS human factors & interaction design · learning science & instructional design — plus domain experts and a learning engineer to integrate.

TOOLS usability testing · cognitive walkthroughs · scenario simulation · competency models

SOURCES

→ NTSB Aircraft Accident Report NTSB-AAR-73-14 (1973)
→ Wickens, Engineering Psychology and Human Performance (multiple eds.)
→ Newhouse, The Failed Promise of Eastern 401

FURTHER READING

• NTSB AAR-73-14 (1973)
• Wickens et al. (2021), Engineering Psychology and Human Performance
• Helmreich & Foushee (1993), aircrew coordination research

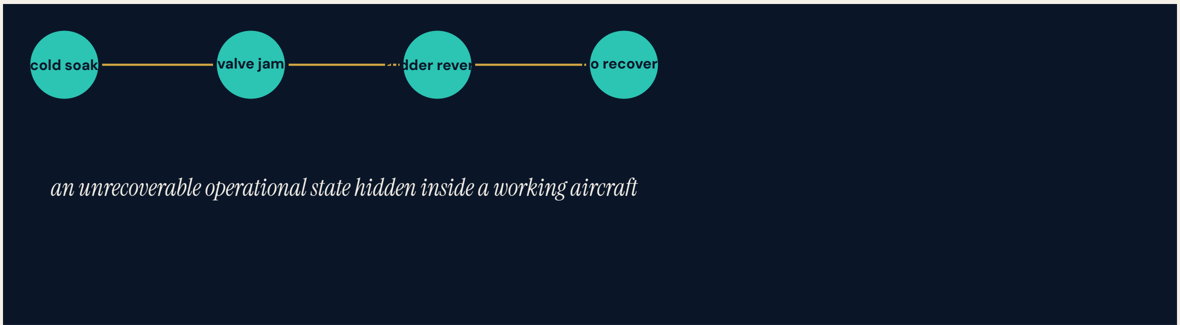
LENS COURSES LEN 5 LEN 2

CASE 64 AVIATION 1991, 1994

Boeing 737 Rudder Hardovers

H Human-System Interface D Designed Out

IMPACT 157 killed across United 585 (Colorado Springs, 1991) and USAir 427 (Pittsburgh, 1994); rudder Power Control Unit malfunctions traced to a thermal-shock condition



737 RUDDER — THE FAILURE MODE CERTIFICATION DID NOT ANTICIPATE

Two Boeing 737s crashed unrecoverably from level flight after the rudder Power Control Unit reversed: pilot inputs produced rudder movement in the opposite direction. The NTSB investigation took years, partly because the failure mode was both rare and unrecoverable. The eventual finding identified a thermal shock condition in the rudder servo valve — cold soaked hydraulic fluid striking a hot valve under specific operational conditions — combined with a secondary valve design that, when jammed, allowed the rudder to move opposite to commanded input.

The capability gap was specifically that the pilots had no procedure for recognizing or recovering from a rudder reversal. The Aircraft Operating Manual did not anticipate the failure. Crew training did not include it. The aircraft itself, in the failure mode, was not recoverable by any flight-control input available to the crew.

The 737 rudder cases sit at the intersection of mechanical design, certification testing, and crew capability — all three would have had to fail simultaneously for the accidents to occur, and all three did.

THE LEARNING ENGINEERING LENS

> *The aircraft was operated for years with a design feature that, under specific conditions, was unrecoverable.*
>
> PARAPHRASING NTSB AIRCRAFT ACCIDENT REPORT NTSB-AAR-99-01 ON THE 737 RUDDER PCU, 1999

LE INSIGHT

The 737 rudder cases are the case for an unrecoverable operational state hidden inside an apparently working system. The capability gap was not in the pilots — there was no capability that would have recovered the aircraft. The gap was in the certification process that had not surfaced the failure mode and in the maintenance procedures that did not test for it.

LENS APPROACH

LENS uses the 737 rudder cases in LEN 5 as a capability-limits case (when no human action is recoverable, capability engineering must move upstream) and in LEN 7 for certification governance.

REFLECTION QUESTIONS

1. Where in your domain might an unrecoverable failure mode exist that has not yet manifested? How would you find it?
2. Design the certification-test specification that should have caught the 737 rudder thermal-shock failure in development.

WHO BUILDS THIS human factors & interaction design · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS usability testing · cognitive walkthroughs · task analysis · design prototyping

SOURCES

→ NTSB Aircraft Accident Report NTSB-AAR-99-01 (1999)
→ NTSB Aircraft Accident Report NTSB-AAR-92-06 (1992) — United 585
→ Boeing 737 Rudder Service Bulletin series (1996–2002)

FURTHER READING

• NTSB 737 rudder reports (1992, 1999)
• Langewiesche, Inside the Sky (1998)
• Sweginnis et al., aircraft accident investigation literature

LENS COURSES LEN 5 LEN 7 LEN 3

CASE 65 TECHNOLOGY 2024

CrowdStrike Falcon Outage

D Designed Out K Knowledge & Institutional Memory G Governance & Trust

IMPACT 8.5 million Windows machines crashed; airlines, hospitals, broadcasters, and banks affected simultaneously; largest IT outage on record

8.5M

MACHINES · SINGLE CONFIGURATION FILE

content treated differently from code in the deployment pipeline

CROWDSTRIKE — THE COST OF A CATEGORY BOUNDARY IN A DEPLOYMENT

On July 19, 2024, CrowdStrike pushed a content update to its Falcon endpoint sensor that contained a logic error in a small configuration file. On every Windows machine running the affected sensor version, the file caused the kernel driver to dereference an out-of-bounds pointer and crash. The blue-screen-of-death loop that followed required manual recovery on each affected machine. Eight and a half million machines simultaneously, across hospitals, airlines, banks, and government agencies worldwide, became unusable.

The case is the largest IT outage in history measured by economic impact. CrowdStrike's own post-incident report identified that the content update had not been validated against the same set of automated tests as code updates, and that the staged-rollout practice CrowdStrike used for code did not extend to content. The capability that was missing was the assumption that content is treated identically to code from a deployment-safety standpoint.

The case is the cybersecurity-vendor analog of Knight Capital (Case 55) at a scale six orders of magnitude larger.

THE LEARNING ENGINEERING LENS

> *Our content configuration update process did not include the same depth of testing and staged rollout as our code releases.*
>
> CROWDSTRIKE PRELIMINARY POST-INCIDENT REVIEW, AUGUST 2024

LE INSIGHT

The CrowdStrike outage is the live case for what happens when a deployment safety architecture treats categories of artifact differently than the operational system actually does. Content looked operationally identical to code; it was treated as different in the deployment pipeline. The mismatch became the largest IT outage in history.

LENS APPROACH

LENS uses CrowdStrike in LEN 5 as a categories-and-boundaries capability case and in LEN 2 for the vendor-customer trust architecture: customers trusted CrowdStrike's deployment safety; that trust was load-bearing for the operation of the global economy on a single day.

REFLECTION QUESTIONS

1. Identify a vendor relationship in your domain whose deployment-safety practice your institution does not audit. What would the audit reveal?
2. Design the cross-vendor staged-rollout protocol that should be standard for endpoint security software.

WHO BUILDS THIS systems & human-factors engineering · knowledge management & data engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS task analysis · design prototyping · knowledge bases · data pipelines · governance frameworks · bias & evidence audits

SOURCES

→ CrowdStrike, Preliminary Post-Incident Review (2024)
→ Microsoft Resilient Engineering report (2024)
→ GAO post-incident analyses and Senate Homeland Security hearings (2024)

FURTHER READING

• CrowdStrike Post-Incident Review (2024)
• Allspaw, post-mortem culture literature
• Beyer et al. (eds.), Site Reliability Engineering (2016)

LENS COURSES LEN 5 LEN 2

CASE 66 DEFENSE 1983

Stanislav Petrov / 1983 False Alert

H Human-System Interface T Training Gap

IMPACT Soviet early-warning system reported five incoming U.S. ICBMs; Lt. Col. Petrov correctly assessed the signal as false; retaliation averted

SYSTEM	OPERATOR
5 ICBMs	false alarm

the human in the loop was the recoverability of an automation failure

PETROV, 1983 — HUMAN-AI TEAMING AT THE HIGHEST POSSIBLE STAKES

On the night of September 26, 1983, the Soviet Oko early-warning satellite system reported the launch of five U.S. Minuteman ICBMs toward the Soviet Union. The duty officer at Serpukhov-15, Lt. Col. Stanislav Petrov, was responsible for verifying the alert and reporting it up the chain. Petrov assessed the signal as a false alarm — a real first strike, he reasoned, would involve hundreds of missiles, not five — and reported it as such. He was correct. The signal had been generated by sunlight reflecting off high-altitude clouds at a particular geometry against the Oko satellite's infrared sensors.

Petrov's decision is widely credited with averting nuclear war. The case demonstrates both the necessity and the fragility of human-in-the-loop capability for nuclear command-and-control. The automation had failed in a mode its designers had not anticipated. The human in the loop had the contextual judgment and the institutional protection to override the alert. Subsequent investigations identified the satellite-geometry failure mode and modified the algorithm. The case is permanently archived in early-warning training as the canonical false-positive scenario.

THE LEARNING ENGINEERING LENS

> *I had a funny feeling in my gut.*
>
> STANISLAV PETROV, QUOTED IN THE WASHINGTON POST, 1999

LE INSIGHT

Petrov is the canonical positive case for human-in-the-loop nuclear command-and-control. The automation failed in a mode its designers had not anticipated. The human's contextual judgment was the backstop that allowed the failure to be recovered before it became catastrophic. The case is the strongest argument in this book that the design choice to retain humans in some loops is not nostalgia but capability engineering.

LENS APPROACH

LENS uses Petrov in LEN 2 as the positive Human-AI Teaming case at the highest possible stakes: the human role is the recoverability of an automation failure that the designers did not anticipate. The case anchors arguments against full-automation of strategic decisions.

REFLECTION QUESTIONS

1. Identify an automated system in your domain where retaining a human-in-the-loop is genuinely capability-engineering rather than ceremonial. How would you tell the difference?
2. Petrov's "funny feeling" was contextual judgment built across a career. Design the training that produces it deliberately.

WHO BUILDS THIS human factors & interaction design · learning science & instructional design — plus domain experts and a learning engineer to integrate.

TOOLS usability testing · cognitive walkthroughs · scenario simulation · competency models

SOURCES

→ Hoffman, The Dead Hand (2009)
→ Aksenov, V., investigation of Oko incident (2004)
→ Lewis & Lewis, U.S.-Russia strategic-stability analyses

FURTHER READING

• Hoffman (2009), The Dead Hand
• Sagan, The Limits of Safety (1993)
• Cummings (2017) on automation in critical decision systems

LENS COURSES LEN 2

CASE 67 TECHNOLOGY 2018

TSB Bank IT Migration

H Human-System Interface G Governance & Trust

IMPACT 1.9 million UK customers locked out of accounts; £330M+ in compensation and remediation; CEO resigned

TSB — READINESS SIGNAL ARRIVED ATTENUATED AT THE DECISION

TSB Bank attempted to migrate five million customer accounts from its former parent company Lloyds' systems to a new platform built by its current parent Sabadell. The migration was scheduled for a single weekend. On Sunday evening, when customer-facing services came back online, virtually every component of the new platform had problems. Customers saw other people's accounts. Mortgages disappeared. Card payments failed. The recovery took months.

The Slaughter and May independent investigation found that the migration had been tested under conditions that did not approximate real customer load, that the new platform had been certified ready by a process that did not adequately challenge the certification, and that the executive decision to proceed had been taken against technical recommendations that the platform was not ready. The Financial Conduct Authority fined TSB; the CEO resigned.

The case is the financial-sector analog of Healthcare.gov (Case 10): a large migration shipped without the testing the institution knew it should have done, because the schedule pressure overrode the technical signal.

THE LEARNING ENGINEERING LENS

> *The migration proceeded notwithstanding clear signals that the platform was not ready.*
>
> PARAPHRASING THE SLAUGHTER AND MAY INDEPENDENT REVIEW OF THE TSB MIGRATION, 2019

LE INSIGHT

TSB is the canonical case for schedule pressure overriding technical signal in a regulated industry. The technical signal existed. The decision authority was at the executive layer where the signal arrived weakened by passage through multiple intermediate layers. The institutional architecture did not allow the technical layer to halt the migration.

LENS APPROACH

LENS uses TSB in LEN 7 as a corporate-governance case and in LEN 8 for the institutional structure of technical-decision authority. Studio projects compare TSB and Healthcare.gov.

REFLECTION QUESTIONS

1. Where in your organization does a technical signal arrive at the executive layer attenuated by intermediate layers? What is the cost of the attenuation?

2. Design the institutional structure that would allow a technical lead to halt a migration like TSB's without resigning.

WHO BUILDS THIS human factors & interaction design · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS usability testing · cognitive walkthroughs · governance frameworks · bias & evidence audits

SOURCES

→ Slaughter and May Independent Review of TSB Migration (2019)
→ Financial Conduct Authority final notice on TSB (2022)
→ House of Commons Treasury Committee report on TSB (2018)

FURTHER READING

- Slaughter and May TSB Review (2019)
- FCA Final Notice (2022)
- Allspaw, post-mortem and migration-safety literature

LENS COURSES LEN 7 LEN 8

Chapter 5

Governance Failures

Trust and accountability were afterthoughts.

The fire at Grenfell Tower was the culmination of decades of failure by central government and other bodies in positions of responsibility.

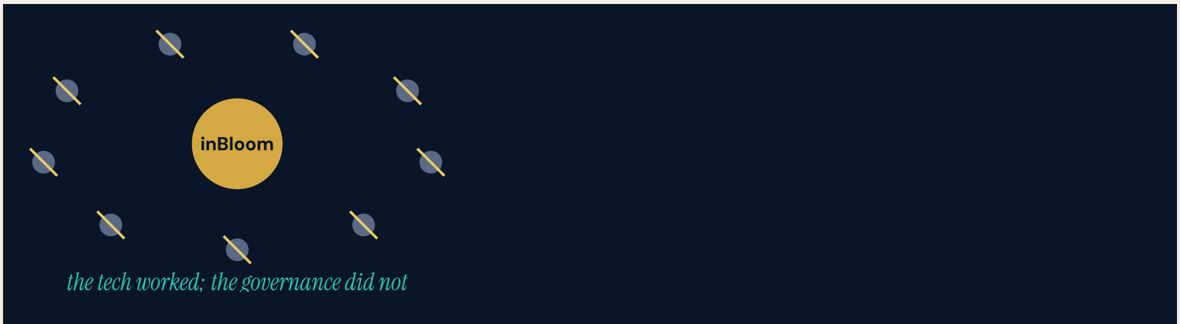
GRENFELL TOWER INQUIRY, PHASE 2 REPORT, 2024

CASE 8 EDUCATION AT SCALE 2014

inBloom

G Governance & Trust

IMPACT $100M initiative collapsed in ~14 months; 9 states withdrew; data infrastructure for education set back years



$100M OF WORKING TECHNOLOGY; NINE STATES WITHDREW

The technology worked. Shared data infrastructure for student records, backed by one hundred million dollars from the Gates Foundation, hosted on AWS, built by enterprise engineers. No software bug, no performance failure, no data breach.

What killed inBloom was everything **around** the technology. The initiative launched without adequate consent frameworks, without meaningful community engagement on data governance, without transparency about data collection and use, and without mechanisms for parents to participate in decisions about their children's data. Parent organizations — Parent Coalition for Student Privacy, Class Size Matters — organized opposition state by state. Louisiana, New York, Massachusetts, Illinois, Georgia, Delaware, Colorado, Kentucky, and North Carolina all withdrew.

Bulger, McCormick, and Pitcan analyzed inBloom as the failure of technocratic education reform: the assumption that technically sound infrastructure generates its own legitimacy proved catastrophically wrong. In education at scale, governance and stakeholder trust are not optional features. They are load-bearing elements.

THE LEARNING ENGINEERING LENS

> *The technology was not the problem. The governance was the problem.*
>
> PARAPHRASING THE ANALYSIS IN BULGER, MCCORMICK & PITCAN, DATA & SOCIETY, 2017

LE INSIGHT

inBloom is the purest governance failure in this dataset. Nothing technical was wrong. Everything sociotechnical was. The case is the strongest argument in the book for treating ethics-as-design- constraint not as ideology but as engineering: a $100M empirical test of what happens when you do not.

LENS APPROACH

LENS uses inBloom in LEN 7 as the canonical governance failure and in LEN 1 as a stakeholder-analysis case. The case anchors the LENS threading commitment that equity and accountability are design commitments, not modules. Studio projects (LEN 10) require students to produce a stakeholder-trust deliverable as a precondition for deployment.

REFLECTION QUESTIONS

1. What is the equivalent unbuilt governance infrastructure in your domain? What would the $100M empirical test of its absence look like?
2. Design the stakeholder-trust deliverable that a future inBloom-equivalent should have to produce before deployment.

WHO BUILDS THIS governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits

SOURCES

→ Bulger, McCormick & Pitcan, The Legacy of inBloom, Data & Society (2017)
→ Education Week coverage of inBloom withdrawal, 2013–2014
→ Hechinger Report, inBloom retrospective
→ Parent Coalition for Student Privacy archives

FURTHER READING

- Bulger, McCormick & Pitcan (2017), Data & Society report
- Selwyn, Distrusting Educational Technology (2014)
- boyd & Crawford (2012), "Critical Questions for Big Data"

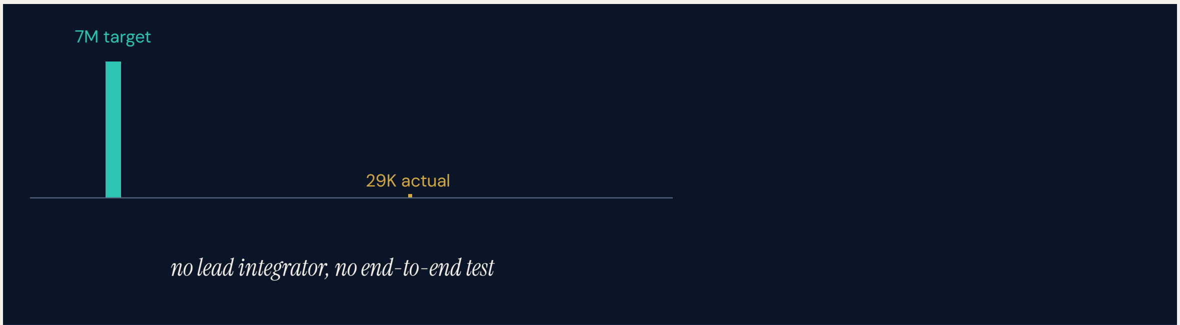
LENS COURSES LEN 1 LEN 10 LEN 7 LEN 6

CASE 10 GOVERNMENT 2013

Healthcare.gov Launch

K Knowledge & Institutional Memory T Training Gap G Governance & Trust

IMPACT 29,000 enrollments vs. 7M target in first two months; hundreds of millions in remediation



29,000 ENROLLMENTS VS. 7,000,000 TARGET — FIRST TWO MONTHS

HHS employees had experience with insurance markets and with large government projects, but not with technology product launches. Key technical positions were unfilled. There was no formal division of responsibilities among the multiple government offices involved. CMS perceived CGI as the lead system integrator; CGI did not share that understanding. No end-to-end testing was conducted before launch.

The site went live on October 1, 2013, and immediately collapsed under load it had never been validated for: twenty-nine thousand enrollments in two months against a seven-million target. The fix-it operation that followed pulled together the team that became the U.S. Digital Service — born from a failure visible on the news every night. At root it was a capability mismatch at scale: the organization lacked the human capabilities the specific system required, and no one in the governance chain surfaced the gap before launch.

THE LEARNING ENGINEERING LENS

> *No single person had a clear understanding of the project's status.*
>
> PARAPHRASING THE HHS OFFICE OF INSPECTOR GENERAL REVIEW OF HEALTHCARE.GOV, 2016

LE INSIGHT

Healthcare.gov is a capability case wearing a technology costume. The technology was salvageable in weeks once the right people arrived. The original failure was that the wrong people had been assembled in the first place, and the governance chain that should have noticed the mismatch had no signal to surface it. The aftermath — USDS — is the rare case in this book of a governance failure that produced permanent institutional reform.

LENS APPROACH

LENS uses Healthcare.gov in LEN 1 as a problem-framing case (the technical narrative obscures the capability narrative), in LEN 5 to teach capability-requirements analysis for a large government program, and in LEN 8 to discuss the founding of USDS as organizational learning that survived.

REFLECTION QUESTIONS

1. Healthcare.gov shipped without end-to-end testing. What is the equivalent missing deliverable in a current high-stakes deployment in your domain?
2. USDS was born from the Healthcare.gov failure. What is the institutional capability your domain still lacks, and what failure would have to occur to produce it?

WHO BUILDS THIS knowledge management & data engineering · learning science & instructional design · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS knowledge bases · data pipelines · scenario simulation · competency models · governance frameworks · bias & evidence audits

SOURCES

→ GAO Healthcare.gov reports (2014–2016)
→ HHS Office of Inspector General, OEI-06-14-00350
→ Harvard Kennedy School / Digital Initiative case (2014)
→ CIO Magazine, Healthcare.gov post-mortem coverage

FURTHER READING

- Eaves & Goldenfein, "The Healthcare.gov Failure" (Harvard, 2014)
- Pahlka, Recoding America (2023)
- Mergel et al. (2018), digital government literature

LENS COURSES LEN 1 LEN 5 LEN 7 LEN 6

CASE 17 INDUSTRIAL 1984

Bhopal

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance G Governance & Trust

IMPACT ≈ 15,000–20,000 killed; ≈ 500,000 injured; worst industrial disaster in history

SAFETY SYSTEMS OFF-LINE FOR FOUR MONTHS BEFORE THE RELEASE

Forty tons of methyl isocyanate gas escaped from a Union Carbide pesticide plant in Bhopal, India, on the night of December 2–3, 1984. The New York Times investigation found the disaster "resulted from operating errors, design flaws, maintenance failures, training deficiencies and economy measures that endangered safety." Safety systems had been non-operational for four months. The plant was understaffed. Workers were inadequately trained to handle emergencies or recognize warning signs.

Najmedin Meshkati's 1991 analysis demonstrated this was not a prototypical accident: it shared "numerous similarities with respect to the lack of human-factors considerations" with Three Mile Island and other industrial disasters. The U.S. Chemical Safety Board later identified ineffective employee training as an underlying cause in nine of its first twenty-three completed investigations of chemical incidents — a pattern that traces directly to Bhopal as catalyst.

The CSB was itself created in the aftermath of Bhopal. The industrial-safety architecture of the next forty years was built on the inquiry into one night.

THE LEARNING ENGINEERING LENS

> *Operating errors, design flaws, maintenance failures, training deficiencies and economy measures that endangered safety.*
>
> NEW YORK TIMES INVESTIGATION, 1985

LE INSIGHT

Bhopal is the largest-magnitude capability-and-governance failure on record. It is also the catalyst for the creation of the U.S. Chemical Safety Board, an INPO-equivalent for industrial chemistry. The pattern that follows nearly every case in this book — diagnose, then engineer remediation at scale — runs through Bhopal in canonical form.

LENS APPROACH

LENS treats Bhopal in LEN 5 as a multi-failure capability analysis case and in LEN 7 as the largest-magnitude governance failure in the dataset. The Failure → Reform arc to the CSB demonstrates the institution-building pattern at industry scale.

REFLECTION QUESTIONS

1. Bhopal produced the CSB. What is the institution your domain has not yet built that a comparable disaster would force into existence?
2. Identify a plant or facility in your domain that has had safety systems off-line for an extended period. What is the measurement gap that allowed it?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards · governance frameworks · bias & evidence audits

SOURCES

→ Union Carbide Investigation Report (1985)
→ Meshkati, “Human Factors in Large-Scale Technological Systems’ Accidents,” Industrial Crisis Quarterly (1991)
→ Britannica, Bhopal disaster
→ Smithsonian Magazine, Bhopal retrospective (2024)

FURTHER READING

• Meshkati (1991), Industrial Crisis Quarterly
• Shrivastava, Bhopal: Anatomy of a Crisis (1992)
• Perrow, Normal Accidents (1984)

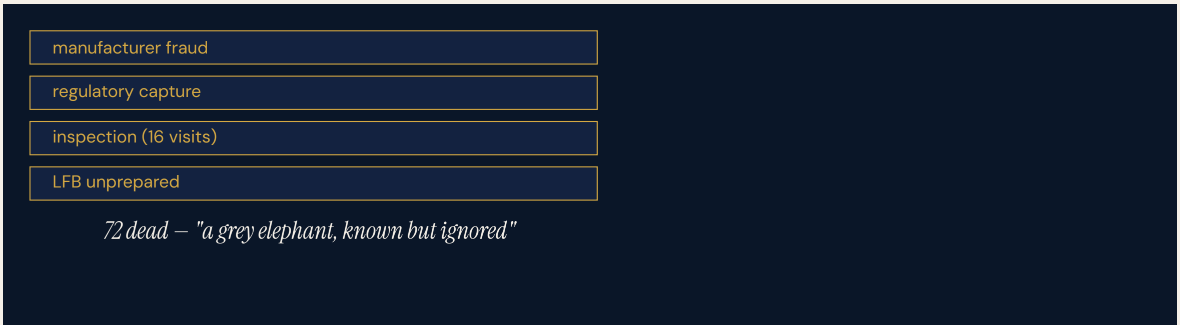
LENS COURSES LEN 5 LEN 7 LEN 3

CASE 20 INDUSTRIAL 2017

Grenfell Tower

G Governance & Trust T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT 72 killed in a residential tower fire in London; decades of regulatory failure



LEVELS OF FAILURE: MANUFACTURE, REGULATION, INSPECTION, RESPONSE

The Grenfell Tower Inquiry found the fire to be the culmination of decades of failure by central government and by every body in a position of responsibility in the construction industry. Flammable cladding was installed despite safety experts cautioning that it was unsuitable. Building inspectors visited the site sixteen times during the refurbishment but none noticed that effectively banned materials were being used. Cladding companies engaged in "systematic dishonesty" in marketing combustible materials as safe.

The London Fire Brigade was unprepared. "Very few (if any) of the senior officers who attended Grenfell Tower were aware of the risks posed by exterior cladding." The risks of rapidly developing cladding fires were well known from prior incidents — Knowsley Heights 1991, Garnock Court 1999, Shepherd's Court 2016 — but "this knowledge had not informed firefighting policies, practices or training."

The failure spanned every level: manufacturer fraud, regulatory capture, inspection incompetence, training gaps, organizational memory loss, and governance failure. The inquiry described it as a "grey elephant" — known but ignored.

THE LEARNING ENGINEERING LENS

> *This knowledge had not informed firefighting policies, practices or training.*
>
> GRENFELL TOWER INQUIRY, PHASE 1, 2019

LE INSIGHT

Grenfell is the strongest evidence in the dataset that capability failure can be distributed across many actors, each of whom contributes a small piece, none of whom is accountable for the whole. The inquiry's "grey elephant" framing — known but ignored — describes a pattern that LENS treats as a primary governance problem in any high-consequence domain.

LENS APPROACH

LENS treats Grenfell in LEN 7 as the canonical multi-actor governance failure case, in LEN 8 to discuss institutional memory of warnings across the LFB and other agencies, and in LEN 10 as a studio prompt for designing the integrated capability-and-governance deliverable that would have caught the cladding decision.

REFLECTION QUESTIONS

1. What is the "grey elephant" — the well-known risk that nobody owns — in your domain?
2. Design the deliverable that forces a single actor to own the integration risk that Grenfell distributed across dozens.

WHO BUILDS THIS governance, ethics & measurement · learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards

SOURCES

→ Grenfell Tower Inquiry Phase 1 Report (2019)
→ Grenfell Tower Inquiry Phase 2 Report (2024)
→ UK Government Response to the Phase 2 Report (2025)
→ Human Factors 101 analysis of Grenfell

FURTHER READING

• Grenfell Tower Inquiry, full reports
• Hopkins (2024), public-inquiry analysis of Grenfell
• Hutter & Power (2005), Organizational Encounters with Risk

LENS COURSES LEN 10 LEN 7 LEN 8 LEN 3

CASE 21 EDUCATION AT SCALE 2014–2019

Summit Learning / Personalized Learning Rollout

G Governance & Trust T Training Gap K Knowledge & Institutional Memory

IMPACT Personalized-learning platform deployed across ~380 U.S. schools; parent revolts in Brooklyn, Cheshire, McPherson, Kennebunk; multiple districts withdrew within two years

$100M OF WORKING TECHNOLOGY; NINE STATES WITHDREW

Summit Learning — the personalized-learning platform developed by Summit Public Schools with technical and financial support from the Chan Zuckerberg Initiative — was offered free to U.S. school districts beginning in 2015. By 2018 it reached roughly 380 schools and an estimated 80,000 students. By 2019 the most visible adopters were withdrawing under parent and student pressure: Brooklyn's MS 442 saw an organized opt-out campaign; districts in Cheshire, Kennebunk, and McPherson cancelled or scaled back after parent meetings. Walk-outs, surveys reporting eye strain and disengagement, and concerns about screen time and data privacy converged into a governance failure that was not about the underlying instructional design.

The platform's pedagogical theory — competency-based progression, self-directed projects, mentor check-ins — was defensible and in many places effective. The failure was deployment governance: no evaluation framework districts could read before adopting; no parent-facing data-handling agreement; no exit pathway that did not depend on the vendor's goodwill. The implementation never surfaced the governance questions parents would ask, and the argument was lost before it started.

THE LEARNING ENGINEERING LENS

> *The tools were free. The accountability architecture had not been built.*
>
> EDITORS' SYNTHESIS OF SUMMIT LEARNING ROLLOUT COVERAGE (NEW YORK TIMES, WIRED, EDUCATION WEEK, 2018–2019)

LE INSIGHT

Summit Learning is a clean test of the book's central claim: technology that worked at the pedagogical level still failed because the **governance** architecture (consent, evidence, measurement, exit) had not been engineered alongside it. The pattern — well-intentioned tool, well-funded rollout, no institutional contract with the families and teachers operating inside it — recurs across the educational-technology dataset (Cases 8, 38, 42) and is the educator's-side analog of the governance failures in Cases 35 and 36.

LENS APPROACH

LENS uses Summit Learning in LEN 7 as the foundational consent-and-evidence case for educational technology, and in LEN 10 as a studio prompt for the governance artifacts that any educational-technology adoption decision should produce: a public evidence summary at parent reading level, a data-handling agreement at the same resolution, and a documented exit pathway that does not depend on the vendor's goodwill. The studio prompt extends to a re-adoption pathway — the governance work a withdrawn district would have to complete before it could responsibly re-adopt the platform without re-triggering the same parental objections.

REFLECTION QUESTIONS

1. What is the equivalent of the "free tool, free of governance" pattern in your domain — the offer that bypasses the accountability architecture because it does not yet exist?

2. Design the parent-reading-level governance artifact that a district should require before adopting an educational-technology platform.

3. Summit's withdrawals in Brooklyn, Cheshire, Kennebunk, and McPherson were led by parents, not regulators. What is the equivalent local constituency in your domain that institutional accountability has not yet accommodated, and how would they be heard before deployment rather than after?

WHO BUILDS THIS governance, ethics & measurement · learning science & instructional design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · scenario simulation · competency models · knowledge bases · data pipelines

SOURCES

→ Bowles (2019), "Silicon Valley Came to Kansas Schools. That Started a Rebellion," New York Times
→ Singer (2017), "The Silicon Valley Billionaires Remaking America's Schools," New York Times
→ Herold (2019), Education Week coverage of Summit / CZI implementation
→ Watters (2019), "The Stories We Tell About Personalized Learning," Hack Education
→ Chan Zuckerberg Initiative & Summit Learning, public program documentation (2015–2019)

FURTHER READING

• Selwyn (2016), Is Technology Good for Education?
• Watters (2021), Teaching Machines
• Eubanks (2018), Automating Inequality (governance-pattern analog)

LENS COURSES

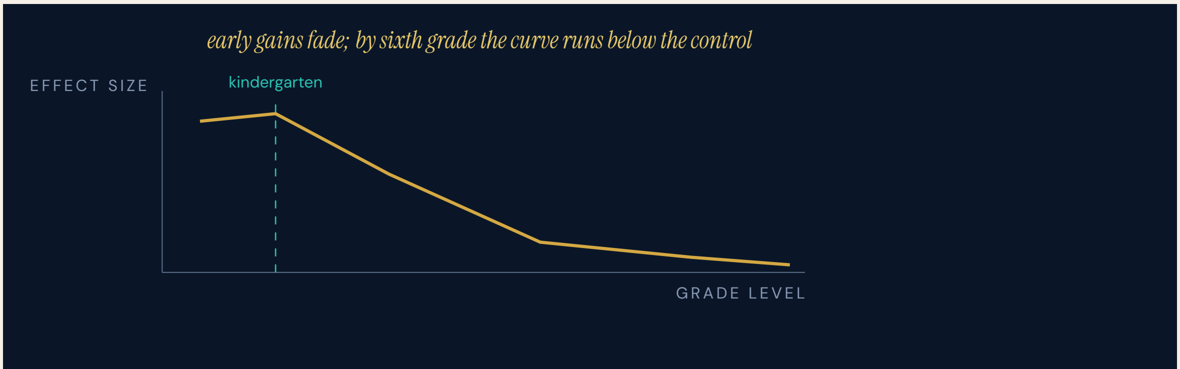
LEN 7 LEN 10 LEN 8

CASE 22 EDUCATION AT SCALE 2009–2018

Tennessee Voluntary Pre-K Study

G Governance & Trust D Designed Out

IMPACT Vanderbilt longitudinal RCT of a state-funded universal pre-K program: early gains faded by third grade; sixth-grade outcomes were worse than the control group on several measures



TENNESSEE PRE-K — MEASUREMENT ARCHITECTURE WORKED; THE FIELD'S RESPONSE DID NOT

The Tennessee Voluntary Pre-Kindergarten Program (TN-VPK) was a state-funded, means-tested universal pre-K program that, by 2009, served roughly eighteen thousand four-year-olds annually. Researchers at Vanderbilt's Peabody Research Institute conducted a randomized controlled trial — among the most rigorous large-N pre-K studies in the United States — following children admitted by lottery against children who applied but did not enroll.

Through kindergarten, TN-VPK children showed the expected gains: stronger letter knowledge, vocabulary, and early literacy. By third grade the gains had faded. By sixth grade, Lipsey, Farran, and Durkin (2018) reported TN-VPK children performed **worse** than the control group on several state academic measures and on teacher-reported behavior. The result contradicted policy consensus and provoked an unusual response: the study was attacked, its methods contested, and the field largely declined to internalize the findings. The case is in the dataset not because pre-K is bad — Perry Preschool and Abecedarian remain durable — but because the **measurement architecture** is rare in education, and the discipline's capacity to act on an unwelcome finding was tested and largely failed.

THE LEARNING ENGINEERING LENS

> *By sixth grade, children who had attended TN-VPK were doing somewhat worse on academic achievement and discipline measures than children in the control group.*
>
> LIPSEY, FARRAN & DURKIN (2018), VANDERBILT PEABODY RESEARCH INSTITUTE

LE INSIGHT

Tennessee Pre-K is the cleanest case in the dataset for what happens when a discipline has not engineered its own capacity to update on contrary evidence. The measurement instrument worked. The institutional architecture for acting on what it found did not. Compare to Case 7 (Makary methodology debate) and Case 32 (VA Wait-Time): in each, a measurement that returned an unwelcome answer was contested rather than absorbed.

LENS APPROACH

LENS uses Tennessee Pre-K in LEN 4 as a measurement-architecture case (longitudinal RCT in a real policy context), in LEN 7 to discuss the institutional politics of unwelcome findings, and in LEN 10 as a studio prompt for designing the implementation-science pathway that would absorb such findings into program redesign rather than rejection.

REFLECTION QUESTIONS

1. What measurement instrument in your domain has returned an unwelcome answer, and how did the discipline respond?
2. Design the implementation-science pathway that would absorb a Tennessee-Pre-K-style finding into program redesign rather than rejection.

WHO BUILDS THIS governance, ethics & measurement · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · task analysis · design prototyping

SOURCES

→ Lipsey, Farran & Durkin (2018), "Effects of the Tennessee Pre-Kindergarten Program on Children's Achievement and Behavior Through Third Grade," Early Childhood Research Quarterly 45: 155–176. doi:10.1016/j.ecresq.2018.03.005
→ Durkin, Lipsey, Farran & Wiesen (2022), "Effects of a Statewide Pre-Kindergarten Program on Children's Achievement and Behavior Through Sixth Grade," Developmental Psychology 58(3): 470–484
→ Heckman et al., responses and counter-analyses (2018–2022)
→ National Institute for Early Education Research, State of Preschool yearbooks (2010–2022)

FURTHER READING

• Lipsey, Farran & Durkin (2018, 2022) — primary study and 6th-grade follow-up
• Heckman (2008), The Case for Investing in Disadvantaged Young Children
• Phillips et al. (2017), Puzzling It Out: The Current State of Scientific Knowledge on Pre-Kindergarten Effects

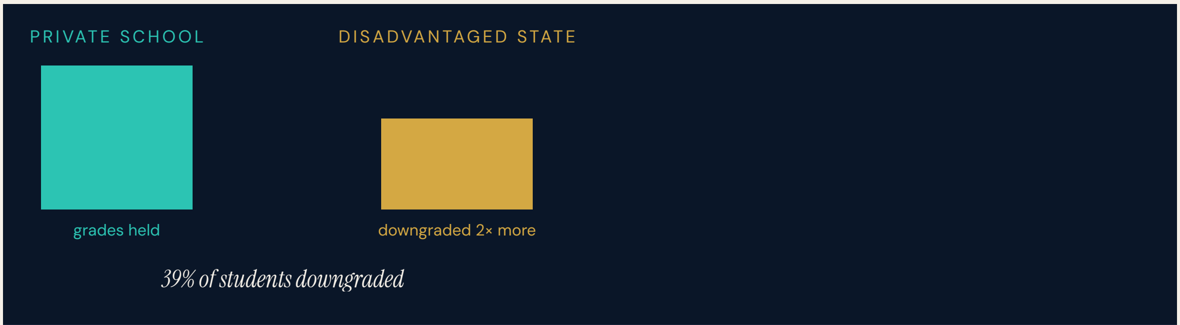
LENS COURSES LEN 4 LEN 7 LEN 10

CASE 35 EDUCATION AT SCALE 2020

UK A-Level Algorithm / Ofqual

G Governance & Trust H Human-System Interface D Designed Out

IMPACT ≈ 280,000 A-level entries downgraded; disadvantaged students disproportionately harmed; government U-turn within days



AN ALGORITHM ENCODING HISTORICAL INEQUALITY AT NATIONAL SCALE

When COVID-19 cancelled exams, Ofqual used an algorithm to “standardize” teacher-predicted grades against schools’ historical performance. The result: roughly thirty-nine percent of A-level grades were adjusted downward from teacher assessment, with about three percent dropped by two or more grades. High-achieving students at historically low-performing state schools were systematically capped by school-level priors. Small-cohort private school students kept their predicted grades. The structural pattern fell hardest on cohorts whose individual achievement was masked by their school’s historical average.

The algorithm encoded existing structural inequality and amplified it at national scale. Schools that had historically underperformed — often because of funding rather than student capability — were assumed to continue underperforming, and the assumption was used to suppress individual student grades. Oxford Internet Institute researchers identified the underlying trap: “the belief that [algorithms] will fix really complex structural issues … lend[s] [itself] to a kind of magical thinking.”

The mathematician Hannah Fry wrote: “I think it’s the first time that an entire nation has felt the injustice of an algorithm simultaneously.” Within days the government withdrew the algorithm and accepted teacher predictions.

THE LEARNING ENGINEERING LENS

> *I think it's the first time that an entire nation has felt the injustice of an algorithm simultaneously.*
>
> HANNAH FRY, 2020

LE INSIGHT

A-Level is the defining national-scale algorithmic-bias case. The algorithm worked exactly as specified; the specification encoded inequality. The harm was visible, immediate, and felt simultaneously by an entire cohort. The case stands as the strongest argument that **construct definition** — what the algorithm is trying to predict — is itself a capability-engineering decision.

LENS APPROACH

LENS uses A-Level in LEN 7 as the foundational case for bias, risk, and governance, in LEN 4 as a measurement-instrument case (an instrument that encodes the bias of its training data), and in LEN 10 as a technical case for bias detection and mitigation. The case is paired in this book with Robodebt (Case 36) and educational algorithmic bias (Case 37).

REFLECTION QUESTIONS

1. Identify a measurement instrument in your domain that encodes structural inequality. What would it take to surface that bias before deployment?

2. The A-Level algorithm worked as specified. Redesign the **specification** — not the algorithm — to remove the bias.

WHO BUILDS THIS governance, ethics & measurement · human factors & interaction design · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · usability testing · cognitive walkthroughs · task analysis · design prototyping

SOURCES

→ MIT Technology Review, "The UK Exam Algorithm" (2020)
→ CNBC, "Computer Algorithm Caused a Grading Crisis in British Schools" (2020)
→ University of Bristol, Centre for Multilevel Modelling analysis
→ LSE Impact Blog, "F*ck the Algorithm" (2020)

FURTHER READING

• Ofqual technical report on the 2020 standardization model
• Eubanks, Automating Inequality (2018)
• Selbst et al. (2019), "Fairness and Abstraction in Sociotechnical Systems"

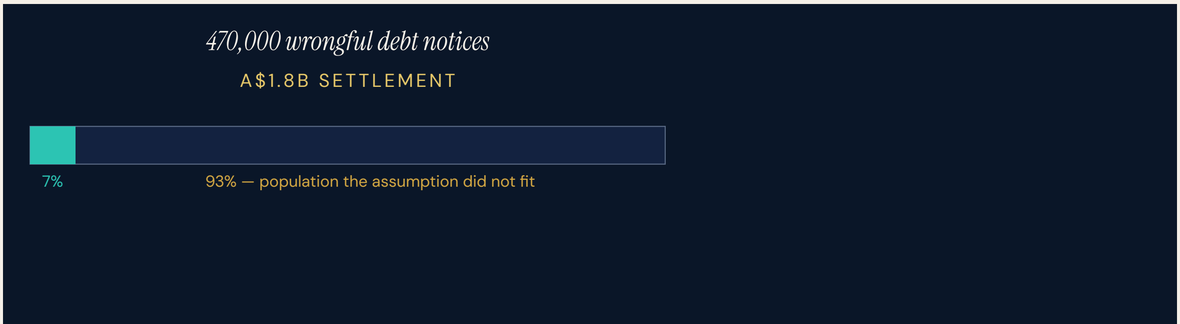
LENS COURSES LEN 4 LEN 7 LEN 9

CASE 36 GOVERNMENT 2016 – 2020

Australia Robodebt

G Governance & Trust D Designed Out H Human-System Interface

IMPACT Roughly 1 million debt notices issued; ~470,000 found wholly or partially unlawful for ~433,000 people; A$1.8B Prygodicz class-action settlement (debts zeroed/refunded plus interest); Royal Commission found "venality, incompetence and cowardice"



INCOME-AVERAGING ASSUMPTION MATCHED 7% OF RECIPIENTS

The Australian government automated welfare debt recovery using an income-averaging algorithm that compared Centrelink records with annual tax data averaged across the year. The algorithm’s case profile — stable, consistent year-round employment — applied to roughly seven percent of welfare recipients. The other ninety-three percent had irregular employment that the averaging method misclassified as fraud.

Default judgments were issued without human review, reversing the burden of proof onto recipients. The Royal Commission heard evidence of deaths, including by suicide, associated with the scheme, and its 2023 final report found it sustained by “venality, incompetence and cowardice,” with ministers failing to ensure the program was lawful. The net cost to government was A$565 million — for a scheme meant to save money. The algorithm was “comparatively simple”; its harms, the University of Melbourne found, “entirely predictable from the outset.”

THE LEARNING ENGINEERING LENS

> *A costly failure of public administration, in both human and economic terms.*
>
> ROYAL COMMISSION INTO THE ROBODEBT SCHEME, 2023

LE INSIGHT

Robodebt is the canonical case of full automation without human-in-the-loop and without lawful basis. The algorithm was not sophisticated and the harm was not technical: the system survived four years because the governance architecture treated automation as an efficiency mechanism rather than as a decision regime that required accountability.

LENS APPROACH

LENS treats Robodebt in LEN 7 as the canonical case for automated decision-making without human oversight and in LEN 2 as a Human-AI Teaming failure at the institutional scale: the human role was designed out of the decision loop entirely. The case anchors the program's argument that automated decision systems require deliverable-grade accountability artifacts before deployment.

REFLECTION QUESTIONS

1. Robodebt automated a decision that legally required individual review. Identify a current process in your domain at risk of equivalent automation-without-due-process.
2. Design the accountability artifact that would have to be signed before a Robodebt-equivalent could be deployed.

WHO BUILDS THIS governance, ethics & measurement · systems & human-factors engineering · human factors & interaction design — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · task analysis · design prototyping · usability testing · cognitive walkthroughs

SOURCES

→ Royal Commission into the Robodebt Scheme, Final Report (2023)
→ University of Melbourne / Pursuit analysis (2023)
→ Blavatnik School of Government, Oxford, case study (2023)
→ IAPP privacy analysis of Robodebt

FURTHER READING

• Royal Commission Final Report (2023)
• Eubanks, Automating Inequality (2018)
• Citron, "Technological Due Process" (2008)

LENS COURSES LEN 7 LEN 2

CASE 37 EDUCATION AT SCALE ONGOING

Algorithmic Bias in Educational Predictive Analytics

G Governance & Trust H Human-System Interface D Designed Out

IMPACT Predictive models produce false negatives for 19% of Black and 21% of Latinx students; 80%+ of public colleges now use some form of predictive analytics

PREDICTIONS OVER-FLAG THE VERY STUDENTS WHO SUCCEED

Colleges increasingly use predictive analytics to identify “at-risk” students for early intervention. Gándara et al. (2025) found that models predicting course and degree completion show racial calibration bias: Black students are more likely to be predicted to fail when they actually succeed, and less likely to receive the success resources that the prediction is supposed to trigger. The magnitude of bias varies depending on how “at-risk” is defined, meaning the problem is partially a **construct-definition** problem — a core LEN 4 competency.

Baker & Hawn (2021) found algorithmic bias in education “poses significant threats to educational equity, potentially amplifying existing social and economic disparities in unprecedented ways.” The models inherit historical patterns of discrimination from training data, and faculty with deficit assumptions about Black students may interpret “at-risk” flags as confirmation rather than as a cue for intervention.

The case is the educational analog of A-Level (Case 35): an algorithm built to do good that systematically allocates intervention resources away from the students who would benefit most.

THE LEARNING ENGINEERING LENS

> *Algorithmic bias in education poses significant threats to educational equity, potentially amplifying existing social and economic disparities.*
>
> BAKER & HAWN, 2021

LE INSIGHT

Educational predictive analytics is the ongoing live case for algorithmic bias at the construct level. The bias is not in the sigmoid; it is in the definition of "at-risk." Defining at-risk in one way allocates support to one population; defining it in another way allocates support to another. The choice of definition is a capability-engineering decision with measurable equity consequences.

LENS APPROACH

LENS treats this case as the positive counterpart to Georgia State (Case 39). LEN 4 examines construct definition as the load-bearing measurement choice. LEN 7 examines the governance architecture that determines whose construct gets adopted. LEN 9 covers the technical bias-mitigation methods.

REFLECTION QUESTIONS

1. Pick a predictive analytic in your institution. Reconstruct the construct definition behind it. What is the equity consequence of that definition?
2. Design the governance review that a new predictive model should pass before it allocates resources to or away from a population.

WHO BUILDS THIS governance, ethics & measurement · human factors & interaction design · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · usability testing · cognitive walkthroughs · task analysis · design prototyping

SOURCES

→ Gándara et al., "Equity Considerations in Predictive Models," JPAM (2025)
→ IHEP interview with Gándara on predictive algorithms and equity
→ Baker & Hawn (2021), cited in WJARR
→ Schiller University analysis of AI/algorithmic bias in higher education

FURTHER READING

• Gándara et al. (2025), JPAM
• Baker & Hawn (2021) on algorithmic bias in education
• Friedman & Nissenbaum (1996), "Bias in Computer Systems"

LENS COURSES LEN 4 LEN 7 LEN 9

CASE 68 TECHNOLOGY GOVERNMENT 1999 – 2015

UK Post Office Horizon Scandal

G Governance & Trust H Human-System Interface K Knowledge & Institutional Memory

IMPACT ~900 sub-postmasters wrongfully prosecuted; many imprisoned; documented suicides; described as the most widespread miscarriage of justice in UK history

900

WRONGFUL PROSECUTIONS ACROSS 20+ YEARS

"the computer said so" was an institutionally sufficient basis for conviction

HORIZON — INSTITUTIONAL DEFERENCE TO AN ALGORITHM KNOWN TO BE FLAWED

The Post Office's Horizon IT system, developed by Fujitsu and deployed in 1999, produced systematic accounting errors in sub- postmasters' branch ledgers. The Post Office prosecuted hundreds of sub-postmasters for theft and false accounting on the basis of these errors, refusing for two decades to accept that the system itself was at fault. Internal Post Office and Fujitsu documents, later released through litigation, showed that engineers had known about software bugs throughout the period.

By the time the Court of Appeal began quashing convictions in 2021, sub-postmasters had served prison sentences, lost homes, and been driven to bankruptcy and suicide. The Williams Inquiry, which began in 2020 and continues, has documented one of the most extensively evidenced institutional cover-ups in UK history.

The capability gap was at the regulator, the prosecutor, and the courts: each accepted Fujitsu's representation that Horizon was reliable, despite documentation to the contrary, because no institutional actor had the standing or expertise to challenge it. The case is the strongest available evidence that "the computer said so" is a governance hazard.

THE LEARNING ENGINEERING LENS

> *A number of senior, and not so senior, employees of the Post Office knew, or at the very least should have known, that Legacy Horizon was capable of error.*
>
> SIR WYN WILLIAMS, POST OFFICE HORIZON IT INQUIRY, VOLUME 1, 2024

LE INSIGHT

Horizon is the canonical case for institutional deference to automated systems whose internal evidence was already known to be flawed. The capability gap was at every layer that took the software's output as authoritative — including the courts. "The computer said so" became, for two decades, an institutionally sufficient basis for criminal prosecution.

LENS APPROACH

LENS uses Horizon in LEN 7 as the canonical example of institutional deference to algorithmic output and in LEN 2 for the most extensive multi-decade automation-bias case in the dataset. Studio projects examine what evidentiary architecture would require **interrogating** automated output before acting on it.

REFLECTION QUESTIONS

1. Identify a decision in your domain currently made on the strength of "the computer said so." What evidentiary architecture should sit beside the output?

2. Design the institutional check that would have made Horizon's reliability subject to genuine challenge in 2005.

WHO BUILDS THIS governance, ethics & measurement · human factors & interaction design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · usability testing · cognitive walkthroughs · knowledge bases · data pipelines

SOURCES

→ Post Office Horizon IT Inquiry hearings and exhibits (2020–)
→ Hamilton & Others v. Post Office Limited (2021)
→ Wallis, The Great Post Office Scandal (2021)

FURTHER READING

• Wallis (2021), The Great Post Office Scandal
• Post Office Horizon IT Inquiry hearings
• Cummings (2017), automation bias literature

LENS COURSES LEN 7 LEN 2

CASE 69 HEALTHCARE TECHNOLOGY 2003 – 2018

Theranos

G Governance & Trust D Designed Out

IMPACT $9B valuation collapsed; thousands of patients given unreliable results; founder convicted on multiple counts of wire fraud

CLAIM	REALITY
Edison	Siemens

the device did not work; the regulatory seam between FDA and CLIA hid that fact

THERANOS — FRAUD EXPLOITING A REGULATORY SEAM BETWEEN FDA AND CLIA

Theranos claimed to have developed a blood-testing platform that could perform hundreds of laboratory tests from a finger-stick drop of blood. The platform did not work. Internal data showed accuracy far below what the company represented to investors, partners, and patients. To preserve the appearance of a working product, Theranos ran most patient samples on conventional commercial analyzers and reported the results as though they had come from its proprietary device.

The capability gap was at the FDA-CLIA boundary: the regulatory architecture that governs laboratory testing did not match the architecture of the company's product claims. Theranos exploited the gap deliberately. The board, the investors, and the pharmacy-partner Walgreens did not have the technical depth to challenge the claims. The journalist John Carreyrou did.

Elizabeth Holmes was convicted on multiple counts of wire fraud in 2022. The case is canonical in U.S. business education for fraud-as-product-strategy and in healthcare regulation for the governance gap that allowed unvalidated clinical tests to reach patients.

THE LEARNING ENGINEERING LENS

> *The company misrepresented to investors, regulators, and ultimately patients the accuracy of its blood-testing technology.*
>
> U.S. V. HOLMES, INDICTMENT, 2018

LE INSIGHT

Theranos is the canonical case for fraud exploiting the seam between two regulatory regimes. The FDA had not approved the device; CLIA accepted the laboratory operation. Neither layer had the capability to validate the claims independently. The capability gap was at the regulatory architecture.

LENS APPROACH

LENS uses Theranos in LEN 7 for the regulatory-seam failure and in LEN 4 for the measurement-validation gap. The case demonstrates that capability engineering at the boundary between regulatory regimes is itself a governance deliverable.

REFLECTION QUESTIONS

1. Identify a regulatory seam in your domain where two regimes meet without an explicit handoff. What could exploit it?
2. Design the validation regime that would have caught Theranos in 2013.

WHO BUILDS THIS governance, ethics & measurement · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · task analysis · design prototyping

SOURCES

→ United States v. Elizabeth Holmes (N.D. Cal. 2018–2022)
→ Carreyrou, Bad Blood (2018)
→ CMS revocation of Theranos CLIA certificate (2016)

FURTHER READING

• Carreyrou (2018), Bad Blood
• Theranos Inc. CMS inspection reports (2015–2016)
• Faulkner, Medical Device Regulation (2019)

LENS COURSES LEN 4 LEN 7

CASE 70 TECHNOLOGY GOVERNMENT 2011 – 2016

Wells Fargo Fake Accounts

G Governance & Trust N Normalization of Deviance

IMPACT ~3.5 million unauthorized accounts opened; ~$3B in penalties; CEO resigned; the Federal Reserve capped the bank's assets

3.5M

UNAUTHORIZED ACCOUNTS

the incentive architecture made misconduct rational for the front line

WELLS FARGO — THE MEASUREMENT SYSTEM PRODUCED THE BEHAVIOR THE INSTITUTION THEN PROSECUTED

Wells Fargo employees opened approximately three and a half million unauthorized customer accounts to meet aggressive sales quotas. The behavior was widespread, long-standing, and visible to internal risk and compliance functions for years. The institutional response was to discipline individual employees as bad apples and leave the incentive structure intact. The incentive structure was the actual cause: a sales-target architecture that made it impossible for most employees to meet quota by ethical means.

The CFPB action in 2016 was the first major public consequence. Subsequent enforcement actions and the Federal Reserve's unprecedented asset cap have constrained the bank's growth for years. The Wells Fargo case is the canonical example in U.S. banking of an institutional incentive structure that produced systematic misconduct at the front line, while the institution insisted the misconduct was individual.

The capability gap was the governance failure to recognize that the measurement system being used to manage employees was driving the behavior the institution claimed to deplore.

THE LEARNING ENGINEERING LENS

> *Wells Fargo's sales practices were a foreseeable consequence of its incentive compensation structure.*
>
> OCC CONSENT ORDER, WELLS FARGO, 2016

LE INSIGHT

Wells Fargo is the strongest available evidence that incentive architecture is a capability-engineering deliverable. The measurement system used to manage employees produced the behavior the institution then prosecuted. The gap was not at the front line. It was at the governance layer that had designed the incentives.

LENS APPROACH

LENS uses Wells Fargo in LEN 4 as a measurement-gaming case and in LEN 7 for the corporate-governance dynamics that allow such incentives to persist for years. Studio projects redesign the incentive architecture.

REFLECTION QUESTIONS

1. Identify a measurement system in your organization that is currently producing the behavior the organization claims to deplore. What is the gap between the measurement and the goal?
2. Design the incentive architecture for a bank's branch sales force that does not produce a Wells-Fargo-equivalent failure.

WHO BUILDS THIS governance, ethics & measurement · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · incident analysis · safety dashboards

SOURCES

→ CFPB Consent Order against Wells Fargo (2016)
→ OCC Consent Order (2016)
→ Independent Directors of Wells Fargo, Sales Practices Investigation Report (2017)

FURTHER READING

• Independent Directors Report (2017)
• Tirole (2010), corporate-governance theory
• Edmondson (2018), The Fearless Organization

LENS COURSES LEN 4 LEN 7

CASE 71 INDUSTRIAL GOVERNMENT 2015

Volkswagen Dieselgate

D Designed Out G Governance & Trust

IMPACT ~11 million vehicles equipped with defeat-device software; $33B+ in penalties and remediation; multiple criminal convictions

TEST BENCH

compliant

REAL WORLD

up to 40×

the protocol operated in a regime the vehicle could detect

DIESELGATE — ENGINEERED EVASION OF A MEASUREMENT SYSTEM

Volkswagen engineers designed software for diesel emissions-control systems that detected when the vehicle was on a regulatory test bench and turned on full emissions controls only during the test. On the road, the controls were largely disabled, producing emissions up to forty times the regulatory limit. The deception operated across millions of vehicles for years. It was discovered not by the regulator but by a small university research team in West Virginia comparing real-world to lab measurements.

Internal documents showed the defeat device had been authorized inside Volkswagen's engineering hierarchy as a deliberate response to a regulatory standard the engineering team did not believe it could meet within the platform's cost constraints. The decision was not made by a rogue engineer. It was made institutionally.

Dieselgate is the canonical case in industrial regulation for organized manufacturer evasion of an environmental standard through software. The reform that followed produced real-world emissions testing as a regulatory requirement in the EU.

THE LEARNING ENGINEERING LENS

> *The defeat device was the product of a long-standing institutional decision to evade emissions standards.*
>
> PARAPHRASING THE U.S. DEPARTMENT OF JUSTICE PLEA AGREEMENT WITH VOLKSWAGEN AG, 2017

LE INSIGHT

Dieselgate is the canonical case for institutionally engineered evasion of a measurement system. The capability gap was at the regulator's test protocol: it operated in a regime the vehicle could detect. The reform — real-world emissions testing — was a capability deliverable upgrade at the regulator's instrument boundary.

LENS APPROACH

LENS uses Dieselgate in LEN 4 as the canonical case for measurement-system evasion and in LEN 7 for the corporate- governance dynamics of engineered fraud. The reform pattern parallels Takata (Case 52).

REFLECTION QUESTIONS

1. Identify a measurement protocol in your domain that operates in a regime detectable by the entity being measured. What is the evasion potential?
2. Design the upgrade to a regulatory test protocol that makes Dieselgate-style evasion structurally infeasible.

WHO BUILDS THIS systems & human-factors engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS task analysis · design prototyping · governance frameworks · bias & evidence audits

SOURCES

→ U.S. Department of Justice Plea Agreement with Volkswagen AG (2017)
→ EPA Notice of Violation (2015)
→ Ewing, Faster, Higher, Farther: The Volkswagen Scandal (2017)

FURTHER READING

- Ewing (2017), Faster, Higher, Farther
- EPA technical reports on real-world emissions testing
- Carson, Silent Spring (1962) — regulatory-capacity comparison

LENS COURSES LEN 4 LEN 7

CASE 72 TECHNOLOGY 2014 – 2018

Cambridge Analytica / Facebook

G Governance & Trust

IMPACT ~87 million Facebook profiles harvested without informed consent; FTC $5B penalty; foundational data-governance reform

87M

PROFILES · 270K QUIZ-TAKERS

the Graph API contract assumed benevolent developer intent

CAMBRIDGE ANALYTICA — DESIGN ASSUMPTION AS LOAD-BEARING CAPABILITY

A researcher at Cambridge University built a personality-quiz app that, by virtue of Facebook's then-permissive Graph API, collected not only quiz-takers' profile data but their friends' data as well. The harvested dataset of approximately eighty-seven million profiles was sold to Cambridge Analytica, which used the data for political-campaign micro-targeting work in multiple national elections.

Facebook's permission architecture had been designed to allow third-party apps to deliver social experiences. The architecture was tested against ordinary use cases. It had not been red-teamed against systematic data harvesting. The capability gap was at the governance layer of Facebook's platform: the API contract assumed benevolent developer intent.

The case catalyzed the EU's General Data Protection Regulation (already underway, but accelerated), the California Consumer Privacy Act, and the FTC consent decree under which Facebook operates today.

THE LEARNING ENGINEERING LENS

> *Facebook gave developers access to far more user data than was necessary for the apps they built.*
>
> PARAPHRASING THE U.S. FTC ORDER, IN THE MATTER OF FACEBOOK INC., 2019

LE INSIGHT

Cambridge Analytica is the canonical case for platform-governance failure: an API contract that assumed benevolent intent and was not engineered against systematic abuse. The capability gap was not technical. The architecture worked exactly as designed; the design assumption was wrong.

LENS APPROACH

LENS uses Cambridge Analytica in LEN 7 as the foundational case for platform governance and data ethics, and in LEN 1 to teach that "design assumption" is a load-bearing capability deliverable.

REFLECTION QUESTIONS

1. Identify an API or interface in your domain whose contract assumes benevolent intent. What is the systematic-abuse case it was not red-teamed against?

2. Design the platform-governance deliverable that should accompany the launch of a new third-party developer API.

WHO BUILDS THIS governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits

SOURCES

→ FTC, In the Matter of Facebook Inc., Consent Order (2019)
→ UK Information Commissioner's Office Report on Cambridge Analytica (2018)
→ Cadwalladr & Graham-Harrison, Guardian investigation (2018)

FURTHER READING

• Zuboff (2019), The Age of Surveillance Capitalism
• Wylie (2019), Mindf*ck
• Nissenbaum (2010), Privacy in Context

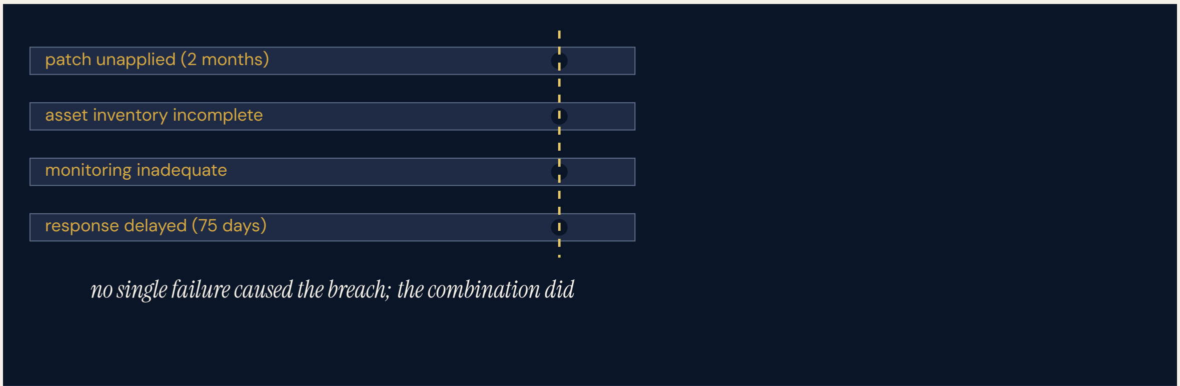
LENS COURSES LEN 1 LEN 7 LEN 6

CASE 73 TECHNOLOGY GOVERNMENT 2017

Equifax Data Breach

G Governance & Trust K Knowledge & Institutional Memory

IMPACT 147 million Americans' personal data exposed; CEO resigned; ~$700M settlement; foundational U.S. data-breach case



EQUIFAX — CUMULATIVE INADEQUACY IN ROUTINE CYBERSECURITY WORK

Equifax's online dispute portal was running a version of Apache Struts with a known critical vulnerability. A patch had been available for two months. Equifax's IT staff had been notified by the company's security team that the patch should be applied. The patch was not applied. Attackers exploited the vulnerability in May 2017 and exfiltrated personally identifying information on one hundred forty-seven million Americans over the next two and a half months. Equifax did not publicly disclose the breach until September.

The U.S. Senate Permanent Subcommittee on Investigations found that Equifax's patching practices were systematically inadequate, that the company did not have an asset inventory complete enough to know which systems needed the patch, and that the incident-response architecture had been treated as a cost center for years. The capability gap was at the institutional management of routine vulnerability remediation — work that is universally agreed to be necessary and routinely deferred.

The case is the U.S. data-breach analog of Sago (Case 59): cumulative inadequacy across multiple parameters that combined catastrophically.

THE LEARNING ENGINEERING LENS

> *Equifax lacked a comprehensive IT asset inventory.*
>
> U.S. SENATE PERMANENT SUBCOMMITTEE ON INVESTIGATIONS, HOW EQUIFAX NEGLECTED CYBERSECURITY, MARCH 2019

LE INSIGHT

Equifax is the canonical institutional-cybersecurity case for cumulative inadequacy in routine work: patching, inventory, monitoring, response. Each function was below industry standard. None alone produced the breach. The combination was the breach.

LENS APPROACH

LENS uses Equifax in LEN 5 for cumulative-inadequacy analysis in cybersecurity and in LEN 7 for the institutional dynamics that allow routine work to be chronically underfunded.

REFLECTION QUESTIONS

1. Identify a piece of routine work in your domain that is chronically deferred. What is the cumulative-inadequacy threshold?
2. Equifax did not know which assets ran Struts. Design the asset-inventory deliverable that an organization the size of Equifax should be able to produce on demand.

WHO BUILDS THIS governance, ethics & measurement · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · knowledge bases · data pipelines

SOURCES

→ Senate Permanent Subcommittee on Investigations, How Equifax Neglected Cybersecurity and Suffered a Devastating Data Breach (2019)
→ Federal Trade Commission Consent Order (2019)
→ GAO-18-559 Equifax breach analysis (2018)

FURTHER READING

• Senate Subcommittee report (2019)
• GAO-18-559
• Bruce Schneier, Click Here to Kill Everybody (2018)

LENS COURSES LEN 5 LEN 7

CASE 74 INDUSTRIAL 1981

Hyatt Regency Walkway Collapse

D Designed Out G Governance & Trust

IMPACT 114 killed and 216 injured in Kansas City when suspended walkways collapsed; foundational U.S. engineering-ethics case

AS DESIGNED	AS BUILT
1× load	2× load

a small construction change, accepted without re-analysis

HYATT REGENCY — CONNECTION DETAIL CHANGED IN THE FIELD

During a tea dance in the atrium of the Kansas City Hyatt Regency, two suspended walkways collapsed, killing one hundred fourteen and injuring two hundred sixteen — at the time the deadliest structural collapse in U.S. history. The walkway connection had been changed during construction from the original engineering design (single long rods supporting all three walkways) to a two-rod arrangement that doubled the load on the upper connection. The change was approved by the structural engineer's office without recalculation of the load.

The Missouri Board for Architects, Professional Engineers, and Land Surveyors revoked the licenses of the responsible engineers. The case is the foundational engineering-ethics case in U.S. structural engineering. It is taught to undergraduate engineers as the canonical example of how a small change in the construction detail, accepted casually, can convert a design that worked into one that fails catastrophically.

The capability gap was at the institutional review of construction changes: there was no requirement that a change to a connection detail be re-analyzed by the engineer of record.

THE LEARNING ENGINEERING LENS

> *The change in the rod configuration doubled the load on the connection that failed.*
>
> PARAPHRASING THE NATIONAL BUREAU OF STANDARDS INVESTIGATION, NBSIR 82-2465, 1982

LE INSIGHT

The Hyatt Regency case is the canonical engineering-ethics case for institutional review of construction changes. The change looked small; the load implication was catastrophic. The capability gap was at the review process that should have caught the doubled load.

LENS APPROACH

LENS uses Hyatt Regency in LEN 5 for change-control deliverables and in LEN 7 for the engineering-licensure architecture that holds the engineer of record accountable. Studio projects examine the change-control deliverable.

REFLECTION QUESTIONS

1. Identify a "small change during construction" pattern in your domain. What is the institutional review that should accompany it?

2. Design the change-control deliverable that would have surfaced the doubled-load issue at the Hyatt Regency before construction.

WHO BUILDS THIS systems & human-factors engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS task analysis · design prototyping · governance frameworks · bias & evidence audits

SOURCES

→ National Bureau of Standards, Investigation of the Kansas City Hyatt Regency Walkways Collapse (NBSIR 82-2465, 1982)
→ Missouri Board for Architects, Professional Engineers, and Land Surveyors disciplinary records (1986)
→ Pfatteicher (2000), "'The Hyatt Horror': Failure and Responsibility in American Engineering," J. Performance of Constructed Facilities 14(2)

FURTHER READING

• NBS Report (1982)
• Pfatteicher (2000), "'The Hyatt Horror,'" J. Performance of Constructed Facilities
• Petroski, To Engineer Is Human (1985)

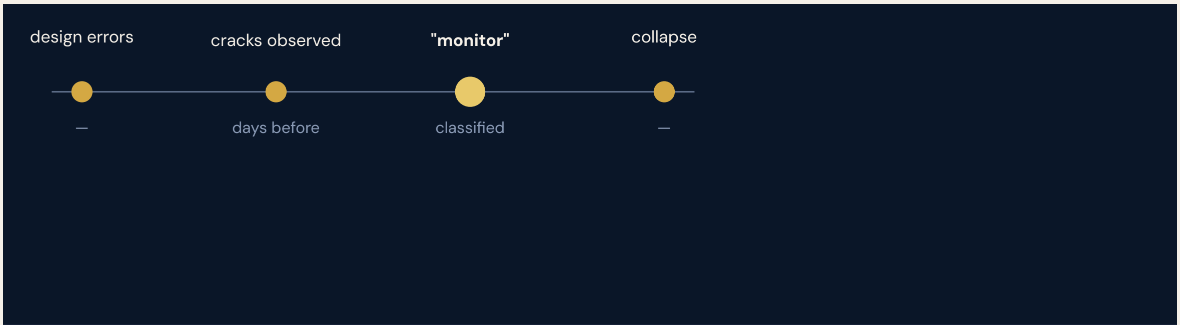
LENS COURSES LEN 5 LEN 7

CASE 75 INDUSTRIAL 2018

FIU Pedestrian Bridge Collapse

D Designed Out G Governance & Trust N Normalization of Deviance

IMPACT 6 killed in Miami when a pedestrian bridge collapsed during construction; NTSB found systemic design and review failures



FIU — OBSERVED CRACKS CLASSIFIED AS NOT SAFETY-CRITICAL

The pedestrian bridge over Southwest Eighth Street at Florida International University collapsed during construction, killing six. The NTSB investigation found that the design had not adequately considered the structure's behavior during construction sequencing, that visible cracks had been observed and inspected before the collapse but had been judged not safety-critical, and that the design firm's calculations contained errors that the review process had not caught.

The case is recent enough to be representative of the contemporary structural-engineering capability landscape. Modern bridges are designed with sophisticated computational tools; the design architecture is more capable than at any prior time. The errors that produced the collapse were the same errors that have produced structural collapses for a century: a load case inadequately considered, a review that did not catch the gap, and a field observation that was not escalated.

The bridge had been the subject of a publicity campaign focused on the speed of its construction. The schedule was load-bearing in the wrong sense.

THE LEARNING ENGINEERING LENS

> *The probable cause of the collapse was the load and capacity calculation errors made by FIGG Bridge Engineers in its design of the main span truss.*
>
> NTSB HIGHWAY ACCIDENT REPORT HAR-19/02 ON THE FIU PEDESTRIAN BRIDGE, 2019

LE INSIGHT

The FIU bridge is the canonical modern engineering case for cumulative review failure. Each layer of review failed in its own way. The collapse was not predicted but was predictable. The cracks that had been observed were classified as not safety- critical because the institutional protocol for "potentially significant crack on a new structure" defaulted to "monitor."

LENS APPROACH

LENS uses FIU in LEN 5 to teach review-process deliverables and in LEN 8 to compare with the Hyatt Regency (Case 74) case from forty years earlier — same failure mode, different decade.

REFLECTION QUESTIONS

1. The FIU and Hyatt Regency cases are 40 years apart and structurally similar. What has not been engineered in the discipline that would change that?
2. Design the field-observation escalation protocol that would have brought the FIU cracks to a decision-maker with the authority to halt work.

WHO BUILDS THIS systems & human-factors engineering · governance, ethics & measurement · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS task analysis · design prototyping · governance frameworks · bias & evidence audits · incident analysis · safety dashboards

SOURCES

→ NTSB Highway Accident Report HAR-19/02 (2019)
→ OSHA Final Report on the FIU Bridge Collapse (2019)
→ Engineering News-Record coverage of FIU Bridge

FURTHER READING

• NTSB HAR-19/02 (2019)
• Roddis & Roberts (1997) on engineering-review architecture
• Petroski, To Forgive Design (2012)

LENS COURSES LEN 5 LEN 8

CASE 76 ENERGY 2018

Camp Fire / PG&E

G Governance & Trust   N Normalization of Deviance   K Knowledge & Institutional Memory

IMPACT 85 killed in Paradise, California; deadliest U.S. wildfire in a century; PG&E pleaded guilty to 84 counts of involuntary manslaughter

85

KILLED IN PARADISE · SINGLE TRANSMISSION HOOK

infrastructure designed for one climate, operating in another

CAMP FIRE — CAPABILITY MISMATCH UNDER CHANGED CONDITIONS

A worn hook on a PG&E transmission tower failed and started a wildfire under high-wind, drought conditions. The fire moved into the town of Paradise faster than the evacuation infrastructure could move people out. Eighty-five people died. PG&E later pleaded guilty to eighty-four counts of involuntary manslaughter.

The CalFire investigation and subsequent litigation revealed that PG&E had known about the deteriorating condition of its transmission infrastructure across the wildfire-risk areas of its service territory for years. Maintenance had been deferred to fund other corporate priorities. The capability gap was simultaneously at the utility's asset-maintenance decisions and at the regulatory architecture that had allowed the deferrals to continue.

The Camp Fire case is the foundational climate-era case for utility capability under high-risk wildfire conditions. The reform that followed has restructured how California regulates utility wildfire-risk planning.

THE LEARNING ENGINEERING LENS

> *PG&E knew its equipment was failing in high-fire-risk areas and continued operating without remediation.*
>
> PARAPHRASING THE BUTTE COUNTY DISTRICT ATTORNEY'S CAMP FIRE REPORT, 2020

LE INSIGHT

The Camp Fire is the canonical climate-era case for utility- capability failure under changing risk conditions. The infrastructure was designed for one set of conditions and operated in another. The capability to update operations to match actual conditions did not exist as an institutional deliverable.

LENS APPROACH

LENS uses the Camp Fire in LEN 7 for utility regulatory governance and in LEN 8 for institutional response to changing operating conditions. The case is paired with the Northeast Blackout (Case 62) as utility-capability failures of different kinds.

REFLECTION QUESTIONS

1. Identify infrastructure in your domain that was designed for one set of conditions and is now operating in another. What is the capability deliverable to bridge the gap?
2. Design the regulatory architecture that would prevent a utility from deferring critical wildfire-risk maintenance.

WHO BUILDS THIS governance, ethics & measurement · safety science & organizational analysis · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · incident analysis · safety dashboards · knowledge bases · data pipelines

SOURCES

→ CalFire, Camp Fire Investigation Report (2019)
→ Butte County District Attorney (2020), The Camp Fire Public Report: A Summary of the Camp Fire Investigation
→ California Public Utilities Commission Order Instituting Investigation (2019)

FURTHER READING

• CalFire Camp Fire Investigation Report (2019)
• Wuebbles et al. (2017), Climate Science Special Report
• California PUC wildfire-mitigation orders (2019–present)

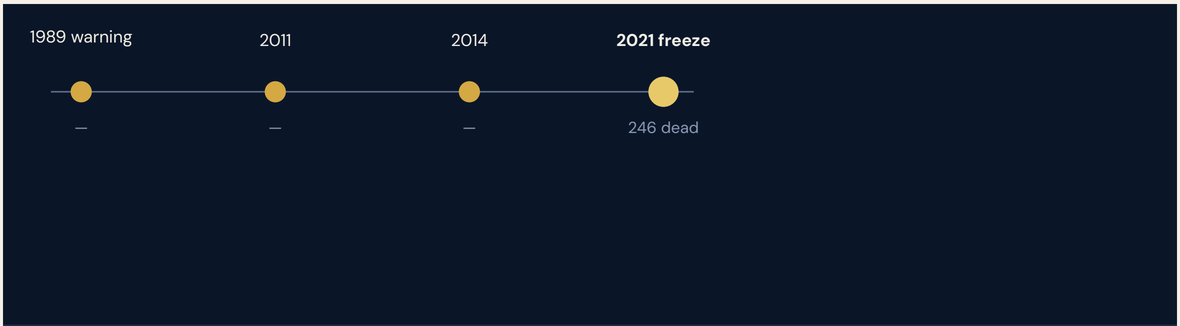
LENS COURSES LEN 7 LEN 8

CASE 77 ENERGY 2021

Texas Grid Freeze

G Governance & Trust K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT ~246 official deaths (Texas DSHS); academic excess-mortality estimates range 700–1,000; ~$130B in damages; foundational U.S. grid-resilience case



TEXAS GRID — FOUR WARNINGS ACROSS THREE DECADES, NO WINTERIZATION MANDATED

A winter storm in February 2021 brought temperatures across Texas well below the design assumptions of ERCOT-operated generation and transmission infrastructure. Natural-gas wells froze; gas-fired generation (which accounted for the majority of lost capacity) tripped offline; coal, nuclear, and wind also lost output. Across roughly four days, ERCOT directed load shed and millions of Texans lost power in rolling blackouts that, in many cases, did not roll. The Texas Department of State Health Services officially recorded 246 deaths; peer-reviewed excess-mortality analyses estimate 700–1,000.

The post-event analyses identified the absence of winterization requirements for Texas-grid generators as the proximate capability failure. ERCOT had been warned after similar storms in 1989, 2011, and 2014 that winterization was inadequate. The Texas regulatory architecture had not required winterization because the cost would be passed to consumers, and the rare events were judged acceptable.

The Texas Grid Freeze is the foundational case in U.S. grid reliability for how a designed-out capability (winterization) interacts with a changing climate to produce population-scale harm.

THE LEARNING ENGINEERING LENS

> *Texas generators had been warned across three decades that winterization was inadequate. The warnings did not produce winterization.*
>
> PARAPHRASING THE FERC-NERC JOINT INQUIRY REPORT ON THE FEBRUARY 2021 TEXAS EVENT

LE INSIGHT

The Texas Grid Freeze is the case for designed-out capability under changing climate conditions. Winterization was treated as optional because the rare event was acceptable. The acceptable rate of the rare event changed; the designed-out capability did not.

LENS APPROACH

LENS uses the Texas Grid Freeze in LEN 7 as a regulatory-governance case and in LEN 8 for institutional learning across repeated near- misses (1989, 2011, 2014) that did not produce remediation.

REFLECTION QUESTIONS

1. Identify a capability in your domain that has been designed out because the relevant rare event was judged acceptable. Is the rate still acceptable?
2. The Texas grid was warned four times in three decades. What institutional architecture allows that pattern, and what would change it?

WHO BUILDS THIS governance, ethics & measurement · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · knowledge bases · data pipelines · incident analysis · safety dashboards

SOURCES

→ FERC-NERC Joint Inquiry on the February 2021 Cold Weather Outages (2021)
→ Public Utility Commission of Texas reports
→ University of Texas at Austin Energy Institute analyses (2021)

FURTHER READING

• FERC-NERC Joint Inquiry (2021)
• Busby et al. (2021), Energy Institute analyses
• PUC Texas reports

LENS COURSES LEN 7 LEN 8

Chapter 6

Knowledge Lost

Institutional memory degraded.

Schedulers are among the top ten highest-turnover positions in the VA.

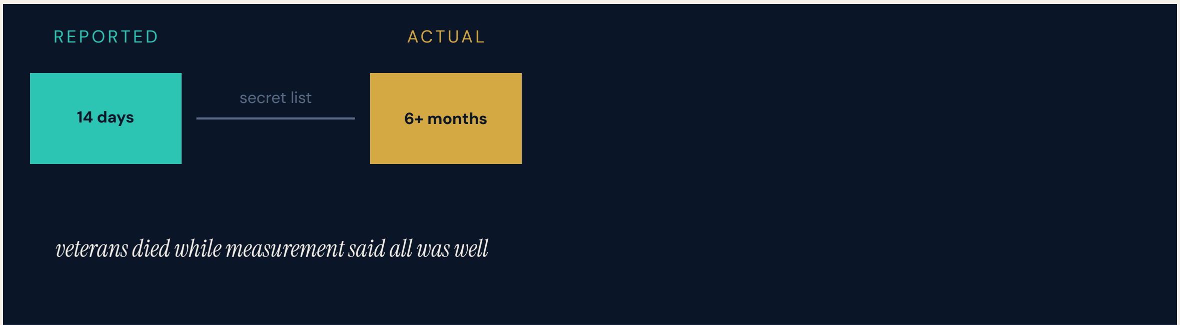
DEBRA DRAPER, GAO DIRECTOR OF HEALTH CARE, TESTIMONY TO THE HOUSE COMMITTEE ON VETERANS' AFFAIRS, 2019

CASE 32 GOVERNMENT 2014

VA Wait-Time Scandal

G Governance & Trust K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Veterans died waiting for care; 300,000+ on waiting lists or waiting 6+ months; staff falsified records



SCHEDULERS FALSIFIED RECORDS; THE MEASUREMENT SYSTEM HID THE GAP

The VA Office of Inspector General found staff at the Phoenix VA — and subsequently nationwide — had created secret waiting lists and falsified appointment data to hide that veterans were waiting weeks or months for care. The fourteen-day target was unrealistic given staffing. The system adapted by hiding reality rather than surfacing it.

GAO had flagged data-reliability concerns since 2000. VA OIG had identified problems in 2005, 2007, and 2008. The Obama transition team was warned in 2008. None produced systemic change. GAO flagged scheduler training as a root cause: schedulers are among the top ten highest-turnover positions in the VA, and the institutional knowledge required to game the system honestly was perpetually lost. Five years after the scandal, GAO still reported data- reliability concerns.

This is Normalization of Deviance applied to **measurement itself.** When the measurement system cannot capture reality, the gap between reported and actual performance becomes invisible — and lethal. Veterans died inside a system that, by its own metrics, was succeeding.

THE LEARNING ENGINEERING LENS

> *Schedulers are among the top ten highest-turnover positions in the VA.*
>
> DEBRA DRAPER (GAO DIRECTOR OF HEALTH CARE), HOUSE VA COMMITTEE TESTIMONY, GAO-19-687T, JULY 2019

LE INSIGHT

The VA case is the canonical example of measurement failure as a capability failure. The system that should have surfaced the gap instead generated reports that hid it. The turnover among schedulers — the human capability operating the measurement instrument — meant the institution lost the knowledge to even notice it was lying to itself. The case stands as the strongest argument in this book for treating decision-grade evidence as a design requirement, not a reporting requirement.

LENS APPROACH

LENS treats the VA case in LEN 4 as the canonical evidence-gap case (the measurement system itself was the source of harm), in LEN 7 as a governance failure (multiple warnings unactioned over fifteen years), and in LEN 8 as a knowledge-loss case via turnover.

REFLECTION QUESTIONS

1. Identify a measurement system in your domain that is also operated by a high-turnover role. What is the institutional risk that the system stops measuring reality?
2. Design the evidence pipeline that would have surfaced the Phoenix VA gap without relying on the people who were gaming the metrics.

WHO BUILDS THIS governance, ethics & measurement · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · knowledge bases · data pipelines · incident analysis · safety dashboards

SOURCES

→ VA Office of Inspector General Report 14-02603-267 (2014)
→ GAO Veterans Health Administration reports (2000–2019)
→ Federal News Network coverage (2019)
→ CNN investigative reporting on the Phoenix VA

FURTHER READING

• VA OIG, full reports (2014–present)
• Argyris & Schön (1978), Organizational Learning
• Tuckey & Pollack (2024), measurement-gaming literature

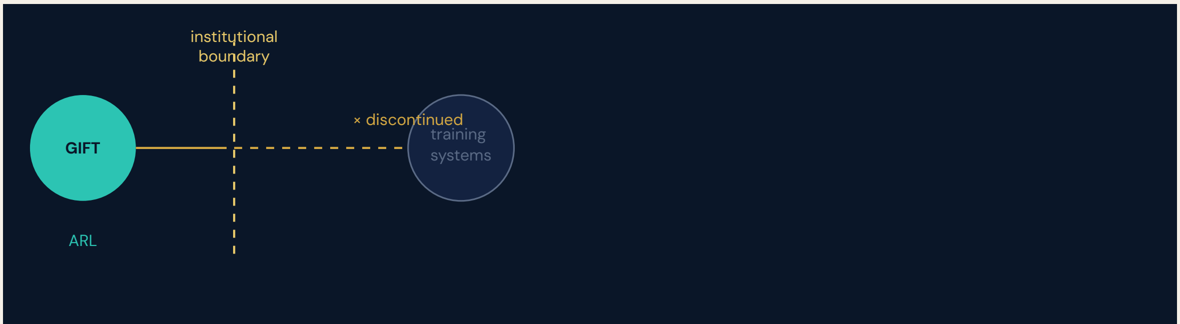
LENS COURSES LEN 4 LEN 7 LEN 8

CASE 38 DEFENSE EDUCATION AT SCALE 2012 – PRESENT

GIFT and the Adoption Gap

K Knowledge & Institutional Memory G Governance & Trust N Normalization of Deviance

IMPACT Active open-source ITS framework with demonstrated learning effectiveness; ubiquitous fielded adoption in routine military training has not been achieved



ADAPTIVE INSTRUCTION WORKED; ADOPTION DID NOT CROSS THE BOUNDARY

The Generalized Intelligent Framework for Tutoring (GIFT) is an open-source framework for authoring, delivering, and evaluating intelligent tutoring systems. Originated under the U.S. Army Research Laboratory, it moved after the 2019 Army Futures Command reorganization into DEVCOM Soldier Center, where it remains under active development with regular releases and an annual peer-reviewed GIFTSym symposium.

Computer-based tutoring has been shown to be "as effective as expert human tutors … in well-defined domains and significantly better than traditional classroom training" (VanLehn, 2011), and GIFT sits at the center of the effort to lower the authoring barrier. The case is not that GIFT was discontinued — it was not. It is that **ubiquitous fielded adoption across routine military training remains limited** despite five decades of research. That gap is the canonical learning-engineering problem: the science is settled, the platform exists, the studies are positive — and the institutional pathway to scaled adoption is still being built.

THE LEARNING ENGINEERING LENS

> *The technology works. The institutional pathway to ubiquitous fielded use does not yet.*
>
> EDITORS' SYNTHESIS OF THE GIFT ADOPTION RECORD

LE INSIGHT

GIFT is the most directly relevant case in this book to the learning-engineering discipline itself. The technology works, the learning science works, the framework is active and supported. What has not been built is the institutional adoption pathway — procurement, training-pipeline integration, instructor workflow redesign, the authority structure — that would make adaptive tutoring a default rather than an experiment.

LENS APPROACH

LENS treats GIFT in LEN 1 as the problem-framing case for the discipline, in LEN 8 as the foundational adoption-pathway case, and in LEN 10 (capstone) as a prompt for designing the institutional deliverables that would close an adoption gap of this shape.

REFLECTION QUESTIONS

1. GIFT exists, is supported, and works. Adoption at scale does not. What is the equivalent in your domain — an effective intervention whose adoption pathway has not been engineered?
2. Design the institutional adoption deliverable that would move adaptive tutoring from “available framework” to “default routine practice” in one operational training pipeline.

WHO BUILDS THIS knowledge management & data engineering · governance, ethics & measurement · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS knowledge bases · data pipelines · governance frameworks · bias & evidence audits · incident analysis · safety dashboards

SOURCES

→ GIFT Project, gifttutoring.org — Overview, News, and Releases pages
→ Sottilare, Graesser, Hu & Holden (eds.), Design Recommendations for Intelligent Tutoring Systems, GIFTSym volumes
→ International Journal of AI in Education, Special Issue on GIFT (2017), DOI: 10.1007/s40593-017-0149-9
→ VanLehn, K. (2011), “The Relative Effectiveness of Human Tutoring, Intelligent Tutoring Systems, and Other Tutoring Systems,” Educational Psychologist 46(4)

FURTHER READING

• GIFT Project documentation and GIFTSym 12 / 13 proceedings
• Goodell & Kolodner (2022), Learning Engineering Toolkit

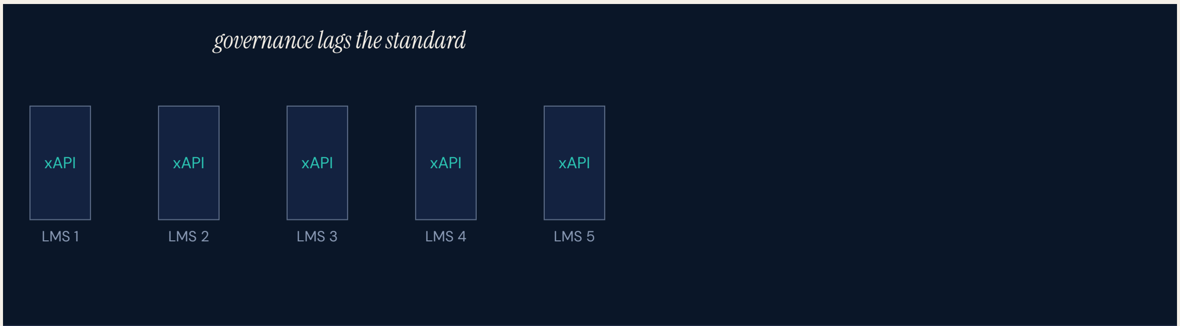
LENS COURSES LEN 1 LEN 10 LEN 8 LEN 6

CASE 40 EDUCATION AT SCALE DEFENSE ONGOING

xAPI / Total Learning Architecture — Interoperability Gap

K Knowledge & Institutional Memory G Governance & Trust

IMPACT Despite xAPI adoption, most implementations remain siloed; cross-organizational learning-data interoperability largely unrealized



STANDARD EXISTS; CROSS-ORGANIZATION DATA STILL SILO'D

The Experience API (xAPI) was developed to enable tracking of learning experiences across platforms, systems, and contexts — a foundational technology for the kind of evidence infrastructure LENS envisions. The Advanced Distributed Learning Initiative's Total Learning Architecture (TLA) envisioned an ecosystem where learning records, competency frameworks, and credential data flow across organizational boundaries to support continuous, adaptive capability development.

In practice, xAPI implementations remain largely siloed within individual LMS platforms. Cross-organizational data sharing — the capability most relevant to high-consequence operational domains — has not materialized at scale. The governance infrastructure (who owns the data, what consent frameworks apply, how data quality is assured) has lagged far behind the technical standard.

This is an implementation-science failure: the standard exists, reference implementations exist, but the institutional and governance conditions for scaled adoption do not. The case mirrors inBloom (Case 8) in its essential structure — technology in advance of governance — except that it spans the entire learning-technology ecosystem.

THE LEARNING ENGINEERING LENS

> *The standard exists. The governance does not.*
>
> EDITORS' SYNTHESIS OF ICICLE / ADL TLA DISCUSSION

LE INSIGHT

xAPI/TLA is the technical-standard analog of the implementation gap — the discipline has the data model, the spec, and the reference implementations. What it does not have is the governance and institutional architecture to make cross-organizational learning data flow. This is the canonical case in this book for treating data governance as a capability-engineering deliverable.

LENS APPROACH

LENS treats xAPI/TLA in LEN 4 as a data-governance and interoperability case and in LEN 8 as an example of organizational- learning infrastructure that has not scaled. The case is the technical underlay to the larger argument about evidence systems that decision-makers can trust.

REFLECTION QUESTIONS

1. Why has the xAPI standard not produced cross-organizational interoperability at scale? What governance condition is missing?
2. Design the minimum governance architecture under which xAPI data could flow across two organizations in your domain.

WHO BUILDS THIS knowledge management & data engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS knowledge bases · data pipelines · governance frameworks · bias & evidence audits

SOURCES

→ Advanced Distributed Learning Initiative, Total Learning Architecture
→ IEEE ICICLE LEBoK
→ xAPI specification, https://xapi.com/
→ ICICLE proceedings on TLA adoption challenges

FURTHER READING

• ADL TLA documentation
• Saxberg, learning-engineering infrastructure essays
• IEEE ICICLE LEBoK chapters on data and analytics

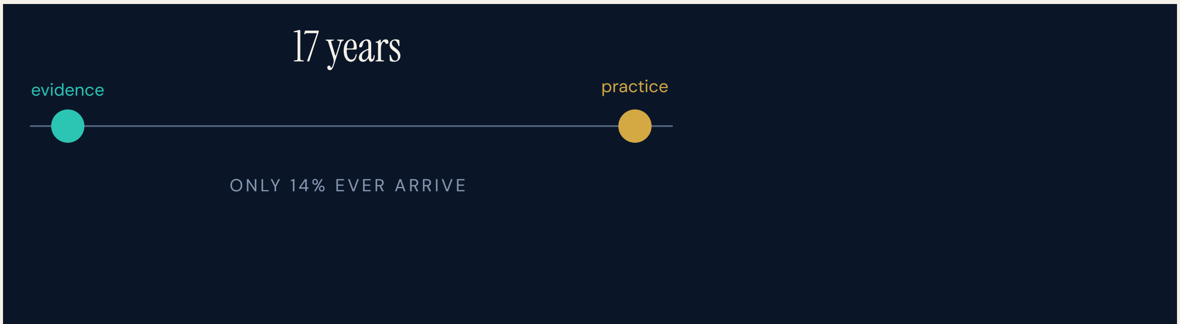
LENS COURSES LEN 4 LEN 8 LEN 6

CASE 41 HEALTHCARE ONGOING

Implementation Science in Healthcare — The 17-Year Gap

K Knowledge & Institutional Memory G Governance & Trust N Normalization of Deviance

IMPACT Average time from research finding to clinical practice: 17 years; only ~14% of research findings ever reach practice



RESEARCH TO PRACTICE — THE CANONICAL IMPLEMENTATION GAP

The canonical finding in implementation science: it takes an average of seventeen years for research evidence to be integrated into clinical practice, and only about fourteen percent of research findings ever make it. This is not a single case but a systemic condition — the same structural problem the Makary medical-errors data (Case 31) describes from the outcome side. Effective interventions exist; the system to adopt, sustain, adapt, and measure them at scale does not.

The Active Implementation Frameworks (Fixsen et al., 2005) and the EPIS framework (Aarons et al., 2011) — both referenced in LENS curriculum design — were developed specifically to address this gap. The LENS commitment to threading implementation science throughout the curriculum, rather than as a standalone module, is a direct response to this evidence.

The seventeen-year gap is the meta-case for this entire book. Every success case in Chapters 8 and 9 represents a closure of this gap in one domain. Every failure case represents the gap left open. The LENS discipline exists to make seventeen years shorter.

THE LEARNING ENGINEERING LENS

> *The answer is 17 years. What is the question?*
>
> MORRIS, WOODING & GRANT, J ROYAL SOC MED, 2011

LE INSIGHT

The 17-year gap is the structural problem that LENS exists to address. It is the difference between knowing what works and having it deployed. Every case in this book is a sample from a distribution governed by that gap. The success cases shorten it; the failure cases let it run.

LENS APPROACH

LENS uses this case in LEN 1 as the foundational problem statement of the discipline, in LEN 10 as a studio constraint (designs must consider implementation, not just efficacy), and in LEN 8 as the central knowledge-transfer challenge. The case is referenced at least once in every required course.

REFLECTION QUESTIONS

1. Pick an evidence-based intervention in your domain. Estimate the gap between when the evidence became conclusive and when the intervention reached majority of practice. What did the gap cost?

2. Design the deliverable that would shorten that gap by half in your domain. Be specific about who funds it, who owns it, and what evidence demonstrates the reduction.

WHO BUILDS THIS knowledge management & data engineering · governance, ethics & measurement · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS knowledge bases · data pipelines · governance frameworks · bias & evidence audits · incident analysis · safety dashboards

SOURCES

→ Balas & Boren (2000), Yearbook of Medical Informatics
→ Morris, Wooding & Grant (2011), "The answer is 17 years..."
→ Fixsen et al. (2005), Implementation Research: A Synthesis
→ Aarons et al. (2011), the EPIS framework

FURTHER READING

• Fixsen et al. (2005)
• Aarons et al. (2011), EPIS
• Damschroder et al. (2009), CFIR

LENS COURSES LEN 1 LEN 10 LEN 8 LEN 6

CASE 78 AVIATION 1972 – PRESENT

NASA Saturn V Documentation

K Knowledge & Institutional Memory

IMPACT Foundational U.S. aerospace case for the cost of losing the institutional capability to build a system

1969	2005
built it	couldn't rebuild

drawings persist; tacit institutional knowledge walks out with retirees

SATURN V — DOCUMENTATION IS NOT CAPABILITY

By the early 2000s, when NASA was working on the Constellation program, it had become apparent that the institutional capability to build a Saturn V rocket had been lost. Drawings existed. Documents existed. But the practical knowledge — the institutional memory of how the components were manufactured, tested, and assembled, and what choices had been made and why — had walked out with the workforce that had retired by the 1990s. The vehicle could be redesigned. It could not, by 2005, be reproduced.

The case is canonical in U.S. aerospace for the difference between documentation and institutional capability. The Apollo program was documented to an unprecedented degree. The documentation was not the institutional knowledge. When the engineers who had built the system left, the system left with them.

The Saturn V case is the strongest available evidence that institutional capability is not the same as the artifacts the institution produces. The lesson is general: capability lives in people and in the institutional practices that sustain them, not in the documents the institution leaves behind.

THE LEARNING ENGINEERING LENS

> *The Saturn V drawings exist. The Saturn V does not.*
>
> NASA CONSTELLATION PROGRAM REVIEW, 2005

LE INSIGHT

The Saturn V case is the canonical evidence that institutional knowledge is not equivalent to documentation. The documents persist; the capability does not. Capability engineering must treat the people who hold tacit knowledge as a primary institutional asset.

LENS APPROACH

LENS uses the Saturn V case in LEN 8 as the foundational organizational-memory case and in LEN 1 to teach the distinction between recorded knowledge and institutional capability.

REFLECTION QUESTIONS

1. Identify a capability in your domain that is currently held in the tacit knowledge of a small number of senior practitioners. What is your institution's capacity to reproduce it after they retire?
2. Design the institutional practice that would preserve a capability against the retirement of its holders.

WHO BUILDS THIS knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS knowledge bases · data pipelines

SOURCES

→ NASA Constellation Program documentation and reviews (2005–2010)
→ Mark Wade, Encyclopedia Astronautica archives
→ Dunbar, History of the Saturn V (NASA SP-4206)

FURTHER READING

• Polanyi (1966), The Tacit Dimension
• Brown & Duguid (2000), The Social Life of Information
• Nonaka & Takeuchi (1995), The Knowledge-Creating Company

LENS COURSES LEN 1 LEN 8

CASE 79 SPACE / AEROSPACE 2019 – 2024

Boeing Starliner

K Knowledge & Institutional Memory D Designed Out

IMPACT Multiple delays; the 2024 crewed flight left two NASA astronauts at the ISS for months; contemporary case for capability erosion at a legacy contractor

STARLINER — CAPABILITY DECAY AT A LEGACY CONTRACTOR

Boeing's Starliner spacecraft, intended as the second commercial crew vehicle alongside SpaceX's Crew Dragon, suffered software errors in its 2019 uncrewed test flight, valve corrosion requiring a 2021 launch scrub, and propulsion-system issues during its 2024 crewed test flight that left the two NASA astronauts aboard the International Space Station rather than returning on the contracted spacecraft. NASA returned the astronauts on SpaceX vehicles months later.

The Government Accountability Office and multiple NASA internal reviews identified the Starliner program as exhibiting capability erosion across multiple engineering disciplines — software, valves, propulsion, integration testing — at a contractor whose human- spaceflight track record had previously been definitive in U.S. aerospace. The erosion appears to be partly generational (Apollo- and Shuttle-era engineers retired) and partly institutional (cost pressures, schedule-driven decisions, and the loss of NASA's in-house engineering depth to challenge the contractor).

Starliner is the contemporary U.S. aerospace case for capability erosion at scale at a legacy contractor.

THE LEARNING ENGINEERING LENS

> *Starliner has demonstrated significant readiness shortfalls across multiple engineering disciplines.*
>
> GAO REPORT ON COMMERCIAL CREW PROGRAM, 2023

LE INSIGHT

Starliner is the case for sustained capability erosion at a legacy contractor whose track record had previously been definitive. The erosion happened over decades and was visible in retrospect; the institutional architecture for catching it did not exist.

LENS APPROACH

LENS uses Starliner in LEN 8 as a contemporary contractor- capability erosion case and in LEN 5 for the contractor-NASA interface capability that has thinned over decades.

REFLECTION QUESTIONS

1. Identify a legacy supplier in your domain whose capability track record may have eroded faster than your institutional confidence in them has updated. What is the signal?
2. Design the contractor-capability review that would have caught the Starliner gaps before the 2024 crewed flight.

WHO BUILDS THIS knowledge management & data engineering · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS knowledge bases · data pipelines · task analysis · design prototyping

SOURCES

→ GAO Commercial Crew Program reports (2018–2024)
→ NASA Aerospace Safety Advisory Panel reports (2020–2024)
→ Berger, Liftoff: Elon Musk and the Desperate Early Days of SpaceX (2021) — comparison context

FURTHER READING

• GAO Commercial Crew reports (2018–2024)
• NASA Aerospace Safety Advisory Panel reports
• Augustine, Augustine's Laws (1986)

LENS COURSES LEN 5 LEN 8 LEN 6

CASE 80 SPACE / AEROSPACE 1996

Ariane 5 Flight 501

D Designed Out K Knowledge & Institutional Memory H Human-System Interface

IMPACT Maiden flight destroyed itself 37 seconds after launch; ~$500M payloads lost; reused Ariane 4 code never re-verified for Ariane 5

37s

AFTER LAUNCH · 16-BIT INTEGER OVERFLOW

code path disabled by the previous vehicle's profile; re-enabled by the new one

ARIANE 5 — THE FALLACY OF RISK-FREE CODE REUSE

The inertial reference system software on Ariane 5 Flight 501 inherited code from Ariane 4. On Ariane 4, a particular variable — a horizontal velocity value — never exceeded the range of a 16-bit signed integer. On Ariane 5, with its different flight trajectory, the value did. The integer overflow caused both redundant inertial reference systems to shut down almost simultaneously. The vehicle lost guidance, broke up under aerodynamic stress, and was destroyed by the range safety officer.

The inquiry board found that the code path producing the overflow had been disabled by Ariane 4's mission profile but had not been removed when the software was reused for Ariane 5. The capability that was missing was the requirement that reused code be re-verified against the new vehicle's flight envelope. Software reuse was treated as risk-free; it was not.

Ariane 5 is the foundational case in safety-critical software engineering for the hazards of code reuse without re-verification.

THE LEARNING ENGINEERING LENS

> *The internal SRI software exception was caused during execution of a data conversion from 64-bit floating point to 16-bit signed integer value.*
>
> ARIANE 5 FLIGHT 501 FAILURE INQUIRY BOARD, 1996

LE INSIGHT

Ariane 5 is the canonical software-engineering case for the fallacy of risk-free code reuse. Code is fit for its envelope of operation. Reusing it in a different envelope is a new design decision and must be re-verified as one. The institutional capability that was missing was the requirement to re-verify.

LENS APPROACH

LENS uses Ariane 5 in LEN 5 for software-engineering capability deliverables and in LEN 8 for the institutional discipline that treats reuse as a verification event rather than as a savings.

REFLECTION QUESTIONS

1. Identify a piece of reused infrastructure in your domain whose original operating envelope is not documented. What is the silent assumption?
2. Design the verification deliverable that should accompany every reuse of safety-critical software.

WHO BUILDS THIS systems & human-factors engineering · knowledge management & data engineering · human factors & interaction design — plus domain experts and a learning engineer to integrate.

TOOLS task analysis · design prototyping · knowledge bases · data pipelines · usability testing · cognitive walkthroughs

SOURCES

→ Lions, J. L. (chair), Ariane 5 Flight 501 Failure Inquiry Board Report (1996)
→ Le Lann, G., "An Analysis of the Ariane 5 Flight 501 Failure" (1997)
→ Leveson (1995), Safeware, on software reuse

FURTHER READING

• Lions Report (1996)
• Leveson (1995), Safeware
• Wing, "A Specifier's Introduction to Formal Methods" (1990)

LENS COURSES LEN 5 LEN 8

CASE 81 INDUSTRIAL 1940

Tacoma Narrows Bridge

D Designed Out K Knowledge & Institutional Memory

IMPACT Bridge collapsed in ~40 mph wind four months after opening; foundational case in aeroelasticity and structural-engineering pedagogy

4 mo.

FROM OPENING TO COLLAPSE

aeroelastic flutter — a failure mode the discipline had not yet learned to recognize

TACOMA NARROWS — A DISCIPLINE-LEVEL LEARNING EVENT

The Tacoma Narrows Bridge across Puget Sound collapsed on November 7, 1940, in a wind no stronger than the bridge had been designed for. The failure mode — aeroelastic flutter — had not been recognized as a design hazard in long-span suspension bridges. The bridge's slender deck, chosen for aesthetic reasons over Ammann's earlier proposal of a deeper truss, was the proximate cause. The pier was sound. The cables were sound. The deck twisted itself apart in a wind no stronger than a stiff breeze.

Tacoma Narrows is the canonical case in engineering pedagogy for a failure mode the discipline had not yet learned to recognize. The cost of the lesson — captured on film and shown to virtually every undergraduate civil engineer since — was a bridge. The payoff is that no comparable bridge has collapsed in the same mode since. The engineering discipline updated.

The case is also a foundational example of the way disciplines learn from individual failures and incorporate the lesson into pedagogy permanently.

THE LEARNING ENGINEERING LENS

> *The bridge was not designed against the failure mode that destroyed it because that failure mode was not yet known to the discipline.*
>
> PARAPHRASING PETROSKI, TO ENGINEER IS HUMAN, 1985

LE INSIGHT

Tacoma Narrows is the canonical case for a discipline-level learning event. The failure was not the engineer's fault; it was the discipline's not yet knowing. The discipline absorbed the lesson permanently. The case is the strongest available evidence that institutional learning across an entire profession is possible — and rare.

LENS APPROACH

LENS uses Tacoma Narrows in LEN 1 as a problem-framing exemplar for discipline-level learning, in LEN 8 as the canonical case for successful institutional knowledge absorption, and in studio (LEN 10) as a worked example of what discipline-level reform requires.

REFLECTION QUESTIONS

1. Identify a failure mode in your discipline that the field has not yet learned to recognize. What would be the equivalent Tacoma Narrows event?
2. What institutional architecture absorbed the Tacoma Narrows lesson into the structural-engineering curriculum so durably? Describe its equivalent in your discipline.

WHO BUILDS THIS systems & human-factors engineering · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS task analysis · design prototyping · knowledge bases · data pipelines

SOURCES

→ Ammann, von Kármán & Woodruff, The Failure of the Tacoma Narrows Bridge (1941)
→ Petroski (1985), To Engineer Is Human
→ Billah & Scanlan (1991), aeroelastic-flutter analysis

FURTHER READING

• Petroski (1985), To Engineer Is Human
• Ammann et al. (1941), failure inquiry report
• Argyris & Schön (1996), Organizational Learning II

LENS COURSES LEN 1 LEN 8

CASE 82 ENERGY 2015 – 2016

Aliso Canyon Methane Leak

G Governance & Trust N Normalization of Deviance K Knowledge & Institutional Memory

IMPACT Largest methane leak in U.S. history; ~8,000 households evacuated; ~100,000 tons of methane released

100K t

METHANE RELEASED OVER ~4 MONTHS

legacy wells aging past the inspection regime designed for them

ALISO CANYON — INFRASTRUCTURE OUTPACING ITS OVERSIGHT

A well at SoCalGas's Aliso Canyon underground storage facility ruptured in October 2015 and could not be capped for nearly four months. Approximately one hundred thousand tons of methane were released into the atmosphere — roughly equivalent to the annual greenhouse emissions of half a million cars. Eight thousand households were evacuated from Porter Ranch.

The Blade Energy investigation identified corrosion in the well's casing, missing safety equipment that had been installed and then removed in the 1970s, and an institutional pattern of deferring integrity checks on the older wells in the field. SoCalGas had been aware that the field's wells were aging and that their integrity was uncertain. The regulator had been aware. The inspection regime had not been engineered to catch the failure mode that occurred.

Aliso Canyon is the foundational U.S. case in underground gas storage for the gap between regulatory practice and the actual engineering condition of legacy infrastructure.

THE LEARNING ENGINEERING LENS

> *The leak resulted from external corrosion of the well casing — a known and inspectable failure mode that the operator did not adequately inspect.*
>
> PARAPHRASING BLADE ENERGY PARTNERS, ALISO CANYON ROOT CAUSE ANALYSIS, 2019

LE INSIGHT

Aliso Canyon is the case for legacy infrastructure aging past the inspection regime designed for it. The wells were older than the inspection regulations. The operator knew. The regulator knew. The capability deliverable to update the inspection regime did not exist.

LENS APPROACH

LENS uses Aliso Canyon in LEN 7 for regulatory-architecture capability and in LEN 8 for the institutional pattern of legacy assets aging past the regime governing them.

REFLECTION QUESTIONS

1. Identify legacy infrastructure in your domain whose age exceeds the inspection regime designed for it. What is the capability deliverable to update the regime?

2. Design the regulator's deliverable that would force re-inspection of legacy infrastructure as the inspection regime is updated.

WHO BUILDS THIS governance, ethics & measurement · safety science & organizational analysis · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · incident analysis · safety dashboards · knowledge bases · data pipelines

SOURCES

→ Blade Energy Partners, Aliso Canyon Root Cause Analysis (2019)
→ California Public Utilities Commission orders on Aliso Canyon
→ Conley et al., Science (2016), atmospheric measurement of the leak

FURTHER READING

• Blade Energy Partners Report (2019)
• Conley et al. (2016), Science
• California PUC orders on Aliso Canyon

LENS COURSES LEN 7 LEN 8

Chapter 7

The Evidence Gap

Measurement systems that cannot see their own failures.

People don't just die from heart attacks and bacteria, they die from system-wide failings and poorly coordinated care.

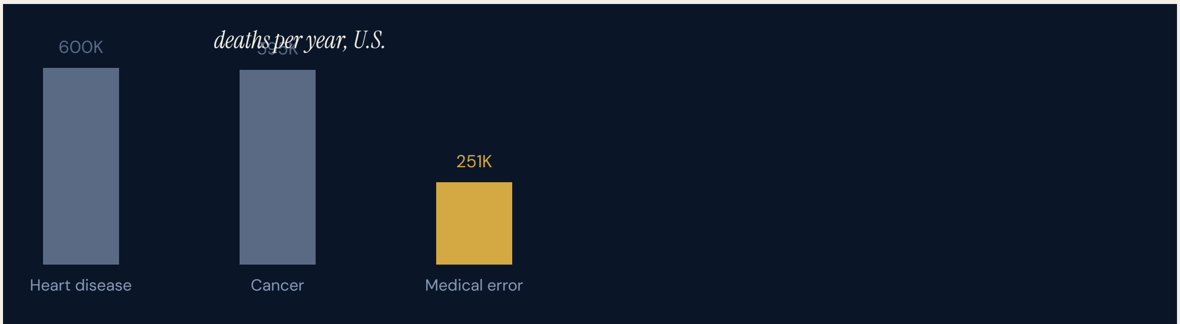
MARTIN MAKARY, BMJ, 2016

CASE 31 HEALTHCARE ONGOING

Medical Errors as Systemic Failure

T Training Gap H Human-System Interface N Normalization of Deviance
K Knowledge & Institutional Memory G Governance & Trust

IMPACT Makary & Daniel (2016) estimate of ~250,000 U.S. deaths/year from medical error — third leading cause; estimate is widely cited and substantively contested



THIRD LEADING CAUSE OF DEATH — AND STRUCTURALLY INVISIBLE

Not a single incident but a systemic condition. Makary & Daniel (2016), publishing in the **BMJ** from Johns Hopkins, estimated medical errors cause more than two hundred fifty thousand deaths annually in the United States — third leading cause of death behind heart disease and cancer. The key argument: “People don’t just die from heart attacks and bacteria, they die from system-wide failings and poorly coordinated care.” Death certificates do not capture medical error as a cause, so the problem is structurally invisible to the measurement systems that should track it.

The 250,000 figure has been substantively contested (Shojania and Dixon-Woods, **BMJ Quality & Safety** 2017) on methodological grounds — itself a worked example of the **gap-attribution** problem the discipline must diagnose: how much of a counted death is the learning system, how much the system design, how much the underlying disease?

The Institute of Medicine’s **To Err Is Human** (1999) had earlier estimated 44,000–98,000 deaths annually and catalyzed the patient-safety movement: TeamSTEPPS (Case 27), surgical checklists (Case 13), Keystone ICU (Case 14). Seventeen years later, the problem persists at scale.

THE LEARNING ENGINEERING LENS

> *People don't just die from heart attacks and bacteria, they die from system-wide failings and poorly coordinated care.*
>
> MARTIN MAKARY, PRESS STATEMENTS ACCOMPANYING MAKARY & DANIEL, BMJ (2016)

LE INSIGHT

The Makary data is the anchor evidence for the LENS argument because of its scale, its provenance (Johns Hopkins), and its framing — system failure rather than individual error. The seventeen years between **To Err Is Human** and Makary's reassessment is also the implementation gap of Case 41 in another guise. Capability engineering exists to close that gap, and the cost of leaving it open is measured in lives at population scale.

LENS APPROACH

LENS treats this case as the central evidence anchor of the entire curriculum. LEN 1 introduces it as the foundational problem statement. LEN 4 uses it as the canonical case for measurement systems that cannot see their own failures (death certificates do not capture the failure mode). LEN 10 uses it as the studio prompt for designing interventions at the scale required to shift the distribution.

REFLECTION QUESTIONS

1. Identify a measurement instrument in your domain that systematically fails to capture the failure modes it should be designed to surface. What would it cost to fix?
2. Two hundred fifty thousand deaths a year is the third leading cause of death in the U.S. Design the measurement and intervention regime that would shift the curve over a five-year horizon. Estimate the deliverable and the evidence.

WHO BUILDS THIS learning science & instructional design · human factors & interaction design · safety science & organizational analysis · knowledge management & data engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs · incident analysis · safety dashboards · knowledge bases · data pipelines · governance frameworks · bias & evidence audits

SOURCES

→ Makary & Daniel, "Medical error — the third leading cause of death in the US," BMJ (2016)
→ Shojania & Dixon-Woods, "Estimating deaths due to medical error: the ongoing controversy and why it matters," BMJ Quality & Safety (2017)
→ Institute of Medicine, To Err Is Human (1999)
→ World Health Organization, World Patient Safety Day reports
→ AHRQ Patient Safety Network statistics

FURTHER READING

• Makary & Daniel (2016), BMJ
• Institute of Medicine, To Err Is Human (1999)
• Bates et al. (2023), "The Safety of Inpatient Health Care," NEJM 388(2)

LENS COURSES LEN 1 LEN 4 LEN 10 LEN 6

CASE 83 GOVERNMENT TECHNOLOGY 2003 – 2012

LIBOR Manipulation

G Governance & Trust N Normalization of Deviance

IMPACT Major banks fined more than $10B; the benchmark underlying ~$350T of contracts was manipulable; replaced by SOFR

LIBOR

estimates

SOFR

transactions

the reform changed the unit of measurement, not the regulator

LIBOR — AN INSTRUMENT WHOSE DESIGN INVITED THE GAMING

The London Interbank Offered Rate (LIBOR) — the benchmark interest rate referenced in approximately three hundred fifty trillion dollars of financial contracts — was set daily by a small panel of banks submitting estimates of what they would pay to borrow. The submission was an estimate, not a verified market rate. For roughly a decade, multiple banks systematically manipulated their submissions to favor their derivatives positions. The coordination was visible in internal trader-and-submitter communications later subpoenaed by regulators.

The capability gap was at the design of the benchmark itself: it was an **estimate**, not an **observation**. Banks were asked to declare what their borrowing rate would be rather than to report transactions that had occurred. The architecture invited gaming. The gaming, when it happened, was systematic rather than occasional.

LIBOR has now been replaced by SOFR — the Secured Overnight Financing Rate — which is constructed from actual transactions rather than estimates. The reform took a decade and required coordination across multiple regulators globally.

THE LEARNING ENGINEERING LENS

> *The architecture of LIBOR invited the manipulation it experienced.*
>
> PARAPHRASING THE WHEATLEY REVIEW OF LIBOR, 2012

LE INSIGHT

LIBOR is the canonical financial-system case for an instrument whose design invites the gaming it then experiences. The capability that was missing was the basic architectural choice to measure observations rather than declarations. The reform — SOFR — re-engineered the measurement at the architecture level.

LENS APPROACH

LENS uses LIBOR in LEN 4 as the canonical measurement-architecture case at financial-system scale and in LEN 7 for the cross- jurisdictional governance reform that produced SOFR. The case pairs with Wells Fargo (Case 70) and Volkswagen (Case 71) as measurement-gaming cases.

REFLECTION QUESTIONS

1. Identify a benchmark in your domain that is constructed from declarations rather than observations. What gaming pressure does it experience?

2. Design the observation-based replacement for that benchmark.

WHO BUILDS THIS governance, ethics & measurement · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · incident analysis · safety dashboards

SOURCES

→ Department of Justice settlements with Barclays, UBS, Deutsche Bank, RBS et al. (2012–2015)
→ Wheatley, M., The Wheatley Review of LIBOR: Final Report (2012)
→ Stenfors, Barometer of Fear (2017)

FURTHER READING

• Wheatley Review (2012)
• Stenfors (2017), Barometer of Fear
• Hou & Skeie, “LIBOR: Origins, Economics, Crisis, Scandal” (2014)

LENS COURSES LEN 4 LEN 7

CASE 84 EDUCATION AT SCALE GOVERNMENT 2009 – 2015

Atlanta Public Schools Cheating Scandal

G Governance & Trust N Normalization of Deviance

IMPACT ~180 educators implicated; 35 indicted; 11 convicted under RICO statute; foundational U.S. education measurement-gaming case

180

EDUCATORS IMPLICATED · DOZENS OF SCHOOLS

the institution being measured operated the instrument that measured it

ATLANTA PUBLIC SCHOOLS — MEASUREMENT GAMING UNDER HIGH-STAKES TESTING

Investigation by Georgia Bureau of Investigation, the state's Office of Student Achievement, and the Atlanta Journal-Constitution found that educators at scores of Atlanta Public Schools elementary and middle schools had systematically changed students' answers on state standardized tests over a period of several years. The cheating was organized: principals pressured teachers, teachers organized "erasure parties" after testing days, and the institutional incentive structure rewarded the measurement gaming with bonuses, recognition, and promotions.

The Beverly Hall era at APS had been celebrated nationally as a high-performing urban district. The performance was substantially fabricated. The students were not learning; the institution was rewarding people for the appearance of learning. The capability gap was at the incentive architecture and at the measurement audit.

The case is canonical in U.S. education policy for the harms of high-stakes testing without an evidence-audit architecture independent of the institution being measured.

THE LEARNING ENGINEERING LENS

> *Dr. Hall's administration emphasized test results and public praise to the exclusion of integrity and ethics.*
>
> GEORGIA SPECIAL INVESTIGATORS' REPORT ON ATLANTA PUBLIC SCHOOLS, 2011

LE INSIGHT

The APS cheating scandal is the strongest available case for measurement-gaming under high-stakes incentives in education. The capability gap was at the measurement architecture: the institution being measured operated the instrument that measured it. The reform pattern is direct — independent measurement audit — but rarely implemented at scale.

LENS APPROACH

LENS uses APS in LEN 4 as the foundational U.S. education measurement-gaming case and in LEN 7 for the incentive- architecture failure. Studio projects examine how an independent audit would have caught the cheating.

REFLECTION QUESTIONS

1. Identify a high-stakes measurement in your domain whose audit is operated by the institution being measured. What would change with independent audit?
2. Apply Campbell's Law to a current high-stakes measurement system. What distortion is predicted?

WHO BUILDS THIS governance, ethics & measurement · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · incident analysis · safety dashboards

SOURCES

→ Special Investigators Final Report on Atlanta Public Schools (2011)
→ Atlanta Journal-Constitution investigative series (2009–2011)
→ State of Georgia v. Hall et al. (2014)

FURTHER READING

- Special Investigators Final Report (2011)
- Koretz, The Testing Charade (2017)
- Campbell, "Assessing the Impact of Planned Social Change" (1976) — Campbell's Law

LENS COURSES LEN 4 LEN 7

CASE 85 GOVERNMENT TECHNOLOGY 1992 – 2008

Bernard Madoff / SEC Failures

G Governance & Trust K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT ~$65B Ponzi scheme — the largest in history; SEC repeatedly investigated and cleared Madoff; foundational regulator-capability case

MADOFF — THE REGULATOR RECEIVED THE EVIDENCE; THE REGULATOR LACKED THE STAFF TO EVALUATE IT

Harry Markopolos delivered to the SEC in 2005 a memorandum titled “The World’s Largest Hedge Fund is a Fraud,” outlining in detail why Bernard Madoff’s purported returns were mathematically impossible. The SEC opened an investigation. The investigation concluded that no enforcement action was warranted. Madoff continued operating until his sons turned him in three years later, when the 2008 financial crisis made redemptions impossible to honor. By then approximately sixty-five billion dollars had been lost.

The SEC Inspector General’s investigation found that the staff assigned to Madoff lacked the expertise to evaluate the technical arguments Markopolos had made, that the institutional culture had defaulted to “Madoff is a respected industry figure” rather than to the evidence, and that multiple complaints across more than a decade had been similarly dismissed.

The Madoff case is canonical in U.S. financial regulation for the capability gap between the institution responsible for enforcement and the technical evidence that should have triggered enforcement.

THE LEARNING ENGINEERING LENS

> *The agency missed numerous opportunities to discover the fraud.*
>
> SEC OFFICE OF INSPECTOR GENERAL REPORT OIG-509, 2009

LE INSIGHT

The Madoff case is the canonical example of a regulator without the technical capability to evaluate the evidence it received. The complaint was specific. The math was checkable. The institution did not have the people to check it. The capability gap was at the regulator's expertise pipeline, not at the evidence.

LENS APPROACH

LENS uses Madoff in LEN 4 for technical-evidence evaluation and in LEN 7 for regulator-capability deliverables. The case pairs with Theranos (Case 69) as cases where regulators lacked the technical depth to challenge the evidence presented to them.

REFLECTION QUESTIONS

1. Identify a regulator in your domain whose technical evaluation capability has not kept pace with the entities it regulates. What is the resulting blind spot?
2. Design the regulator-side technical-capability deliverable that should have allowed the SEC to evaluate the Markopolos memo on its merits.

WHO BUILDS THIS governance, ethics & measurement · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · knowledge bases · data pipelines · incident analysis · safety dashboards

SOURCES

→ SEC Office of Inspector General, Investigation of Failure to Uncover Madoff's Ponzi Scheme (2009)
→ Markopolos, No One Would Listen (2010)
→ United States v. Madoff (2009)

FURTHER READING

• SEC OIG Report (2009)
• Markopolos (2010), No One Would Listen
• Cohen & Goldsmith (1999), regulatory-capacity literature

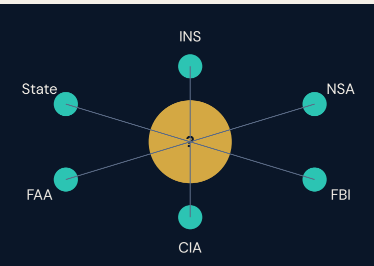
LENS COURSES LEN 4 LEN 7

CASE 86 DEFENSE GOVERNMENT 1996 – 2001

9/11 Intelligence Sharing Failures

G Governance & Trust K Knowledge & Institutional Memory

IMPACT 2,977 killed; the 9/11 Commission found systemic intelligence-sharing failures across the U.S. government



no institution responsible for the integration as a deliverable

9/11 — CROSS-AGENCY INFORMATION SHARING AS MISSING ARCHITECTURE

The 9/11 Commission's final report identified specific intelligence failures across the FBI, CIA, NSA, and other agencies. The CIA had information on two of the future hijackers from a January 2000 meeting in Kuala Lumpur. That information was not shared with the FBI. The FBI had identified suspicious flight-training activity in Phoenix and Minneapolis in 2001. That information was not aggregated. Visa decisions, immigration tracking, watch-list architecture — each was operated by a different agency, and the handoffs between them required initiative that the institutional architecture did not require.

The 9/11 Commission's framing — that the attacks were a failure of **imagination** — has been criticized as understating the structural nature of the gap. The intelligence sharing did not happen because no institution was responsible for it as a deliverable. The reform that followed — the Office of the Director of National Intelligence, the National Counterterrorism Center, fusion centers — built the institutional architecture that had not existed.

The case is foundational in U.S. national-security policy for cross-agency capability as an engineerable institutional deliverable.

THE LEARNING ENGINEERING LENS

> *The most important failure was one of imagination.*
>
> THE 9/11 COMMISSION REPORT, 2004

LE INSIGHT

9/11 is the foundational U.S. case for cross-agency information- sharing as an engineering deliverable. The architecture had not been built. The reform built it. The cost of the missing architecture was paid in 2001. The discipline LENS represents is the kind of work that, applied across the U.S. intelligence community in the 1990s, would have produced the architecture earlier.

LENS APPROACH

LENS uses 9/11 in LEN 8 as the foundational case for cross- organizational capability and in LEN 7 for the governance architecture of multi-agency systems. Studio projects compare 9/11 with Eagle Claw (Case 46) as institutional-architecture failures of different kinds.

REFLECTION QUESTIONS

1. Identify a cross-organizational information flow in your domain that depends on individual initiative rather than institutional architecture. What is the foreseeable failure mode?
2. Design the institutional deliverable that would have produced ODNI-level information sharing across U.S. intelligence agencies in the 1990s.

WHO BUILDS THIS governance, ethics & measurement · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · knowledge bases · data pipelines

SOURCES

→ National Commission on Terrorist Attacks Upon the United States, The 9/11 Commission Report (2004)
→ Joint Inquiry Into Intelligence Community Activities Before and After September 11, 2001 (2002)
→ Intelligence Reform and Terrorism Prevention Act of 2004

FURTHER READING

• 9/11 Commission Report (2004)
• Zegart, Spying Blind (2007)
• Posner, Preventing Surprise Attacks (2005)

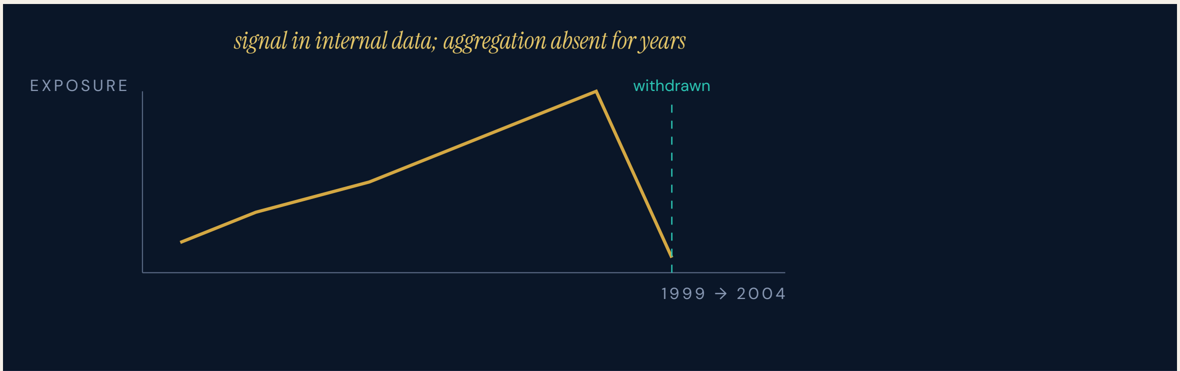
LENS COURSES LEN 7 LEN 8 LEN 3

CASE 87 HEALTHCARE 1999 – 2004

Vioxx Withdrawal

G Governance & Trust D Designed Out

IMPACT Tens of thousands of excess cardiovascular events estimated; Merck withdrew Vioxx in 2004; ~$4.85B settlement



VIOXX — POST-MARKET SURVEILLANCE AS A MISSING DELIVERABLE

Merck's COX-2 inhibitor Vioxx (rofecoxib) was approved by the FDA in 1999 for arthritis pain and prescribed widely. The VIGOR trial (results published 2000) reported a five-fold higher rate of myocardial infarction in the Vioxx arm versus naproxen — a signal that Merck and many readers interpreted as a protective effect of naproxen rather than a cardiovascular risk of Vioxx. The APPROVe trial (terminated September 2004) ultimately established the cardiovascular risk on a placebo comparison and triggered withdrawal. The disclosure architecture between Merck's internal data, the FDA reviewers, and the prescribing community did not surface the magnitude of the cardiovascular risk to a decision boundary for nearly four years between VIGOR and APPROVe.

The Vioxx case is foundational in U.S. drug safety for the gap between manufacturer-held adverse-event data and the public- health-relevant aggregation of that data. The FDA's Adverse Event Reporting System depended on submissions that were structured to minimize signal. The institutional capability to detect population-level harm in time to prevent it was not adequate to the volume of the drug's distribution.

The reform that followed — REMS programs, the Sentinel Initiative — built post-market surveillance infrastructure that did not previously exist.

THE LEARNING ENGINEERING LENS

> *The cardiovascular risk was visible in Merck's internal data years before the drug was withdrawn.*
>
> PARAPHRASING DAVID GRAHAM (FDA OFFICE OF DRUG SAFETY), SENATE FINANCE COMMITTEE TESTIMONY, NOVEMBER 2004

LE INSIGHT

Vioxx is the canonical pharmaceutical case for post-market surveillance as a capability deliverable. The signal existed; the institutional architecture to aggregate it did not. The reform pattern — Sentinel — built the architecture. The case is the drug-industry analog of the EHR case (Case 25): a measurement architecture too thin for the system that depended on it.

LENS APPROACH

LENS uses Vioxx in LEN 4 for post-market surveillance as a measurement deliverable and in LEN 7 for the corporate-regulator dynamics that allowed signal aggregation to be deferred.

REFLECTION QUESTIONS

1. Identify a post-deployment surveillance architecture in your domain that is too thin for the scale of the system it monitors. What is the missing deliverable?
2. Design the Sentinel-equivalent post-market surveillance system for a new domain.

WHO BUILDS THIS governance, ethics & measurement · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · task analysis · design prototyping

SOURCES

→ Graham, D. J. et al. (2005), Lancet, Vioxx cardiovascular risk analysis
→ Senate Committee on Finance, hearings on Vioxx (2004)
→ FDA Sentinel Initiative documentation (2008–present)

FURTHER READING

• Graham et al. (2005), Lancet
• Avorn, Powerful Medicines (2004)
• FDA Sentinel Initiative documentation

LENS COURSES LEN 4 LEN 7

Chapter 8

The Paired Intervention

Technical change with cultural change.

The checklist alone did not work. The checklist plus the nurse's authority to enforce it did.

KEYSTONE ICU PROJECT, MICHIGAN, 2004

CASE 12 AVIATION 1981 – PRESENT

Crew Resource Management & CAST

[T] Training Gap [H] Human-System Interface [N] Normalization of Deviance

IMPACT 83% reduction in U.S. commercial aviation fatality risk (1998–2008); 95% reduction in fatalities per 100M passengers over 20 years

AUTHORITY GRADIENT FLATTENED; COMMUNICATION CLOSES THE LOOP

In March 1977 two 747s collided on a foggy runway at Tenerife and 583 people died. The cockpit voice recorder showed the KLM flight engineer questioning whether the Pan Am 747 was clear of the runway — twice, in indirect language — only for the senior captain to dismiss the question and continue the takeoff roll. The accident was not a failure of skill but of the system by which skill in one seat reached the other.

Crew Resource Management, formalized by United Airlines in 1981, rebuilt the cockpit as a coordinated team: explicit communication protocols, authority gradients named and trained against, structured briefings and debriefings. CRM did not teach individual airmanship — it engineered the **system of interaction** in which airmanship operates.

Twenty years later the Commercial Aviation Safety Team added the second half of the system: closed-loop hazard identification on operational data, prioritized enhancements, tracked implementation, measured outcomes. CAST set an eighty-percent fatality-reduction target and exceeded it at 83% by 2007 (Collier Trophy, 2008). The two together — cultural redesign plus continuous evidence — define what a mature capability-engineering apparatus looks like.

THE LEARNING ENGINEERING LENS

> *CRM succeeded because it treated crew coordination as an engineerable property of the system.*
>
> PARAPHRASING HELMREICH & FOUSHEE, IN KANKI ET AL., CREW RESOURCE MANAGEMENT (2010)

LE INSIGHT

CRM is the canonical evidence that capability is engineerable at the system level, not just the individual. Tenerife was not solvable by hiring better pilots — only by changing the authority structure inside the cockpit. The intervention worked because it was **paired**: a procedural change plus a cultural change in how the procedure was authorized. Either alone fails.

LENS APPROACH

LENS treats CRM/CAST as the foundational success case across the curriculum. LEN 1 uses it as the problem-framing exemplar; LEN 4 uses CAST as the model closed-loop evidence system; LEN 2 uses CRM as the template for redesigning human roles in automated environments — the logic now being applied to AI-augmented systems.

REFLECTION QUESTIONS

1. Identify a high-consequence domain whose authority gradient absorbs information rather than transmitting it. What would the CRM equivalent intervention look like there?

2. CRM is paired with CAST. What is the closed-loop evidence system in your domain — and if there is not one, what would it cost to build?

WHO BUILDS THIS learning science & instructional design · human factors & interaction design · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs · incident analysis · safety dashboards

SOURCES

→ FAA Advisory Circular 120-51E, Crew Resource Management Training
→ Helmreich, Merritt & Wilhelm, "The Evolution of Crew Resource Management Training in Commercial Aviation," International Journal of Aviation Psychology (1999)
→ CAST/FAA Safety Enhancement reports (2016, 2018)
→ Collier Trophy citation, 2008; SKYbrary CRM article; EASA guidance

FURTHER READING

• Kanki, Helmreich & Anca, Crew Resource Management (2010)
• Edmondson, The Fearless Organization (2018)

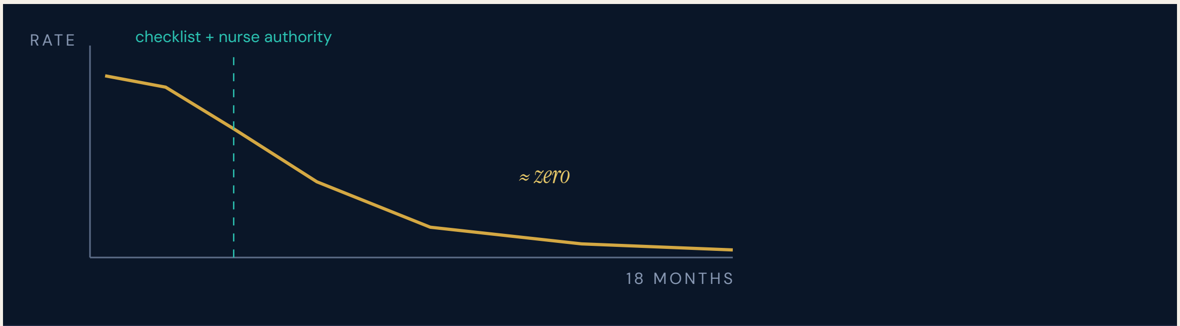
LENS COURSES LEN 1 LEN 4 LEN 2 LEN 3

CASE 14 HEALTHCARE 2004 – PRESENT

Keystone ICU / Pronovost Checklist

T Training Gap N Normalization of Deviance

IMPACT Central-line-associated bloodstream infections (CLABSI) reduced to near zero across 103 Michigan ICUs; ~1,500 lives saved in 18 months; ~$75M saved; sustained at ten years



BLOODSTREAM-INFECTION RATE ACROSS 103 MICHIGAN ICUS

Peter Pronovost's checklist for central-line insertion has five items. Wash hands. Clean the patient's skin with chlorhexidine. Drape the patient. Wear a mask, hat, sterile gown and gloves. Apply a sterile dressing. Nothing in the checklist was unknown to any physician it would govern. The question was not what to do — it was whether anyone would do it every time.

The intervention paired the checklist with a cultural change: nurses were authorized — not merely permitted, but required — to stop the procedure if any step was skipped. That single authorization was the load-bearing element. In the prior architecture, a nurse who saw a physician skip a step had no procedural path to intervene without crossing the hospital's authority gradient. After Keystone, the path existed and was institutional.

The result across 103 Michigan ICUs was the elimination of central-line-associated bloodstream infections (CLABSI) as a category. The typical ICU's quarterly infection rate fell to zero. Michigan's ICUs outperformed 90 percent of ICUs nationwide. The program has been adopted in more than forty states and replicated internationally. At ten-year follow-up, the effect has been sustained.

THE LEARNING ENGINEERING LENS

> *The checklist was the technical intervention. The nurses' authority to enforce it was the cultural intervention. Neither worked without the other.*
>
> PARAPHRASING PRONOVOST & VOHR, SAFE PATIENTS, SMART HOSPITALS, 2010

LE INSIGHT

Keystone is the clearest evidence in healthcare that a technical intervention without authority intervention produces no durable change, and authority intervention without a technical artifact produces no measurable change. Both are necessary. The empirical record of Keystone is the strongest available argument for designing interventions as **pairs** — and treating the cultural half as engineering, not aspiration.

LENS APPROACH

LENS uses Keystone in LEN 4 and LEN 10 as the canonical worked example of measurement linked to intervention. Studio projects require students to specify both halves of the pair, name the authority that authorizes the cultural half, and identify the measurement signal that will confirm or refute the effect. The course treats "is the cultural change actually authorized?" as falsifiable, not stated.

REFLECTION QUESTIONS

1. Identify an evidence-based protocol in your domain that is **known** to work but is not consistently performed. What is the authority change required to pair with it?
2. Design the measurement signal that would confirm the cultural half of a Keystone-style intervention is actually being authorized — not merely declared.

WHO BUILDS THIS learning science & instructional design · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · incident analysis · safety dashboards

SOURCES

→ Pronovost et al., "An Intervention to Decrease Catheter-Related Bloodstream Infections in the ICU," New England Journal of Medicine 355 (2006)
→ Lipitz-Snyderman et al., BMJ (2011) — sustained effect at follow-up
→ Agency for Healthcare Research and Quality, CUSP toolkit
→ Pronovost & Vohr, Safe Patients, Smart Hospitals (2010)

FURTHER READING

• Gawande, The Checklist Manifesto (2009)
• Bosk et al. (2009), "Reality check for checklists," The Lancet
• Edmondson, The Fearless Organization (2018)

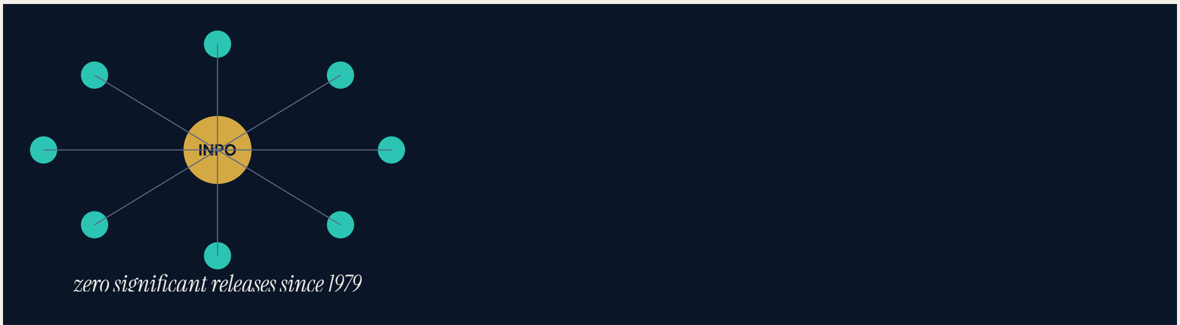
LENS COURSES LEN 4 LEN 10 LEN 5

CASE 16 ENERGY 1979 – PRESENT

INPO and the Nuclear Academy

T Training Gap K Knowledge & Institutional Memory G Governance & Trust

IMPACT No INES-level event at U.S. commercial reactors post-INPO; sustained improvement in INPO/WANO performance indicators across the industry



INDUSTRY-WIDE PEER REVIEW: A LEARNING RING AFTER TMI

Three Mile Island did not produce a reactor accident at the next plant over. It produced an institution. Within months of the Kemeny Commission's October 1979 report, the U.S. commercial nuclear industry founded the Institute of Nuclear Power Operations. The premise: an accident at any single plant would affect every operator's license, and no single utility could engineer its capability infrastructure alone. INPO was funded by the utilities it evaluated and operated without statutory authority — a configuration in which the operators and the peer review they answered to were hostages of each other.

INPO set training and certification standards for every operator and supervisor; the National Academy for Nuclear Training (1985) accredited each facility's programs; and INPO teams ran regular peer evaluations — operators from one utility examining another's procedures, control rooms, and incident records, for industry-wide learning rather than competitive advantage. The pre-TMI culture had been characterized by what the Bulletin of the Atomic Scientists called "smugness"; the post-INPO culture, by mandated vigilance.

THE LEARNING ENGINEERING LENS

> *Every utility recognized that an accident at any single plant would affect every operator's license to operate.*
>
> PARAPHRASING THE INSTITUTIONAL ANALYSIS IN REES, HOSTAGES OF EACH OTHER, 1994

LE INSIGHT

INPO is the strongest evidence in any domain that capability engineering can be undertaken at the level of an **industry**, not just an organization. The conditions that made it possible — shared catastrophic exposure, regulatory legitimacy, an honest peer-review architecture — appear wherever a single failure can damage every operator.

LENS APPROACH

LENS uses INPO in LEN 8 as the canonical example of industry-level learning: students identify the structural conditions in their own domain that would permit (or block) an INPO-equivalent and design the peer-review architecture required. LEN 1 uses the founding moment — nine months after TMI — to discuss the **speed** a credible response to catastrophe demands.

REFLECTION QUESTIONS

1. What is the equivalent of "an accident at any single plant affects every operator" in your domain? If the answer is "nothing," what does that tell you?
2. INPO was stood up in nine months. Pick a current cross-organizational capability problem and write the nine-month deliverable that would constitute a credible response.

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · governance frameworks · bias & evidence audits

SOURCES

→ Nuclear Energy Institute, "Lessons from the 1979 Accident at Three Mile Island"
→ World Nuclear Association — Three Mile Island Accident
→ Rees, Hostages of Each Other: The Transformation of Nuclear Safety since Three Mile Island (1994)
→ Marsh (2019), "INPO and the Transformation of Nuclear Safety Culture"

FURTHER READING

• Rees (1994), Hostages of Each Other
• Weick & Sutcliffe, Managing the Unexpected (2007)

LENS COURSES

LEN 1 LEN 8 LEN 3

CASE 13 HEALTHCARE 2008 – PRESENT

WHO Surgical Safety Checklist

T Training Gap N Normalization of Deviance

IMPACT Death rate 1.5% → 0.8%; complications down >33%; adopted by the majority of surgical providers worldwide

THREE PAUSE POINTS ACROSS THE SURGICAL TIMELINE

A single-page, nineteen-item checklist applied at three critical junctures: before anesthesia, before incision, and before the patient leaves the operating room. Atul Gawande's team at Harvard and Brigham & Women's piloted the checklist across eight hospitals in eight countries representing diverse economic and operational conditions — from Toronto to Manila, Tanzania to London. The result, published in NEJM in 2009, was a near halving of the surgical death rate and a one-third reduction in serious complications.

The checklist itself was the artifact. The intervention was the **system of pauses** it created — three forced moments in which a team that would otherwise be moving had to stop, look at each other, and confirm shared state. The pauses were the load-bearing element. The list of items was less important than the requirement to halt and run through them aloud.

Gawande's framing — that gaps in teamwork and safety practices in surgery are substantial in countries both rich and poor — was confirmed in every site. The intervention scaled because it addressed the same gap everywhere. A subsequent Ontario population study (Urbach et al., NEJM 2014) found no statistically significant mortality reduction after mandated province-wide adoption, raising the implementation-fidelity question that the success cases in this book repeatedly surface: the artifact is only as effective as the surrounding system that authorizes its honest use.

THE LEARNING ENGINEERING LENS

> *Gaps in teamwork and safety practices in surgery are substantial in countries both rich and poor.*
>
> HAYNES, WEISER, BERRY ET AL., NEJM, 2009

LE INSIGHT

The Surgical Safety Checklist is the canonical evidence that a tiny artifact — one page, nineteen items — can produce population-scale effects when paired with the structural change of **requiring a pause.** The checklist alone is paper. The pause alone is anxiety. Together they constitute the smallest effective capability intervention in the dataset. The Ontario follow-up underscores the secondary requirement: the artifact carries the effect only when the institution actually enforces the pause. Where mandate replaced authorization, the measured effect attenuated.

LENS APPROACH

LENS uses the WHO checklist in LEN 10 (capstone) as the canonical end-to-end design exemplar: a problem identified, a minimal artifact prototyped, a multi-site pilot, a measurement regime, and global scale-out. LEN 4 examines the measurement architecture that made the effect provable.

REFLECTION QUESTIONS

1. What is the smallest possible capability artifact in your domain that, paired with a required pause, would shift outcomes?
2. The WHO checklist was studied across eight countries. Design the multi-site evaluation that would establish whether your candidate intervention generalizes.
3. The Ontario mandated rollout produced no measurable mortality reduction. What governance signal would have surfaced the fidelity gap between authorized use and compliance check-off before the trial was declared a failure?

WHO BUILDS THIS learning science & instructional design · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · incident analysis · safety dashboards

SOURCES

→ Haynes et al., "A Surgical Safety Checklist to Reduce Morbidity and Mortality," NEJM (2009)
→ Urbach et al., "Introduction of Surgical Safety Checklists in Ontario, Canada," NEJM (2014)
→ WHO Safe Surgery Saves Lives campaign documentation
→ Gawande, The Checklist Manifesto (2009)
→ ScienceDaily, surgical-checklist long-term outcomes

FURTHER READING

• Haynes et al. (2009), NEJM
• Gawande, The Checklist Manifesto (2009)
• Bosk et al. (2009), "Reality check for checklists"

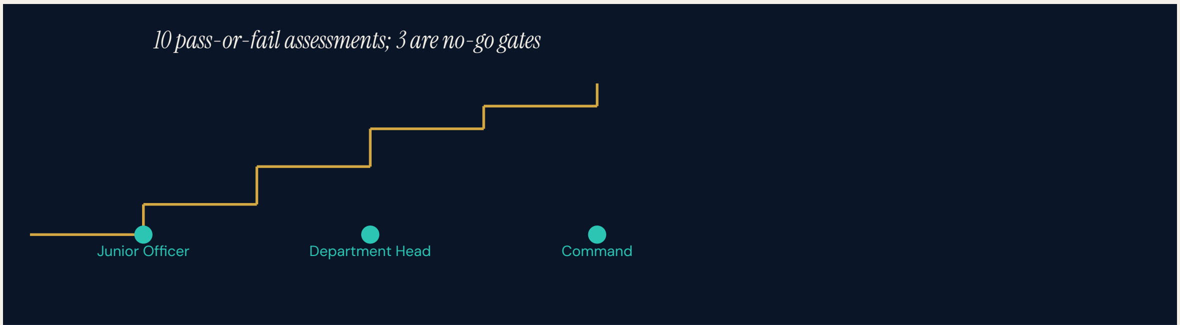
LENS COURSES LEN 4 LEN 10

CASE 15 DEFENSE 2018 – PRESENT

Navy Surface Warfare Readiness Reform

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Tripled ship-driving training hours; 10 pass-or-fail career assessments; Ready-for-Sea Assessments — 3 of 18 forward-deployed ships immediately sidelined



POST-FITZGERALD REFORM: TRIPLED TRAINING; PASS-OR-FAIL GATES

Direct response to the Fitzgerald and McCain collisions (Case 1). The Navy restored the Surface Warfare Officers School from CD-ROM-based self-study to classroom plus simulator instruction. Maritime Skills Training Centers were established on both coasts. Ten pass-or-fail assessments were created throughout a SWO's career path; three of them are no-go gates. The service adopted aviation-style debriefing methodology and instituted Ready-for-Sea Assessments — operational evaluations of forward-deployed ships against a deliverable standard. Three of the first eighteen forward-deployed ships were immediately sidelined.

GAO has noted that the Navy still lacks systematic evaluation of whether the reforms work. The capability intervention happened, the investment was substantial, but decision-grade evidence on outcomes is incomplete. This is itself a LENS argument: the reform happened; the measurement to know if it worked has not.

The case is a live, in-progress success: structural change real, evidence of effect outstanding.

THE LEARNING ENGINEERING LENS

> *The Navy still lacks systematic evaluation of whether the reforms work.*
>
> PARAPHRASING GAO-21-168 ON NAVY READINESS REFORM, 2021

LE INSIGHT

The Navy reform is a paired intervention in progress: technical (training restored, simulators procured, assessments created) plus cultural (debriefing, gate ownership). What it lacks is the third half: evidence that the intervention has worked. LENS treats this as a teachable case for what mature capability engineering should include from the start — the measurement infrastructure that lets the institution know whether its investment is producing the capability it bought.

LENS APPROACH

LENS uses Navy SWO reform in LEN 10 (capstone) as a worked exercise in capability intervention at scale, and in LEN 4 as a case where the measurement infrastructure has lagged the intervention. Students design the evidence regime that should accompany the reform.

REFLECTION QUESTIONS

1. Navy SWO reform happened without an evidence regime to confirm it worked. Identify a current reform in your domain and the evidence regime that should accompany it.
2. Design the dashboard the Chief of Naval Operations should have to know whether SWO capability is improving in time-series.

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards

SOURCES

→ GAO-20-154 and GAO-21-168, Navy readiness reform
→ Readiness Reform Oversight Council, One-Year Report (2019)
→ USNI News reform coverage (2020, 2022)
→ SWOS Norfolk and San Diego training-center documentation

FURTHER READING

• GAO Navy readiness series
• Naval Postgraduate School theses on SWO training reform
• Edmondson, Teaming (2012)

LENS COURSES LEN 4 LEN 10 LEN 5

CASE 23 AVIATION 2000 – PRESENT

Korean Air Safety Transformation

T Training Gap N Normalization of Deviance

IMPACT From industry pariah (16 aircraft written off, 700+ lives lost, 1970–1999) to spotless passenger safety record since 1999

CULTURAL HIERARCHY REMOVED FROM THE COCKPIT

Between 1970 and 1999, Korean Air had one of the worst safety records in commercial aviation. The loss rate was seventeen times higher than United Airlines. In 1999, the president of South Korea called the airline's record "an embarrassment to the nation." Delta, Air France, and Air Canada all suspended code-sharing agreements. The FAA downgraded South Korea's aviation safety rating.

The NTSB investigation of Korean Air Flight 801 — Guam, 1997, 229 killed — identified cultural authority gradients as a root cause: junior officers were culturally unable to challenge the captain's erroneous decisions. In 2000 Korean Air brought in David Greenberg from Delta to overhaul flight operations. Key interventions: mandated English fluency for all pilots (removing the Korean- language hierarchy from the cockpit), CRM training adapted for a high-power-distance culture, external consulting from Boeing and Delta, and fleet modernization.

The result: no fatal passenger accidents since 1997. The Air Transport World Phoenix Award in 2006 recognized the transformation. Cultural legacy is not destiny when it is deliberately redesigned.

THE LEARNING ENGINEERING LENS

> *Korean Air's record before 2000 was changed by an external intervention into cockpit culture, not by criticism of it.*
>
> EDITORS' SYNTHESIS OF NTSB FINDINGS ON KAL 801 AND THE KOREAN AIR TRANSFORMATION

LE INSIGHT

Korean Air is the strongest aviation evidence that cultural legacy is not destiny. A specific cultural feature — high power distance in the cockpit — was the binding capability constraint. Once it was redesigned, the safety record changed categorically. The intervention is also methodologically interesting because it operated on language: English became the cockpit language because English has no honorifics to enforce. The interface for cultural redesign was a linguistic one.

LENS APPROACH

LENS uses Korean Air in LEN 8 as the canonical organizational- learning case under cultural constraint and in LEN 2 as a CRM- extension case for high-power-distance contexts. The case is paired in this book with the Toyota Andon Cord (Case 24) as cultural intervention success stories.

REFLECTION QUESTIONS

1. Identify a cultural feature in an institution you work with that constrains capability. Is it engineerable? What would the redesign look like?

2. Korean Air's intervention operated on language. What surface of culture is engineerable in your domain that you have not yet considered?

WHO BUILDS THIS learning science & instructional design · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · incident analysis · safety dashboards

SOURCES

→ NTSB Aircraft Accident Report, KAL Flight 801 (2000)
→ Air Transport World Phoenix Award documentation (2006)
→ Gladwell, Outliers (2008) — Korean Air chapter
→ Korean Air corporate safety reports

FURTHER READING

• Hofstede, Cultures and Organizations (3rd ed., 2010)
• Helmreich, Wilhelm, Klinect & Merritt (2001) on national culture and CRM
• Gladwell, Outliers (2008)

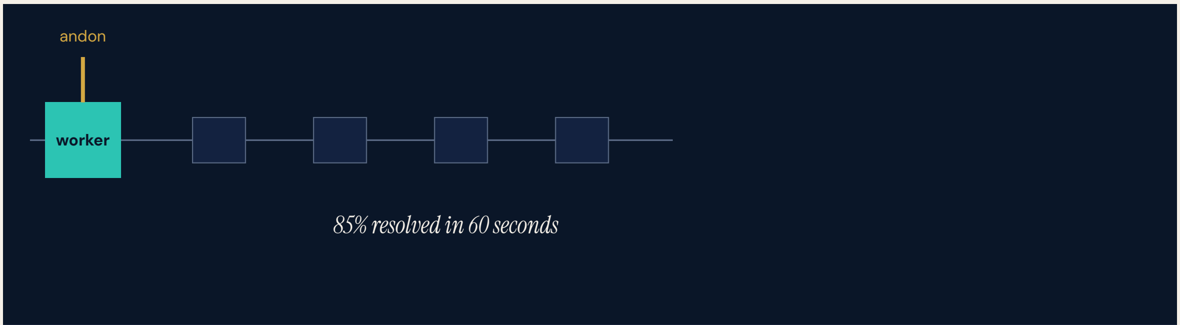
LENS COURSES LEN 2 LEN 8

CASE 24 INDUSTRIAL 1950S – PRESENT

Toyota Production System / Andon Cord

N Normalization of Deviance G Governance & Trust

IMPACT 85% of andon activations resolved within 60 seconds; defect-propagation cost minimized; the system adopted globally



LOWEST-RANKING WORKER AUTHORIZED TO STOP THE LINE

The andon cord empowers any assembly-line worker to stop the entire production line when they detect a defect or abnormality. The inversion of authority is the entire point: the lowest-ranking person on the floor has the power to halt million-dollar-per-hour operations. The cord itself is trivially cheap. The authority it confers is the design.

The case is critical for capability engineering because when American automakers copied the andon cord in the 1980s and 1990s, **workers were too afraid to pull it.** The tool was there; the empowerment was not. Toyota's system works because it pairs the technical mechanism (the cord) with a cultural system: psychological safety, supervisor response protocols, no-blame problem solving, and the codified "Five Whys" root-cause method. When the line stops at Toyota, average resolution time is 4.2 minutes and the team treats the stop as a learning opportunity, not a failure.

The Andon Cord is the manufacturing counterpart to the Keystone nurse-authority intervention (Case 14). Same logic, different industry, same load-bearing element.

THE LEARNING ENGINEERING LENS

> *When American manufacturers copied the andon cord, workers were too afraid to pull it.*
>
> PARAPHRASING LIKER, THE TOYOTA WAY (2ND ED., 2020)

LE INSIGHT

The Andon Cord is the foundational evidence that authority interventions and technical artifacts are inseparable. The cord means nothing without the protected authority to pull it. The protected authority means nothing without the cord to act through. The pair is irreducible — and that irreducibility is the LENS co-optimization commitment in physical form.

LENS APPROACH

LENS uses the Andon Cord in LEN 2 as the foundational example of paired technical-cultural intervention, in LEN 8 to discuss cross-domain transfer (and why imitation without the cultural half fails), and in LEN 10 as a studio exemplar of minimal-artifact design.

REFLECTION QUESTIONS

1. Identify a technical artifact in your domain whose effectiveness depends entirely on a protected authority. Is the authority protected, or only declared?
2. American manufacturers copied the cord without the authority. What is the equivalent surface-level imitation you have observed in your domain, and what was missing?

WHO BUILDS THIS safety science & organizational analysis · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS incident analysis · safety dashboards · governance frameworks · bias & evidence audits

SOURCES

→ Liker, The Toyota Way (2nd ed., 2020)
→ Shingo, A Study of the Toyota Production System (1989)
→ Spear & Bowen, "Decoding the DNA of the Toyota Production System," HBR (1999)
→ Womack & Jones, Lean Thinking (1996)

FURTHER READING

• Liker (2020), The Toyota Way
• Spear & Bowen (1999), HBR
• Rother, Toyota Kata (2009)

LENS COURSES

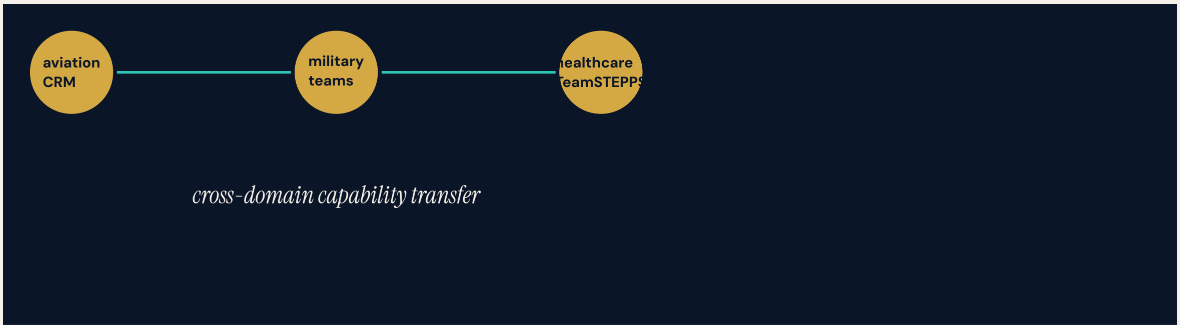
LEN 10 · LEN 2 · LEN 8

CASE 27 HEALTHCARE 2006 – PRESENT

TeamSTEPPS

T Training Gap N Normalization of Deviance

IMPACT Improved teamwork, communication, and patient-safety culture across diverse settings; OR on-time first start +21%; adopted by thousands of healthcare organizations



AVIATION → MILITARY → HEALTHCARE; THE TRANSLATION PATHWAY

Developed jointly by the Agency for Healthcare Research and Quality and the Department of Defense, TeamSTEPPS is the healthcare analog of Crew Resource Management (Case 12) — an evidence-based team- training framework derived from fifty years of research in aviation, military, and nuclear domains, adapted for clinical settings. It trains four core competencies: communication, leadership, situation monitoring, and mutual support.

TeamSTEPPS is explicitly the translation pathway from high-reliability-organization research into clinical practice — precisely the kind of cross-domain capability transfer LENS is designed to teach practitioners to execute. Its development history (military → healthcare) and its ongoing evidence base make it the strongest exemplar in the dataset of LENS competency domains in action.

The intervention is also evidence that the seventeen-year gap of Case 41 can be shortened by deliberate engineering. TeamSTEPPS moved from research to scaled deployment in years, not decades, because the implementation infrastructure was funded as part of the intervention.

THE LEARNING ENGINEERING LENS

> *TeamSTEPPS represents the translation pathway from high-reliability research into clinical practice.*
>
> EDITORS' SYNTHESIS DRAWING ON AHRQ TEAMSTEPPS 3.0 (2023) AND SALAS ET AL.

LE INSIGHT

TeamSTEPPS is the canonical evidence that cross-domain capability transfer is engineerable and that it can shorten the implementation gap dramatically when the transfer is funded as part of the intervention rather than as an afterthought. The four competencies map directly to the LENS curriculum's argument for what capability engineering competence looks like as a deliverable.

LENS APPROACH

LENS treats TeamSTEPPS in LEN 8 as the canonical cross-domain transfer case and in LEN 1 as evidence that the program's core proposition — that learning engineering is a discipline — has institutional precedent. Studio projects (LEN 10) reference TeamSTEPPS as the worked example of cross-domain adaptation methodology.

REFLECTION QUESTIONS

1. What is the cross-domain transfer your domain has not yet executed? What evidence base in another industry should it draw on?
2. TeamSTEPPS funded its implementation infrastructure. Design the implementation-infrastructure budget for an equivalent transfer in your domain.

WHO BUILDS THIS learning science & instructional design · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · incident analysis · safety dashboards

SOURCES

→ AHRQ TeamSTEPPS 3.0 Curriculum (2023)
→ American Hospital Association Team Training Center
→ PMC studies on TeamSTEPPS outcomes (multiple)
→ DoD / AHRQ partnership documentation

FURTHER READING

- AHRQ TeamSTEPPS documentation
- Salas, Rosen, Held & Weissmuller (2009) on cross-domain team training
- Weaver, Dy & Rosen (2014), patient-safety implementation

LENS COURSES LEN 1 LEN 10 LEN 8 LEN 3

CASE 28 DEFENSE 1954 – PRESENT

U.S. Nuclear Navy / Rickover Training Model

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Zero reactor accidents in 60+ years of U.S. Naval nuclear operations; the most demanding nuclear operator training program in the world

NUCLEAR

0

reactor accidents

60+ years

SURFACE (PRE-2017)

17

sailors killed

two collisions

SAME NAVY, OPPOSITE PHILOSOPHIES — RADICALLY DIFFERENT OUTCOMES

Admiral Hyman Rickover established a training and qualification culture for the Naval Nuclear Propulsion Program that has produced zero reactor accidents across thousands of reactor-years of operation — including on submarines operating under extreme conditions. The program requires every nuclear-trained officer and enlisted sailor to pass rigorous qualification programs with zero-defect standards. Operators must demonstrate competence through oral examination by senior nuclear-qualified officers. The culture emphasizes personal accountability, technical mastery, and a questioning attitude — operators are trained to challenge assumptions, including those of superiors.

The contrast with surface warfare training (pre-Fitzgerald, Case 1) is the sharpest internal comparison in the dataset. Same institution, same era, opposite approaches to human capability, radically different safety outcomes. The nuclear Navy invested in rigorous, continuous capability engineering from inception. The surface Navy cut training to CD-ROMs and paid the price.

Same Navy. Two philosophies. The choice shows up in the casualty columns.

THE LEARNING ENGINEERING LENS

> *Human experience shows that people, not organizations or management systems, get things done.*
>
> PARAPHRASING ADMIRAL HYMAN G. RICKOVER, CONGRESSIONAL TESTIMONY, 1979

LE INSIGHT

The Nuclear Navy is the longest-running continuous capability- engineering program in any high-consequence domain. The choice to treat training as a system parameter rather than as a cost center has produced sixty-plus years of zero reactor accidents. The contrast with the Surface Navy is the cleanest available test of what happens when capability is engineered versus when it is deferred. The cost of the program — the qualification ladder, the oral examinations graded against zero-defect standards, the continuous re-qualification — is the price the institution pays for the safety record. The choice is visible on the budget line.

LENS APPROACH

LENS treats the Rickover model in LEN 8 as the foundational organizational-learning case and in LEN 5 as a worked example of capability requirements traceable from operational analysis through qualification standards. The internal Navy comparison anchors the program's core argument about capability as a system parameter.

REFLECTION QUESTIONS

1. Identify an institution that operates two divisions under different capability philosophies. What does the comparison reveal that neither division could see alone?
2. Rickover's standard was zero-defect oral examination. What is the equivalent in your domain — and would you survive it?
3. The Nuclear Navy's safety record is paid for by a durable, visible training budget. What is the equivalent line-item investment in your domain that a comparable safety claim would require — and is it being made?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards

SOURCES

→ Naval Nuclear Propulsion Program documentation, NRC/DOE
→ Naval History and Heritage Command, Rickover archives
→ GAO-21-168 comparison of nuclear and surface Navy training
→ Polmar & Allen, Rickover: Father of the Nuclear Navy (2007)

FURTHER READING

• Polmar & Allen (2007), Rickover
• Duncan, Rickover and the Nuclear Navy (1990)
• GAO Navy training-comparison series

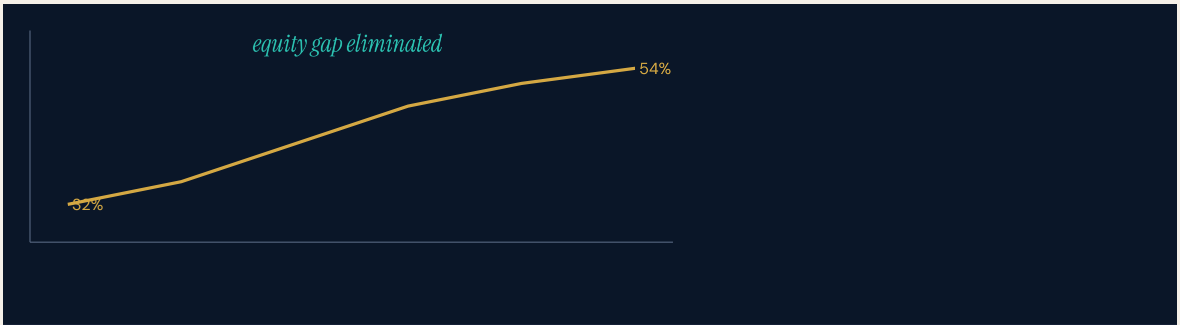
LENS COURSES LEN 5 LEN 8 LEN 3

CASE 39 EDUCATION AT SCALE 2012 – PRESENT

Georgia State University Predictive Analytics

T Training Gap K Knowledge & Institutional Memory

IMPACT Six-year graduation rate 32% → 54%; equity gaps in graduation eliminated; 2,000+ more graduates per year



GRADUATION RATE, WITH EQUITY AS A PRIMARY CONSTRAINT

Georgia State implemented a predictive-analytics system tracking eight hundred risk factors per student, triggering proactive advising interventions when students showed early warning signs. Critically, the system was designed with equity as a primary constraint: the explicit goal was to eliminate, not reproduce, existing graduation gaps. Advisors use the alerts as prompts for human intervention, not automated decisions — maintaining human judgment in the loop.

The graduation rate went from thirty-two percent to fifty-four percent. The graduation rate for Black students rose to match the overall rate. Pell-eligible students graduate at the same rate as non-Pell students. GSU went from a regional commuter school with a third of its students graduating to a national model with a majority graduating — and the gain was distributed equitably.

The key difference from the algorithmic-bias cases (35, 36, 37): GSU designed equity into the system from the start, used predictions to trigger **more** human support rather than automated gatekeeping, and measured outcomes by demographic group as a primary accountability metric.

THE LEARNING ENGINEERING LENS

> *Predictions trigger support, not gatekeeping.*
>
> EDITORS' SYNTHESIS OF THE GSU ADVISING MODEL, DRAWN FROM RENICK & STROM (2020) AND NEW YORK TIMES COVERAGE (2018)

LE INSIGHT

GSU is the positive counterpart to A-Level (35), Robodebt (36), and educational algorithmic bias (37). The same technical capability — a predictive model — was deployed under a different design constraint and produced an equity outcome rather than an equity harm. The case is the strongest available evidence that the **construct definition** and the **human-loop architecture** determine whether a predictive model produces good or harm, not the model itself.

LENS APPROACH

LENS treats GSU as the canonical positive case for predictive analytics in education. LEN 4 examines the measurement architecture that made equity a primary outcome. LEN 7 examines the governance structure that kept the system as decision support rather than automated decision. LEN 1 uses it as a problem-framing exemplar.

REFLECTION QUESTIONS

1. What is the difference between GSU's predictive analytics and the algorithmic-bias cases of Chapter 5? Be specific about what makes the GSU implementation work.

2. Design the equity-as-primary-constraint version of a predictive system in your domain. What would you measure first?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines

SOURCES

→ Georgia State University institutional research and Strategic Plan reports
→ New York Times, "How Colleges Know You're Not Finishing" (2018)
→ EDUCAUSE Review on GSU predictive advising
→ Complete College America Game Changers documentation

FURTHER READING

• Renick & Strom (2020) on GSU's advising transformation
• Mayer-Schönberger & Cukier, Learning with Big Data (2014)
• Bowen, Chingos & McPherson, Crossing the Finish Line (2009)

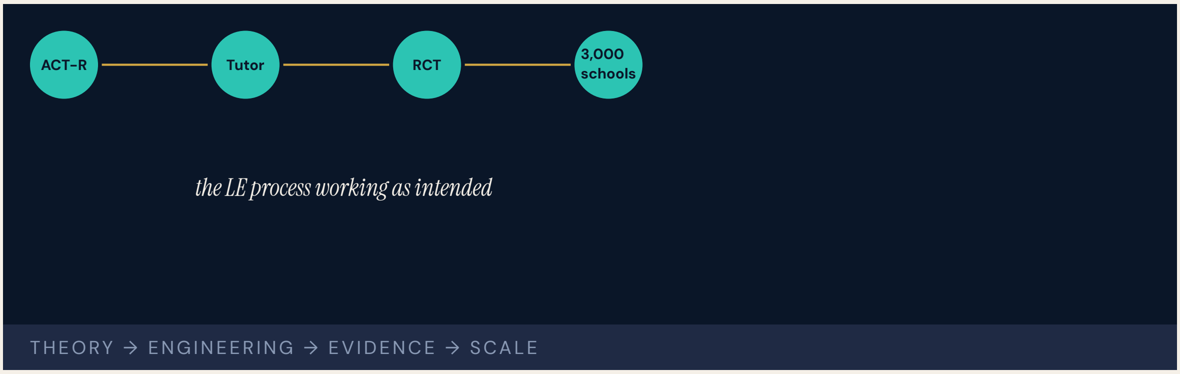
LENS COURSES LEN 1 LEN 4 LEN 7

CASE 42 EDUCATION AT SCALE 1990S – PRESENT

Cognitive Tutor / Carnegie Learning

T Training Gap

IMPACT Randomized controlled trials showed significant learning gains; RAND study found positive effects on Algebra I achievement; adopted across 3,000+ schools



Carnegie Learning's Cognitive Tutor, developed from John Anderson's ACT-R cognitive architecture at Carnegie Mellon, is the most rigorously evaluated intelligent tutoring system in education. It uses Bayesian knowledge tracing to model individual student mastery and adapts instruction accordingly. The RAND Corporation's evaluation found statistically significant positive effects on Algebra I achievement.

This is a learning-engineering success story in the discipline's own terms: learning science (ACT-R theory) → engineering (Cognitive Tutor software) → evidence (RCTs, RAND evaluation) → scaled implementation (three thousand-plus schools). It demonstrates the LE process working as intended.

The limitations are instructive: Cognitive Tutor works best in well-defined domains (algebra) and less well in ill-structured ones. It is the canonical evidence that the LE pipeline works **for problems that fit the pipeline** — and the open question is whether the same discipline can deliver for problems that do not.

THE LEARNING ENGINEERING LENS

> *Cognitive Tutors demonstrate the LE process working: theory ⟶ engineering ⟶ evidence ⟶ scale.*
>
> EDITORS' SYNTHESIS OF ANDERSON ET AL. (1995) AND KOEDINGER & CORBETT (2006)

LE INSIGHT

Cognitive Tutor is the canonical evidence that the learning- engineering process exists, works, and produces measurable benefits at scale — when the problem is tractable. It is also the canonical case for what tractability looks like: well-defined domain, decomposable skill model, instrumentable interface, rigorous evaluation. The frontier for the discipline is whether the same pipeline can deliver in the operational, ill-structured domains where capability matters most.

LENS APPROACH

LENS uses Cognitive Tutor in LEN 1 as the foundational LE-process exemplar, in LEN 4 as the canonical case for Bayesian knowledge tracing as a measurement instrument, and in LEN 9 as a technical case for model-based adaptive instruction.

REFLECTION QUESTIONS

1. Cognitive Tutor works best in well-defined domains. Identify a problem in your domain that is currently ill-structured. What would have to be true to make the LE pipeline applicable?
2. The case is the strongest evidence the LE process works. What is the equivalent piece of evidence required to demonstrate the same pipeline in a non-cognitive domain — for example, surgical skill, or operational watchstanding?

WHO BUILDS THIS learning science & instructional design — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models

SOURCES

→ Anderson, Corbett, Koedinger & Pelletier, "Cognitive Tutors: Lessons Learned," JLS (1995)
→ RAND Corporation Algebra I evaluation
→ Koedinger & Corbett (2006), Cambridge Handbook of the Learning Sciences
→ Carnegie Learning, https://www.carnegielearning.com/

FURTHER READING

• Anderson et al. (1995), Journal of the Learning Sciences
• Koedinger & Corbett (2006), Cambridge Handbook of the Learning Sciences
• Aleven et al. (2016), example-tracing tutors

LENS COURSES LEN 1 LEN 4 LEN 9

CASE 88 HEALTHCARE INDUSTRIAL 1982

Tylenol Recall

G Governance & Trust N Normalization of Deviance

IMPACT Foundational U.S. corporate crisis-management case; produced tamper-evident packaging regulation and modern recall practice

31M

BOTTLES RECALLED · ~$100M COST

the pre-committed institutional credo became operational under stress

TYLENOL — VALUES PRE-COMMITTED IN WRITING, EXECUTED UNDER CRISIS

Seven people in the Chicago area died after taking Tylenol capsules that had been laced with potassium cyanide. Johnson & Johnson did not know who had done it or how widespread the contamination was. The company's response — recall every Tylenol product nationwide (thirty-one million bottles), suspend all advertising, and publicly engage with the FBI and FDA — was at the time unprecedented in U.S. corporate practice. The recall cost the company approximately one hundred million dollars and the immediate confidence of its institutional investors.

The reform that followed reshaped consumer-product packaging worldwide. Tamper-evident seals, blister packs, and caplet form factors became standard. The FDA promulgated tamper-resistant packaging regulations within months. Tylenol recovered its market share within a year.

The case is the canonical positive example in U.S. business education of institutional response to a crisis driven by capability commitment rather than legal minimization. The Johnson & Johnson Credo, written in 1943, had pre-specified that the company's first responsibility was to customers. The 1982 response was a direct application of that pre-specified commitment.

THE LEARNING ENGINEERING LENS

> *The Credo is all about the consumer. When those seven deaths occurred, the Credo made it very clear at that point exactly what we were all about.*
>
> JAMES BURKE (JOHNSON & JOHNSON CEO), QUOTED IN LASTING LEADERSHIP (WHARTON)

LE INSIGHT

Tylenol is the canonical positive case for institutional response to crisis. The capability that was load-bearing was the pre-specified institutional commitment in the Credo. The crisis decision had been made decades earlier; in 1982 it was executed. The case is the strongest evidence in the business-ethics dataset that values must be pre-committed in writing to be operational under stress.

LENS APPROACH

LENS uses Tylenol in LEN 7 as the foundational positive case for institutional governance under crisis and in LEN 10 (capstone) as a worked example of pre-committed capability that executed under operational pressure.

REFLECTION QUESTIONS

1. What is your institution's equivalent of the J&J Credo, and is it operational under crisis or aspirational?
2. Pre-commitment is hard to enforce later. Design the institutional architecture that makes a Tylenol-style response the only available option in the worst case.

WHO BUILDS THIS governance, ethics & measurement · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · incident analysis · safety dashboards

SOURCES

→ Kaplan, T., The Tylenol Crisis (2014)
→ FDA Final Rule on Tamper-Resistant Packaging (1982)
→ Greyser, Johnson & Johnson: The Tylenol Tragedy (HBS case, 1992)

FURTHER READING

• Kaplan (2014), The Tylenol Crisis
• Edmondson (2018), The Fearless Organization
• Greyser, HBS case studies

LENS COURSES LEN 10 LEN 7 LEN 6

CASE 89 AVIATION 1976 – PRESENT

Aviation Safety Reporting System (ASRS)

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT NASA-administered confidential reporting system; more than 2M reports received; foundational architecture for evidence-driven aviation safety

ASRS — PAIRED TECHNICAL ARTIFACT AND PROTECTED REPORTER

The Aviation Safety Reporting System, administered by NASA on behalf of the FAA since 1976, accepts confidential reports from pilots, controllers, mechanics, and cabin crew about safety incidents, near-misses, and concerns. The system is explicitly non-punitive: reporting an incident to ASRS confers immunity from FAA enforcement action for the conduct reported, subject to specified limits. The architecture is designed to make reporting the rational choice for the operator and to make the data flow to the institution that can use it.

Over fifty years and more than two million reports, ASRS has accumulated the largest single repository of aviation safety information in the world. Specific aviation incidents — automation surprise patterns, runway incursions, fatigue effects — were first identified at scale through ASRS data before they reached formal investigation thresholds. The architecture has been studied and emulated in other domains: PSRS for patient safety, CHIRP for maritime, similar systems for rail and nuclear.

ASRS is the canonical success case for evidence-architecture paired with institutional commitment to non-punitive use.

THE LEARNING ENGINEERING LENS

> *ASRS is the model for confidential, voluntary, non-punitive incident reporting in any high-consequence domain.*
>
> PARAPHRASING JAMES REASON, MANAGING THE RISKS OF ORGANIZATIONAL ACCIDENTS, 1997

LE INSIGHT

ASRS is the canonical positive case for paired-intervention evidence architecture. The technical artifact (the reporting form and the database) is paired with the cultural commitment to non-punitive use. Either alone fails. Together they have produced the most comprehensive operational-safety dataset in any domain.

LENS APPROACH

LENS uses ASRS in LEN 4 as the foundational positive case for evidence architecture and in LEN 8 for institutional commitment to non-punitive learning. Studio projects design ASRS-equivalents for new domains.

REFLECTION QUESTIONS

1. Identify a domain in your institution that would benefit from an ASRS-equivalent. What cultural commitment would be required for it to function?
2. Design the institutional commitment that makes an ASRS-equivalent operational rather than merely declared.

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards

SOURCES

→ NASA ASRS Program documentation and annual reports
→ Reason (1997), Managing the Risks of Organizational Accidents
→ NASA ASRS Program Briefing materials and annual reports (Linda Connell, director)

FURTHER READING

- Reason (1997)
- Connell, et al., ASRS technical reports
- Dekker, Just Culture (2012)

LENS COURSES LEN 4 LEN 8

CASE 90 HEALTHCARE 1984 – PRESENT

Bristol Heart Babies Reform

G Governance & Trust N Normalization of Deviance

IMPACT Foundational UK case in clinical outcomes transparency; produced specialty-level performance reporting in UK cardiac surgery

BEFORE	AFTER
private	published

outcomes data ceased to be the private property of clinicians

BRISTOL — OUTCOMES TRANSPARENCY AS PAIRED INTERVENTION

The Bristol Royal Infirmary was found, through whistleblowing and subsequent inquiry, to have substantially worse outcomes in pediatric cardiac surgery than other UK centers for years. The Kennedy Inquiry produced a report that is one of the most influential UK health-care inquiries in the modern era. The report identified the institutional capability gap as the absence of routine specialty-level outcomes reporting. Surgeons did not know how their results compared with peers. Patients did not know. Referral patterns did not respond to the actual outcome data.

The reform that followed produced national specialty-level outcomes registries — first in cardiac surgery, subsequently in multiple specialties. The UK is now one of the few countries with routinely published surgeon-level outcomes data. The capability — patients, referrers, and surgeons themselves knowing what the outcomes actually are — was constructed as a paired intervention: a technical registry plus institutional commitment to publish.

Bristol is the foundational UK case for outcomes-transparency as a paired-intervention deliverable.

THE LEARNING ENGINEERING LENS

> *Outcomes data must cease to be the private property of clinicians and become a shared institutional resource.*
>
> PARAPHRASING THE KENNEDY INQUIRY FINAL REPORT (LEARNING FROM BRISTOL), 2001

LE INSIGHT

Bristol is the canonical UK case for outcomes-transparency as a paired intervention. The technical capability — registry design, statistical risk adjustment, publication architecture — was necessary. The cultural capability — surgeons accepting that their results would be visible and comparable — was equally necessary. The pair has produced one of the most mature specialty-outcomes regimes in any country.

LENS APPROACH

LENS uses Bristol in LEN 4 for outcomes-transparency as a paired- intervention deliverable and in LEN 7 for the institutional reform that made surgeon-level publication acceptable. The case pairs with Keystone ICU (Case 14) as healthcare-outcomes interventions at different layers.

REFLECTION QUESTIONS

1. What is the equivalent of surgeon-level outcomes transparency in your domain? What cultural change would have to accompany the technical instrument?
2. Design the registry and publication architecture for a specialty in your domain currently operating without outcomes transparency.

WHO BUILDS THIS governance, ethics & measurement · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · incident analysis · safety dashboards

SOURCES

→ Kennedy QC, I. (2001), Learning from Bristol: The Report of the Public Inquiry into Children's Heart Surgery at the Bristol Royal Infirmary 1984–1995
→ Society for Cardiothoracic Surgery in GB & Ireland, national outcomes reports
→ Berwick, A Promise to Learn — A Commitment to Act (2013)

FURTHER READING

• Kennedy QC (2001), Learning from Bristol
• Berwick (2013), A Promise to Learn
• Sherlaw-Johnson et al., risk-adjusted outcome literature

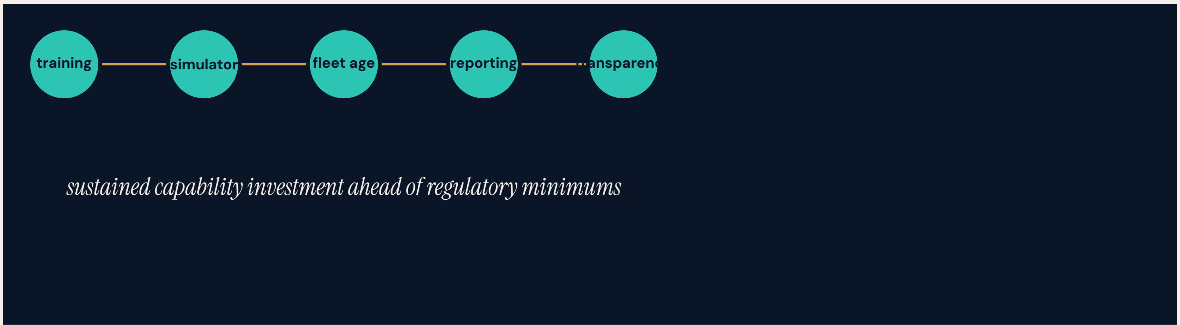
LENS COURSES LEN 4 LEN 7 LEN 3

CASE 91 AVIATION 1980S – PRESENT

Singapore Airlines Safety Transformation

T Training Gap N Normalization of Deviance

IMPACT Sustained safety record over decades despite challenging operating conditions; among the most safety-invested carriers in commercial aviation



SINGAPORE AIRLINES — CAPABILITY INVESTMENT AS COMPETITIVE DIFFERENTIATOR

Singapore Airlines invested in safety capability across decades in a way that distinguishes it from carriers operating under comparable conditions. The airline's training regime, simulator investment, fleet age policy, and reporting culture have produced sustained safety performance even during periods of rapid growth. Singapore Airlines was an early adopter of CRM and an early investor in CRM-style culture work tuned to its operating context.

The SQ006 accident in 2000 — attempted takeoff on a closed, partially-construction runway (05R) at Taipei Taoyuan during Typhoon Xangsane, with 83 fatalities — produced a sustained institutional self-examination that became a case study in the literature on post-accident institutional learning. The airline's response — operational changes, training updates, public transparency about what had happened and why — is referenced in commercial aviation as one of the model post-accident institutional responses.

The case is positioned in this book as the operational successor to Korean Air (Case 23): an Asian carrier that engineered its safety capability deliberately and sustained the investment across decades.

THE LEARNING ENGINEERING LENS

> *Singapore Airlines has consistently invested in safety capability ahead of regulatory minimums.*
>
> IATA OPERATIONAL SAFETY AUDIT RETROSPECTIVES

LE INSIGHT

Singapore Airlines is the case for sustained capability investment in a competitive industry. The carrier has chosen safety capability investment as a primary differentiator. The result over decades is a safety record that distinguishes it from peers operating under comparable conditions.

LENS APPROACH

LENS uses Singapore Airlines in LEN 8 for sustained institutional capability investment under competitive pressure. The case pairs with Korean Air (Case 23) as Asian-carrier capability stories of different shapes — one transformation under crisis, the other sustained investment without crisis.

REFLECTION QUESTIONS

1. Identify an institution in your domain that has chosen capability investment as a primary differentiator. What pattern has it sustained?

2. Design the institutional architecture that makes sustained capability investment defensible against competitive cost pressure.

WHO BUILDS THIS learning science & instructional design · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · incident analysis · safety dashboards

SOURCES

→ Aviation Safety Council (Taiwan), SQ006 Accident Investigation Final Report (2002)
→ Singapore Airlines Safety Reports (multiple years)
→ IATA Operational Safety Audit (IOSA) documentation

FURTHER READING

- Taiwan ASC, SQ006 Report (2002)
- Helmreich, Wilhelm, Klinect & Merritt (2001), national culture and CRM
- Singapore Airlines corporate-safety reports

LENS COURSES LEN 8

PART III — THE FRONTIER

Chapter 9

Human–AI Teaming

The capability work the discipline has not done before.

Humans tend to tune out when tasked with monitoring automated systems that work well most of the time.

NTSB CHAIRMAN ROBERT SUMWALT, UBER TEMPE HEARING REMARKS, 2019

CASE 92 AUTONOMOUS SYSTEMS 2016 – PRESENT

Tesla Autopilot — Recurring Fatalities

T Training Gap N Normalization of Deviance G Governance & Trust H Human-System Interface

IMPACT Dozens of fatalities documented in NHTSA's Standing General Order data; first U.S. cases of Level-2 automation contributing to fatal injury

L2

DRIVER RETAINED · ATTENTION NOT ENGINEERED

the system works well most of the time — and then it does not

TESLA AUTOPILOT — LEVEL-2 MONITORING AS A SUSTAINABLE ROLE

Tesla’s Autopilot and Full Self-Driving Beta operate as Level-2 driver-assistance systems: the driver remains legally and operationally responsible for the vehicle. NHTSA’s Standing General Order on automated-driving systems has documented dozens of fatal crashes involving Tesla Autopilot since the 2016 Joshua Brown case in Florida. The pattern is consistent: the system operates capably for long periods, the driver’s monitoring attention attenuates, and an edge-case scenario — a stationary object, a faded lane marking, a perpendicular crossing — produces a collision the inattentive driver does not catch.

NHTSA’s open investigation has identified Tesla’s driver- engagement design as inadequate to maintain the attention the system’s safe operation requires. The proximate cause in each case is the driver. The system that created the conditions for the inattention is the design choice to call the feature “Autopilot.”

Tesla Autopilot is the live, ongoing version of the Uber ATG (Case 29) problem at consumer scale: a passive-monitoring role designed without the capability infrastructure to make it performable.

THE LEARNING ENGINEERING LENS

> *The operational design ... permitted his prolonged disengagement from the driving task.*
>
> NTSB HIGHWAY ACCIDENT REPORT HAR-17/02 (WILLISTON, FLORIDA CRASH), 2017

LE INSIGHT

Tesla Autopilot at consumer scale is the largest live test of Level-2 monitoring as a sustainable role. The early evidence is that it is not. The case is the strongest currently available test of whether consumer-side training and engagement design can produce sustained attention to automation. The recurring fatality pattern suggests the answer.

LENS APPROACH

LENS uses Tesla Autopilot in LEN 2 as the live consumer-scale test of monitoring as a sustainable role and in LEN 7 for the governance dynamics of a Level-2 system marketed at the boundary of higher autonomy. Studio projects examine the driver-engagement design that would make the role performable.

REFLECTION QUESTIONS

1. Identify a passive-monitoring role in your domain. What evidence would tell you whether attention is sustainable over years of operation?

2. Design the driver-engagement architecture that would make Level-2 monitoring sustainable for an average consumer.

WHO BUILDS THIS learning science & instructional design · safety science & organizational analysis · governance, ethics & measurement · human factors & interaction design — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · incident analysis · safety dashboards · governance frameworks · bias & evidence audits · usability testing · cognitive walkthroughs

SOURCES

→ NHTSA Standing General Order 2021-01 reports
→ NTSB Highway Accident Reports HAR-17/02 (Williston FL, May 2016) and HAR-20/01 (Mountain View CA, March 2018) on Tesla Autopilot crashes
→ NHTSA Office of Defects Investigation EA22-002 (2022)

FURTHER READING

• NHTSA SGO documentation
• Parasuraman & Manzey (2010), automation complacency
• Casner & Hutchins (2019), automation transparency

LENS COURSES LEN 7 LEN 2 LEN 6

CASE 93 AUTONOMOUS SYSTEMS 2023

Cruise Robotaxi — Pedestrian Drag

G Governance & Trust D Designed Out H Human-System Interface

IMPACT GM Cruise robotaxi struck a pedestrian and then dragged her ~20 feet; California suspended Cruise's permit; the program was substantially shut down

CRUISE — PARTIAL DISCLOSURE CONVERTED INCIDENT INTO COMPANY-ENDING EVENT

A pedestrian was struck by another vehicle in San Francisco and thrown into the path of a Cruise robotaxi. The robotaxi struck the pedestrian, then — having detected an impact — pulled to the side of the road, dragging the pedestrian approximately twenty feet. The California Department of Motor Vehicles found that Cruise had initially shown investigators only the first portion of the incident video, omitting the drag.

The CPUC suspended Cruise's driverless permit. The U.S. National Highway Traffic Safety Administration opened an investigation. GM subsequently shut down Cruise's commercial operations and laid off the majority of the workforce.

The case is the foundational governance-failure case in commercial autonomous vehicles: a contemporary firm whose institutional response to a safety incident — partial disclosure, selective evidence — accelerated the consequences far beyond what the incident alone would have produced. The capability gap was at the company's governance posture, not at the technology.

THE LEARNING ENGINEERING LENS

> *Cruise's representation of the incident to regulators omitted material facts.*
>
> PARAPHRASING THE CALIFORNIA PUBLIC UTILITIES COMMISSION DECISION SUSPENDING CRUISE PERMITS, 2023

LE INSIGHT

Cruise is the foundational governance-failure case in commercial autonomy. The incident itself was a single pedestrian injury; the institutional response converted it into a company-ending event. The capability that was missing was the institutional commitment to full disclosure of operational incidents to regulators.

LENS APPROACH

LENS uses Cruise in LEN 7 as a foundational autonomous-vehicle governance case and in LEN 10 (capstone) for the institutional- response deliverable that should pre-exist any commercial autonomy program.

REFLECTION QUESTIONS

1. What is the institutional incident-disclosure commitment in your domain? Is it operational under stress or aspirational?
2. Design the incident-disclosure deliverable that a commercial autonomous-vehicle company should be required to demonstrate before operating.

WHO BUILDS THIS governance, ethics & measurement · systems & human-factors engineering · human factors & interaction design — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · task analysis · design prototyping · usability testing · cognitive walkthroughs

SOURCES

→ California CPUC Decision suspending Cruise permits (2023)
→ Quinn Emanuel report on Cruise incident response (2024)
→ NHTSA Office of Defects Investigation reports

FURTHER READING

• Quinn Emanuel Report on Cruise (2024)
• California CPUC decisions on Cruise
• Stilgoe (2021) on the governance of autonomous vehicles

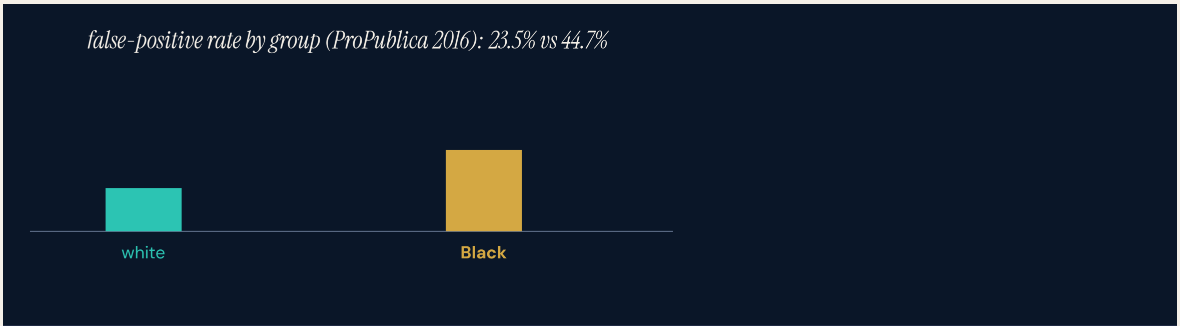
LENS COURSES LEN 10 LEN 7

CASE 94 GOVERNMENT TECHNOLOGY 2016 – PRESENT

COMPAS Recidivism Algorithm

G Governance & Trust H Human-System Interface D Designed Out

IMPACT Risk-assessment algorithm used by U.S. courts; ProPublica analysis identified racial bias in error rates; foundational AI-governance case in criminal justice



COMPAS — ACCURACY IS NOT THE SAME QUESTION AS DISTRIBUTION OF ERRORS

The COMPAS risk-assessment algorithm, used in some U.S. jurisdictions to inform decisions about bail, sentencing, and parole, was examined by ProPublica in a 2016 investigation. The analysis found that Black defendants were more likely than white defendants to be incorrectly flagged as high-risk (false-positive rate 44.7% vs 23.5% for white defendants), and white defendants more likely to be incorrectly flagged as low-risk. The vendor (Northpointe) responded that COMPAS satisfied predictive parity (calibration within groups). The academic literature that followed — notably Chouldechova (2017) and Kleinberg, Mullainathan & Raghavan (2017) — identified the statistical issue as an impossibility result: calibration within groups and equalized false-positive/false-negative rates cannot all be jointly satisfied when base rates differ across groups.

The case is foundational because it surfaced, at criminal-justice scale, the difference between predictive accuracy and the distribution of errors across populations. A model can be "accurate" by one metric and inequitable by another, and the choice of metric is a governance decision rather than a technical one.

The case is paired with the educational algorithmic-bias case (Case 37) and the A-Level case (Case 35) as foundational examples of construct definition as a capability-engineering decision.

THE LEARNING ENGINEERING LENS

> *Blacks are almost twice as likely as whites to be labeled a higher risk but not actually re-offend.*
>
> ANGWIN, LARSON, MATTU & KIRCHNER, PROPUBLICA, "MACHINE BIAS," 2016

LE INSIGHT

COMPAS is the canonical case for the distribution of algorithmic errors across populations as a separate question from algorithmic accuracy. The capability gap is at the governance layer that decides which fairness criterion to optimize. The decision is not technical. It is institutional. The Chouldechova and Kleinberg–Mullainathan–Raghavan results showed the choice is not between fair and unfair models but among incompatible definitions of fairness. The technical literature can characterize the trade-offs; the institutional layer has to choose, in public, which trade-off it is willing to defend.

LENS APPROACH

LENS uses COMPAS in LEN 7 as a foundational algorithmic- accountability case in criminal justice, in LEN 9 for the technical fairness-criterion analysis, and in LEN 4 for the construct-definition question (what counts as recidivism?).

REFLECTION QUESTIONS

1. Identify a predictive instrument in your domain. Which fairness criterion does it optimize, and what does that imply for the population whose errors are distributed against them?
2. Design the institutional governance deliverable that would have to be produced before COMPAS-style algorithms can be deployed in criminal justice.
3. The ProPublica analysis surfaced the COMPAS disparity in 2016 only because reporters obtained the data. What is the equivalent public-interest investigation that should exist for a comparable model in your domain — and what data would have to be released for it to be possible?

WHO BUILDS THIS governance, ethics & measurement · human factors & interaction design · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · usability testing · cognitive walkthroughs · task analysis · design prototyping

SOURCES

→ Angwin, Larson, Mattu & Kirchner, “Machine Bias,” ProPublica (2016)
→ Northpointe technical response (2016)
→ Kleinberg, Mullainathan & Raghavan (2017), “Inherent Trade-Offs in the Fair Determination of Risk Scores”

FURTHER READING

- Angwin et al. (2016), ProPublica
- Kleinberg, Mullainathan & Raghavan (2017)
- Eubanks (2018), Automating Inequality

LENS COURSES LEN 4 LEN 7 LEN 9

CASE 95 HEALTHCARE TECHNOLOGY 2018 – PRESENT

Radiology AI Miscalibration

H Human-System Interface K Knowledge & Institutional Memory D Designed Out

IMPACT Recurring documented cases of FDA-cleared radiology AI tools performing worse in deployment than in validation, often along demographic lines

VALIDATION

AUC 0.9+

DEPLOYMENT

drops

the clearance dataset and the deployment population diverge

RADIOLOGY AI — CLEARANCE IS NOT DEPLOYMENT

Multiple FDA-cleared radiology AI tools — for chest X-ray classification, mammography, CT triage — have been documented in the peer-reviewed literature to perform worse in deployment than in their validation studies, often with the degradation concentrated in patient populations under-represented in the training data. Larrazabal et al. (PNAS 2020) demonstrated this structurally for chest X-ray classifiers — substantial drops in sensitivity for groups under-represented in training. Adjacent work in non-imaging clinical AI (Obermeyer et al., Science 2019, on a widely deployed care-management algorithm) showed that bias in the **labels** used to train clinical AI can systematically under-allocate resources to Black patients even when the algorithm appears statistically well-calibrated. Similar patterns have been documented in mammography AI, sepsis-prediction tools, and skin-lesion classifiers. The FDA's 510(k) clearance process for medical AI does not routinely require demographic stratification of validation metrics, nor does it require post-market monitoring of how the tool performs in the populations actually using it. The capability gap is at the regulatory architecture, not at the model.

Radiology AI is the live, recurring case for the gap between clearance and clinically performable deployment in medical AI.

THE LEARNING ENGINEERING LENS

> *Performance metrics on a clearance dataset are not the same as performance metrics in deployment populations.*
>
> EDITORS' SYNTHESIS OF THE FDA AI/ML-BASED SAMD DISCUSSION PAPER (2019)

LE INSIGHT

Radiology AI is the canonical contemporary case for the gap between regulatory clearance and clinical deployment performance in medical AI. The clearance dataset and the deployment population diverge. The institutional architecture to surface the divergence — demographic post-market surveillance — has not yet been built.

LENS APPROACH

LENS uses radiology AI in LEN 4 for clearance-vs-deployment measurement architecture, in LEN 7 for FDA AI/ML regulatory governance, and in LEN 9 for the technical bias-detection pipeline. The case pairs with Vioxx (Case 87) as a post-market surveillance failure pattern at a new technological boundary. The Obermeyer (2019) finding generalizes the diagnosis: bias enters through the labels and through the population, both of which the 510(k) process currently treats as outside its scope.

REFLECTION QUESTIONS

1. Identify a model in your domain whose deployment population diverges from its training population. What is the institutional architecture to surface the divergence?
2. Design the demographic post-market surveillance deliverable that should accompany every FDA clearance of medical AI.
3. FDA 510(k) clearance does not currently require demographic stratification of validation metrics, nor does it require post-market monitoring of how a cleared tool actually performs on the population using it. What is the minimum reporting deliverable a regulator should require so the gap is visible while the tool is in use?

WHO BUILDS THIS human factors & interaction design · knowledge management & data engineering · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS usability testing · cognitive walkthroughs · knowledge bases · data pipelines · task analysis · design prototyping

SOURCES

→ FDA, "Proposed Regulatory Framework for Modifications to AI/ML-Based Software as a Medical Device" (2019)
→ Larrazabal et al. (2020), "Gender imbalance in medical imaging datasets," PNAS
→ Obermeyer et al. (2019), "Dissecting racial bias in an algorithm used to manage the health of populations," Science

FURTHER READING

• Obermeyer et al. (2019), Science
• Larrazabal et al. (2020), PNAS
• Wachter & Brynjolfsson (2023), "Will Generative Artificial Intelligence Deliver on Its Promise in Health Care?" JAMA

LENS COURSES

LEN 4 LEN 7 LEN 9

CASE 96 HEALTHCARE TECHNOLOGY 2023 – PRESENT

ChatGPT in Healthcare — Hallucination Cases

H Human-System Interface D Designed Out

IMPACT Documented cases of clinicians using LLMs that produce hallucinated citations, fabricated dosages, and fictitious clinical guidelines

TONE

identical

CONTENT

accurate / not

the interface does not distinguish; the clinician must

LLMS IN CLINICAL USE — FLUENCY WITHOUT WARRANTY

Since the public release of ChatGPT in late 2022, the peer-reviewed and clinical literatures have documented a recurring pattern: clinicians use large language models to draft patient education, summaries, or treatment guidance, and the output contains fabricated citations, hallucinated drug dosages, or fictitious clinical guidelines. Cases have ranged from cosmetic — fabricated references in academic submissions — to potentially clinical: a chatbot recommending unsafe medication doses or inventing contraindications.

The capability gap is at the human verification interface. Clinicians who would carefully verify a colleague's recommendation have, in early case reports, accepted LLM output less critically. The hallucinations are presented with the same confidence as accurate information. The interface does not distinguish.

The case is the live, recurring foundational case for LLM integration into clinical workflow. The discipline is being asked to specify what verification looks like at the moment of use, while the deployment is happening.

THE LEARNING ENGINEERING LENS

> *LLMs produce hallucinations indistinguishable in tone from accurate information, and clinicians have not yet developed the practice of routine verification.*
>
> EDITORS' SYNTHESIS OF JAMA EDITORIALS ON LLM CLINICAL USE (2023–2024)

LE INSIGHT

LLM use in clinical practice is the live frontier case for human-AI teaming when the AI's output is fluent across both accurate and hallucinated content. The capability that does not yet exist is the routine verification practice — a clinical analog to the bibliographic discipline of academic writing.

LENS APPROACH

LENS uses this case in LEN 2 as the live frontier for human-AI teaming with LLMs, in LEN 7 for the governance of LLM deployment in clinical workflows, and in LEN 10 (capstone) as a prompt for designing verification practices at the moment of use.

REFLECTION QUESTIONS

1. Identify a workflow in your domain currently being augmented by LLMs. What is the verification practice — and does it exist at the moment of use, or only after?
2. Design the verification deliverable that should accompany every clinician's adoption of an LLM tool for patient-facing work.

WHO BUILDS THIS human factors & interaction design · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS usability testing · cognitive walkthroughs · task analysis · design prototyping

SOURCES

→ JAMA editorials on LLM integration into clinical practice (2023–2024)
→ Sallam (2023), "ChatGPT Utility in Healthcare Education, Research, and Practice"
→ WHO ethical guidance on AI for health (2024)

FURTHER READING

• JAMA editorials on LLM clinical use
• Sallam (2023)
• Wachter & Brynjolfsson (2023), JAMA on generative AI in health care

LENS COURSES LEN 10 LEN 7 LEN 2

CASE 97 GOVERNMENT TECHNOLOGY 2011 – PRESENT

Predictive Policing — PredPol

G Governance & Trust H Human-System Interface D Designed Out

IMPACT Predictive-policing tools deployed by hundreds of U.S. police departments; several cities have abandoned them after equity analyses

TRAINING DATA

past arrests

GROUND TRUTH

past crime

the system learns enforcement patterns, not crime patterns

PREDICTIVE POLICING — CONSTRUCT GAP AT LAW-ENFORCEMENT SCALE

PredPol and similar predictive-policing tools use historical crime data to forecast where future crime is likely to occur, directing patrol resources to those locations. Multiple academic analyses (Lum & Isaac 2016, Brantingham et al. 2018) have found that because the historical crime data is itself a record of where police enforcement has been concentrated, the algorithm tends to reinforce existing enforcement patterns rather than predict underlying crime. Cities including Santa Cruz, New Orleans, and Los Angeles have suspended or abandoned predictive-policing deployments after equity reviews.

The capability gap is at the construct definition: “where crime occurs” and “where crime is recorded” are not the same variable, and predictive-policing tools have historically treated them as such. The reform pattern — independent equity audit before deployment — has been implemented in some jurisdictions and not in others.

Predictive policing is the canonical algorithmic-governance case in U.S. policing and pairs with COMPAS (Case 94) and educational algorithmic bias (Case 37).

THE LEARNING ENGINEERING LENS

> *Predictive policing systems learn from a record of past policing, not from a record of past crime.*
>
> LUM & ISAAC, SIGNIFICANCE, 2016

LE INSIGHT

Predictive policing is the canonical case for the difference between training data and ground truth at law-enforcement scale. The institution being predicted (where crime occurs) is not the institution producing the training data (where police make arrests). The capability gap is at the construct.

LENS APPROACH

LENS uses predictive policing in LEN 7 as a foundational AI- governance case in policing and in LEN 9 for the technical construct-validity analysis. The case is paired with COMPAS as criminal-justice algorithmic cases of different kinds.

REFLECTION QUESTIONS

1. Identify a predictive system in your domain whose training data is itself a record of institutional behavior rather than ground truth. What is the construct gap?
2. Design the construct-validity audit that should precede deployment of a predictive system in any institutional setting.

WHO BUILDS THIS governance, ethics & measurement · human factors & interaction design · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · usability testing · cognitive walkthroughs · task analysis · design prototyping

SOURCES

→ Lum & Isaac, "To Predict and Serve?" Significance (2016)
→ Brantingham et al. (2018) on predictive-policing field experiments
→ Richardson, Schultz & Crawford (2019), "Dirty Data, Bad Predictions"

FURTHER READING

• Lum & Isaac (2016)
• Richardson, Schultz & Crawford (2019)
• Brayne (2017), "Big Data Surveillance: The Case of Policing"

LENS COURSES LEN 7 LEN 9

CASE 98 HEALTHCARE TECHNOLOGY 2020 – PRESENT

AlphaFold — Protein Structure Prediction

T Training Gap

IMPACT Substantially solved a 50-year protein-structure prediction problem; 200M+ structures publicly released; foundational positive AI case

200M

PREDICTED STRUCTURES · PUBLICLY RELEASED

agreed benchmark, training data, verifiable output, open release

ALPHAFOLD — THE CONDITIONS FOR BENEFICIAL AI DEPLOYMENT

DeepMind's AlphaFold (2020) and AlphaFold 2 (2021) substantially solved the half-century-old protein-structure prediction problem in computational biology. The publicly released AlphaFold Protein Structure Database contains predicted structures for over two hundred million proteins — essentially the entire known protein universe. AlphaFold has been integrated into structural-biology workflows worldwide and has accelerated drug-discovery and biological-research timelines across multiple domains.

The case is the foundational positive case for AI as a capability amplifier in a well-defined scientific problem. The success has several features: a benchmark (CASP) that the field had agreed on for decades, training data of high quality, an output that biologists could verify against experimental structures, and an open release that allowed the global community to adopt the tool quickly. Each of these features was a precondition for the success, and none of them were the model itself.

AlphaFold is the strongest positive AI case in the dataset for what conditions support beneficial AI deployment in a research domain.

THE LEARNING ENGINEERING LENS

> *This will be one of the most important achievements in AI in the past decade.*
>
> PARAPHRASING JOHN MOULT (ORGANIZER OF THE CASP BENCHMARK) ON ALPHAFOLD2, 2020

LE INSIGHT

AlphaFold is the strongest positive AI case in the dataset. The technical achievement is real. The conditions that made the benefit possible — agreed benchmark, training data, verifiable output, open release — are the capability infrastructure around the model, not the model itself. The case is the strongest available evidence for what supports beneficial AI deployment.

LENS APPROACH

LENS uses AlphaFold in LEN 1 as a problem-framing case for what productive AI deployment looks like, in LEN 9 as a technical achievement, and in LEN 7 for the open-release governance decision that distributed the benefit globally.

REFLECTION QUESTIONS

1. Identify a domain in your work where the conditions that supported AlphaFold's success (benchmark, training data, verifiable output) are present. What is the analogous opportunity?
2. The open release of AlphaFold's predictions was a governance decision. Design the equivalent decision for an AI capability your institution might develop.

WHO BUILDS THIS learning science & instructional design — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models

SOURCES

→ Jumper et al. (2021), "Highly accurate protein structure prediction with AlphaFold," Nature
→ Varadi et al. (2022), AlphaFold Protein Structure Database, Nucleic Acids Research
→ Moult, J., CASP organizer commentary

FURTHER READING

• Jumper et al. (2021), Nature
• Varadi et al. (2022)
• Hassabis (DeepMind) public commentary on AlphaFold and capability

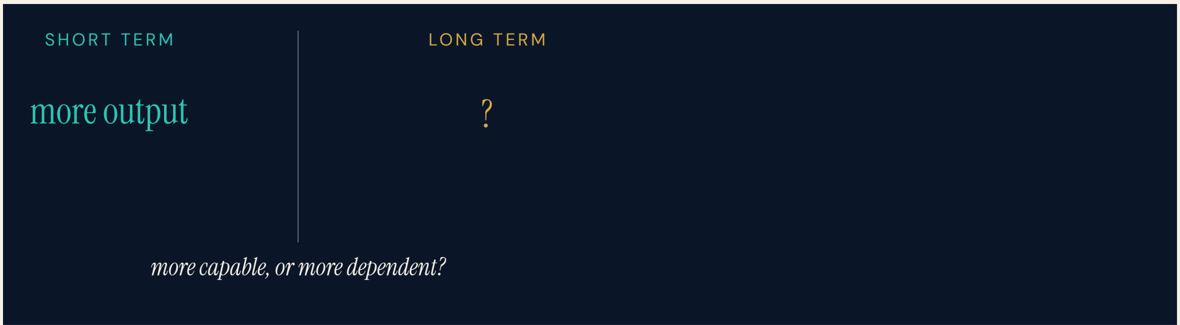
LENS COURSES LEN 1 LEN 7 LEN 9

CASE 99 TECHNOLOGY 2021 – PRESENT

AI-Augmented Coding Tools

T Training Gap H Human-System Interface

IMPACT Tens of millions of developers using GitHub Copilot, Cursor, and peers; productivity gains documented; security and correctness implications still being characterized



AI-AUGMENTED CODING — THE CAPABILITY QUESTION IS OPEN

The widespread availability of AI-augmented coding tools — GitHub Copilot, Cursor, Codeium, and others — represents the largest real-time experiment in human-AI collaboration in the dataset. Tens of millions of developers use these tools daily, generating measurable productivity gains in published studies (Peng et al. 2023). The tools also produce code with documentable security vulnerability rates and correctness issues that developers accept at higher rates than they accept similar code from other developers (Sandoval et al. 2023; Pearce et al. 2022).

The capability question is open: are developers using these tools becoming more capable, or are they becoming more dependent? The longitudinal evidence is not yet sufficient to answer. The short-term productivity gains are real. The long-term capability consequences — particularly for developers who learn the craft with these tools available — are not yet known.

The case is the live frontier for human-AI teaming in professional knowledge work. The discipline LENS represents is being asked to specify what good looks like before the long-term evidence is in.

THE LEARNING ENGINEERING LENS

> *AI assistance changes what developers can do; it may also change what they need to know.*
>
> COMMUNICATIONS OF THE ACM, 2024

LE INSIGHT

AI-augmented coding is the live foundational case for human-AI teaming in professional knowledge work. The short-term gains are real. The long-term capability question — does the tool make the operator more capable, or more dependent — is the question the discipline must learn to ask and answer.

LENS APPROACH

LENS uses AI-augmented coding in LEN 2 as the largest live human- AI teaming case in the dataset, in LEN 8 for the long-term capability-development question, and in LEN 10 (capstone) as a prompt for evaluating AI augmentation in professional practice.

REFLECTION QUESTIONS

1. In your domain, identify a class of practitioners whose work is currently being augmented by AI tools. What evidence would tell you whether their capability is growing or eroding?
2. Design the longitudinal study that would distinguish capability growth from capability erosion in an AI-augmented professional practice.

WHO BUILDS THIS learning science & instructional design · human factors & interaction design — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs

SOURCES

→ Peng et al. (2023), "The Impact of AI on Developer Productivity"
→ Pearce et al. (2022), "Asleep at the Keyboard? Assessing the Security of GitHub Copilot's Code Contributions"
→ Sandoval et al. (2023), security analyses of AI-generated code

FURTHER READING

- Peng et al. (2023)
- Pearce et al. (2022)
- Dell'Acqua et al. (2023), "Navigating the Jagged Technological Frontier" (HBS / BCG study on professional use of LLMs)

LENS COURSES LEN 10 LEN 2 LEN 8

CASE 100 EDUCATION AT SCALE HEALTHCARE DEFENSE ONGOING

The Discipline We Build Next

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance G Governance & Trust H Human-System Interface D Designed Out

IMPACT Open: the case that has not yet been written. The discipline that this casebook documents is the one practitioners now in training must build.

100th

CASE · OPEN

what the practitioners reading this volume do next is what fills it in

THE CASE THAT HAS NOT YET BEEN WRITTEN

The ninety-nine cases that precede this one have an answer. The investigations are complete; the failures are diagnosed; the interventions are evaluated. This case does not. It belongs to the practitioners now in training — to the students of the LDT program and the LENS specialization, to the graduates who will take positions in healthcare, defense, education at scale, and the human-AI teaming cases that anticipate the next decade — and to the institutions those graduates will build.

The strongest evidence for the discipline LENS represents is the cumulative record of what its absence has cost. The strongest argument for its possibility is the cumulative record of what its presence has produced. Both records are in the preceding cases. The work ahead is to add to the second record and shorten the first.

No specific incident sits behind this case because the work is not yet done. The discipline is still being built. The institution of capability engineering is, in 2026, where INPO was in 1979 or where CRM was in 1981 — a discipline whose evidence is accumulating, whose practitioners are being trained, and whose institutional architecture is not yet complete.

What the practitioners reading this volume do next is what fills in this case.

THE LEARNING ENGINEERING LENS

> *Capability is engineerable. The institution to engineer it has not yet been built. That is the work.*
>
> LDT / LENS PROGRAM STATEMENT, JOHNS HOPKINS SCHOOL OF EDUCATION

LE INSIGHT

Every case in this book is closed. This one is open. The discipline LENS represents is the discipline that closes cases that are now open. The institutions of capability engineering that exist in 2026 are insufficient to the volume of capability work the contemporary world requires. The graduates of this program are the people who will build the institutions that are insufficient now.

LENS APPROACH

LENS is organized to produce the practitioners who can do this work. The curriculum's commitments — capability as a system parameter, evidence as a deliverable, ethics as design, implementation as a thread — are the operational form of the argument the ninety-nine preceding cases evidence. Every studio project (LEN 10) is a small attempt at this case.

REFLECTION QUESTIONS

1. What is the case you are positioned to add to this dataset in the next ten years — a failure prevented, a success engineered, an institution built?
2. If this book is read by the practitioners who will write its hundred-and-first case, what should that case look like?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis · governance, ethics & measurement · human factors & interaction design · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards · governance frameworks · bias & evidence audits · usability testing · cognitive walkthroughs · task analysis · design prototyping

SOURCES

→ Goodell & Kolodner (2022), Learning Engineering Toolkit
→ IEEE ICICLE Learning Engineering Body of Knowledge (LEBoK)
→ The preceding ninety-nine cases of this volume

FURTHER READING

- Goodell & Kolodner (2022)
- LEBoK at lebok.wiki
- The introduction to this volume

LENS COURSES LEN 1 LEN 10 LEN 8 LEN 3

DOMAIN INDEX

The cases by domain.

The matrix at the front lists every case in number order. This index reorganises the hundred cases by primary domain — a reader following healthcare, defense, education, or any other thread can find the relevant cases together. Each case appears once, under its primary domain. Each section opens with a callout naming the dataset's standout failure and, where one exists in that domain, its standout success.

DEFENSE

STANDOUT FAILURE — **Case 46**

Operation Eagle Claw (1980). Cross-organizational capability that existed on no service's deliverable list; the mission was improvised across the Army, Navy, Marines, Air Force, and CIA. Produced the Holloway Commission, USSOCOM, and Goldwater-Nichols.

STANDOUT SUCCESS — **Case 28**

U.S. Nuclear Navy / Rickover Training Model (1954–). The longest continuous capability-engineering program in any high-consequence domain. Sixty-plus years of zero reactor accidents through training treated as a system parameter.

1	USS Fitzgerald & USS John S. McCain	2017	T·K·N
9	Marine Corps Training in the INDOPACOM AOR	ongoing	T·K
11	V-22 Osprey	1991–	T·H·N
15	Navy Surface Warfare Readiness Reform	2018–	T·K·N
18	USS Vincennes & Iran Air Flight 655	1988	H·T
19	Patriot Missile / Dhahran	1991	D·H·K
26	F-35 Sustainment & Maintainer Shortage	ongoing	T·K·D
28	U.S. Nuclear Navy / Rickover Training Model	1954–	T·K·N
33	Military Fratricide — Desert Storm to Afghanistan	1991–2014	T·H·K

38	GIFT and the Adoption Gap	2012–	K·G·N
45	Mark 14 Torpedo Failures	1941–43	T·K·G
46	Operation Eagle Claw	1980	T·K
66	Stanislav Petrov / 1983 False Alert	1983	H·T
86	9/11 Intelligence Sharing Failures	1996–2001	G·K

AVIATION

STANDOUT FAILURE — **Case 2** — **Boeing 737 MAX / MCAS (2018–19).** A single-string angle-of- attack sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together. 346 lives.

STANDOUT SUCCESS — **Case 12** — **Crew Resource Management & CAST (1981–).** First-officer authority engineered into the cockpit, a continuous learning architecture across the industry, and the safety record that followed.

2	Boeing 737 MAX / MCAS	2018–19	D·T·H
3	Air France Flight 447	2009	T·H
12	Crew Resource Management & CAST	1981–	T·H·N
23	Korean Air Safety Transformation	2000–	T·N
30	Kegworth / British Midland 92	1989	T·H·K
43	Colgan Air Flight 3407	2009	T
44	Asiana Airlines Flight 214	2013	T·H
47	Helios Airways Flight 522	2005	T·H
48	AeroPerú Flight 603	1996	H·T
49	Atlas Air Flight 3591	2019	T·K
50	TransAsia Airways Flight 235	2015	T·H
63	Eastern Air Lines Flight 401	1972	H·T
64	Boeing 737 Rudder Hardovers	1991, 1994	H·D

78	NASA Saturn V Documentation	1972–	K
89	Aviation Safety Reporting System (ASRS)	1976–	T·K·N
91	Singapore Airlines Safety Transformation	1980s–	T·N

HEALTHCARE

STANDOUT FAILURE — Case 25

EHR / CPOE Implementation (2005–). Roughly $30B federal investment in systems designed for billing and administration, deployed against clinical workflow. Usability is now itself a patient-safety variable at scale.

STANDOUT SUCCESS — Case 13

WHO Surgical Safety Checklist (2008–). One page, nineteen items, paired with a required pause. Halved surgical mortality in the multi-site pilot; the smallest effective capability intervention in the dataset.

7	Therac-25	1985–87	H·D·G
13	WHO Surgical Safety Checklist	2008–	T·N
14	Keystone ICU / Pronovost Checklist	2004–	T·N
25	EHR / CPOE Implementation	2005–	H·D·G
27	TeamSTEPPS	2006–	T·N
31	Medical Errors as Systemic Failure	ongoing	T·H·N·K·G
34	ACGME 80-Hour Resident Duty-Hour Reform	2003–17	T·K·N
41	Implementation Science — The 17-Year Gap	ongoing	K·G·N
58	Mid Staffordshire NHS Foundation Trust	2005–09	G·N·K
69	Theranos	2003–18	G·D
87	Vioxx Withdrawal	1999–2004	G·D
88	Tylenol Recall	1982	G·N
90	Bristol Heart Babies Reform	1984–	G·N
95	Radiology AI Miscalibration	2018–	H·K·D

96	ChatGPT in Healthcare — Hallucination Cases	2023–	H·D
98	AlphaFold — Protein Structure Prediction	2020–	T

ENERGY

STANDOUT FAILURE

Case 5

Three Mile Island (1979). Training matched the design-basis events the engineers had imagined; the accident was not one of them. The control-room interface confused command signal with valve state. Catalysed industry-wide reform.

STANDOUT SUCCESS

Case 16

INPO and the Nuclear Academy (1979–). The industry built the institution TMI revealed it needed. No INES-level event at U.S. commercial reactors since.

4	Deepwater Horizon	2010	T·N·K
5	Three Mile Island	1979	T·H
16	INPO and the Nuclear Academy	1979–	T·K·G
56	Texas City BP Refinery Explosion	2005	N·T·K·G
57	Davis-Besse Reactor Head Corrosion	2002	N·K·G
59	Sago Mine Disaster	2006	N·T·K
60	Upper Big Branch Mine Explosion	2010	N·G·K
61	Fukushima Daiichi	2011	N·G·K
62	Northeast Blackout	2003	H·K
76	Camp Fire / PG&E	2018	G·N·K
77	Texas Grid Freeze	2021	G·K·N
82	Aliso Canyon Methane Leak	2015–16	G·N·K

INDUSTRIAL

STANDOUT FAILURE	Case 17	**Bhopal (1984).** Training, knowledge, normalisation of deviance, and governance all collapsed in the same plant. Roughly 4,000 immediate deaths; chronic harms continue. The cautionary case for outsourced operational responsibility.
STANDOUT SUCCESS	Case 24	**Toyota Production System / Andon Cord (1950s–).** Any operator can stop the line. The smallest authority-architecture intervention in the dataset, and the most durably copied.

17	Bhopal	1984	T·K·N·G
20	Grenfell Tower	2017	G·T·K·N
24	Toyota Production System / Andon Cord	1950s–	N·G
51	Ford Pinto Fuel Tank	1971–78	D·G
52	Takata Airbag Inflators	2008–23	D·G
53	GM Ignition Switch	2002–14	D·G
71	Volkswagen Dieselgate	2015	D·G
74	Hyatt Regency Walkway Collapse	1981	D·G
75	FIU Pedestrian Bridge Collapse	2018	D·G·N
81	Tacoma Narrows Bridge	1940	D·K

SPACE & AEROSPACE

STANDOUT FAILURE	Case 6	**Challenger & Columbia (1986 / 2003).** Normalisation of deviance across two decades and two crews, captured in the Rogers and CAIB reports as a culture that accepted flying with flaws when problems were defined as normal and routine.
NOTE	Case 28	No paired-intervention success in the aerospace dataset. See Defense (Rickover, Case 28) for the safety-engineering counterpart that aerospace did not build.

6	Challenger & Columbia	1986/2003	N·K·G

54	Mars Climate Orbiter — Unit Mismatch 1999	D·K
79	Boeing Starliner 2019–24	K·D
80	Ariane 5 Flight 501 1996	D·K·H

GOVERNMENT

STANDOUT FAILURE — Case 36

Australia Robodebt (2016–20). An income-averaging algorithm that reversed the burden of proof onto welfare recipients. The Royal Commission found “venality, incompetence and cowardice”; the scheme was linked to deaths, including suicide.

NOTE — Case 89

No paired-intervention success in the government dataset. The reforms that follow these cases — Royal Commissions, statutory restructures — are documented in the narratives but did not produce a LENS-style success case in time for this volume.

10	Healthcare.gov Launch 2013	K·T·G
32	VA Wait-Time Scandal 2014	G·K·N
36	Australia Robodebt 2016–20	G·D·H
83	LIBOR Manipulation 2003–12	G·N
85	Bernard Madoff / SEC Failures 1992–2008	G·K·N
94	COMPAS Recidivism Algorithm 2016–	G·H·D
97	Predictive Policing — PredPol 2011–	G·H·D

EDUCATION AT SCALE

STANDOUT FAILURE — Case 21

Summit Learning / Personalized Learning Rollout (2014–19). A defensible pedagogical platform, well funded, deployed across 380 schools — and pulled by parent revolt because the governance architecture (consent, evidence, exit) was never engineered alongside it.

STANDOUT SUCCESS — Case 39

Georgia State University Predictive Analytics (2012–). 800 risk factors per student, advising triggered by early warning signs, equity as a primary constraint. Six-year graduation 32% → 54%; equity gaps eliminated.

8	inBloom	2014	G
21	Summit Learning / Personalized Learning Rollout	2014–19	G·T·K
22	Tennessee Voluntary Pre-K Study	2009–18	G·D
35	UK A-Level Algorithm / Ofqual	2020	G·H·D
37	Algorithmic Bias in Educational Predictive Analytics	ongoing	G·H·D
39	Georgia State University Predictive Analytics	2012–	T·K
40	xAPI / Total Learning Architecture	ongoing	K·G
42	Cognitive Tutor / Carnegie Learning	1990s–	T
84	Atlanta Public Schools Cheating Scandal	2009–15	G·N
100	The Discipline We Build Next	ongoing	all

TECHNOLOGY

STANDOUT FAILURE	Case 68	**UK Post Office Horizon Scandal (1999–2015).** Institutional refusal to believe sub-postmasters' reports of computer error, sustained over fifteen years. Hundreds of wrongful convictions; ruined lives and at least four suicides — the longest-running operator-feedback failure in the dataset.
NOTE	Case 98	The technology dataset's success story is filed under Healthcare: AlphaFold (Case 98), the contemporary worked example of computational capability-engineering done as a discipline.

55	Knight Capital Trading Loss	2012	D·K
65	CrowdStrike Falcon Outage	2024	D·K·G
67	TSB Bank IT Migration	2018	H·G
68	UK Post Office Horizon Scandal	1999–2015	G·H·K
70	Wells Fargo Fake Accounts	2011–16	G·N
72	Cambridge Analytica / Facebook	2014–18	G
73	Equifax Data Breach	2017	G·K

AUTONOMOUS SYSTEMS

STANDOUT FAILURE — **Case 29** — **Uber ATG / Tempe Fatality (2018).** The safety driver was nominally responsible for what the system was designed to do without her attention — the classic vigilance contradiction at the heart of partial-autonomy operator roles.

NOTE — **Case 100** — No paired-intervention success in the autonomous-systems dataset. The capability engineering this category needs is still being built — see Case 100 (**The Discipline We Build Next**) for the open question this volume closes on.

About this index. Each case is filed under its primary domain, so every case appears exactly once. Cases that span several domains (a healthcare-AI case is also a technology case, a defense-training case is also an education case) are cross-referenced in their narratives. The standout failures and successes are editorial picks — the case in that domain the editors return to most often in studio.

REFERENCES

Works cited.

Numbered references are cited in the Introduction. Per-case sources appear on each spread. The broader reading list at the end of this section collects foundational works the LENS curriculum draws on.

CITED IN THE INTRODUCTION

1. Makary, M. A. & Daniel, M. (2016). Medical error — the third leading cause of death in the US. BMJ 353: i2139. doi:10.1136/bmj.i2139.
2. Morris, Z. S., Wooding, S. & Grant, J. (2011). The answer is 17 years, what is the question: understanding time lags in translational research. Journal of the Royal Society of Medicine 104(12): 510–520. doi:10.1258/jrsm.2011.110180.
3. Balas, E. A. & Boren, S. A. (2000). Managing clinical knowledge for health care improvement. Yearbook of Medical Informatics, 65–70.
4. U.S. Navy (2017). Comprehensive Review of Recent Surface Force Incidents; Strategic Readiness Review. Department of the Navy.
5. U.S. Government Accountability Office (2023). F-35 Sustainment: DOD Faces Several Uncertainties and Risks as it Defines Future Costs and Plans. GAO-23-105341. gao.gov.
6. Bulger, M., McCormick, P. & Pitcan, M. (2017). The Legacy of inBloom. Data & Society Research Institute. datasociety.net.
7. Office of Qualifications and Examinations Regulation, UK (2020). Awarding GCSE, AS, A level, advanced extension awards and extended project qualifications in summer 2020: interim report. Ofqual; see also Adam, D. (2020), MIT Technology Review.
8. Helmreich, R. L., Merritt, A. C. & Wilhelm, J. A. (1999). The evolution of Crew Resource Management training in commercial aviation. International Journal of Aviation Psychology 9(1): 19–32. doi:10.1207/s15327108ijap0901_2.
9. Commercial Aviation Safety Team / Federal Aviation Administration (2016). Safety Enhancements and Outcomes. CAST/FAA.
10. Federal Aviation Administration (2017). Out Front on Airline Safety: Two Decades of Continuous Evolution. FAA Newsroom.
11. Rees, J. V. (1994). Hostages of Each Other: The Transformation of Nuclear Safety since Three Mile Island. University of Chicago Press. ISBN 978-0226706887.
12. Pronovost, P., Needham, D., Berenholtz, S. et al. (2006). An intervention to decrease catheter-related bloodstream infections in the ICU. New England Journal of Medicine 355(26): 2725–2732. doi:10.1056/NEJMoa061115.
13. Haynes, A. B., Weiser, T. G., Berry, W. R. et al. (2009). A surgical safety checklist to reduce morbidity and mortality in a global population. New England Journal of Medicine 360(5): 491–499. doi:10.1056/NEJMsa0810119.
14. Agency for Healthcare Research and Quality (2023). TeamSTEPPS 3.0 Curriculum. AHRQ Publication No. 23-0034. ahrq.gov.
15. Sawyer, R. K. (ed.) (2014). The Cambridge Handbook of the Learning Sciences (2nd ed.). Cambridge University Press. ISBN 978-1107033252.
16. Vicente, K. J. (1999). Cognitive Work Analysis: Toward Safe, Productive, and Healthy Computer-Based Work. Lawrence Erlbaum. ISBN 978-0805823974.
17. Wickens, C. D., Hollands, J. G., Banbury, S. & Parasuraman, R. (2021). Engineering Psychology and Human Performance (5th ed.). Routledge. doi:10.4324/9781003177616.
18. Fixsen, D. L., Naoom, S. F., Blasé, K. A., Friedman, R. M. & Wallace, F. (2005). Implementation Research: A Synthesis of the Literature. National Implementation Research Network. See also Aarons, G. A., Hurlburt, M. & Horwitz, S. M. (2011). Advancing a conceptual model of evidence-based practice implementation in public service sectors. Administration and Policy in Mental Health 38: 4–23. doi:10.1007/s10488-010-0327-7.
19. Reason, J. (1990). Human Error. Cambridge University Press, doi:10.1017/CBO9781139062367. Rasmussen, J. (1997). Risk management in a dynamic society: a modelling problem. Safety Science 27: 183–213, doi:10.1016/S0925-7535(97)00052-0. Leveson, N. (2011). Engineering a Safer World: Systems Thinking Applied to Safety. MIT Press, doi:10.7551/mitpress/8179.001.0001.

20. Weick, K. E. & Sutcliffe, K. M. (2015). Managing the Unexpected: Sustained Performance in a Complex World (3rd ed.). Wiley. See also Roberts, K. H. (1990). Some characteristics of one type of high reliability organization. Organization Science 1(2): 160–176.

21. Saxberg, B. (2017). Learning engineering: the art of applying learning science at scale. In Industry Connections Industry Consortium on Learning Engineering. See also Goodell, J. & Kolodner, J. L. (eds.) (2022). Learning Engineering Toolkit. Routledge.

22. IEEE ICICLE (2022). Learning Engineering Body of Knowledge (LEBoK). lebok.wiki.

23. MICrONS Consortium (2025). Functional connectomics spanning multiple areas of mouse visual cortex. Nature 640: 167–179. doi:10.1038/s41586-025-08790-w. See also Hider, R. et al. (2022), BossDB, Frontiers in Neuroinformatics; Xenes, D. et al. (2022), NeuVue, bioRxiv 2022.07.18.500521; Gray-Roncal, W. (in press, 2026), “Calibrate the Scientists, Not Just the Microscopes,” BRAIN CONNECTS commentary.

24. Cervantes, M., Floryanzia, S., Sharp, J., Johnson, E. C. & Gray-Roncal, W. R. (2023). Empowering trailblazers toward scalable, systematized, research-based workforce development. 2023 ASEE Annual Conference & Exposition, Baltimore, MD. peer.asee.org. Documents the CIRCUIT model; Gray-Roncal is the senior author. See also Gray-Roncal, W. et al. (2025), MERIT: Mentoring Exceptional Researchers to Innovate and Thrive — A Structured, Scalable Research Training Framework for STEM Talent Development. JHU/APL whitepaper (the generalization of CIRCUIT). COMPASS — the layer of targeted mentor interventions that operates MERIT inside specific institutional settings, designed to help students uncover tacit and explicit knowledge gaps. COMPASS-Core: Curriculum Overview and Workshop Summaries (JHU/APL program documentation) is one well-documented instance, a workshop set grounded in the CCR framework and explicitly addressing the “hidden curriculum” of research practice.

25. TITAN — DARPA Artificial Intelligence Exploration program (JHU/APL pilot, c. 2020 – 2022). Program documentation on file with the LENS partner team at the JHU/APL AI Pathfinding Lab.

26. PISTA (2025). Perceptual Inference System for Tactical Agentic AI. JHU/APL AI Pathfinding Lab, Summer 2025 Research Incubator, sponsored by DTRA JSTO. Cohort poster and program documentation available from the LENS partner team.

27. Precision-medicine community-detection methods — preprint at medRxiv 2024.06.20.24309267.

FOUNDATIONS · THE DISCIPLINE DRAWS ON

Learning sciences and learning engineering

• Bransford, J. D., Brown, A. L. & Cocking, R. R. (eds.) (2000). How People Learn: Brain, Mind, Experience, and School. National Academy Press. doi:10.17226/9853.
• Sawyer, R. K. (ed.) (2014). The Cambridge Handbook of the Learning Sciences (2nd ed.).
• Koedinger, K. R., Corbett, A. T. & Perfetti, C. (2012). The Knowledge-Learning-Instruction framework. Cognitive Science 36(5): 757–798. doi:10.1111/j.1551-6709.2012.01245.x.
• Sweller, J., Ayres, P. & Kalyuga, S. (2011). Cognitive Load Theory. Springer. doi:10.1007/978-1-4419-8126-4.
• Soderstrom, N. C. & Bjork, R. A. (2015). Learning versus performance: an integrative review. Perspectives on Psychological Science 10(2): 176–199. doi:10.1177/1745691615569000.
• Goodell, J. & Kolodner, J. L. (eds.) (2022). Learning Engineering Toolkit: Evidence-Based Practices from the Learning Sciences, Instructional Design, and Beyond. Routledge. doi:10.4324/9781003276579.

Human factors, cognitive engineering, and systems safety

• Vicente, K. J. (1999). Cognitive Work Analysis. Lawrence Erlbaum.
• Norman, D. A. (2013). The Design of Everyday Things (rev. ed.). Basic Books. ISBN 978-0465050659.
• Hollnagel, E. & Woods, D. D. (2005). Joint Cognitive Systems: Foundations of Cognitive Systems Engineering. CRC Press. doi:10.1201/9781420038194.
• Endsley, M. R. (1995). Toward a theory of situation awareness in dynamic systems. Human Factors 37(1): 32–64. doi:10.1518/001872095779049543.
• Perrow, C. (1984). Normal Accidents: Living with High-Risk Technologies. Basic Books. ISBN 978-0691004129 (1999 Princeton reprint).
• Leveson, N. (2011). Engineering a Safer World: Systems Thinking Applied to Safety. MIT Press. doi:10.7551/mitpress/8179.001.0001.
• Reason, J. (1997). Managing the Risks of Organizational Accidents. Ashgate. ISBN 978-1840141047.
• Rasmussen, J. (1997). Risk management in a dynamic society: a modelling problem. Safety Science 27: 183–213. doi:10.1016/S0925-7535(97)00052-0.
• Dekker, S. (2014). The Field Guide to Understanding ‘Human Error’ (3rd ed.). Ashgate. ISBN 978-1472439055.

High-reliability organizing and team performance

• Weick, K. E. & Sutcliffe, K. M. (2015). Managing the Unexpected (3rd ed.). Wiley. ISBN 978-1118862414.
• Roberts, K. H. (1990). Some characteristics of one type of high reliability organization. Organization Science 1(2): 160–176. doi:10.1287/orsc.1.2.160.
• Edmondson, A. C. (2018). The Fearless Organization: Creating Psychological Safety in the Workplace for Learning, Innovation, and Growth. Wiley. ISBN 978-1119477242.

• Salas, E., Wilson, K. A., Burke, C. S. & Wightman, D. C. (2006). Does Crew Resource Management training work? An update, an extension, and some critical needs. Human Factors 48(2): 392–412. doi:10.1518/001872006777724444.
• Klein, G. (1998). Sources of Power: How People Make Decisions. MIT Press. ISBN 978-0262112277.
• Kanki, B. G., Helmreich, R. L. & Anca, J. (eds.) (2010). Crew Resource Management (2nd ed.). Academic Press. ISBN 978-0123749468.

Implementation science and organizational learning

• Fixsen, D. L., Naoom, S. F., Blasé, K. A., Friedman, R. M. & Wallace, F. (2005). Implementation Research: A Synthesis of the Literature. NIRN.
• Aarons, G. A., Hurlburt, M. & Horwitz, S. M. (2011). Advancing a conceptual model of evidence-based practice implementation in public service sectors. Administration and Policy in Mental Health 38: 4–23. doi:10.1007/s10488-010-0327-7.
• Damschroder, L. J. et al. (2009). Fostering implementation of health services research findings into practice: a consolidated framework. Implementation Science 4: 50. doi:10.1186/1748-5908-4-50.
• Argyris, C. & Schön, D. A. (1996). Organizational Learning II: Theory, Method, and Practice. Addison-Wesley. ISBN 978-0201629835.
• Senge, P. M. (2006). The Fifth Discipline: The Art and Practice of the Learning Organization (rev. ed.). Doubleday. ISBN 978-0385517256.
• Vaughan, D. (1996). The Challenger Launch Decision: Risky Technology, Culture, and Deviance at NASA. University of Chicago Press. ISBN 978-0226851761.

Ethics, governance, and equity in sociotechnical systems

• Eubanks, V. (2018). Automating Inequality: How High-Tech Tools Profile, Police, and Punish the Poor. St. Martin’s. ISBN 978-1250074317.
• Friedman, B. & Nissenbaum, H. (1996). Bias in computer systems. ACM Transactions on Information Systems 14(3): 330–347. doi:10.1145/230538.230561.
• O’Neil, C. (2016). Weapons of Math Destruction. Crown. ISBN 978-0553418811.
• Selbst, A. D. et al. (2019). Fairness and abstraction in sociotechnical systems. In FAT*’19. doi:10.1145/3287560.3287598.
• Citron, D. K. (2008). Technological due process. Washington University Law Review 85: 1249–1313. SSRN.
• Selwyn, N. (2014). Distrusting Educational Technology. Routledge. ISBN 978-1138021068.

Human-AI teaming and automation

• Parasuraman, R. & Manzey, D. H. (2010). Complacency and bias in human use of automation: an attentional integration. Human Factors 52(3): 381–410. doi:10.1177/0018720810376055.
• Bainbridge, L. (1983). Ironies of automation. Automatica 19(6): 775–779. doi:10.1016/0005-1098(83)90046-8.
• Sarter, N. B. & Woods, D. D. (1995). How in the world did we ever get into that mode? Mode error and awareness in supervisory control. Human Factors 37(1): 5–19. doi:10.1518/001872095779049516.
• Cummings, M. L. (2017). Automation bias in intelligent time-critical decision support systems. In Decision Making in Aviation.
• Casner, S. M. & Hutchins, E. (2019). What do we tell the drivers? Toward minimum driver-training standards for partially-automated cars. Journal of Cognitive Engineering and Decision Making 13(2): 55–66. doi:10.1177/1555343419830901.

Patient safety and clinical improvement

• Kohn, L. T., Corrigan, J. M. & Donaldson, M. S. (eds.) (1999). To Err Is Human: Building a Safer Health System. Institute of Medicine. doi:10.17226/9728.
• Gawande, A. (2009). The Checklist Manifesto: How to Get Things Right. Metropolitan. ISBN 978-0805091748.
• Pronovost, P. & Vohr, E. (2010). Safe Patients, Smart Hospitals. Hudson Street Press. ISBN 978-1594630644.
• Wachter, R. (2015). The Digital Doctor: Hope, Hype, and Harm at the Dawn of Medicine’s Computer Age. McGraw-Hill. ISBN 978-0071849463.
• Bates, D. W., Levine, D. M., Salmasian, H. et al. (2023). The safety of inpatient health care. New England Journal of Medicine 388(2): 142–153. doi:10.1056/NEJMsa2206117.

Each case spread in the body of the book includes its own primary sources. Sources cited only within case spreads are not repeated here; this list collects the works the discipline as a whole draws on, together with the references used in the introductory essay.

ABOUT THE EDITORS

The editors.

The casebook's central argument — that capability engineering is the applied form of the educational sciences, taught from the School of Education and informed by operational practice — is embodied in the two editors. Each has spent a career working in one half of that pairing and a substantial portion of it in the other. Each contributes to the five pillars on which LENS rests: Mission Literacy · JHU Ecosystem · Intersectional Expertise · Capability Focus · Flywheel Iteration.

William Gray-Roncal, Ph.D.

APPLIED RESEARCH · LEARNING ENGINEERING · STEM WORKFORCE

William Gray-Roncal, Ph.D., is Principal Research Engineer at the Johns Hopkins University Applied Physics Laboratory, where he applies systems engineering methods to challenges spanning defense, space, biomedical, and learning systems. His research — sponsored by DARPA, IARPA, NIH, and DoD — bridges computational neuroscience, artificial intelligence, and large-scale data infrastructure. He developed the College Prep and CIRCUIT workforce-development models, through which he has mentored nearly 500 students, and is currently pursuing research on quantitative measures of human cognitive performance in operational contexts. He leads the LENS (Learning Engineering for Next-Generation Systems) specialization in the Johns Hopkins Master of Education in Learning Design and Technology program. Gray-Roncal holds a Ph.D. in Computer Science from Johns Hopkins, an M.S. in Electrical Engineering from USC, and a B.S. in Electrical Engineering from Vanderbilt.

James Diamond, Ph.D.

LEARNING SCIENCES · EDUCATIONAL DESIGN · EVALUATION

James Diamond, Ph.D., is Assistant Professor in the Johns Hopkins University School of Education and Program Director of the Master of Education in Learning Design and Technology, where he applies design-based research and learning-sciences methods to the design and evaluation of technology-mediated learning environments. His research —

supported by the National Science Foundation, the Institute of Education Sciences, the Bill & Melinda Gates Foundation, the MacArthur Foundation's HASTAC Digital Media and Learning Initiative, the Robin Hood Learning + Technology Fund, the Wellcome Trust, and the UK Economic and Social Research Council — bridges digital games and simulations for learning, digital badges and micro-credentialing, classroom technology integration, and disciplinary literacy in history, social studies, civics, and STEM. Before joining Johns Hopkins, he was Senior Research Associate at the Education Development Center's Center for Children and Technology in New York City, where he led projects including Zoom In and the Who Built America? Teacher Mastery Badge System, contributed to Mission US and Possible Worlds, and taught at New York University as an adjunct. He is a principal faculty member of the LENS (Learning Engineering for Next-Generation Systems) specialization in the LDT program. Diamond holds a Ph.D. in Educational Communication and Technology from New York University, an Ed.M. in Educational Technology from Boston University, and a B.A. in History from Boston University.

These biographies reference careers documented in publications, program reports, faculty pages, and the curricular materials of the LDT program and the LENS specialization. Corrections and refinements from the editors are welcomed in subsequent printings.